GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

INSTRUCTIONS
GEK- 40635

ELECTRIC CONTROL EQUIPMENT
FOR
MECHANICAL TENSION
PAPER MILL WINDER DRIVE

SUPPORTING DRAWINGS FOR USE WITH
DRIVE SYSTEM ELEMENTARY DIAGRAM

GENERAL ELECTRIC

INSTRUCTIONS GEK-40635

PAPER MILL MECHANICAL WINDER

INDEX

DESCRIPTION	ELEMENTARY DIAGRAM	
Tab 1 Card Elementaries		
Operational Amplifier - 1 Stage	IC3600AOAA1	3.0 - FL.
Operational Amplifier - 2 Stage	IC3600AOAA2	3.0 - FL.
Operational Amplifier	IC3600AOAC1	3.0 - FL.
Voltage Sensitive Relay	IC3600AVSA	3.0 - FL.
Reed Relay	IC3600KRSS	3.0 - FL.
Reed Relay	IC3600KRST	3.0 - FL.
Firing Circuit	IC3600TFCA1	3.0 - FL.
Firing Circuit	IC3600TFCU1	3.0 - FL.
Power Supply	IC3600TPSA1	3.0 - FL.
Power Supply	IC3600TPSB1	3.0 - FL.
Power Supply	IC3600TPSC1	3.0 - FL.
Power Supply	IC3600TPSJ1	3.0 - FL.
Gate and Filter Card - 230V	68A993117	3.0 - FL.
Gate and Filter Card - 460V	68A993124	3.0 - FL.
Firing Circuit - 460V-NRX	68A993159	3.0 - FL.
Diode Filter	68A993188	3.0 - FL.
Diode and Load Filter - CPE	68A993189	3.0 - FL.
40A-180V SCR Exciter	68A993671	1 - A
CPE - 22A-250V	68A993690	1 - 4
CPE - 50A-250V	68A993691	1 - 4
CPE - 100A-250V	68A993692	1 - 4
CPE - 150A-250V	68A993693	1 - 4
CPE - 300A-250V	68A993694	1 - 4

INSTRUCTIONS GEK-40635

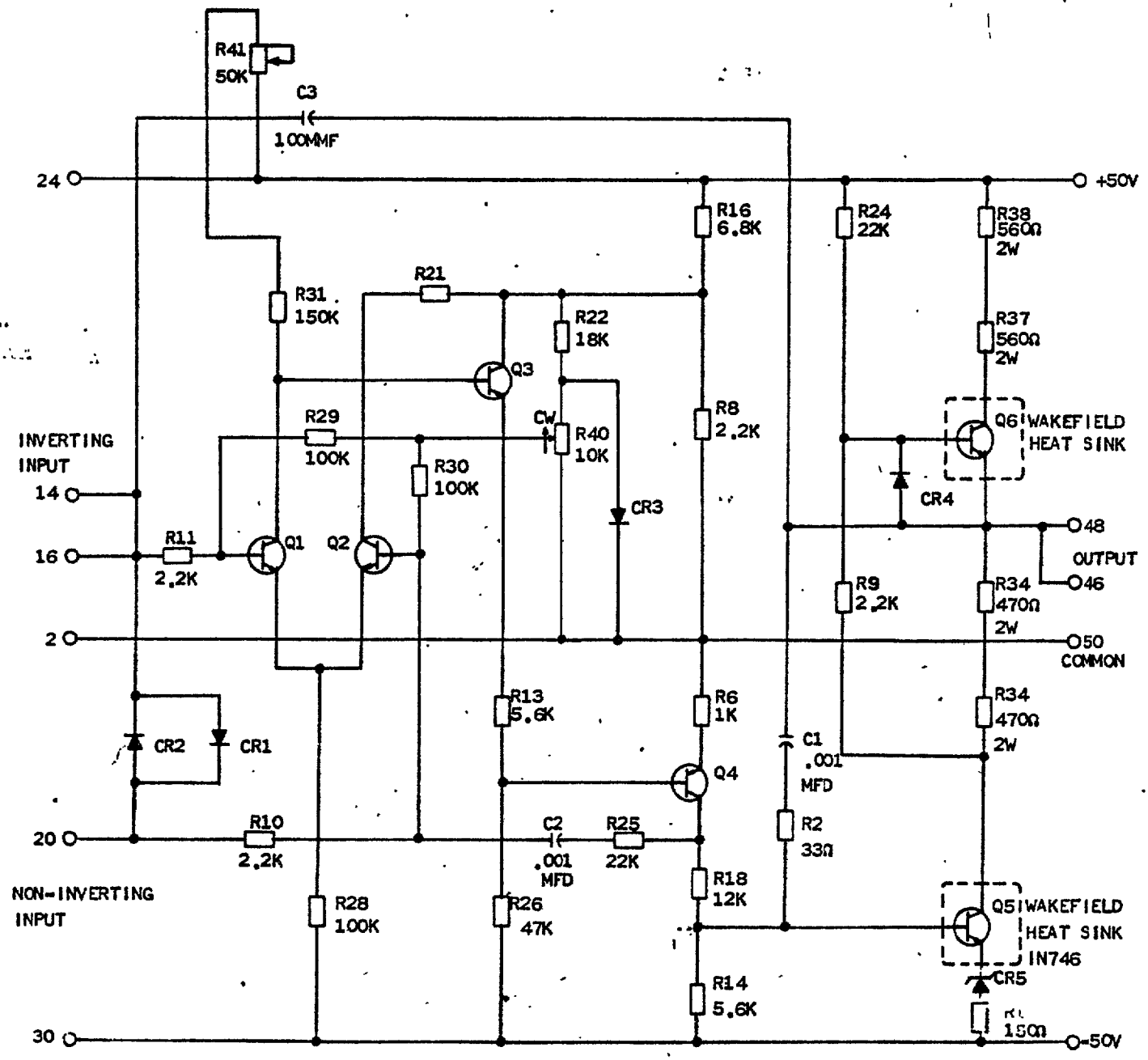
PAPER MILL MECHANICAL WINDER

INDEX

DESCRIPTION	ELEMENTARY DIAGRAM	
Tab 2 Modules		
Rate with Rounding	187B6492	3.0 - FL.
Shunt Isolator	68A993746	3.0 - FL.
Armature Current Limit	171B5875	3.0 - FL.
±67V Power Supply	IC3601A133	3.0 - FL.
Speed Regulator	171B5876	3.0 - FL.
Voltage Regulator	177B6773	3.0 - FL.
Zero Speed Relay	195B3479	3.0 - FL.
Tab 3 Field Exciter		
Field Exciter	GEK-8074	
Unregulated CPE 30A	IC3602A125	3.0 - FL.
Modification for a 22 A CPE	219A3404	0 - 4
Modification for a 50 A CPE	219A3405	0 - 4
Modification for a 100 A CPE	218A6617	0 - 4
Unregulated CPE 100A	IC3602A124	3.0 - FL.
Tab 4 Drive Instructions		
	244A9459	1-30

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

NOTE: POTENTIOMETER R41 IS NOT PROVIDED BY REV. 4 CARDS. ON REV. 4 A SELECTED RESISTOR (R27) IS USED.



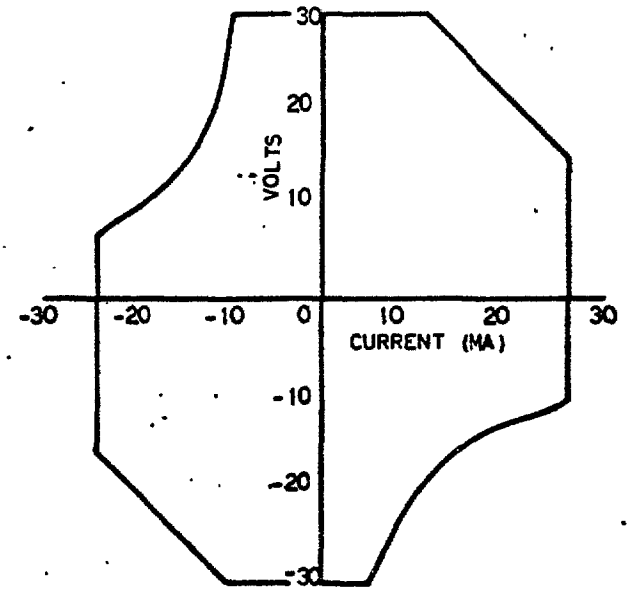
Q1, Q2 - MATCHED PAIR 68A7310G1
Q3 - 68A7373P1
Q4 - 68A7385P1
Q5, Q6 - 68A7261P1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1 BU47037 REH 6-18-64.	REV 2 BD 1/9/73	REV 3	PRINTS TO DL1, DI8	APPROVALS	ELEMENTARY DIAGRAM 3 STAGE OPERATIONAL AMPLIFIER
REV 4 MAC 11-14-76	REISSUED March 18, 71. 2520			FIRST MADE FOR STD. LINE	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 3 3-18-72 REDRAWN ACW	MADE BY S. BEVERLY			I.C. NO.	I C 3 6 0 0 A 0 A A 1 CONT. ON SH. 3.1 SH NO 3.0

CIRCUIT CHARACTERISTICS

1.	POWER INPUT	+50V, 15MA + LOAD CURRENT PER AMPLIFIER -50V, 10MA + LOAD CURRENT PER AMPLIFIER
2.	SIGNAL INPUT	±.67MV FOR FULL OUTPUT.
3.	INPUT IMPEDANCE	25K MIN., 40K TYP.
4.	DRIFT AT INPUT	NANOAMPS | MICROVOLTS
	A. FOR TEMP. CHANGES	-10 ±5/°C | ±50/°C
	B. 24 HOURS CONSTANT TEMP.	±25 (TYPICAL) | ±30 (TYPICAL)
	C. WARMUP (WARMUP TIME CONSTANT 15 MIN)	±25 (TYPICAL) | ± 100 (TYPICAL)
5.	LINEARITY (100K INPUT AND FEEDBACK)	BETTER THAN .015%
6.	GAIN (100K INPUT AND FEEDBACK)	
	A. ACCURACY	.0125% AT DC (DUE TO AMPLIFIER ONLY)
	B. STABILITY	.01% AT DC (DUE TO AMPLIFIER ONLY)
7.	FREQUENCY RESPONSE (100K INPUT AND FEEDBACK)	3 DECIBELS DOWN AT 14KC
8.	SETTLING TIME (15V STEP INPUT) (100K INPUT AND FEEDBACK)	OUTPUT SETTLES WITHIN .1% OF FINAL VALUE IN 100 MICRO-SECONDS.
9.	OVERLOAD RECOVERY (10K INPUT, 100K FEEDBACK)	OUTPUT RECOVERS FROM 50V INPUT IN 1 MILLISECOND. (5 MILLISECONDS FOR 1 MEG FEEDBACK)
10.	NOISE AT OUTPUT (100K INPUT AND FEEDBACK	LESS THAN 10 MV RMS
11.	OUTPUT IMPEDANCE	1.5K MAX WITHOUT FEEDBACK. .5 OHMS MAX. DC TO 60 CPS WITH 100K INPUT AND FEEDBACK
12.	LOAD - OUTPUT VOLTAGE AND CURRENT SHOULD BE MAINTAINED WITHIN AREA SHOWN AT RIGHT TO AVOID SATURATION OR OVERLOAD. (RESISTIVE OR INDUCTIVE LOAD) MAX. CAPACITIVE LOAD 1000PF.	



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

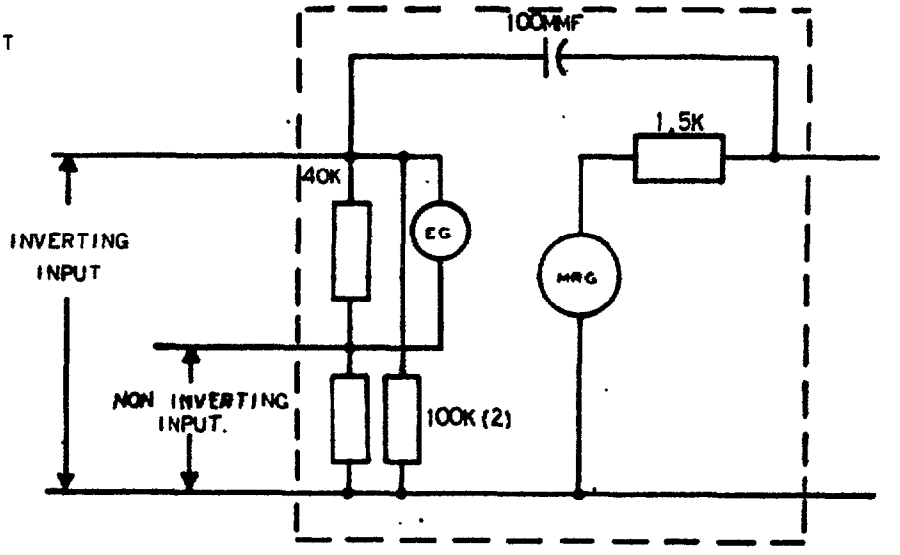
REV. 1 BD 1/9/73	REV. 2	PRINTS TO DL1	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM 3 STAGE OPERATIONAL AMPLIFIER
DRAWN February 25, 1969		DL8	FIRST MADE FOR MFG. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	1C3600A0AA1 / DRAWING NO. 3,2 / SH. NO. 3,1
MADE BY S. BEVERLY			LC NO.		

13. EFFECT OF POWER SUPPLY VARIATIONS

$$\Delta Vo = [1 + |BUS| - 1 - |BUS|] (.035)(1+G)$$

$$G = \frac{\text{FEEDBACK RESISTANCE}}{\text{PARALLEL RESISTANCE OF INPUTS}}$$

14. APPROXIMATE EQUIVALENT CIRCUIT



15. AMPLITUDE-FREQUENCY LIMIT

$$M = \frac{-40{,}000}{(1 + .009P)(1 + 16 \times 10^{-8}P)}$$

SINE WAVE OUTPUT IS DISTORTED IF INPUT SINE WAVE PEAK EXCEEDS

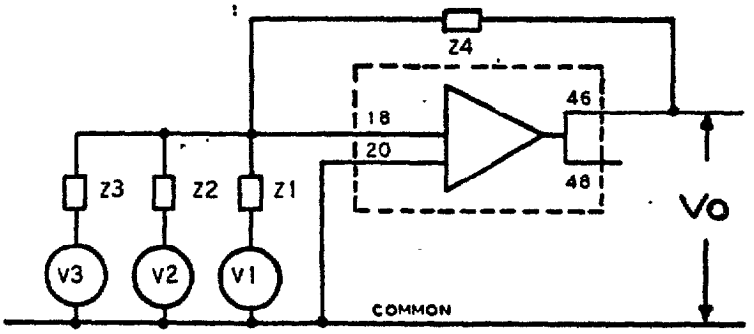
$$\frac{75000}{G(2500+F)}$$ (G=GAIN, F= FREQUENCY)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 BD 1/9/73	REV. 2	PRINTS TO DL1	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM 3 STAGE OPERATIONAL AMPLIFIER
ISSUED February 25 1964		DL8	FIRST MADE FOR DEPT. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	
MADE BY S BEVERLY			LC NO.	IC3600A0AA1 CONT. DOC NO. 3.3	SH. NO. 3.2

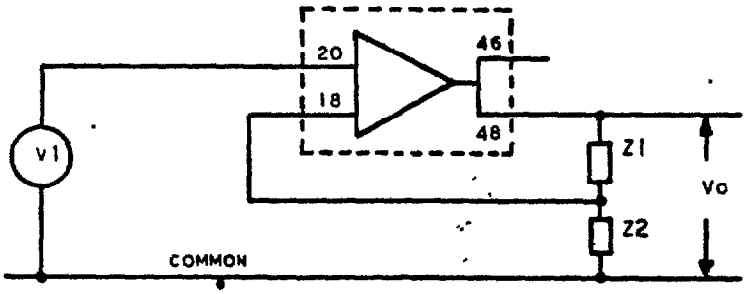
APPLICATION NOTES

1. TRANSFER FUNCTION (INVERTING CONNECTION)



$$V_o = \frac{-V_1 Z_4}{Z_1} - V_2 \frac{Z_4}{Z_2} - V_3 \frac{Z_4}{Z_3}$$

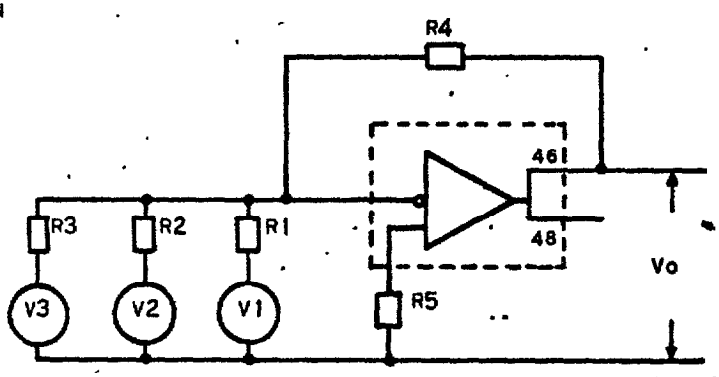
2. TRANSFER FUNCTION (NON-INVERTING CONNECTION)



$$V_o = V_1 \frac{Z_1 + Z_2}{Z_2}$$

$|V_o| = |$ VOLTS MAX.

3. DRIFT CALCULATION



R_{IN} = PARALLEL RESISTANCE OF R1, R2, AND R3

ΔI_{IN} = PREDICTED INPUT CURRENT DRIFT

ΔV_{IN} = PREDICTED INPUT VOLTAGE DRIFT

$\Delta V_C = I_{IN} R4 + \Delta V_{IN} \left(1 + \frac{R4}{R_{IN}}\right)$ (FOR R5 = 0).

INPUT CURRENT DRIFT AT THE INVERTING AND NON-INVERTING INPUTS WILL TEND TO MATCH. IF R5 IS MADE EQUAL TO R_{IN} THE EFFECT OF CURRENT DRIFT IS THEORETICALLY CANCELLED. ACTUALLY THE CURRENT DRIFTS WILL NOT EXACTLY MATCH BUT THE DRIFT DUE TO INPUT CURRENT SHOULD BE REDUCED BY AT LEAST 3 TO 1 IF RESISTOR R6 IS USED.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 BD 1/9/73	REV. 2	PRINTS TO Dil	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM 3 STAGE OPERATIONAL AMPLIFIER
DATE February 25 1969		DL8	FIRST MADE FOR DEPT. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	1C3600A0AA1
MADE AT S. BEVERLY			LC. NO.		CONT. ON SH. 3.4

SH. NO. 3.3

4. EFFECT OF IR DROPS IN COMMON BUS

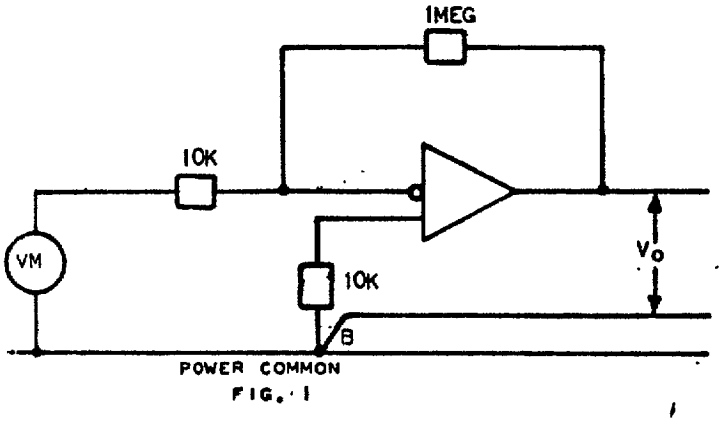


FIG. 1

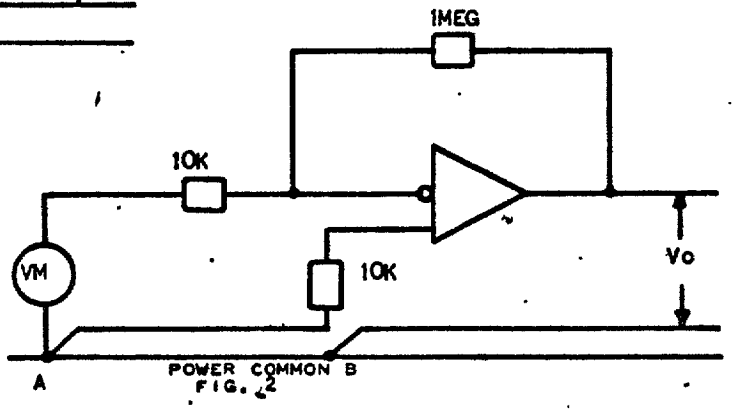
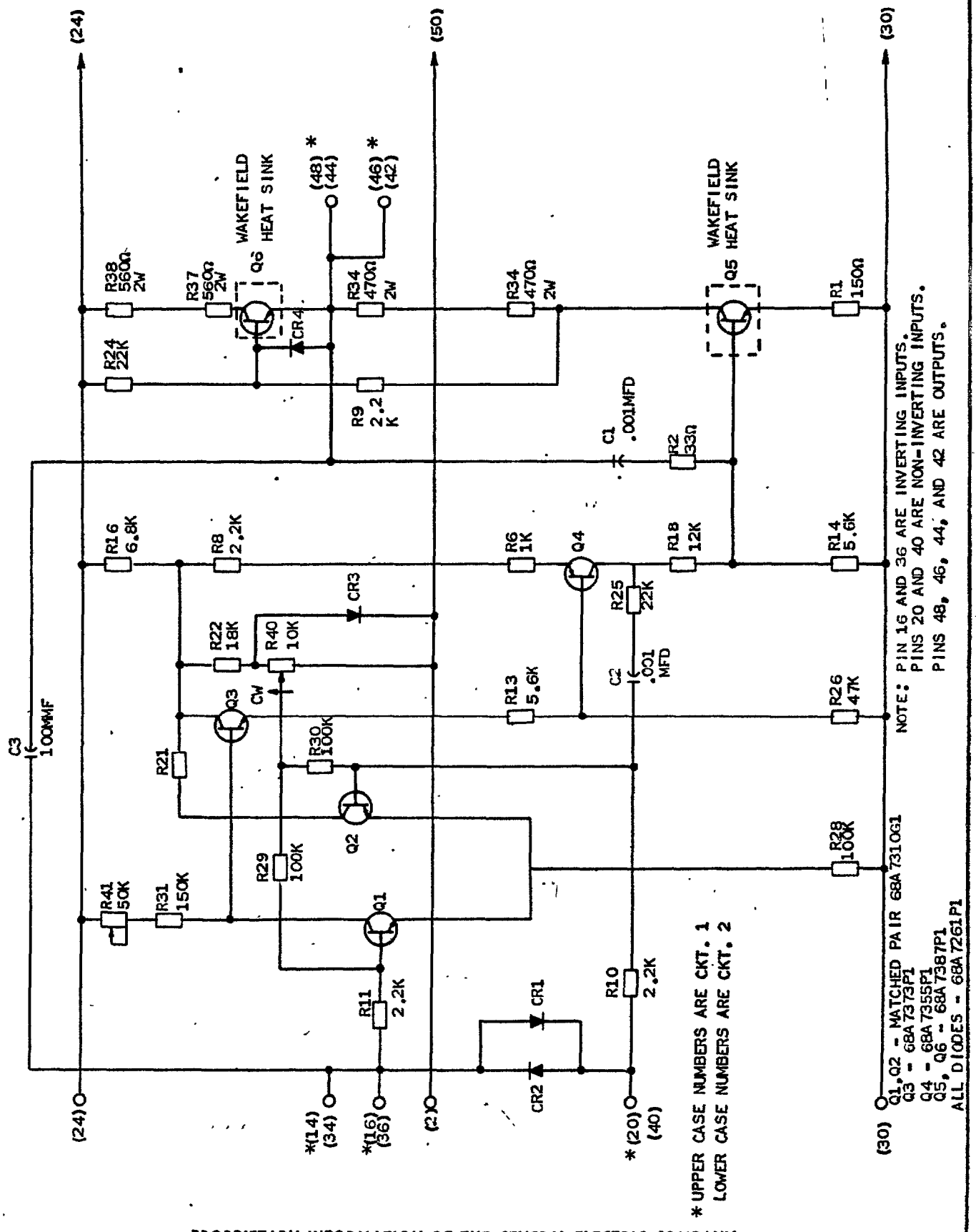


FIG. 2

THE POWER COMMON CARRIES RELATIVELY HIGH AND VARYING CURRENTS SO A VARYING VOLTAGE DROP WILL NORMALLY EXIST BETWEEN POINTS A AND B. IF THE NON-INVERTING INPUT OR SIGNAL COMMON LEAD IS CONNECTED TO POINT B AS IN FIG. 1, THE BUS DROP WILL BE A SIGNAL INPUT TO THE AMPLIFIER. IF THE BUS DROP CHANGED BY 10 MILLIVOLTS THE OUTPUT WOULD CHANGE .01 X 100 OR 1 VOLT. IN FIG. 2, THE BUS DROP DOES NOT APPEAR AS AN INPUT SIGNAL SO THE AMPLIFIER OUTPUT WOULD APPEAR TO CHANGE AS AN INPUT SIGNAL SO THE AMPLIFIER OUTPUT WOULD APPEAR TO CHANGE BY 10 MILLIVOLTS WITH RESPECT TO POINT B. WITH RESPECT TO POINT A THERE WOULD BE NO OUTPUT ERROR DUE TO BUS DROP.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 BD 1/9/73	REV. 2	PRINTS TO DL1	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM 3 STAGE OPERATIONAL AMPLIFIER
DRAWN February 25 1969		DL8	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600A0AA1
MADE BY S. BEVERLY			L.C. NO.		CONST. CODE SEL. FL. SHT. NO. 3.4

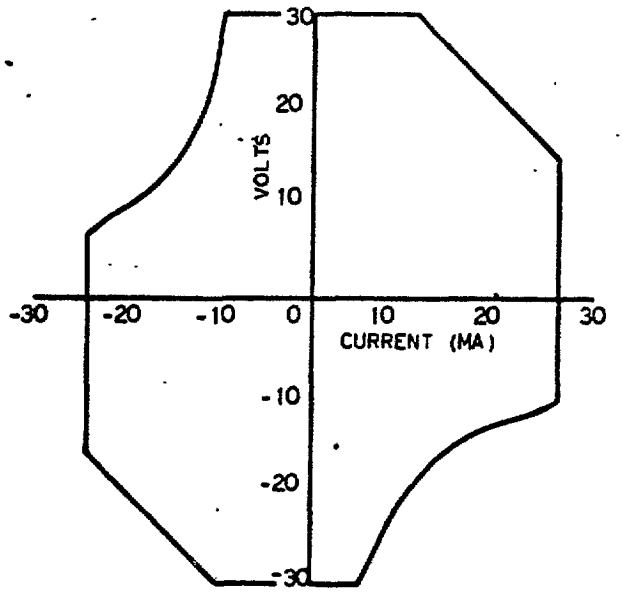


PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1 6-18-64 REH BU47037	REV 3 BD 1/9/73	REV 5	PRINTS TO DL1 DL6	APPROVALS W.B. JARVINEN	ELEMENTARY DIAGRAM DUAL OPERATIONAL AMP.
REV 2 6-16-72 A. WRIGHT	ISSUED 5-5-72			FIRST MADE FOR STD. LINE	
REV 3 5-5-72 REH Per ARE only	MADE BY S. BEVERLY			I.C. NO.	I C 3 6 0 0 A 0 A A 2
GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.					CONT. ON SH 3.1 SH NO 3.0

CIRCUIT CHARACTERISTICS

1.	POWER INPUT	+50V, 15MA + LOAD CURRENT PER AMPLIFIER -50V, 10MA + LOAD CURRENT PER AMPLIFIER
2.	SIGNAL INPUT	±.67MV FOR FULL OUTPUT.
3.	INPUT IMPEDANCE	25K MIN., 40K TYP.
4.	DRIFT AT INPUT	NANOAMPS | MICROVOLTS
	A. FOR TEMP. CHANGES	-10 ±5/°C | ±50/°C
	B. 24 HOURS CONSTANT TEMP.	±25 (TYPICAL) | ±30 (TYPICAL)
	C. WARMUP (WARMUP TIME CONSTANT 15 MIN)	±25 (TYPICAL) | ± 100 (TYPICAL)
5.	LINEARITY (100K INPUT AND FEEDBACK)	BETTER THAN .015%
6.	GAIN (100K INPUT AND FEEDBACK)	
	A. ACCURACY	.0125% AT DC (DUE TO AMPLIFIER ONLY)
	B. STABILITY	.01% AT DC (DUE TO AMPLIFIER ONLY)
7.	FREQUENCY RESPONSE (100K INPUT AND FEEDBACK)	3 DECIBELS DOWN AT 14KC
8.	SETTLING TIME (15V STEP INPUT) (100K INPUT AND FEEDBACK)	OUTPUT SETTLES WITHIN .1% OF FINAL VALUE IN 100 MICRO-SECONDS.
9.	OVERLOAD RECOVERY (50K INPUT, 100K FEEDBACK)	OUTPUT RECOVERS FROM 50V INPUT IN 1 MILLISECOND. (5 MILLISECONDS FOR 1 MEG FEEDBACK)
10.	NOISE AT OUTPUT (100K INPUT AND FEEDBACK	LESS THAN 10 MV RMS
11.	OUTPUT IMPEDANCE	1.5K MAX WITHOUT FEEDBACK. .5 OHMS MAX. DC TO 60 CPS WITH 100K INPUT AND FEEDBACK
12.	LOAD - OUTPUT VOLTAGE AND CURRENT SHOULD BE MAINTAINED WITHIN AREA SHOWN AT RIGHT TO AVOID SATURATION OR OVERLOAD. (RESISTIVE OR INDUCTIVE LOAD) MAX. CAPACITIVE LOAD 1000PF.	



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

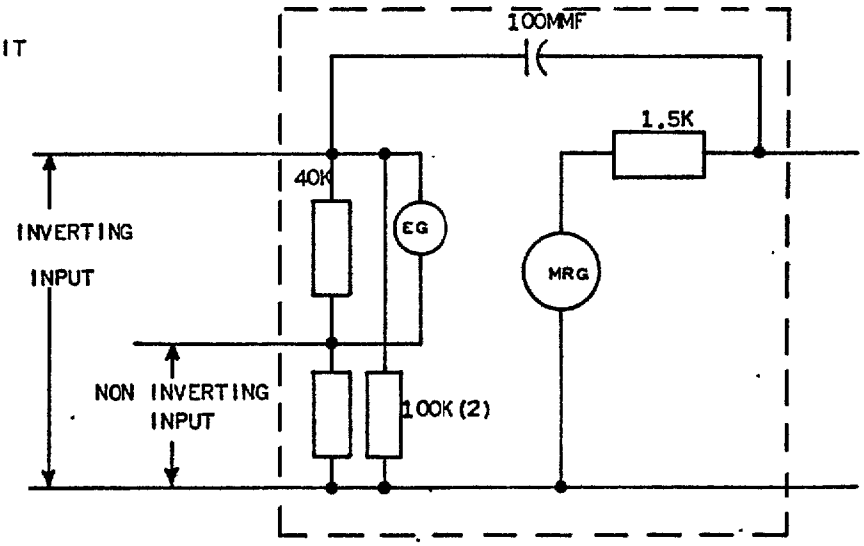
REV. 1 BD 1/9/73	DL1	PRINTS TO DL1	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM DUAL OPERATIONAL AMPLIFIER
DATE February 25, 1969		DL8	FIRST ASSY FOR SER. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	1C3600A0AA2 3.2
DRAWN BY S. BEVERLY					DR. NO. 3.1

13. EFFECT OF POWER SUPPLY VARIATIONS

$$\Delta Vo = [1 + BUS1 - 1 - BUS\ 1]\ (.035)\ (1+G)$$

$$G = \frac{\text{FEEDBACK RESISTANCE}}{\text{PARALLEL RESISTANCE OF INPUTS}}$$

14. APPROXIMATE EQUIVALENT CIRCUIT



15. AMPLITUDE-FREQUENCY LIMIT

$$M = \frac{-40{,}000}{(1 + .009P)\ (1 + 16 \times 10^{-8} P)}$$

SINE WAVE OUTPUT IS DISTORTED IF INPUT SINE WAVE PEAK EXCEEDS

$$\frac{75000}{G\ (2500+F)}$$

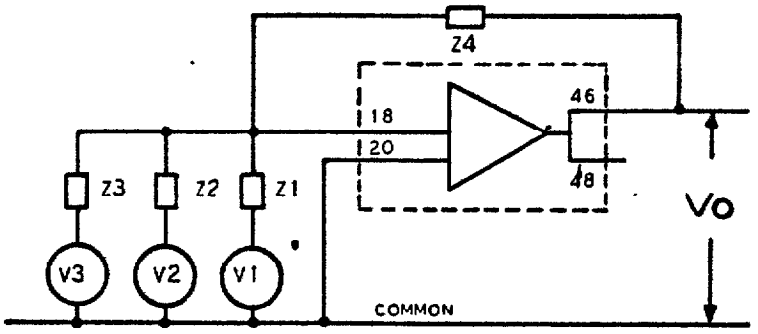
(G=GAIN, F=FREQUENCY)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1 @ 2770 REDRAWN, ORIGINAL REV. LTR. ACW	REV. 4	REV. 5	PRINTS TO DL1, BL8	APPROVALS	ELEMENTARY DIAGRAM DUAL OPERATIONAL AMPLIFIER
REV. 2 BD RE- 1/9/73	ISSUED AUGUST 27, 1970		FIRST MADE FOR STAN. LINE	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600A0AA2
REV 3	MADE BY S. BEVERLY		I.C. NO.		CONT ON SH 3.3 SH. NO. 3.2

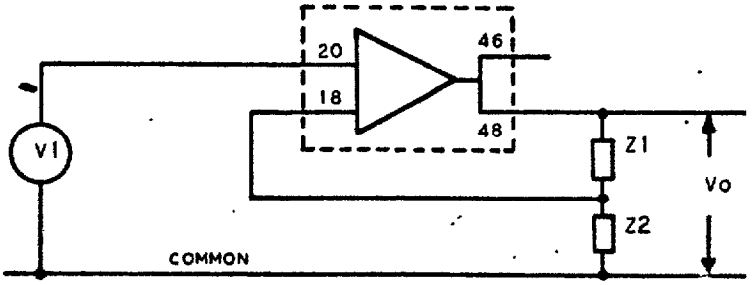
APPLICATION NOTES

1. TRANSFER FUNCTION (INVERTING CONNECTION)



$$V_o = \frac{-V1\ Z4}{Z1} - V2\ \frac{Z4}{Z2} - V3\ \frac{Z4}{Z3}$$

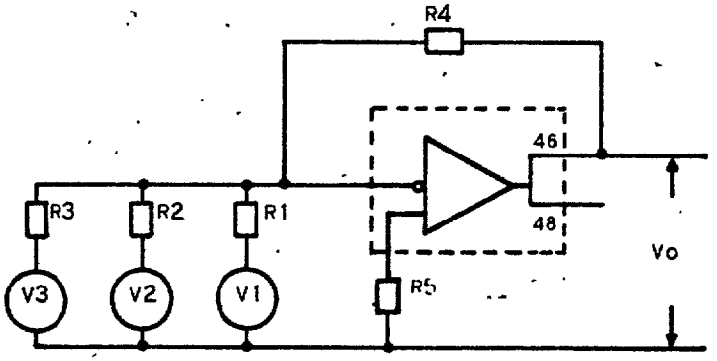
2. TRANSFER FUNCTION (NON-INVERTING CONNECTION)



$$V_o = V1\ \frac{Z1 + Z2}{Z2}$$

$|V_o|$ = | VOLTS MAX.

3. DRIFT CALCULATION



R_{IN} = PARALLEL RESISTANCE OF R1, R2, AND R3

ΔI_{IN} = PREDICTED INPUT CURRENT DRIFT

ΔV_{IN} = PREDICTED INPUT VOLTAGE DRIFT

$\Delta V_C = _\ I_{IN}\ R4 + \Delta V_{IN}\ (1 + \frac{R4}{R_{IN}})$ (FOR R5 = 0)

INPUT CURRENT DRIFT AT THE INVERTING AND NON-INVERTING INPUTS WILL TEND TO MATCH. IF R5 IS MADE EQUAL TO R_{IN} THE EFFECT OF CURRENT DRIFT IS THEORETICALLY CANCELLED. ACTUALLY THE CURRENT DRIFTS WILL NOT EXACTLY MATCH BUT THE DRIFT DUE TO INPUT CURRENT SHOULD BE REDUCED BY AT LEAST 3 TO 1 IF RESISTOR R5 IS USED.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 BD 1/9/73		PRINTS TO DL1	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DUAL OPERATIONAL AMPLIFIER
ISSUED February 25, 1969		DL8	FIRST MADE FOR BEQ. STANDARD LINE		C3600ACAA2 SH. NO. 3.3
MADE BY S. BEVERLY			TC NO		CONT. ON SH. 3.4

4. EFFECT OF IR DROPS IN COMMON BUS

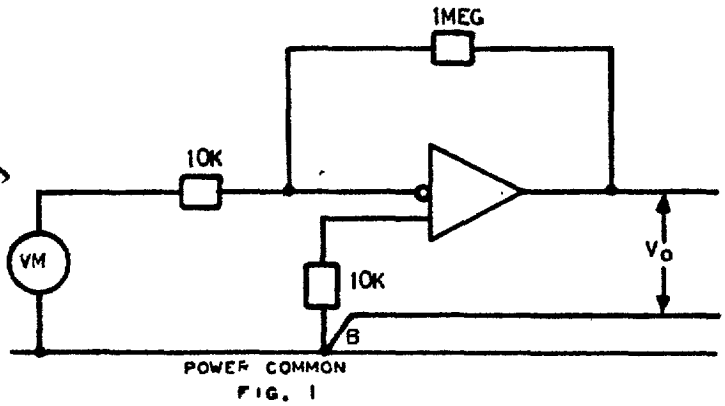


FIG. 1

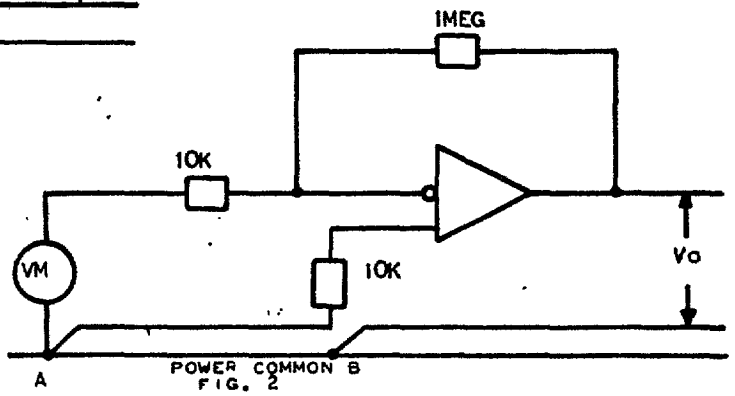
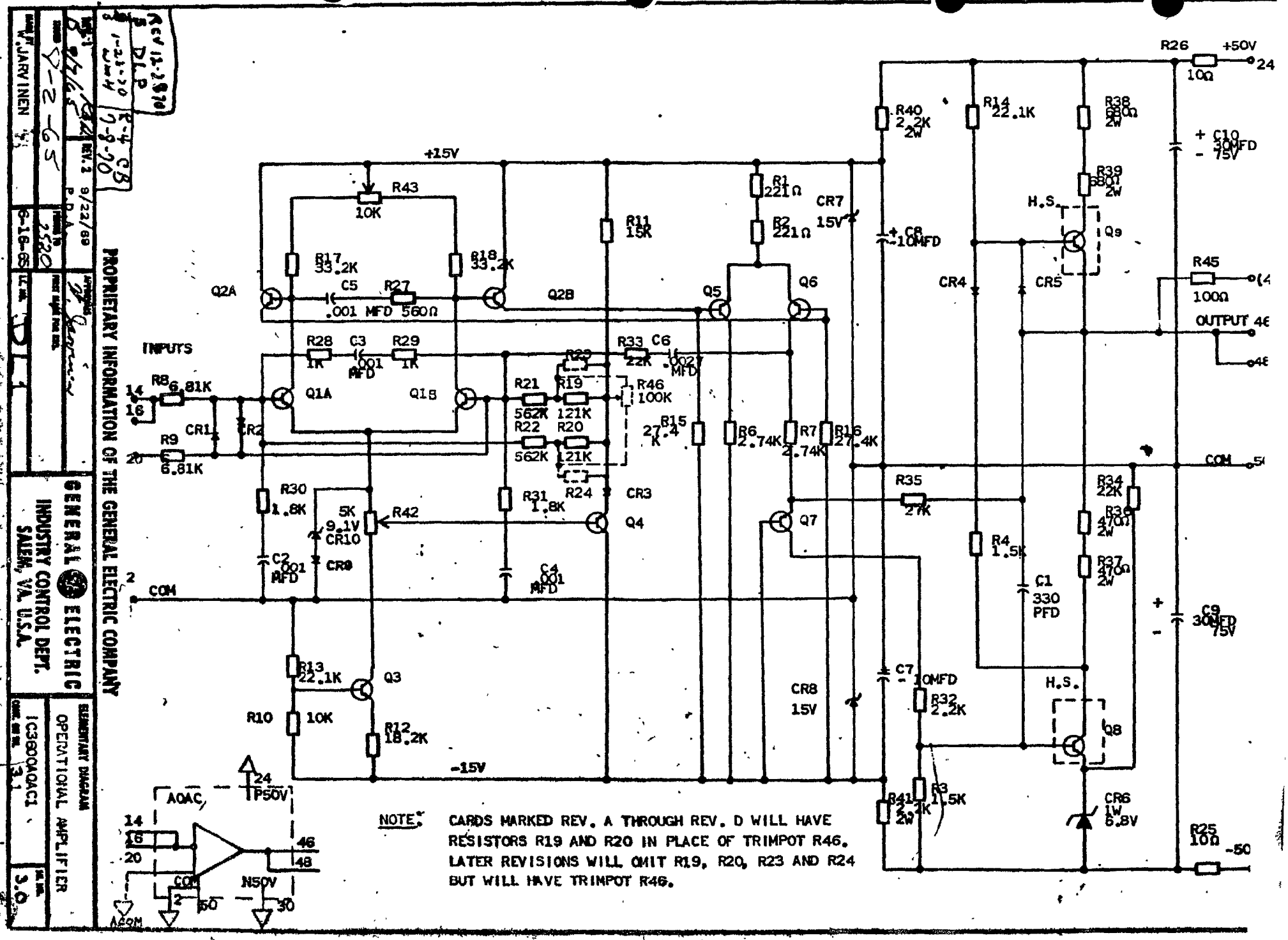


FIG. 2

THE POWER COMMON CARRIES RELATIVELY HIGH AND VARYING CURRENTS SO A VARYING VOLTAGE DROP WILL NORMALLY EXIST BETWEEN POINTS A AND B. IF THE NON-INVERTING INPUT OR SIGNAL COMMON LEAD IS CONNECTED TO POINT B AS IN FIG. 1, THE BUS DROP WILL BE A SIGNAL INPUT TO THE AMPLIFIER. IF THE BUS DROP CHANGED BY 10 MILLIVOLTS THE OUTPUT WOULD CHANGE .01 X 100 OR 1 VOLT. IN FIG. 2, THE BUS DROP DOES NOT APPEAR AS AN INPUT SIGNAL SO THE AMPLIFIER OUTPUT WOULD APPEAR TO CHANGE AS AN INPUT SIGNAL SO THE AMPLIFIER OUTPUT WOULD APPEAR TO CHANGE BY 10 MILLIVOLTS WITH RESPECT TO POINT B. WITH RESPECT TO POINT A THERE WOULD BE NO OUTPUT ERROR DUE TO BUS DROP.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 BD 1/9/73	REV. 2	PRINTS TO DL1	APPROVED W. C. Jamieson	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM DUAL OPERATIONAL AMPLIFIER
DATED February 25, 1964		DL8	FIRST MADE FOR REQ. STANDARD LINE.		IC3600A0AA2
MADE BY BEVERLY			TC NO		CONT ON SH FL.
					SH. NO. 3.4


PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
OPERATIONAL AMPLIFIER
IC3800A0AC1
3.0
W. JARVINEN
6-16-65
REV. 1 9/22/69
P.O.A.
2520
+15V
-15V
+50V
-50V
COM
INPUTS
OUTPUT
R43 10K
R11 15K
R1 221Ω
R2 221Ω
R17 33.2K
R18 33.2K
C5 .001 MFD
R27 560Ω
Q2A
Q2B
Q5
Q6
CR7 15V
R26 10Ω
R40 2.2K 2W
R14 22.1K
R38 680Ω 2W
R39 680Ω 2W
C10 30MFD 75V
C8 10MFD
H.S.
Q9
CR4
CR5
R45 100Ω
R28 1K
C3 .001 MFD
R29 1K
R33 22K
C6 .0027 MFD
R8 6.81K
R9 6.81K
CR1
CR2
Q1A
Q1B
R21 562K
R19 121K
R22 562K
R20 121K
R46 100K
R15 27.4K
R6 2.74K
R7 2.74K
R16 27.4K
R30 1.8K
5K
R42
9.1V
CR10
CR9
C2 .001 MFD
R31 1.8K
R24
CR3
Q4
C4 .001 MFD
R35 27K
Q7
R34 22K
R36 470Ω 2W
R4 1.5K
R37 470Ω 2W
C1 330 PFD
C9 30MFD 75V
R13 22.1K
Q3
R10 10K
R12 18.2K
C7 10MFD
R32 2.2K
CR8 15V
H.S.
Q8
R3 1.5K
R41 2.2K 2W
CR6 1W 6.8V
R25 10Ω
NOTE: CARDS MARKED REV. A THROUGH REV. D WILL HAVE RESISTORS R19 AND R20 IN PLACE OF TRIMPOT R46. LATER REVISIONS WILL OMIT R19, R20, R23 AND R24 BUT WILL HAVE TRIMPOT R46.
A0AC
A0OM

AMPLIFIER CHARACTERISTICS

1.	POWER INPUT	+50V, 24MA + LOAD CURRENT -50V, 30MA + LOAD CURRENT	
2.	SIGNAL INPUT	±.66MV FOR FULL OUTPUT RANGE (DC) ± 6.6×10^{-9} AMPS FOR FULL OUTPUT RANGE (DC)	
3.	OUTPUT	±30V, ±10MA (SEE APPLICATION DATA)	
4.	FREQUENCY RESPONSE		
	A. 10K INPUT AND FEEDBACK	285KC TYPICAL	
	B. 100K INPUT AND FEEDBACK	65KC TYPICAL	
	C. 1MEG INPUT AND FEEDBACK	7.6KC TYPICAL	
	D. GAIN BANDWIDTH (OPEN LOOP)	GREATER THAN 800KC	
5.	TEMP. DRIFT	INPUT CURRENT	INPUT VOLTAGE
	A. 0°C TO 25°C	$-1.5 \pm 2 \times 10^{-9}$ AMPS/°C	$\pm 20 \times 10^{-6}$ VOLTS/°C
	B. 25°C TO 70°C	$\pm 2 \times 10^{-8}$ AMPS/°C	$\pm 20 \times 10^{-6}$ VOLTS/°C
	C. 24 HOUR DRIFT (25°C)	$\pm 25 \times 10^{-9}$ AMPS	$\pm 250 \times 10^{-6}$ VOLTS
6.	RECOVERY FROM 100V INPUT	RECOVERS TO 1% OF FINAL VALUE IN	
	A. 10K INPUT AND FEEDBACK	.06 MILLISEC. (TYPICAL)	
	B. 100K INPUT AND FEEDBACK	.10 MILLISEC. (TYPICAL)	
	C. 1MEG INPUT AND FEEDBACK	.40 MILLISEC. (TYPICAL)	
7.	RECOVERY FROM SHORTED OUTPUT	RECOVERS TO 1% OF FINAL VALUE IN	
	A. 10K INPUT AND FEEDBACK	.2 MILLISEC. (TYPICAL)	
	B. 100K INPUT AND FEEDBACK	.15 MILLISEC. (TYPICAL)	
	C. 1MEG INPUT AND FEEDBACK	.5 MILLISEC. (TYPICAL)	
8.	OUTPUT IMPEDANCE (OPEN LOOP)	TYPICALLY LESS THAN 10 OHMS FROM DC TO BEYOND 1 MEGACYCLE	
9.	MAX. CAPACITIVE LOAD	200PFD (SEE NOTE BELOW)	
10.	FREQ. LIMIT AT MAX. AMPLITUDE		
	A. RESISTIVE LOAD (3.3K)	70 KC (TYPICAL) (12V/MICROSEC. SLEW RATE)	
	B. 680 PFD LOAD	35 KC (TYPICAL) (6V/MICROSEC. SLEW RATE)	
	C. 100 OHM, .01 MFD SERIES LOAD	3.5KC (TYPICAL)	
11.	NOISE AT OUTPUT (100K INPUT AND FEEDBACK)	3 MILLIVOLTS DUE TO AMPLIFIER ALONE.	
12.	COMMON MODE INPUT	±8 VOLTS MAXIMUM	
13.	COMMON MODE REJECTION	CAN BE ADJUSTED FOR MORE THAN 120 DB AT DC. SEE APPLICATION NOTES FOR CIRCUITS TO BE USED.	

NOTE: IF THE AMPLIFIER OUTPUT IS CONNECTED TO A LONG LINE OR CAPACITIVE LOAD A RESISTOR BUFFER MAY BE NEEDED TO PREVENT OSCILLATIONS. A 100Ω RESISTOR IN SERIES WITH THE LOAD IS SUFFICIENT . CARDS MARKED REV. E OR LATER HAVE A 100Ω RESISTOR FROM OUTPUT TO TERM. 40.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 9/22/69 P.D.A.	REV. 2 12-28-70 DLP DLJ	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
ISSUED 8-2-65	PRINTS TO 1338	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600A0AC1	SHT. NO. 3.1
MADE BY J.SMITH	P6A 2570	I.C. NO.		CONT. ON SH. 3.2	

APPLICATION DATA

THEORY OF OPERATION

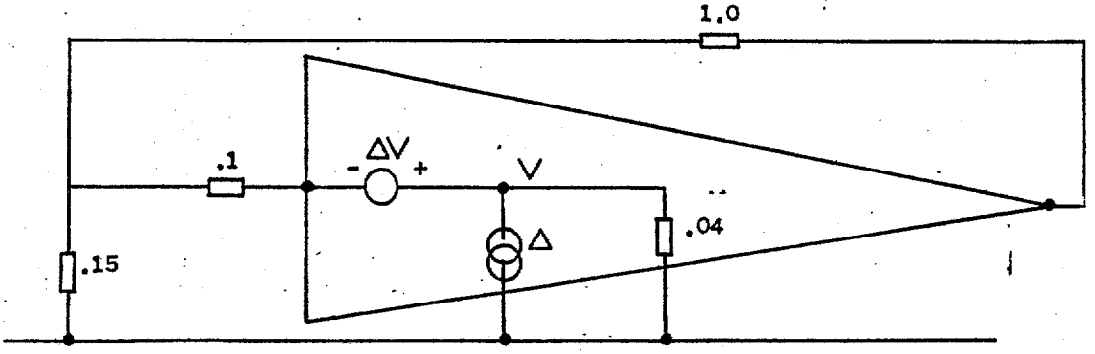
THE OPERATIONAL AMPLIFIER IS A HIGH GAIN AMPLIFIER WITH A CONTROLLED "SINGLE BREAK" FREQUENCY RESPONSE. AN INVERTING INPUT AND A NON-INVERTING INPUT ARE PROVIDED. NORMALLY THE NON-INVERTING INPUT IS TIED TO COMMON OF THE ANALOG POWER SUPPLY. A FEEDBACK RESISTOR, CAPACITOR, OR SERIES RC CIRCUIT FROM OUTPUT TO INVERTING INPUT WILL PROVIDE A STABLE CLOSED LOOP WHICH WILL RESPOND IN A CONTROLLED MANNER TO ANY SIGNAL OR SIGNALS APPLIED TO THE INVERTING INPUT. BECAUSE OF THE HIGH AMPLIFIER GAIN, THE AMPLIFIER OUTPUT VOLTAGE WILL BE VERY NEARLY EQUAL TO THE NEGATIVE OF THE INPUT VOLTAGE TIMES THE RATIO OF THE FEEDBACK IMPEDANCE TO THE INPUT IMPEDANCE. IF THE INPUT OR FEEDBACK CIRCUITS ARE NETWORKS THE INVERTING INPUT CAN BE CONSIDERED AS BEING AT ANALOG COMMON POTENTIAL. THE SUM OF THE CURRENTS FROM INPUT VOLTAGE TO INVERTING INPUT AND FROM OUTPUT VOLTAGE TO INVERTING INPUT CAN BE EQUATED TO ZERO AND THE EQUATION SOLVED FOR $\frac{V_{OUT}}{V_{IN}}$

EFFECT OF TEMPERATURE VARIATIONS

THE AMPLIFIER IS ADJUSTED IN TEST TO HAVE ZERO INPUT VOLTAGE AND CURRENT ERROR AT THE TEMPERATURE PREVAILING WHEN THE ADJUSTMENT IS MADE. THIS WILL NORMALLY BE DONE AT 25 TO 30°C. WHEN THE AMPLIFIER IS OPERATED AT A DIFFERENT AMBIENT TEMPERATURE THERE WILL BE AN ERROR VOLTAGE AND CURRENT AT THE AMPLIFIER INPUT BECAUSE THE TEMPERATURE COMPENSATION CIRCUITS DO NOT EXACTLY COMPENSATE THE TRANSISTOR CHARACTERISTICS . THE AMOUNT OF ERROR CANNOT BE PREDICTED ACCURATELY UNLESS EACH AMPLIFIER IS PRETESTED THROUGH THE ENTIRE TEMPERATURE RANGE. THE DRIFT DATA GIVEN ON THE CHARACTERISTICS SHEET IS BASED ON TESTS PERFORMED ON FOUR PROTOTYPE AMPLIFIERS. THE CALCULATION SHOWN BELOW ILLUSTRATES HOW THE CHARACTERISTIC DATA WOULD BE USED TO CALCULATE THE OFFSET BAND WITHIN WHICH ALL AMPLIFIERS ARE EXPECTED TO OPERATE. FIGURE 1 SHOWS THE OFFSET BAND, THE CALCULATED DRIFT OF FOUR AMPLIFIERS BASED ON INPUT DRIFTS MEASURED IN A PREVIOUS TEST, AND MEASURED OFFSET WITH THIS CIRCUIT.

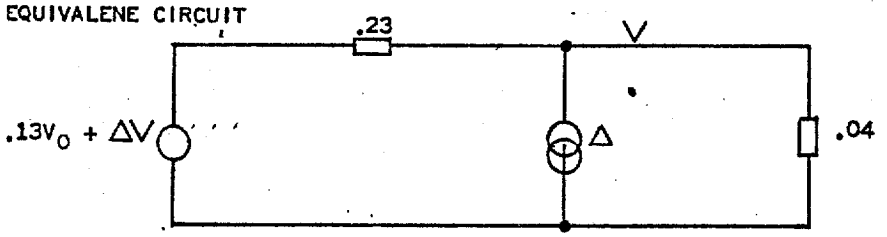
TEMP. DRIFT OF AOAC AMPLIFIER

1. AMPLIFIER INPUT AND FEEDBACK CIRCUIT WITH DRIFT ELEMENTS.



POLARITIES SHOWN FOR V AND I ARE FOR POSITIVE DRIFT QUANTITIES.

2. EQUIVALENE CIRCUIT



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 12-18-70	REV. 4	REV. 5	PRINTS TO DL1	APPROVALS	ELEMENTARY DIAGRAM
REV. 2 REISSUED	DECEMBER 18, 1970 P6A			FIRST MADE FOR STAND. LINE	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 3	MADE BY D L PEARSON	2320 1338	I.C. NO.		I C 3 6 0 0 A 0 A C 1
					CONT. ON SH. 3.3 SH. NO. 3.2

3. EQUATIONS RELATING V_o, ΔV, AND Δi FOR V_i = ZERO

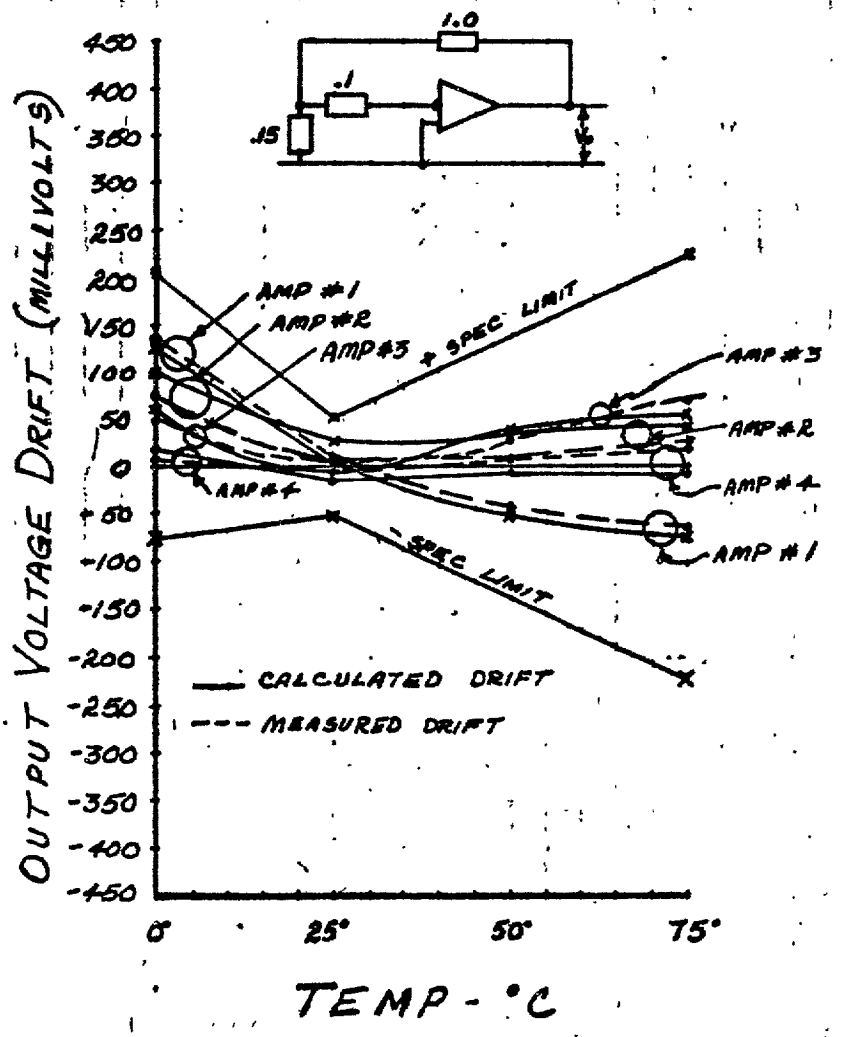
$$\frac{.13\,V_o + \Delta V}{.23} = \Delta i$$

$$V_o = \frac{.23\,\Delta i - \Delta V}{.13} = 1.77\,\Delta i - 7.7\,\Delta V$$

IF Δi IS IN NANO-AMPS AND ΔV IN MILLIVOLTS THEN V_o WILL BE IN MILLIVOLTS.

IF THE AMPLIFIER IS EXPECTED TO OPERATE OVER A TEMP. RANGE OF 25°C TO 50°C THE SPECIFICATION SHEET GIVES $\Delta i = \pm 25 \times 10^{-9} \pm 25 \times 2 \times 10^{-9}$ OR $\pm$ 75 NANO-AMPS. SIMILARLY $\Delta V = \pm 25 \times 20 \times 10^{-6}$ $\pm 250 \times 10^{-6} = \pm 750 \times 10^{-6}$ OR $\pm$ 75 MILLIVOLTS. THESE VALUES SUBSTITUTED IN THE ABOVE EQUATION WOULD GIVE THE MAXIMUM OFFSET. THE TRACKING ERROR FOR TWO DRIVES WOULD BE TWICE THIS VALUE.

TEMP. TEST OF 4 PROTOTYPE AMPLIFIERS



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

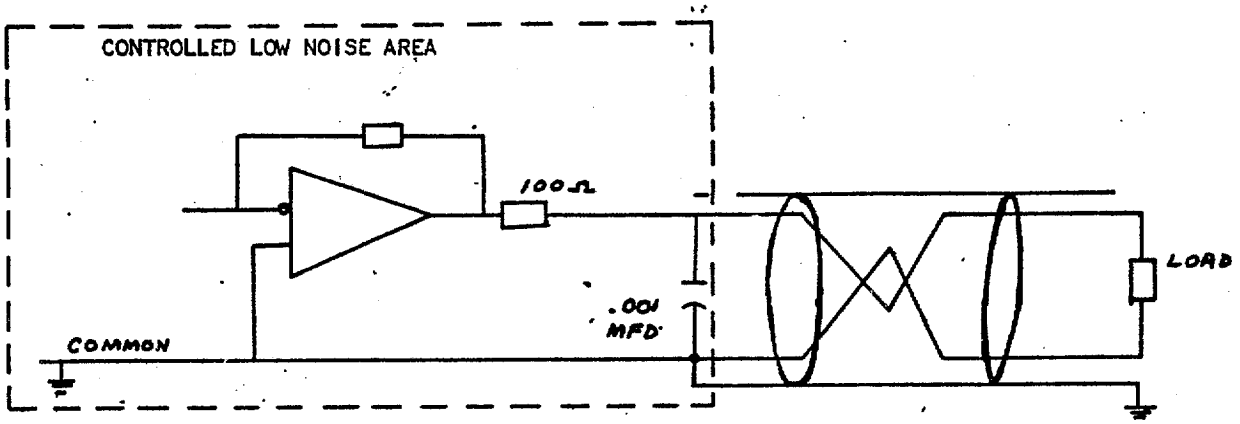
REV. 1	REV. 2 DL1	APPROVALS W. Jarvis	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
ISSUED 8-2-65	PRINTS TO 1338	FIRST MADE FOR ENG. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	[C3600A0AC1	SH. NO. 3.3
MADE BY J.SMITH	2320 P6A	LC NO.		ASY. OR SK. 3.4	

3. NOISE PROTECTION

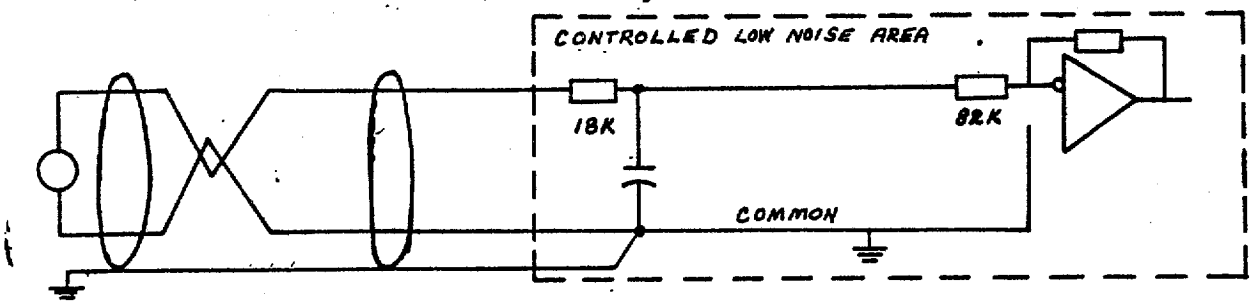
THE AMPLIFIER HAS AN INPUT FILTER, A LOW OUTPUT IMPEDANCE, BUS FILTERS AND A SEPARATE ZENER REGULATED AND FILTERED BUS SYSTEM FOR THE INPUT STAGES OF THE AMPLIFIER. ALL OF THESE FEATURES HELP TO REDUCE THE SUSCEPTABILITY OF THE AMPLIFIER TO NOISE TRANSIENTS. LARGE NOISE INPUTS CAN STILL CAUSE THE AMPLIFIER TO SATURATE. TO PROTECT THE AMPLIFIER FROM THESE LARGE NOISE INPUTS IT IS ESSENTIAL THAT THE ANALOG BUS SYSTEM (ESPECIALLY ANALOG COMMON) MUST BE STABLE AND MAINTAINED VERY CLOSE TO EARTH OR ENCLOSURE POTENTIAL. A DIRECT TIE FROM BUS TO ENCLOSURE SHOULD BE MADE AT ONE POINT ONLY BUT DISTRIBUTED CAPACITY FROM THE COMMON BUS TO ENCLOSURE WILL GREATLY REDUCE THE EFFECT OF HIGH FREQUENCY TRANSIENTS.

IF A STABLE BUS SYSTEM IS MAINTAINED, FILTERS CAN BE ADDED WHICH WILL REDUCE THE EFFECT OF SIGNAL TRANSIENTS AS WELL AS HIGH FREQUENCY GROUND LOOP CURRENTS. SOME RECOMMENDED FILTERS ARE SHOWN BELOW.

A. WHEN THE AMPLIFIER OUTPUT MUST BE WIRED TO A REMOTE LOCATION WHERE IT MAY BE SUBJECTED TO NOISE, IT IS RECOMMENDED THAT A SERIES 100 OHM RESISTOR BE MOUNTED ON THE PAGE, NEAR THE AMPLIFIER, AND A SUITABLE CAPACITOR FILTER BE MOUNTED WHERE THE OUTPUT LEAVES THE CONTROLLED LOW NOISE AREA OF THE PANEL. THE OUTPUT SHOULD BE CARRIED THROUGH A TWISTED SHIELDED PAIR TO THE LOAD. IT WILL FREQUENTLY HELP TO GROUND BOTH ENDS OF THE SHIELD IF THE OUTPUT LINES ARE SUBJECTED TO HIGH FREQUENCY NOISE.



B. PROTECTION FOR INPUT SIGNAL WIRES CAN BE PROVIDED IN A SIMILAR MANNER. IT IS RECOMMENDED THAT 10 TO 20% OF THE INPUT RESISTANCE BE ADDED IN SERIES WITH THE INPUT LINE WHERE IT ENTERS THE CONTROLLED LOW NOISE AREA. ALSO A SUITABLE FILTER CAPACITOR SHOULD BE MOUNTED CLOSE TO THIS RESISTOR. THE FIGURE BELOW ILLUSTRATES THIS FILTERING.

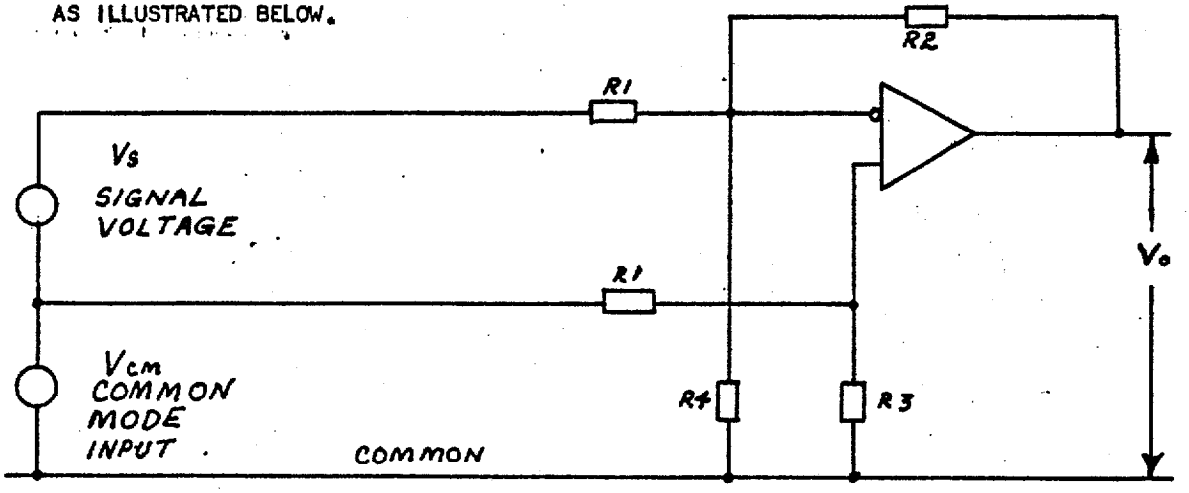


THIS FILTER CAN BE USED TO REDUCE HIGH FREQUENCY TRANSIENTS. IT WILL NOT BE SATISFACTORY FOR NOISE FREQUENCIES WHICH ARE EQUAL OR LOWER THAN THE SIGNAL FREQUENCIES TO BE CONTROLLED.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

4. COMMON MODE REJECTION

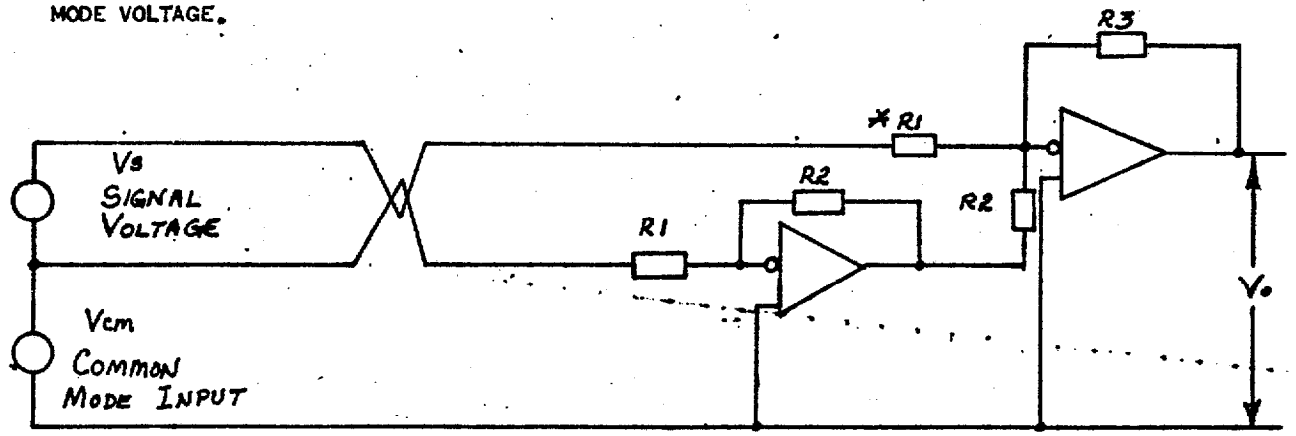
DC OR LOW FREQUENCY COMMON MODE SIGNALS CAN BE APPRECIABLY ATTENUATED WITH A CIRCUIT AS ILLUSTRATED BELOW.



FOR THIS CONFIGURATION $V_o = -V_s \frac{R_2}{R_1}$

ALSO V_{CM} (MAX.) $\times \frac{R_3}{R_1 + R_3} \leq$ 8 VOLTS.

R4 IS APPROXIMATELY $\frac{R_3\ R_2}{R_3 + R_2}$. IT IS RECOMMENDED THAT R4 SHOULD CONSIST OF A FIXED RESISTOR AND A SERIES RHEOSTAT SO THAT A ±10% ADJUSTMENT CAN BE MADE. FOR DC COMMON MODE SIGNALS THE COMMON MODE REJECTION CAN BE SET AS HIGH AS 120 DB. AT 60 CYCLES THE REJECTION RATIO IS REDUCED BECAUSE THE AMPLIFIER STABILIZING IS NOT BALANCED FOR BOTH INPUTS. AN ADDITIONAL RC CIRCUIT IN THE INVERTING INPUT CAN BE USED TO IMPROVE THE HIGH FREQUENCY COMMON MODE REJECTION. THIS REQUIRES CAREFUL SELECTION OF COMPONENTS AND IS NOT RECOMMENDED IF A HIGH DEGREE OF AC COMMON MODE REJECTION IS REQUIRED. THE CIRCUIT SHOWN BELOW CAN BE EASILY ADJUSTED TO GIVE UP TO 72 DB REJECTION OF 60 CYCLE COMMON MODE VOLTAGE.



FOR THIS CONFIGURATION $V_o = -V_s \frac{R_2}{R_1}$

ALSO V_{CM} (MAX.) $\times \frac{R_2}{R_1} \leq$ 30 VOLTS

THE REJECTION RATIO IS INDEPENDENT OF IMPEDANCE IN SERIES WITH THE COMMON MODE INPUT. IMPEDANCE OF THE SIGNAL SOURCE MAY REDUCE THE COMMON MODE REJECTION. THE RESISTOR R1 IDENTIFIED BY AN ASTERISK ABOVE SHOULD BE ADJUSTABLE TO COMPENSATE FOR SIGNAL SOURCE IMPEDANCE.

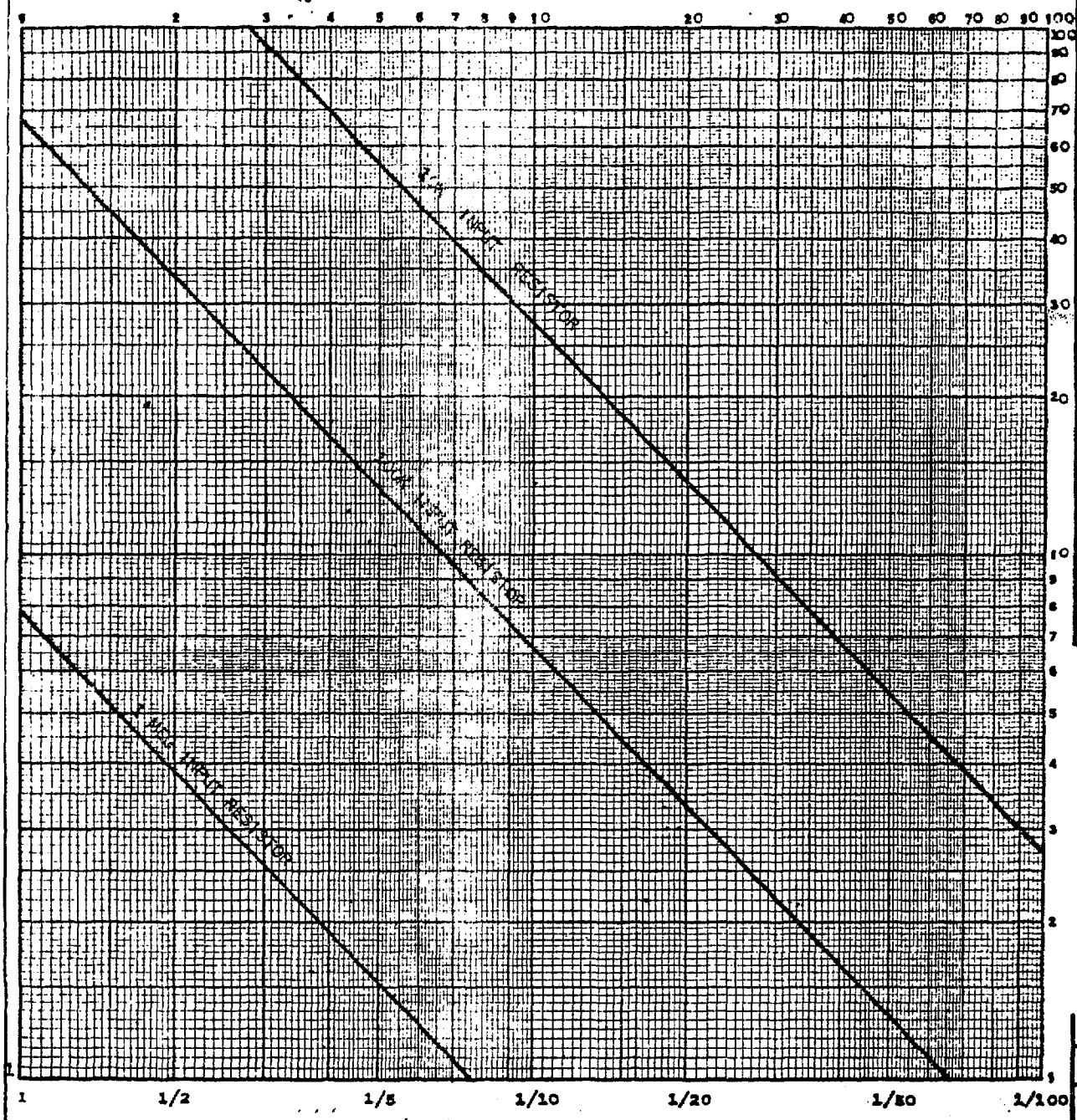
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 2 DL1 2520	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
ISSUED 9-2-65	PRINTS TO 1338	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600A0AC1	SH. NO. 3.5
MADE BY J.SMITH	P6A	I.C. NO.		CONT. ON SH. 3.6	

GENERAL ELECTRIC

REV NO.	TITLE	
A	OPERATIONAL AMPLIFIER	1C3600A0AC1
1C3600A0AC1		CONT ON SHEET FL. SH NO. 3,6

TYPICAL FREQUENCY RESPONSE DATA IS SHOWN ON THE FOLLOWING GRAPH. ALSO RESPONSE TIME FOR 99.0% RECOVERY FROM A STEP INPUT IS INDICATED. THE DATA IS SUBJECT TO CONSIDERABLE ERROR ABOVE 100 KC BECAUSE OF STRAY CAPACITANCE OF WIRING AND CAPACITANCE OF RESISTORS IN THE INPUT AND FEEDBACK CIRCUITS.



REVISIONS

GAIN VOLTS/VOLT

1338

P6A

PXW

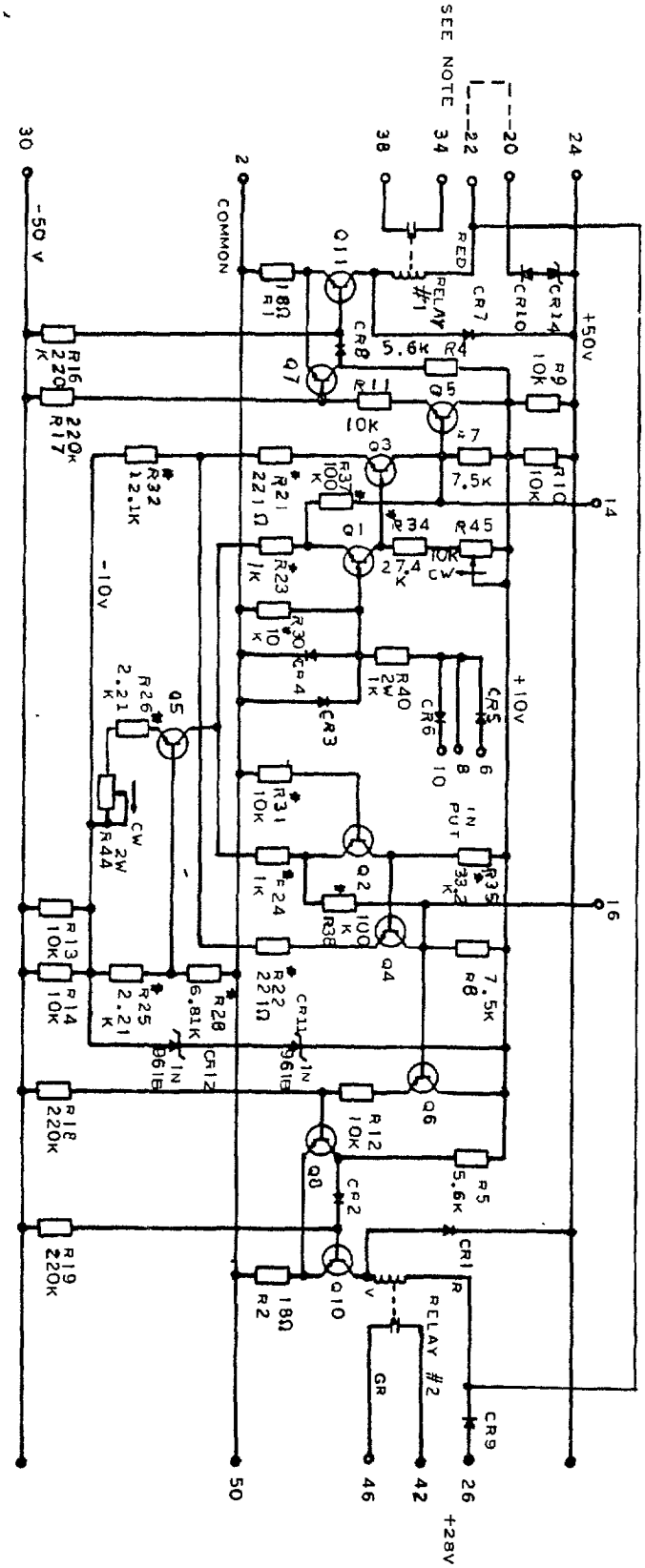
DLI

2520

PRINTS TO

MADE BY	APPROVALS		
J.H. SMITH	JHS.	INDUSTRY CONTROL DIV OR DEPT	1C3600A0AC1
ISSUED 8-21-65		SALEM, VIRGINIA LOCATION	CONT ON SHEET FL SH NO. 3,6
			CODE IDENT NO.

PF8GS-WP (10-64)



*1/2 WATT METAL FILM 100 PPM

Q1, Q2
Q3, Q4 } (MATCHED PAIR ON COMMON HEAT SINK) 68A7310G1

Q5, Q6, Q7, Q8, Q9 - 68A7373P7

Q10, Q11 - 68A8305P1

RELAYS ARE REED SWITCHES WITH 28V COILS

ALL DIODES - 68A726IP1

CR14 - 68A7237P9

NOTE - AN EXTERNAL JUMPER MUST BE CONNECTED FROM PIN 20 TO PIN 22 IF A 28 VOLT POWER SUPPLY IS NOT CONNECTED TO PIN 26.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. #3 7/20/64 R-117264 PSO	REV. 2 /	PRINTS TO 1338	APPROVALS N. B. Jarvinen	GENERAL ELECTRIC	ELEMENTARY DIAGRAM VOLTAGE SENSITIVE RELAY
ISSUED July 20 1964		PRINTS TO K6A	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	1C3600AVSA1
MADE BY S. BEVERLY		PXW CW	I.C. NO.		CONT. ON SH. 3.1 / SH. NO. 3.0

CIRCUIT CHARACTERISTICS

1.	POWER INPUT	+50V, 40MA (PIN 24) -50V, 9MA (PIN 30)
	OR	+50V, 9MA (PIN 24) -50V, 9MA (PIN 30) +28V, 31MA (PIN 26) (POWER SUPPLY COMMON ON PINS 2 AND 50)
2.	SIGNAL INPUT (K2 PICKS UP WITH POSITIVE INPUT. K1 PICKS UP WITH NEGATIVE INPUT).	RELAY PICKS UP ON 4 TO 10 MICROAMP INPUT. RELAY DROPS OUT AT 85% TO 95% OF PICK UP CURRENT.
3.	INPUT INPEDANCE	11K
4.	DRIFT AT INPUT	
	A. VOLTAGE	LESS THAN 1MV/°C
	B. CURRENT	LESS THAN .05 MICROAMPS/°C
5.	FREQUENCY RESPONSE	200 CYCLES/SEC. MAX. PICK UP TIME - LESS THAN 1 MILLISEC. DROP OUT TIME - LESS THAN 1 MILLISEC. CONTACT BOUNCE TIME - 0 TO 2 MILLISEC.
6.	MAX INPUT SIGNAL	±30 VOLTS AT ANY INPUT.
7.	CONTACT RATING (MAX.)	50VA, 250V, 3AMP (CONTACTS MUST BE PROTECTED IN INDUCTIVE CIRCUITS)

NOTE: CONTACTS MUST SEE A MINIMUM OF 4 VA, OR 12 V, OR 15 MA. ANY CONDITION BELOW THESE MINIMUMS MAY RESULT IN MIS-OPERATION.

APPLICATION NOTES

1. THE VOLTAGE SENSITIVE RELAY AMPLIFIER CONSISTS OF TWO CASCADED DIFFERENTIAL AMPLIFIER STAGES WITH FEEDBACK. THE OUTPUTS ARE TAKEN SINGLE ENDED FROM THE COLLECTORS AND DRIVE SCHMITT TRIGGER CIRCUITS THROUGH EMITTER FOLLOWERS.

THE FIRST DIFFERENTIAL AMPLIFIER CONSISTS OF TRANSISTORS Q1 AND Q2 WITH Q9 ACTING AS A CONSTANT CURRENT SOURCE. TRANSISTORS Q3 AND Q4 FORM THE SECOND DIFFERENTIAL PAIR. FEEDBACK IS PROVIDED FROM THE COLLECTORS OF Q3 AND Q4 BACK TO THE EMITTERS OF Q1 AND Q2. THE OUTPUT OF Q3 DRIVES EMITTER FOLLOWER Q5 WHICH DROVES A SCHMITT TRIGGER CONSISTING OF Q7 AND Q11. SIMILARLY THE OUTPUT OF Q4 DRIVES EMITTER FOLLOWER Q6 WHICH DRIVES A TRIGGER UNIT CONSISTING OF Q8 AND Q10. RELAY COILS ARE CONNECTED IN THE COLLECTOR CIRCUITS OF Q10 AND Q11.

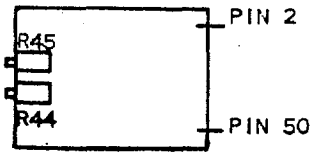
2. THE VOLTAGE SENSITIVE RELAY NORMALLY OPERATES WITHIN PLUS OR MINUS 5 MICRO-AMPS OF ZERO INPUT. IF OPERATION OF THE RELAYS IS DESIRED AT A SIGNAL LEVEL AWAY FROM ZERO A FIXED BIAS CURRENT CAN BE APPLIED TO THE INPUT. FOR EXAMPLE IF A 1 MEG RESISTOR IS CONNECTED FROM THE INPUT TO THE NEGATIVE 50 VOLT BUS RELAY K1 WILL PICK UP. THE RELAY WILL NOT DROP OUT UNTIL A POSITIVE SIGNAL OF APPROX. 45 MICROAMPS IS APPLIED TO THE INPUT. A FURTHER INCREASE OF THE POSITIVE SIGNAL TO APPROX. 55 MICROAMPS IS REQUIRED TO PICK UP RELAY K2.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 NJA 4-8-70	DL8 DL1	1338	APPROVALS K.B. Jamieson	GENERAL ELECTRIC	ELEMENTARY DIAGRAM VOLTAGE SENSITIVE RELAY
ISSUED June 4, 1964	PRINTS TO K6A		FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600AVSA1
MADE BY S. POWERS	PXW CW		I.C. NO.		CONT. ON SH. 3.2 / SH. NO. 3.1

3. THE AMPLIFIER IS DESIGNED FOR FAST OPERATION OF THE RELAYS. ALSO THE AMPLIFIER NEEDS ONLY A LOW LEVEL SIGNAL TO OPERATE THE RELAY. TO PREVENT "CHATTERING" OF THE RELAY, SIGNALS MUST BE FREE OF LOW FREQUENCY NOISE. SIGNAL LEADS MUST HAVE SMALL LOOPS OR A FILTER CAPACITOR MUST BE ADDED AT THE INPUT

4. THE TRIMPOTS ON THIS CARD ARE ADJUSTED IN TEST AND SHOULD NOT BE ADJUSTED UNLESS THE COMPLETE TEST PROCEDURE IS FOLLOWED. THE LOCATION OF THE TRIMPOTS AND THEIR FUNCTION IS GIVEN BELOW.



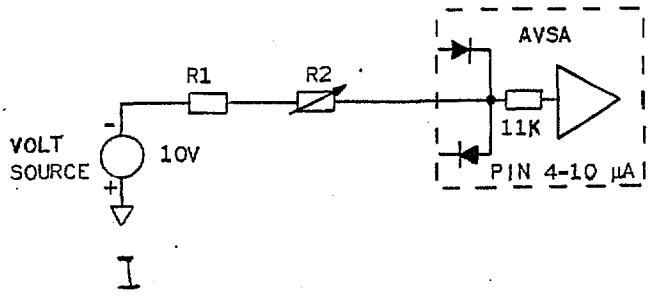
R45 - SETS COLLECTOR VOLTAGES OF Q3 AND Q4 EQUAL WHEN INPUT SIGNAL IS ZERO.

R44 - SETS THE COLLECTOR VOLTAGE OF Q3 AND Q4 AT APPROX. 5.8 VOLTS WHEN INPUT SIGNAL IS ZERO.

CIRCUIT APPLICATIONS

DEPENDING ON THE APPLICATION IT MAY BE IMPORTANT TO CONSIDER THE VARIATION IN PICKUP OF THE VOLTAGE SENSITIVE RELAYS, ESPECIALLY IF A REPLACEMENT BOARD IS REQUIRED AFTER INSTALLATION, DUE TO FAILURE ETC., AND RE-ADJUSTMENT OF CIRCUIT IS NOT DONE. TWO CIRCUITS ARE GIVEN BELOW TO INCLUDE THE VARIATIONS THAT COULD EXIST. ASSUME THAT THE APPLICATION REQUIRES RELAY ENERGIZATION WITH ±10 VOLTS FROM THE VOLTAGE SOURCE.

CURRENT



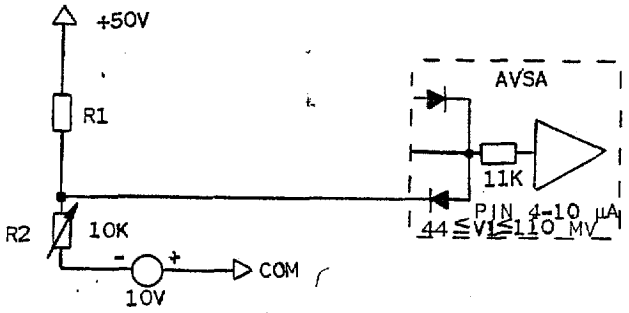
A) AT INITIAL INSTALLATION THE RELAY CHARACTERISTICS SHOW PICK UP AT 4 UA

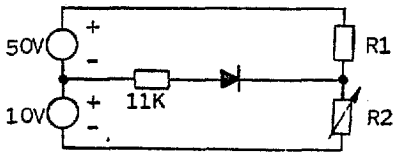
$$\frac{10V}{R1+R2} = 4\mu A \;:\; R1+R2 = 2.5M$$

B) CARD IS REPLACED THAT HAS A 10 µA PICKUP CHARACTERISTIC

THE VOLTAGE SOURCE WOULD NEED TO INCREASE TO 2.5 MX 10 UA = 25V OR A 150% ERROR.

VOLTAGE





A) AT INITIAL INSTALLATION RELAY CHARACTERISTIC SHOULD PICK UP AT 4 µA [44MV] AND DIODE BREAKOVER AT 0.5V

$$10 - \frac{(60)\,10K}{R1 + 10K} = .5V + .044V$$

$$R1 = 53K$$

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 1-10-74 RSN	DL1	DL8	APPROVED W.S. Sarvinen	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
ISSUED June 4, 1964			FIRST MADE FOR DEPT. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	VOLTAGE SENSITIVE RELAY
MADE BY S.POWERS			LC NO.		IC3600AVSA1
					CONT. DWG NO. 3.3
					DWG. NO. 3.2

II.

A) AT INITIAL INSTALLATION THE RELAY CHARACTERISTIC SHOW PICK UP AT 10V μA.

$$\frac{20V}{R1 + R2} = 10\ \mu A \therefore R1 + R2 = 1M$$

B) CARD IS REPLACED THAT HAS A 4 μA PICKUP CHARACTERISTIC THE VOLTAGE SWITCH WOULD NEED TO DECREASE TO 4 μA X 1M = 4 VOLTS OR A 40% ERROR.

B) CARD IS REPLACED THAT HAS A 10 μA PICKUP (110MV) AND DIODE BREAKOVER OF 1V. THE VOLTAGE SOURCE WOULD HAVE TO INCREASE PER THE FOLLOWING TO CAUSE RELAY PICK UP.

$$V - \frac{(50 + V)\ 10K}{63K} = 1 + .11$$

V = 10.7V OR 7 % ERROR

WORST CASE ERROR FOR CURRENT CIRCUIT IS 150 % , WORST CASE ERROR FOR THE VOLTAGE CIRCUIT IS 7%

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, DL8	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM VOLTAGE SENSITIVE RELAY
REV. 2	ISSUED 1-11-74			FIRST MADE FOR STANDARD LINE		IC3600AVSA1
REV. 3	MADE BY S. POWERS			I.C. NO.		CONT. ON SH. FL. SH. NO. 3.3

GENERAL ELECTRIC

IC3600KRSE

CONT ON SHEET 3.1 SH NO. 3.0

REV NO. A

IC3600KRSE

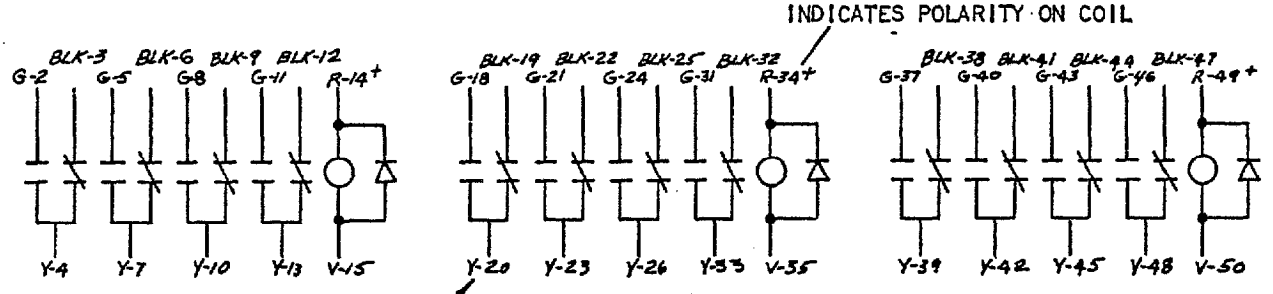
CONT ON SHEET 3.1 SH NO. 3.0

TITLE

REED RELAY CARD ASM.

FIRST MADE FOR

REVISIONS



PIN NUMBERS AND LEAD COLORS

GENERAL INFORMATION:

THIS CARD HAS 3 FORM "C", 10VA REED RELAYS MOUNTED FOR 50V CREEPAGE BETWEEN PINS. EACH RELAY HAS 4 "C" FORM REEDS.

THE CARD IS A STANDARD PRINTED CIRCUIT BOARD DESIGNED TO PLUG INTO I.P.C. SOCKETS.

EACH RELAY ASSEMBLY IS CONTAINED IN A METAL ENCLOSURE FOR SHEILDING PURPOSES. THE ENCLOSURE EXTENDS .75 INCHES ABOVE THE BOARD REQUIRING 2 CARD SLOTS FOR CLEARANCE BETWEEN PRINTED CIRCUIT BOARDS. THE RELAY ASSEMBLY IS HELD TO THE BOARD BY METAL TABS AND THE LEADS TERMI POINTED TO THE CARD PINS.

DIODES ARE FLOW SOLDERED TO THE BOARD AND PROVIDE INDIVIDUAL COIL SUPPRESSION.

THE COILS OPERATE ON 28 VOLTS ONLY.

IC3600KRSE ← FORM NO.

FORM NO.	COIL VOLTS
2	28

2520 PXW CW

DL1

P6A

1338

PRINTS TO

MADE BY R.E. Hannah

ISSUED 4/2/65

APPROVALS R.E.H.

INDUSTRY CONTROL DIV OR DEPT

SALEM, VIRGINIA LOCATION

IC3600KRSE

CONT ON SHEET 3.1 SH NO. 3.0

CODE IDENT NO.

FF-803 WI (6-64)
PRINTED IN U.S.A.

GENERAL ELECTRIC

REV NO. A	TITLE	IC3600KRSE
IC3600KRSE	REED RELAY CARD ASM.	CONT ON SHEET 3.2 SH NO. 3.1
CONT ON SHEET 3.2 SH NO. 3.1	FIRST MADE FOR	

REVISIONS

COIL INFORMATION

DC COILS FOR REED RELAYS ARE DESIGNED TO PROVIDE AT LEAST 20 AMPERE TURNS (NI) MORE THAN MAXIMUM PICKUP NI OF THE REEDS FOR ALL OF THE FOLLOWING CONDITIONS SIMULTANEOUSLY:

1. MIN. VOLTAGE FROM SUPPLY 9.6 (FOR 12V NOM.)
   23.4 (FOR 28V NOM.)
2. MAX. COIL TEMPERATURE 120°C.
3. +10% NOM. COIL RESISTANCE
4. -2% NOM. COIL TURNS

MAX. AMPERE TURNS OCCUR WHEN ALL THE FOLLOWING CONDITIONS OCCUR SIMULTANEOUSLY:

1. MAX. VOLTAGE FROM SUPPLY 14.3 (FOR 12V NOM)
   30.8 (FOR 28V NOM)
2. MIN. COIL TEMPERATURE 0°C
3. -10% NOM. COIL RESISTANCE
4. +2% NOM. COIL TURNS.

2520
1338
PGA
DL1
PXW CW

PRINTS TO

MADE BY R.E. HANNAH	APPROVALS R.E.H.	INDUSTRY CONTROL DIV OR DEPT	IC3600KRSE
ISSUED 4/2/65		SALEM, VIRGINIA LOCATION	CONT ON SHEET 3.2 SH NO. 3.1
			CODE IDENT NO.

FF-803 WF (6-64)
PRINTED IN U.S.A.

GENERAL ELECTRIC

IC3600KRSE
CONT ON SHEET 3.3 SH NO. 3.2

REV NO. A

IC3800KRSE
CONT ON SHEET 3.3 SH NO. 3.2

TITLE
REED RELAY CARD ASM.

FIRST MADE FOR

REVISIONS

4 "C" REED PER COIL

ENG. DATA 68A992066

TURNS 6150

28 VOLT COIL
NOM. VOLTS 28V
MAX. VOLTS 30.8V
MIN. VOLTS 23.4V

NOM. RES. 490 OHMS (25°C)
MIN. RES. 398 OHMS (10% AT 0°C)
MAX. RES. 675 OHMS (+10% AT 0°C)

NOM. AMPS .057
MAX. AMPS .078
MIN. AMPS .035

WATTS
NOM. VOLTS 28X NOM. AMPS .057 = 1.6 NOM. WATTS
COIL TEMP. RISE (CONTINUOUS) 22°C BY RESISTANCE OPEN

AMPERE TURNS
NOM. 350 NI, MAX. 490 NI, MIN. 211 NI

2520
1338
PGA
DL1
PXW CW

PRINTS TO

MADE BY R.E. HANNAH
ISSUED 4/2/65

APPROVALS R.E.H.

INDUSTRY CONTROL DIV OR DEPT
SALEM, VIRGINIA LOCATION

IC3600KRSE
CONT ON SHEET 3.3 SH NO. 3.2

CODE IDENT NO.

FF-603 WF (8-64)
PRINTED IN U.S.A.

GENERAL ELECTRIC

REV NO. A	TITLE	IC3600KRSE
IC3600KRSE	REED RELAY CARD ASM.	CONT ON SHEET 3.4 SH NO. 3.3
CONT ON SHEET 3.4 SH NO. 3.3	FIRST MADE FOR	

REVISIONS

REED INFORMATION

CONTACT ARRANGEMENT	FORM C	
CONTACT RATING	10 WATTS; UP TO 5 AMPS; UP TO 250 VOLTS (D-C RES. LOAD) 20 WATTS; UP TO .5 AMPS; UP TO 250 VOLTS (A-C RES. LOAD)	
CONTACT RES. (MAX.)	80 MILLIOHMS INITIAL	
LIFE	(D-C RESISTIVE LOADS) OVER $100X10^6$ OPERATIONS AT FULL LOAD 10^{12} OPERATIONS ON DRY CIRCUIT APPLICATION.	
CONTACT CAPACITANCE	.6 MMF (MAX.)	
RESONANT FREQUENCY (APPROX.)	ABOVE 800 CPS	
AMPERE TURNS	MUST OPERATE	105 NI
	MUST RELEASE	15 NI
OPERATE TIME	2.5MS (NO) 4.5 MS (NC)	TYPICAL AT EXCESS OF 50 NI
TEMPERATURE RANGE	$-65^{0}C$ TO $+150^{0}C$	

DL1
1338
P6A
PXW CW
2520
PRINTS TO

MADE BY	APPROVALS		
R.E. HANNAH	R.E.H.	INDUSTRY CONTROL — DIV OR DEPT	IC3600KRSE
ISSUED 4/2/65		SALEM, VIRGINIA — LOCATION	CONT ON SHEET 3.4 SH NO 3.3
			CODE IDENT NO.

REVISIONS

RELAY ASM. INFORMATION

4 "C" REED PER COIL

NOM. VOLTS	NOM. RES.	COIL RES. AT 25°C MIN.	COIL RES. AT 25°C MAX.	NO PICKUP I MAX.	NC OPENING	NO MAX. TOTAL OPERATING TIME	NC MAX. CLOSING TIME
28	545	490	600	32MA	8-32MA	4MS	11MS

LOADS OTHER THAN RESISTIVE:

ALL LOADS OTHER THAN RESISTIVE MUST HAVE SUPPRESSION, UNLESS NOTED IN LOAD LIFE INFORMATION.

DL1

1338

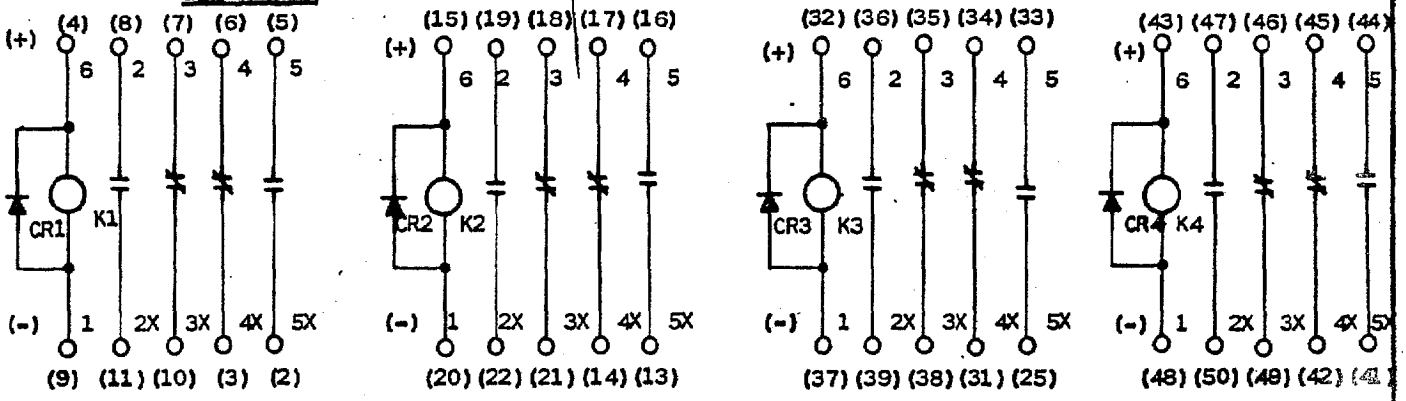
P6A

PXW CW

2520

PRINTS TO

ELEMENTARY



APPLICATION DATA

1. THIS PWB CONTAINS 4 REED RELAYS. EACH RELAY HAS 2 N.O. AND 2 N.C. CONTACT.
2. COIL RATING - 22-29 VOLTS D.C., 740 OHMS. RELAY WILL OPERATE AT 14.3V AND RELEASE AT 1.4V MIN (25°C)
3. CONTACT MATERIAL - RHODIUM OVER GOLD PLATE
4. CONTACT RATING/LIFE - (50 VOLTS MAX AND TRANSIENT SUPPRESSED)
   RESISTANCE: 200 MILLIOHMS (MAX INITIAL)
   FOLLOWING LIFE DATA APPLIES TO A 10VA N.O. CONTACT - SIMILAR LIFE COULD BE EXPECTED FROM 4VA N.C. CONTACT.

LIFE EXPECTANCY TABLE IN MILLIONS OF OPERATIONS

DC CURRENT - AMPS

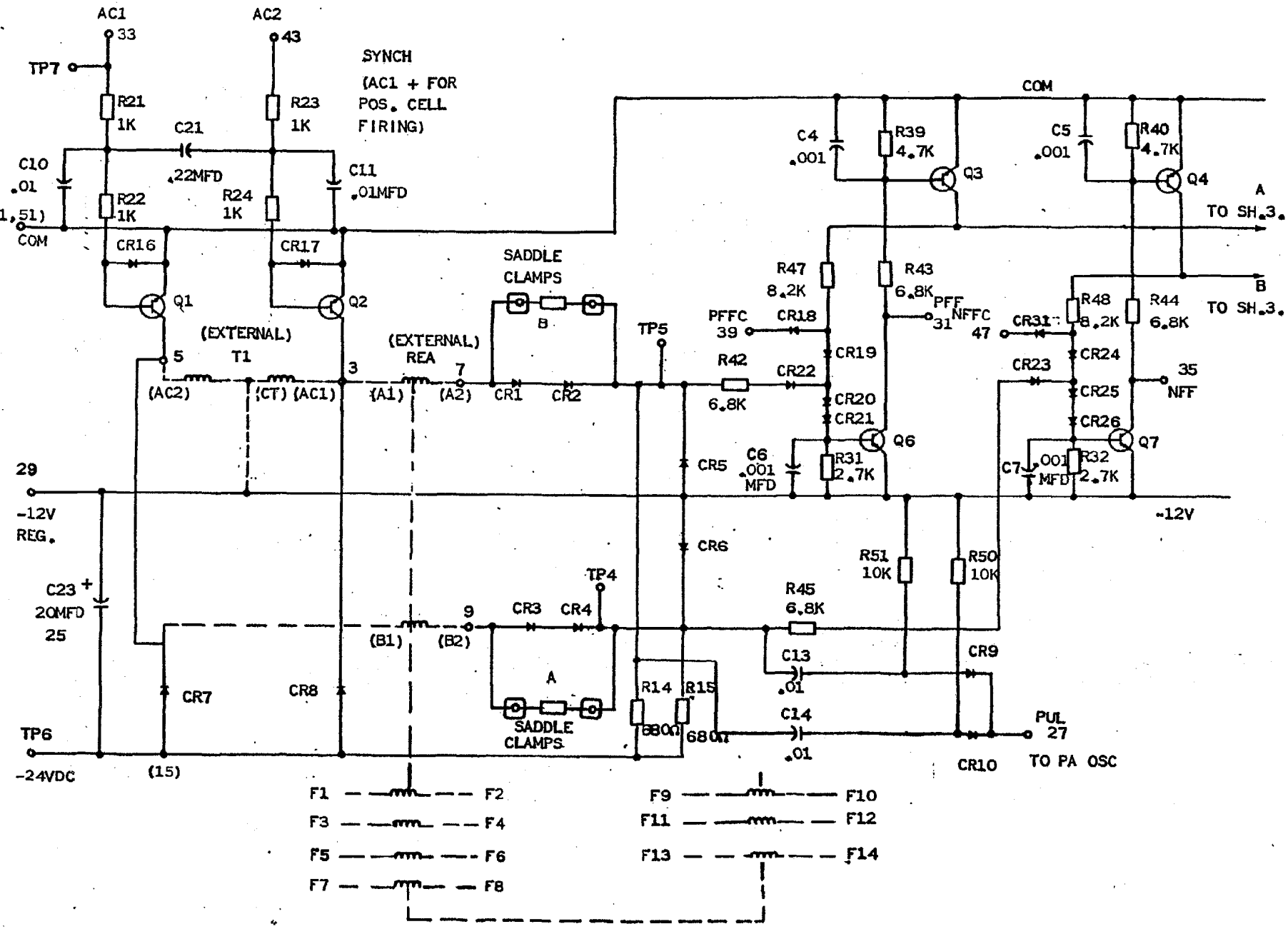
DC - VOLTS	.010	.033	.100	.333	1.0
100	100+	20	1*	2*	
33	100+	100+	100+	100+	1.0*
10	100+	100+	100+	10*	10+
3.3	100+	100+	100+	100+	
1.0	20	100+	5		
.33	10	5			
.10	10				

100+ INDICATES 100% SURVIVAL AT 100×10^6 OPERATIONS. OTHERS GREATER THAN 90% SURVIVAL. FAILURE = WELDED CONTACT OR CONTACT RESISTANCE GREATER THAN 10% OF LOAD RESISTANCE.

* INDICATES WELD FAILURE MODE

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1	APPROVALS 1-13-71 K.S. Nelson	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM REED RELAY
REV. 2	ISSUED 1-14-71. PGA			FIRST MADE FOR STANDARD LINE		IC3600KRST1.
REV. 3	MADE BY W M HURD	1338	I.C. NO.			CONT. ON SH. F1 SH. NO. 3.0



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

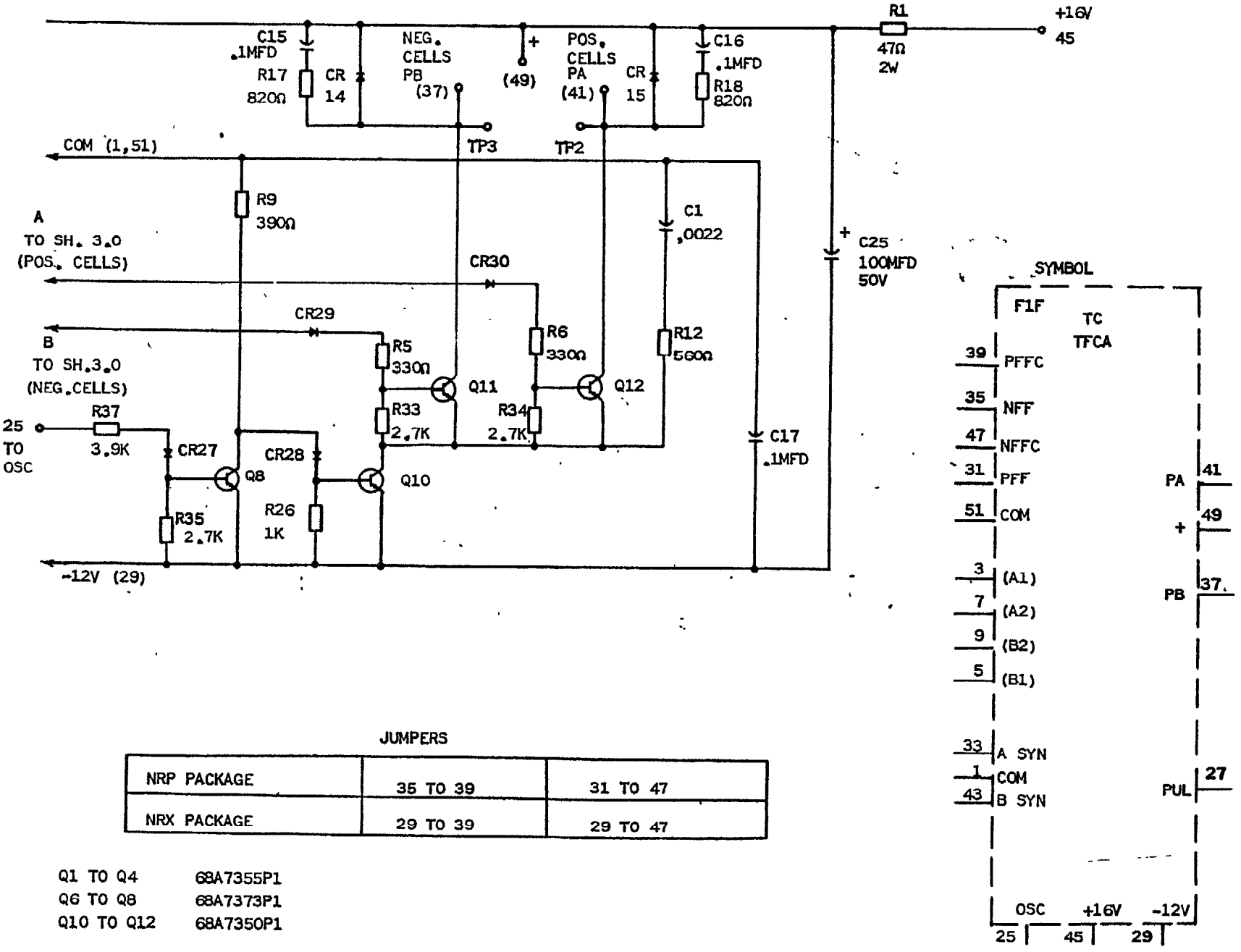
GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
FIRING CIRCUIT AMPLISTAT

IC3600TFCA1
3.1
3.0*

MADE BY J.H.SMITH
ISSUED
REV. LETTER
PRINTS TO 1338
DL1
P6A
FIRST MADE FOR REQ. STANDARD LINE
APPROVALS
E.C. NO.

FORM IC 2 (1-66)



JUMPERS

NRP PACKAGE	35 TO 39	31 TO 47
NRX PACKAGE	29 TO 39	29 TO 47

Q1 TO Q4 68A7355P1
Q6 TO Q8 68A7373P1
Q10 TO Q12 68A7350P1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
FIRING CIRCUIT AMPLISTAT

IC3600TFCA1
3.2

SH. NO. 3.1

MADE BY J.H. SMITH
ISSUED
REV. 1
REV. 2
PGA
DL1
PRINTS TO 1338
I.C. NO. STANDARD LINE
FIRST MADE FOR REQ.
APPROVALS

THIS CARD, IN CONJUNCTION WITH A SATURABLE REACTOR AND TRANSFORMER MODULE, SUPPLIES PHASE CONTROLLED PULSES THAT ARE USED TO FIRE THYRISTORS (SCR,) CONNECTED TO AC POWER LINES, THERE ARE MANY FEATURES THAT ARE PARTICULARLY DESIRABLE WHEN THE SCRS ARE CONNECTED FOR DC OUTPUT INTO INDUCTIVE LOADS. A TYPICAL REACTOR TRANSFORMER MODULE IS 688995018G1 (FOR 60 CPS), G2 (FOR 50 CPS).

THE BASIC TIMING DEVICE USED FOR PHASE CONTROL IS AN AMPLISTAT (THE EXTERNAL SATURABLE REACTOR PLUS SELF-SATURATING DIODES CR1-CR4). AN AMPLISTAT WILL FIRE AT VARIOUS POINTS IN THE CYCLE DEPENDING ON THE TOTAL AMPERE TURN EXCITATION OF THE CONTROL WINDINGS.

TO SYNCHRONIZE THE TIMING OF THE AMPLISTAT, A CENTERTAPPED AC SUPPLY (AC1, COM, AC2) MUST BE DERIVED FROM THE POWER LINES FEEDING THE SCRS. THIS AC IS LAGGED 8 DEGREES BY C21 TO FILTER NOISE AND THEN CONTROLS Q1 AND Q2. Q1 AND Q2 EXCITE TRANSFORMER T1 TO SUPPLY SQUARE WAVE VOLTAGES TO AMPLISTAT WINDINGS A1-A2 AND B1-B2.

WITH AC1 POSITIVE, AC2 IS NEGATIVE AND Q2 SATURATES TO EXCITE A1-A2. DEPENDING ON THE CONTROL WINDING SIGNALS, A1-A2 WILL SATURATE AT SOME POINT IN THIS HALF CYCLE. THIS APPLIES VOLTAGE TO R42 AND RESULTS IN SETTING THE POSITIVE FLIP FLOP Q6-Q3. SIMILARLY DURING THE NEXT HALF CYCLE, B1-B2 WILL SATURATE TO SET THE NEGATIVE FLIP FLOP Q7-Q4.

THE FLIP FLOPS ARE CLEARED BY THE CLEAR TERMINALS 39 AND 47. IF THE TWO FLIP FLOPS ARE CROSS CONNECTED, THEN ONE WILL RESET THE OTHER AND VICE VERSA, WITH THE RESULT THAT EACH STAYS SET 180 DEGREES.

THIS CONNECTION IS USED WHEN THE -12V BUS IS SET AT -12V. THE SATURABLE REACTOR IS BUILT TO WITHSTAND ONLY 10V. THUS THE OVERVOLTAGE WILL FORCE THE AMPLISTAT TO FIRE 25 DEGREES BEFORE THE END OF THE HALF CYCLE PROVIDING A PHASE RETARD LIMIT. THIS IS NECESSARY FOR THE SCRS TO PLUG AN INDUCTIVE LOAD TO ZERO.

IF THE -12V BUS IS REDUCED TO 10 VOLTS, THEN THERE MAY BE NO OUTPUT FROM THE AMPLISTAT TO ALTERNATELY SET THE FLIP FLOPS. IN THIS CASE THE CLEAR TERMINALS ARE TIED TO -12V, AND THEN Q6-Q3 (OR Q7-Q4) ARE MERELY AMPLIFIERS AND WILL RESET THEMSELVES AT THE END OF THE HALF CYCLE WHEN THE AMPLISTAT VOLTAGE GOES TO ZERO. THIS METHOD IS DESIRABLE FOR NON-PLUGGING SCR EXCITERS.

A -24V BUS IS DERIVED FROM T1 BY CR7 AND CR8. THIS BUS CAUSES 16MA TO FLOW THROUGH CR5 AND R14 (ALSO CR6 AND R15). THE AMPLISTAT THEN CANNOT SET A FLIP FLOP UNLESS IT SUPPLIES IN EXCESS OF THESE 16MA. AN AMPLISTAT WILL TURN ON SLIGHTLY WHEN DRIVEN HARD IN THE OFF DIRECTION. THIS CURRENT BIAS CIRCUIT PERMITS NEGATIVE DRIVES OVER 10 TIMES NORMAL.

ALTHOUGH MATCHED CORES ARE USED IN THE AMPLISTAT, BALANCE BETWEEN THE TWO HALVES MAY BE IMPROVED BY ADDING 100K OR LARGER RESISTOR IN ONE OF THE SADDLE CLAMPS ACROSS CR1, 2 (OR CR3(4).

A 50 MICROSECOND LONG PULSE APPEARS AT PUL (27) WHEN EITHER CORE OF THE AMPLISTAT FIRES. THIS IS FED BACK TO THE EXTERNAL OSCILLATOR TO SYNCHRONIZE IT WITH THE OPERATION OF THE FLIP FLOPS.

THE FFS CONTROL POWER TRANSISTORS Q11 AND Q12 TO STEER THE PULSES TO THE CORRECT SCR. IN SERIES WITH THESE IS Q10 WHICH IS DRIVEN BY Q8 AND AN EXTERNAL OSCILLATOR, RUNNING AT APPROXIMATELY 25 MICROSECONDS ON AND 50 MICROSECONDS OFF.

IN THE CASE WHERE EACH FF IS SET 180 DEGREES, PULSES ARE CONTINUOUSLY GOING TO SOME SCR CELL, THE FFS MERELY DETERMINE AT WHAT POINT IN THE CYCLE THE PULSES SWITCH FROM THE POSITIVE CELLS TO THE NEGATIVE CELLS OR VICE VERSA.

THE POWER TRANSISTORS ARE RATED ON THE BASIS THAT EACH CAN DRIVE TWO SCRS IF AN RC CIRCUIT IS USED IN SERIES WITH A PULSE TRANSFORMER. THEN THE TRANSISTORS DELIVER ONE LARGE PULSE FOLLOWED BY SMALLER PULSES AS THE CAPACITOR CHARGES UP.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FORM IC 221 (56)

REV. 1	REV. 2	APPROVAL B. Thomas	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
ISSUED November 21, 1966	PRINTS TO	FIRST MADE FOR REQ.	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	FIRING CIRCUIT	
MADE BY J.H. SMITH		I.C. NO.		IC3600TFCA1 CONT. ON SH. 3.3	SH. NO. 3.2

APPLICATION DATA

CONNECTIONS TO SATURABLE REACTOR - TRANSFORMER MODULE (LIKE 68B995018)

A. CARD TERM 5 TO B1
B. CARD TERM 3 TO A1
C. CARD TERM 9 TO B2
D. CARD TERM 7 TO A2
E. CARD TERM 1 TO CT

PFF (31), NFF (35)

A. CAN ACCEPT ONLY NAND TYPE LOADS (CURRENT INTO Q7 OR Q6) CONNECTED TO -12V (LIKE THE CLEAR TERMINALS BELOW).
B. MAX. LOAD 2 MA.

PFFC (39), NFFC (47)

A. WILL CLEAR RESPECTIVE FLIP FLOP WHEN BROUGHT WITHIN 1 VOLT OF -12V.
B. MIN. 4 VOLTS TO NOT CLEAR FF.
C. NAND TYPE LOAD OF 1.5 MA.

OSC (25)

A. MUST BE WITHIN ONE VOLT OF -12V (A ZERO) TO PROVIDE OUTPUT PULSE.
B. MUST BE 6 VOLTS ABOVE -12V (A ONE) TO STOP OUTPUT PULSE.
C. RECOMMEND 25 MICROSECONDS AT ZERO, 50 MICROSECONDS AT ONE.
D. 4K LOAD, 1.5 MA.

OUTPUTS: PA (41-, 49+) POSITIVE CELLS; PB (37-, 49+) NEGATIVE CELLS

A. NOMINAL 25 V PULSE, VARIES WITH AC VOLTAGES.
B. PEAK CURRENT 500 MA.
C. CONTINUOUS RATING 200 MA FOR 25 MICROSECONDS ON, 50 MICROSECONDS OFF CYCLE.
D. EACH WILL DRIVE TWO GATE CIRCUIT CARDS DESIGNED FOR IC3605 TYPE SCR PACKAGES.

PUL (27)

A. 3V MINIMUM PULSE, 50 MICROSECOND WIDTH AT 1.5V LEVEL.
B. WILL DRIVE 1K LOAD CONNECTED FROM PIN 27 TO -12V.

AC SUPPLY (AC1, 33) (AC2, 43)

A. CENTERTAPPED 24 VOLT SUPPLY WITH CENTERTAP TIED TO COM.
B. MUST BE SYNCHRONOUS WITH SCR SUPPLY AND AC1 IS POSITIVE WHEN THERE IS FORWARD VOLTAGE ON A POSITIVE SCR CELL.
C. 8 MA RMS LOAD ON EACH HALF.

-12V BUS (29)

A. 10 TO 12V VARIABLE BUT WITH $\pm$.1V STABILITY TO MEET ABOVE SPECS.
B. AT 8V, CARD WILL STILL SUPPLY AND PHASE SHIFT PULSES, BUT WITH CONSIDERABLE DRIFT AND NO PHASE RETARD LIMIT.
C. LOAD 60 MA WHEN NOT SUPPLYING PULSES.
90 MA SUPPLYING ONE GATE CIRCUIT.
120 MA SUPPLYING TWO GATE CIRCUITS.

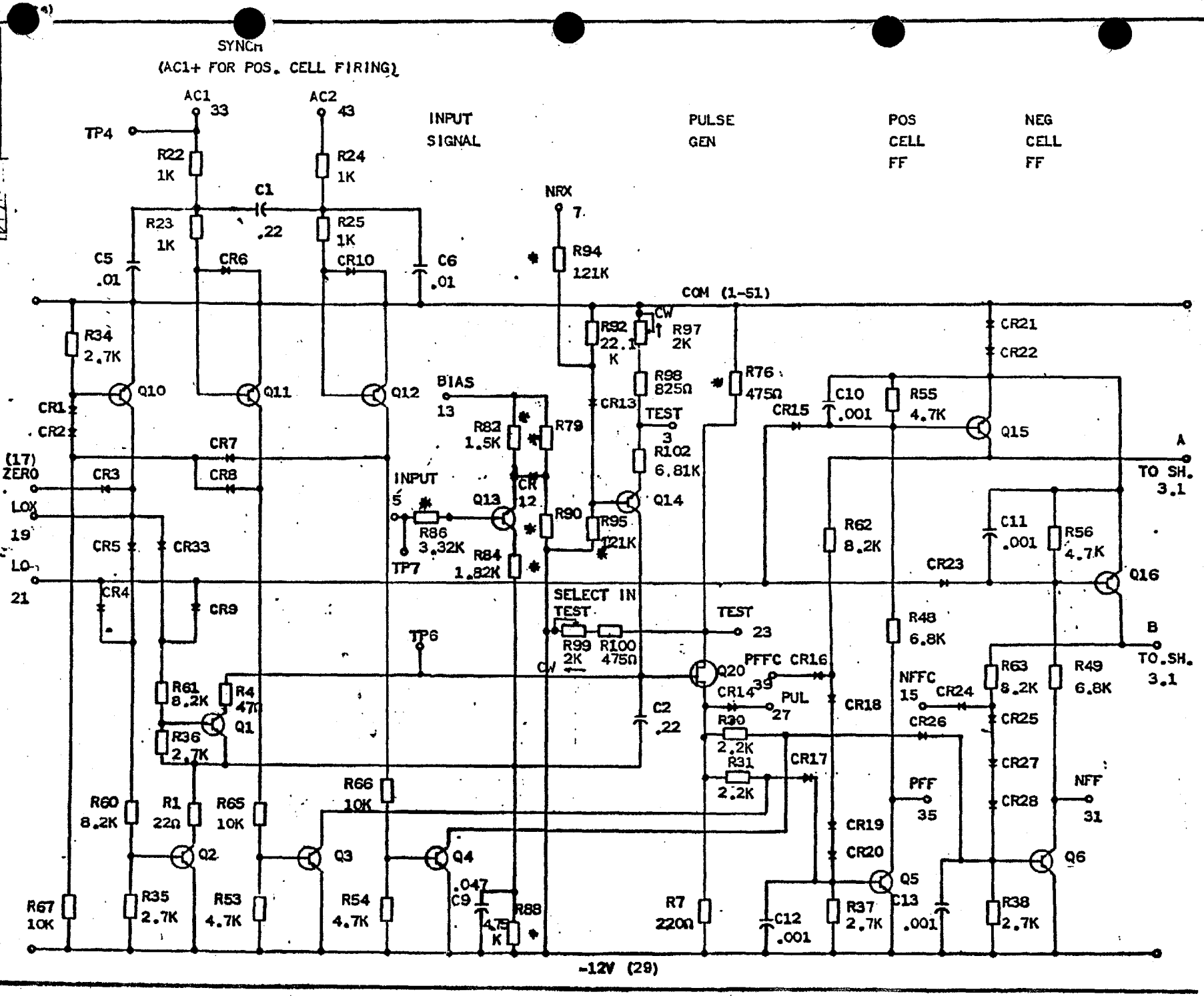
+16V BUS (45)

A. MAX. 20V, MIN. 12V, WITH ONE VOLT MAX. RIPPLE.
B. LOAD 30 MA SUPPLYING ONE GATE CIRCUIT
60 MA SUPPLYING TWO GATE CIRCUITS
0 WHEN NOT SUPPLYING PULSES.

FORM IC 2 (.66)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 2	APPROVALS B.P. Chausse	GENERAL ELECTRIC	ELEMENTARY DIAGRAM FIRING CIRCUIT	
ISSUED November 21, 1966	PRINTS TO	FIRST MADE FOR REQ.	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.		
MADE BY J. H. SMITH		I.C. NO.		IC3600TFCA1 FL CONT. ON SH.	SH. NO. 3.3



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

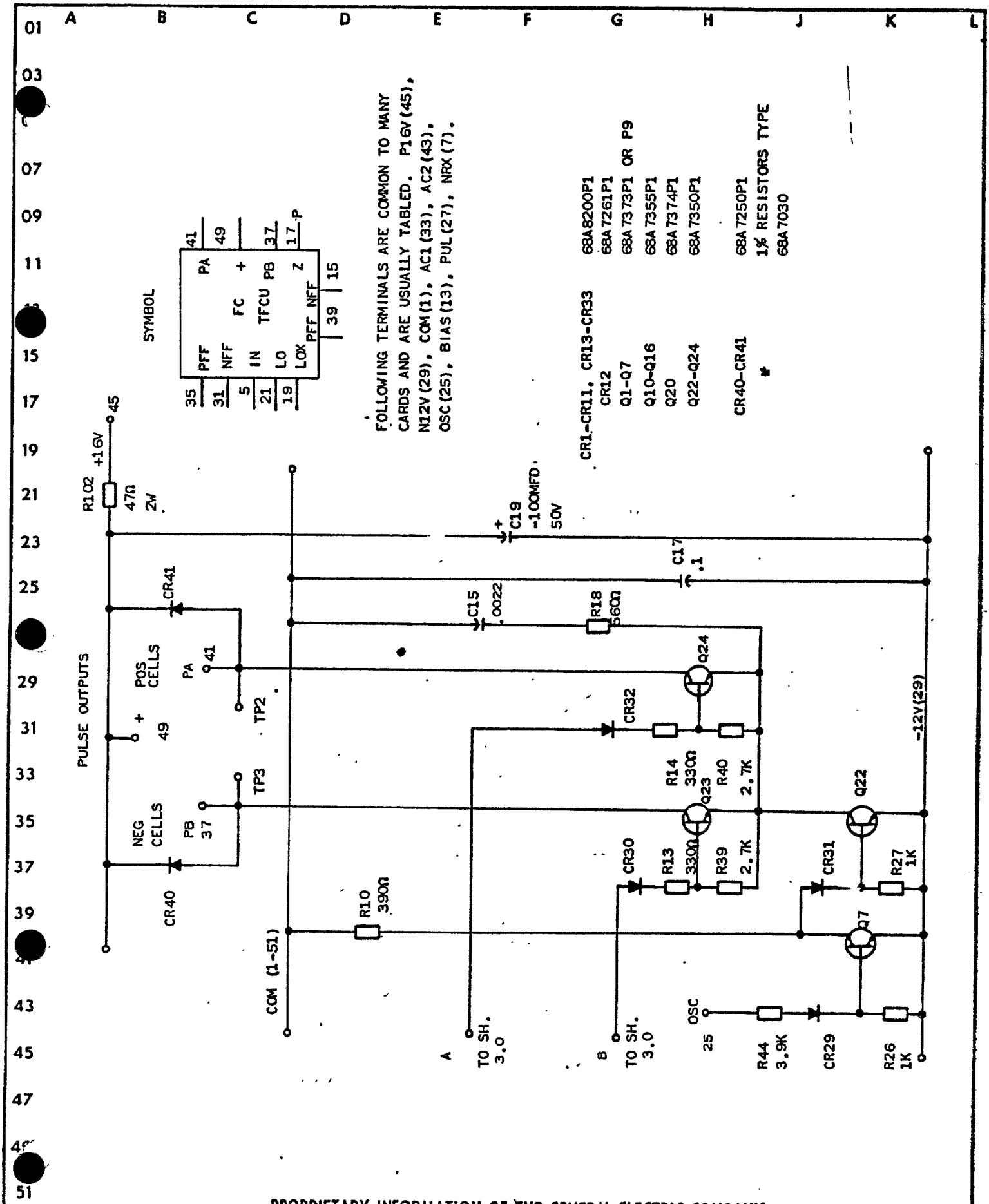
GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
SCR FIRING CIRCUIT
(UNIJUNCTION)

IC3600TFCU1
3.1

3.0

DRAWN BY R.E.HANNAH
ISSUED 7/6/70
REV. 1 2/24/71
REV. 2 JAE 3/22/67 DLL
PRINTS TO 1338
P6A
APPROVALS
STANDARD LINE



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1 12-8-66 RCH	REV 4	REV 5	PRINTS TO DL1	APPROVED BC Mauser	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV 2 8-21-70 ACW REDRAWN ORIGINAL 4-65	ISSUED AUGUST 27, 1970		1338	FIRST MADE FOR STAND. LINE	ELEMENTARY DIAGRAM SCR FIRING CIRCUIT (UNIJUNCTION)
REV 3	MADE BY R. E. HANNAH		PGA	I.C. NO	IC3600TFCU1 3.1
					CONT. ON SH. 3.2 SH. NO. 3.1

THIS CARD SUPPLIES PHASE CONTROLLED PULSES THAT ARE USED TO FIRE THYRISTORS (SCR'S) CONNECTED TO AC POWER LINES. THERE ARE MANY FEATURES THAT ARE PARTICULARLY DESIRABLE WHEN THE SCR'S ARE CONNECTED FOR DC OUTPUT INTO INDUCTIVE LOADS.

THE BASIC TIMING DEVICE USED FOR PHASE CONTROL IS THE UNIJUNCTION TRANSISTOR Q20, WHICH OPERATES AS A RELAXATION OSCILLATOR WHEN THE INPUT VOLTAGE IS ZERO. THE CURRENT SOURCE FROM Q14 CHARGES C2 UNTIL TP6 REACHES 7 VOLTS. AT THIS POINT THE LOWER HALF OF Q20 SWITCHES TO A LOW RESISTANCE AND C2 AND C9 DISCHARGE INTO R7 PROVIDING A THREE VOLT (APPROX.) OUTPUT PULSE AT PUL (27). AFTER THE DISCHARGE Q20 RECOVERS TO ITS HIGH RESISTANCE STATE AND THE TIMING WILL RESTART.

TO SYNCHRONIZE THE TIMING, A CENTERTAPPED AC SUPPLY (AC1, COM, AC2) MUST BE DERIVED FROM THE POWER LINES FEEDING THE SCRs. THIS AC IS LAGGED 8 DEGREES BY C1 TO FILTER NOISE AND THEN CONTROLS Q11 AND Q12. SHORTLY AFTER A VOLTAGE ZERO, BOTH Q11 AND Q12 ARE OFF WHICH SATURATES Q10 PROVIDING A 300μs. PULSE AT ZERO (17). THIS PULSE ALSO SATURATES Q1 AND Q2 TO DISCHARGE C2 AND C9. RELEASE OF Q1 AND Q2 AT THE END OF ZERO INITIATES THE TIMING CYCLE OF Q20 AND Q20 TIMES ON EACH HALF CYCLE.

PHASE CONTROL IS OBTAINED FROM AN INPUT SIGNAL BETWEEN INPUT (5) AND COM. INITIALLY ASSUME BIAS (13) IS CONNECTED TO COM. AS Q13 CIRCUIT IS AN EMITTER FOLLOWER, THE COLLECTOR CURRENT IS PROPORTIONAL TO THE INPUT VOLTAGE. THIS CAUSES THE VOLTAGE ON R88 TO BE PROPORTIONAL TO THE INPUT VOLTAGE (DELAYED LESS THAN 1/2 MILLISECOND BY C9). AS R88 VOLTAGE IS IN SERIES WITH C2, IT CAN IMMEDIATELY FIRE Q20 WITHOUT WAITING TO CHARGE C2, GIVING VERY FAST TRANSIENT RESPONSE. A LINEAR CHANGE ON R88 CAUSES A LINEAR CHANGE IN FIRING ANGLE WHICH PROVIDES LINEAR SCR OUTPUT OVER 2/3 OF THE RANGE. FUNCTION GENERATOR R79, CR12 IMPROVES LINEARITY AT FULL OUTPUT.

THIS CARD PROVIDES A FIXED PHASE RETARD LIMIT AT 25 DEGREES ADVANCE. THIS REQUIRES A FIXED TIMING CHARACTERISTIC ON C2. AS R88 VOLTAGE MAY JUMP AROUND DUE TO RIPPLE ON THE INPUT, CURRENT REGULATOR Q14 IS NECESSARY TO PROVIDE CONSTANT CHARGING ON C2.

TEST SELECTS FOUR RESISTORS SO EACH CARD WILL TRACK A MASTER WITHIN 2 DEGREES. IN EFFECT THEY SELECT R99 AND 100 (APPROXIMATELY 1500 OHMS) SO TP6 IS 7 VOLTS ABOVE -12V WHEN Q20 FIRES, AND R97, R98 (APPROXIMATELY 2K) SO THAT A PULSE IS GENERATED AT 25 DEGREES ADVANCE. NRX (7) MAY BE TIED TO COM SO Q20 WILL NOT FIRE AT THE END OF A HALF CYCLE.

BIAS (13) IS USED TO SET THE SCR OUTPUT WITH ZERO INPUT. NORMAL RANGE IS FROM .5 TO 1.5 VOLTS POSITIVE WITH RESPECT TO COM.

Q20 OUTPUT PULSE IS FED TO TWO FLIP FLOPS, Q5-Q15 AND Q6-Q16. IF AC1 IS POSITIVE, THEN AC2 IS NEGATIVE, SATURATING Q12 AND Q4. Q4 SHORTS THE SIGNAL TO THE NEGATIVE FF, SO ON THIS HALF CYCLE THE POSITIVE FF IS SET.

THE FLIP FLOPS MAY BE INDIVIDUALLY CLEARED BY CLEAR TERMINALS 15 AND 39, OR CLEARED TOGETHER BY LOCKOUT, LO (21), BEING TIED TO COM.

IF THE FLIP FLOPS ARE CROSS CONNECTED, THEN ONE WILL RESET THE OTHER, AND VICE VERSA, WITH THE RESULT THAT EACH STAYS SET 180 DEGREES. IN 3 PHASE SYSTEMS CLEAR WILL COME FROM A FF ON ANOTHER PHASE SO THAT EACH FF IS SET 120 DEGREES. FOR NON-PLUGGING APPLICATIONS THE FFs MAY BE CLEARED AT EACH VOLTAGE ZERO BY TYING LOX (19) TO LO.

THE FFs CONTROL POWER TRANSISTORS Q23 AND Q24 TO STEER THE PULSES TO THE CORRECT SCR. IN SERIES WITH THESE IS Q22 WHICH IS DRIVEN BY Q7 AND AN EXTERNAL OSCILLATOR, RUNNING AT APPROXIMATELY 25 MICROSECONDS ON AND 50 MICROSECONDS OFF.

IN THE CASE WHERE EACH FF IS SET 180 DEGREES, PULSES ARE CONTINUOUSLY GOING TO SOME SCR CELL, THE FFs MERELY DETERMINE AT WHAT POINT IN THE CYCLE THE PULSE SWITCH FROM THE POSITIVE CELLS TO THE NEGATIVE CELLS OR VICE VERSA.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

THE POWER TRANSISTORS ARE RATED ON THE BASIS THAT EACH CAN DRIVE TWO SCRs IF AN RC CIRCUIT IS USED IN SERIES WITH A PULSE TRANSFORMER. THEN THE TRANSISTORS DELIVER ONE LARGE PULSE FOLLOWED BY SMALLER PULSES AS THE CAPACITOR CHARGES UP.

APPLICATION DATA

INPUT (COM) TO (5)

A. 2 VOLT CHANGE TO SWING PULSES 160 DEGREES. MAY BE 0 TO 2 OR ±1 VOLT DEPENDING ON BIAS VOLTAGE.

B. TERM 5 MINUS WITH RESPECT TO COM TO ADVANCE FIRING PULSES.

C. CURRENT 30 MICRO AMPS UNTIL SATURATION, THEN 3.4K LOAD, IF TERM 5 SWUNG WITH RESPECT TO GND AS RECOMMENDED.

D. CURRENT MAY BE 2MA TO SWING COM, THE POWER SUPPLY, ETC. WITH RESPECT TO GND. (NOT RECOMMENDED).

E. SYNCHRONOUS RIPPLE OF 2 VOLTS AS OBTAINED BY VOLTAGE FEEDBACK IS OK. CARD WILL FIRE AT THE PEAKS OF THE RIPPLE.

F. 60 CY OR ANY OTHER UNBALANCING RIPPLE MUST BE UNDER 0.1 VOLTS PEAK TO PEAK.

BIAS (13)

A. APPROXIMATELY .5V POSITIVE FOR ZERO ADVANCE; APPROXIMATELY 1.5V POSITIVE FOR 90 DEGREE ADVANCE; AT ZERO INPUT SIGNAL.

B. BIAS MUST CHANGE 3MV/°C (MORE NEGATIVE AT HIGH TEMP.) FOR ±2 DEGREE TEMPERATURE STABILITY.

C. ABOVE CONDITIONS ARE MET WITH BIAS SUPPLY ON IC3600TPSA CARD.

D. 1.5 MA LOAD.

LOCKOUT (LO) (21)

A. TIE TO COM WITH PNP TRANSISTOR SWITCH TO STOP ALL PULSES.

B. MAXIMUM .8V DROP FOR LOCKOUT, 12 VOLTS NOT LOCKED OUT.

C. LOAD 7MA.

LOX (19)

A. TIE TO LO TO RESET FLIP FLOPS AT EACH VOLTAGE ZERO.

B. THIS IS DESIRABLE ON NON-PLUGGING SCR PACKAGES.

NRX (7)

A. TIE TO COM TO CHANGE PHASE RETARD LIMIT FROM 25 DEGREES TO 0°.

B. THIS IS DESIRABLE ON NON-PLUGGING SCR PACKAGES.

ZERO (17)

A. CAN DRIVE RESISTOR LOAD OF 10K CONNECTED FROM PIN 17 TO -12V.

B. VOLTAGE WILL BE WITHIN 1 VOLT OF COM NEAR EACH VOLTAGE ZERO.

C. WIDTH APPROXIMATELY 300 MS.

PFF (35), NFF (31)

A. CAN ACCEPT ONLY NAND TYPE LOADS (CURRENT INTO Q5 OR Q6) CONNECTED TO -12V (LIKE THE CLEAR TERMINALS BELOW).

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

B. MAXIMUM LOAD 2 MA.

PFFC (39), NFFC (15)

A. WILL CLEAR RESPECTIVE FLIP FLOP WHEN BROUGHT WITHIN 1 VOLT OF -12V.

B. MINIMUM 4 VOLTS TO NOT CLEAR FF.

C. NAND TYPE LOAD OF 1.5 MA.

OS (25)

A. MUST BE WITHIN ONE VOLT OF -12V (A ZERO) TO PROVIDE OUTPUT PULSE.

B. MUST BE 6 VOLTS ABOVE -12V (A ONE) TO STOP OUTPUT PULSE.

C. RECOMMEND 25 MICROSECONDS AT ZERO, 50 MICROSECONDS AT ONE.

D. 4K LOAD, 1.5 MA.

OUTPUTS: PA (41-, 49+); PB (37-, 49+)

A. NOMINAL 25V PULSE, VARIES WITH AC VOLTAGE.

B. PEAK CURRENT 500 MA.

C. CONTINUOUS RATING, 200 MA FOR 25 MICROSECONDS ON, 50 MICROSECONDS OFF CYCLE.

D. EACH WILL DRIVE TWO GATE CIRCUIT CARDS DESIGNED FOR IC3605 TYPE SCR PACKAGES.

PUL (27)

A. 3.5V MINIMUM PULSE, 15 MICROSECOND WIDTH AT 1.5V LEVEL.

B. WILL DRIVE 1K LOAD CONNECTED FROM PIN 27 TO -12V.

AC SUPPLY (AC1, 33) (AC2, 43)

A. CENTERTAPPED 24 VOLT SUPPLY WITH CENTAP TIED TO COM.

B. MUST BE SYNCHRONOUS WITH SCR SUPPLY AND AC1 IS POSITIVE WHEN THERE IS FORWARD VOLTAGE ON A POSITIVE SCR CELL.

C. 8 MA RMS LOAD ON EACH HALF.

-12V BUS (29)

A. 12V ±.2V TO MEET ABOVE SPECS.

B. AT 8V CARD WILL STILL SUPPLY AND PHASE SHIFT PULSES, BUT WITH CONSIDERABLE DRIFT AND NO PHASE RETARD LIMIT.

C. LOAD 60 MA IN LOCKOUT.
90 MA SUPPLYING ONE GATE CIRCUIT.
120 MA SUPPLYING TWO GATE CIRCUITS.

+16V BUS (45)

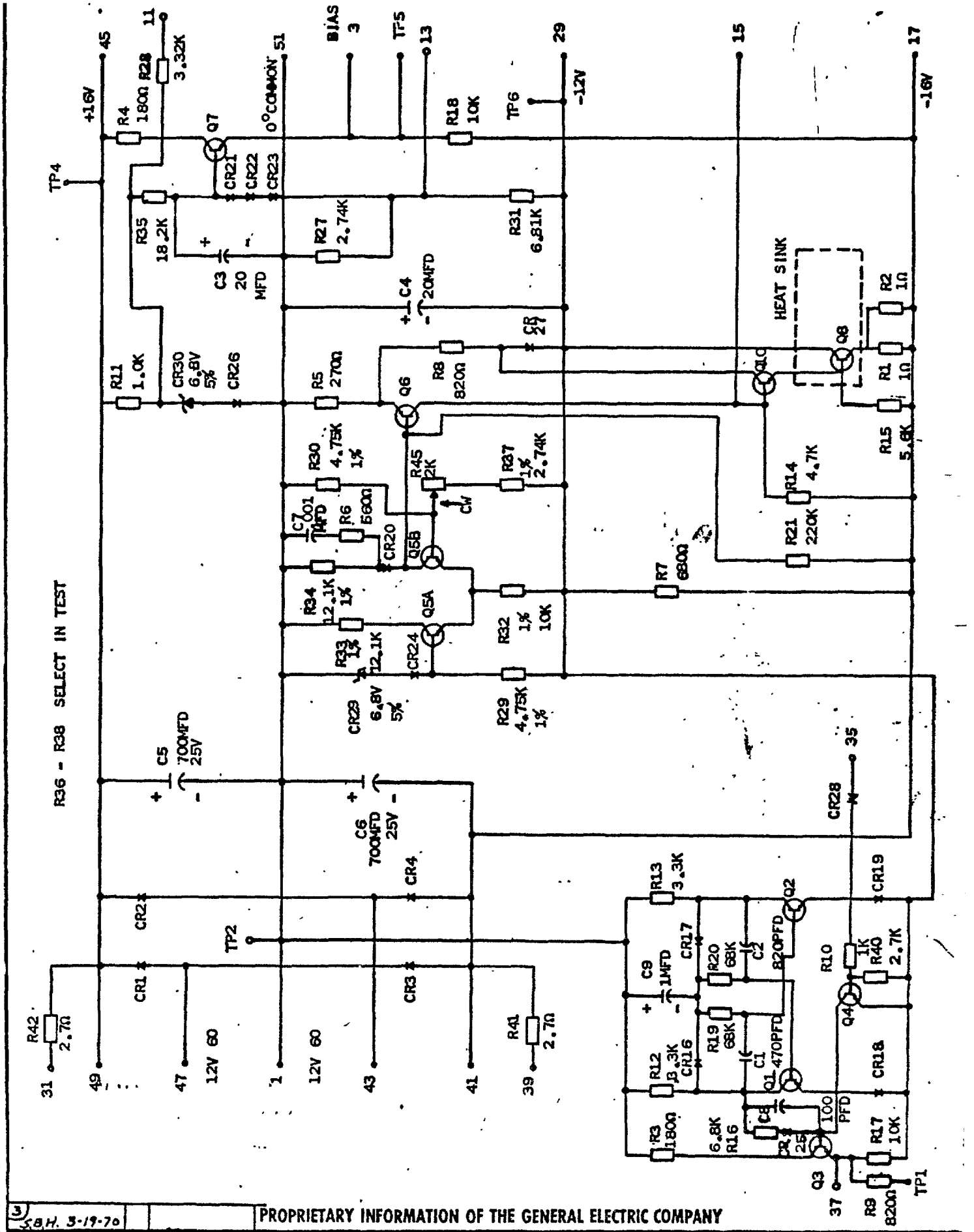
A. MAXIMUM 20V, MINIMUM 12V, WITH ONE VOLT MAXIMUM RIPPLE.

B. LOAD 30MA SUPPLYING ONE GATE CIRCUIT.
60MA SUPPLYING TWO GATE CIRCUITS
0 IN LOCKOUT.

TRACKING:

ONE CARD WILL TRACK ANOTHER CARD WITHIN 5 DEGREES THROUGHOUT PHASE SHIFT RANGE WHEN CONNECTED TO COMMON POWER SUPPLIES.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY



3 S.B.H. 3-19-70

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. B 12/22/66 VDH; C 4/12/67 EC 13908; D 10/9/68 DL1

ISSUED December 21, 1966

MADE BY V.M. STOCKTON

PRINTS TO 1338

P6A

APPROVALS J. B. Jamison

FIRST MADE FOR REQ. STANDARD LINE

I.C. NO.

GENERAL ELECTRIC

INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
1Ø POWER SUPPLY

IC3600TPSA1

CONT. DWG. SH. 3.1

SH. NO. 3.0

CHARACTERISTICS

1. POWER INPUT - 24V CENTER-TAPPED
   50/60 CYCLES, .6 AMPS RMS MAX.

2. OUTPUT - +16V UNREGULATED, .2AMPS
   -16V UNREGULATED,
   -12V REGULATED, .2 AMPS MAX.

3. OUTPUT IMPEDANCE - LESS THAN .4 OHMS
   (REGULATED)

4. TEMP. DRIFT - APPROX. 1 MILLIVOLT/°C
   (REGULATED OUTPUT)

APPLICATION DATA

THIS CARD PROVIDES THE POWER SUPPLY FOR AN SCR EXCITER SYSTEM. IT ALSO PROVIDES A BIAS CIRCUIT WHICH CAN BE USED TO PROVIDE AN OFFSET OR A ZEROING SIGNAL TO THE FIRING CIRCUIT INPUT, AND A MULTIVIBRATOR WHICH IS USED TO GATE THE OUTPUT OF A FIRING CIRCUIT CARD TO PROVIDE A TRAIN OF PULSES.

THE BIAS CIRCUIT IS DESIGNED TO HAVE A NEGATIVE DRIFT OF APPROXIMATELY 3 MILLIVOLTS PER °C. THIS VERY NEARLY COMPENSATES THE DRIFT OF THE INPUT TRANSISTOR AND LINEARIZING NETWORK OF THE FIRING CIRCUIT CARD IC3600TFCU. THE BIAS CIRCUIT HAS AN EMITTER FOLLOWER OUTPUT STAGE WHICH IS CAPABLE OF SUPPLYING A 9 MILLIAMP LOAD.

THE MULTIVIBRATOR HAS AN OPERATING FREQUENCY OF APPROXIMATELY 15KHZ AND CAN DRIVE A 680 OHM LOAD CONNECTED FROM OUTPUT TO THE -12V BUS. THE OUTPUT SWITCHES FROM A NOMINAL -6V OUTPUT TO A -12V OUTPUT.

THEORY OF OPERATION

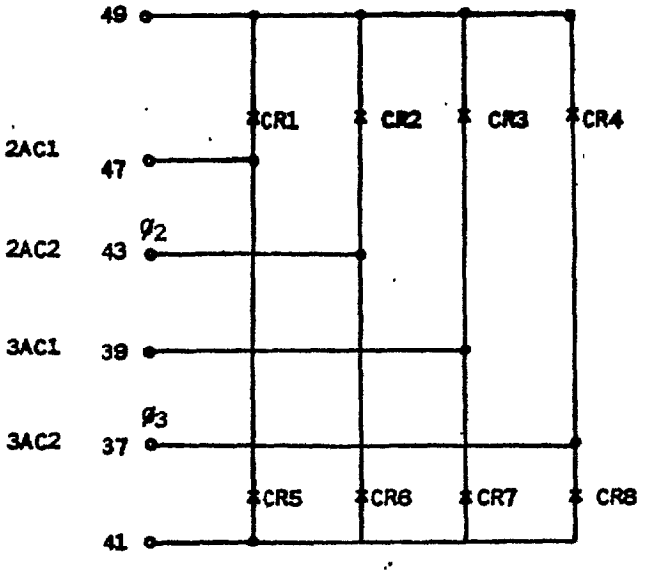
A. RECTIFIER AND FILTER SECTION
A.C. POWER FROM THE PHASING TRANSFORMER IS SUPPLIED TO PINS 47 AND 43 OF THE CARD. THE CENTER TAP OF THE TRANSFORMER SECONDARY IS CONNECTED TO PIN 1. THE A.C. INPUT IS RECTIFIED BY DIODES CR1, 2, 3, AND 4. FILTERING IS PROVIDED BY CAPACITORS C5 AND C6. THE NOMINAL DC OUTPUT IS +16 VOLTS AT PIN 45 WITH RESPECT TO PIN 1 AND -16 VOLTS AT PIN 17 WITH RESPECT TO PIN 1.

B. VOLTAGE REGULATOR
TRANSISTORS Q5A, Q5B, Q6, Q8, AND Q10 ARE THE AMPLIFIERS USED IN THE VOLTAGE REGULATOR. POWER INPUT IS FROM THE UNREGULATED -16 VOLT BUS. RESISTOR R7 IS USED TO START THE CIRCUIT BY PROVIDING ENOUGH POWER FOR THE TRANSISTORS TO BEGIN REGULATING ACTION. TRANSISTORS Q5A AND Q5B ARE CONNECTED IN A COMPARATOR CIRCUIT TO COMPARE THE VOLTAGE OF REFERENCE ZENER CR29 AND COMPENSATING DIODE CR24 WITH THE VOLTAGE DROP ACROSS RESISTOR R30. THE VOLTAGE DROP ACROSS R30 IS PROPORTIONAL TO THE VOLTAGE FROM PIN 1 TO PIN 29. THE PROPORTIONALITY FACTOR IS ADJUSTABLE BY TRIMPOT R45, AND IS SET SO THAT -12 VOLTS AT PIN 29 CAUSES THE CORRECT VOLTAGE DROP ACROSS R30. IF THE VOLTAGE COMPARATOR DETECTS AN ERROR VOLTAGE, TRANSISTORS Q6 AND Q10 WILL AMPLIFY THE DETECTED ERROR AND CAUSE AN INCREASED OR DECREASED AMOUNT OF CURRENT FLOW THROUGH Q8 AS REQUIRED TO RESTORE THE VOLTAGE AT PIN 29 TO -12 VOLTS.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. B 12/22/66 DMS	REV. 2 7-14-70 DLI	APPROVALS S.B. Ingrouse	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
			INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	1Ø POWER SUPPLY	
ISSUED 11/21/66	PRINTS TO 1338	FIRST MADE FOR G.O. STANDARD LINE		IC3600TPSA1	SH. NO. 3.1
MADE BY V.M. STOCKTON	PGA	I.C. NO.		CONT. ON SH. FL.	

FORM IC 2219 (1)

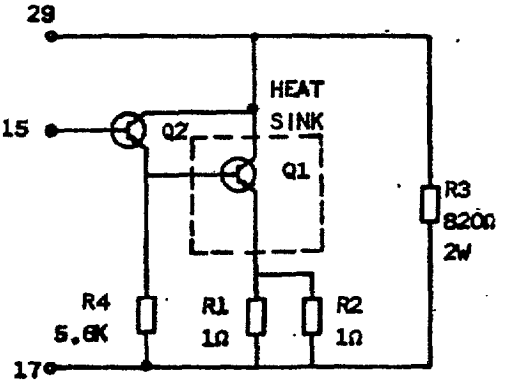


Q1 - 68A7712P1
WITH HEAT SINK
SIMILAR TO 68A997024P1

CR1-8 68A7250P1

Q2 68A7373

NOTE NEW HEAT SINK MUST FIT ON TPSA PRINTED CIRCUIT



APPLICATION DATA

THIS CARD PROVIDES TWO SINGLE PHASE FULL WAVE RECTIFIER BRIDGES. THESE ARE NORMALLY USED WITH THE SINGLE PHASE BRIDGE ON CARD IC3600TPSA TO PROVIDE A 6 PHASE RECTIFIER.

ALSO A DARLINGTON POWER AMPLIFIER STAGE IS PROVIDED WHICH CAN BE CONNECTED IN PARALLEL WITH THE POWER OUTPUT STAGE OF IC3600TPSA. THE COMBINED CARDS WILL PROVIDE UP TO 600 MA OF REGULATED OUTPUT.

FORM IC 2219 (6-66)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. R.J.F. 10/14/68	REV. 2 DL1	APPROVALS M. Garrison	GENERAL ELECTRIC	ELEMENTARY DIAGRAM POWER SUPPLY 3Ø
ISSUED December 11, 1966	PRINTS TO 1338	FIRST MADE FOR MDL STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600TPS81 SH. NO. 3.0
MADE BY R.G. MUCKENFUSS	P6A	LC NO.		

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM
1Ø POWER SUPPLY

IC3600TPSC1
3.1

3.0

R36 - R38 SELECT IN TEST

TP4
+16V
45

49
CR1
CR2
C5
700MFD
25V
47
12V 60
TP2
1
12V 60
0°COMMON
51
43
C6
700MFD
25V
CR3
CR4
41

CR29
6.8V
5%
R33
1%
12.1K
R34
12.1K
1%
C7
.001
MFD
R6
560Ω
CR20
R30
4.75K
1%
R5
270Ω
CR24
Q5A
Q5B
Q6
R45
2K
+ C4
- 20 MFD
R8
820Ω

R29
4.75K
1%
R32
1%
10K
R37
1%
2.74K
CR
27
TP6
29
-12V

R3
180Ω
R12
3.3K
CR16
C9
1MFD
CR17
R13
3.3K
R7
680Ω
6.8K
R16
R19
68K
R20
68K
C8
C1
C2
CR
25
Q1
470PFD
820PFD
Q2
15
Q10
HEAT SINK
R21
220K
R14
4.7K
Q3
37
100
PFD
R10
Q4
1K
R40
2.7K
CR18
CR19
CR28
35
Q8
R9
820Ω
R17
10K
TP1
R15
5.6K
R1
1Ω
R2
1Ω
17
-16V

CHARACTERISTICS

1. POWER INPUT - 24V CENTER-TAPPED
   50/60 CYCLES, .6 AMP RMS. MAX.

2. OUTPUT - +16V UNREGULATED, .2 AMPS
   -16V UNREGULATED, .2 AMPS
   -12V REGULATED, .2 AMPS MAX.
   OUTPUT ADJUSTABLE FROM 10.5 TO 14V BY TRIMPOT R40

3. OUTPUT IMPEDANCE - LESS THAN .4 OHMS
   (REGULATED)

4. TEMP. DRIFT - APPROX. 1MV/°C
   (REGULATED OUTPUT)

APPLICATION DATA

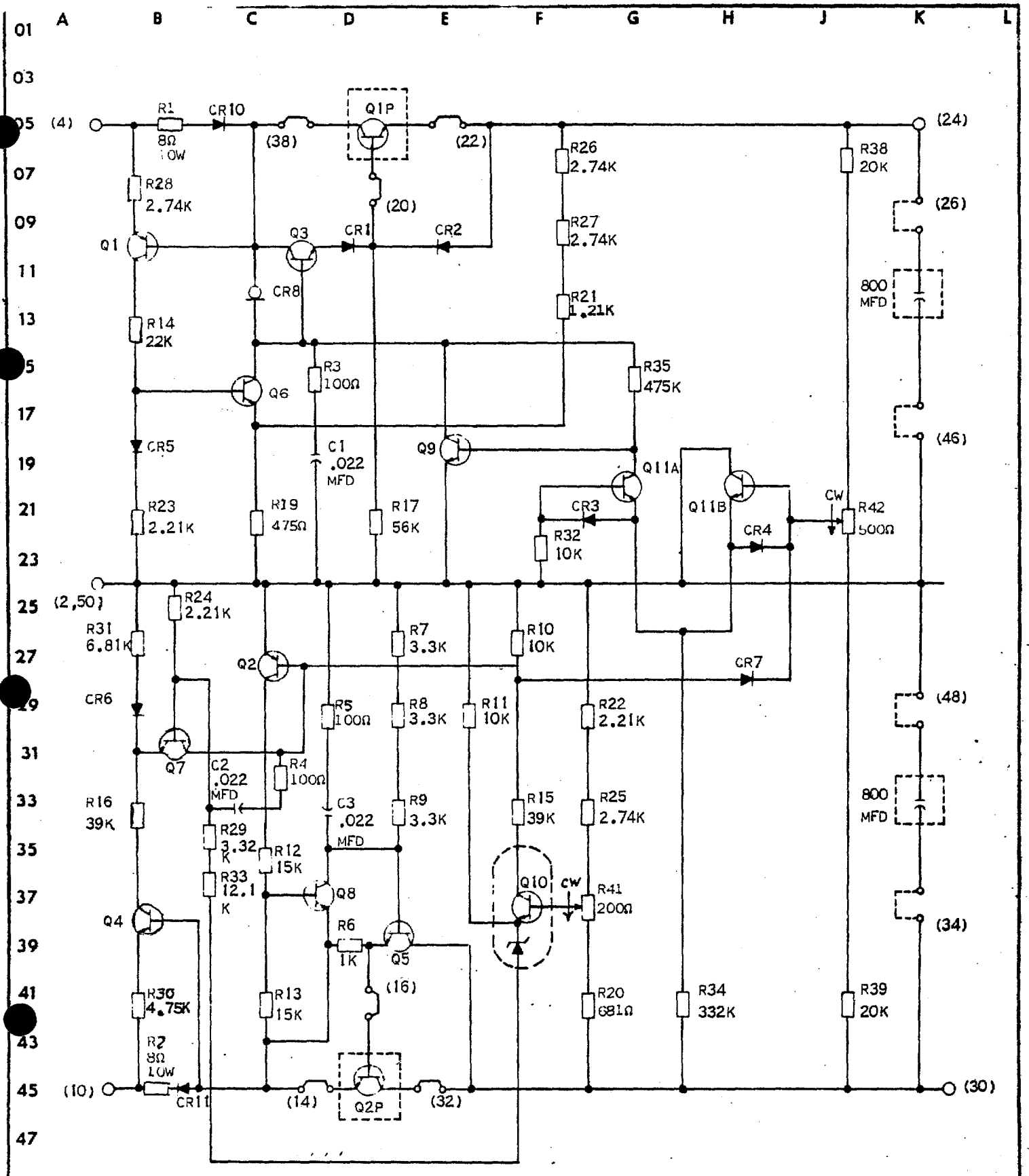
THIS CARD IS NORMALLY USED TO PROVIDE THE DC POWER SUPPLY FOR AN SCR EXCITER WITH AN AMPLISTAT FIRING CIRCUIT. THE AMPLISTAT FIRING CIRCUIT CONSISTS OF AMPLISTAT ASSEMBLY 68B995018 AND PRINTED CIRCUIT BOARD IC3600TFCA.

A MULTIVIBRATOR IS ALSO PROVIDED ON THE CARD. THE OUTPUT OF THE MULTIVIBRATOR IS USED TO GATE THE FIRING CIRCUIT CARD SO A TRAIN OF PULSES WILL BE PROVIDED TO THE SCR GATES. THE MULTIVIBRATOR CAN DRIVE A 680 OHM LOAD CONNECTED FROM OUTPUT TO THE -12V BUS. THE OUTPUT VOLTAGE SWITCHES FROM A NOMINAL -6V OUTPUT TO A -12V OUTPUT.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FORM IC 2219

REV. B 12/22/66 DMS	REV. 2 DLI	APPROVALS E. B. Jamison	GENERAL ELECTRIC	ELEMENTARY DIAGRAM 1Ø POWER SUPPLY
ISSUED November 21, 1966	PRINTS TO 1338	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM VA. U.S.A.	IC3600TPSC1 SH. NO. 34
MADE BY V. M. STOCKTON	PGA	IT NO.		



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

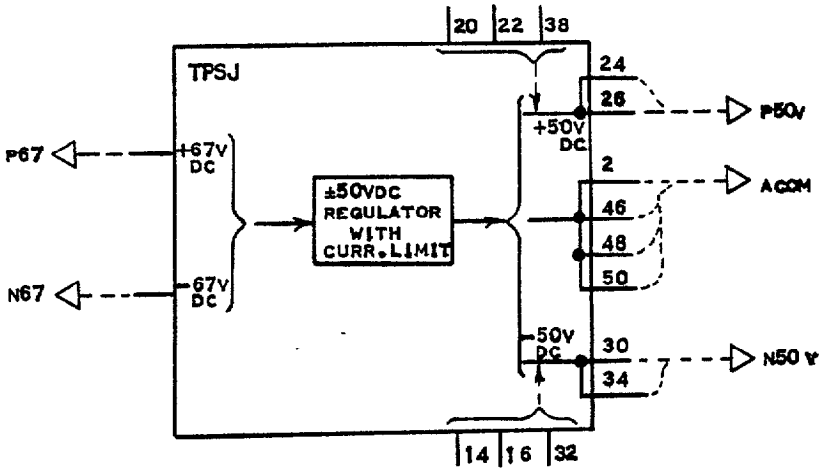
REV. 1	REV. 4	REV. 5	PRINTS TO 2N DL1	APPROVALS	
REV. 2	ISSUED July 14, 69		P6A	FIRST MADE FOR SILTROL	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 3	MADE BY R. G. MUCKENFUSS		1338	I.C. NO.	

ELEMENTARY DIAGRAM ± 50V 0.4A POWER SUPPLY

IC3600TPSJ1

CONT. ON SH. 3.1 SH. NO. 3.0

SYMBOL



REV. 1	REV. 4	REV. 5	PRINTS TO 2N	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ±50V, 0.4A REGULATOR
REV. 2	ISSUED 7-16-70		DL1 P6A	FIRST MADE FOR SILTROL	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600TPSJ1
REV 3	MADE BY A.C.WRIGHT		1338	I.C. NO.		CONT. ON SH. 3.2 SH NO. 3.1

APPLICATION DATA

THIS CARD IS USED TO PROVIDE REGULATED ± VOLTS FROM AN UNREGULATED ±67 VOLT SOURCE. THIS IS A TRACKING TYPE REGULATOR IN THAT THE -50 VOLT REGULATOR IS THE MASTER AND ACTS AS A REFERENCE FOR THE +50 VOLT REGULATOR. CURRENT LIMIT IS PROVIDED ON BOTH OUTPUTS AND THE OUTPUTS WILL TEND TO TRACK (WITHIN APPROX. 5 VOLTS) IF EITHER OUTPUT IS OVERLOADED OR SHORTED TO COMMON.

FOR ELECTRIC POWER SHOVEL DUTY THIS SUPPLY IS POWERED FROM A ±78 VOLT SOURCE - THEREFORE IT IS NECESSARY TO DERATE OUTPUT CAPACITY TO 260 M.A.

ENVIROMENTAL

1. GOOD FOR -550° STORAGE
2. OPERATING TEMPERATURE 0 - 70°C
3. MEETS MIL-STD-167 VIBRATION SPEC.

SPECIFICATIONS (MODULE RATING)

OUTPUT CURRENT - 400 MILLIAMPS

LOAD REGULATION - .05%

TEMP. DRIFT - ±.15% FOR 0 TO 70°C

TRACKING ERROR - ±.05%

OUTPUT RIPPLE - 20 MV PEAK TO PEAK (MAX)

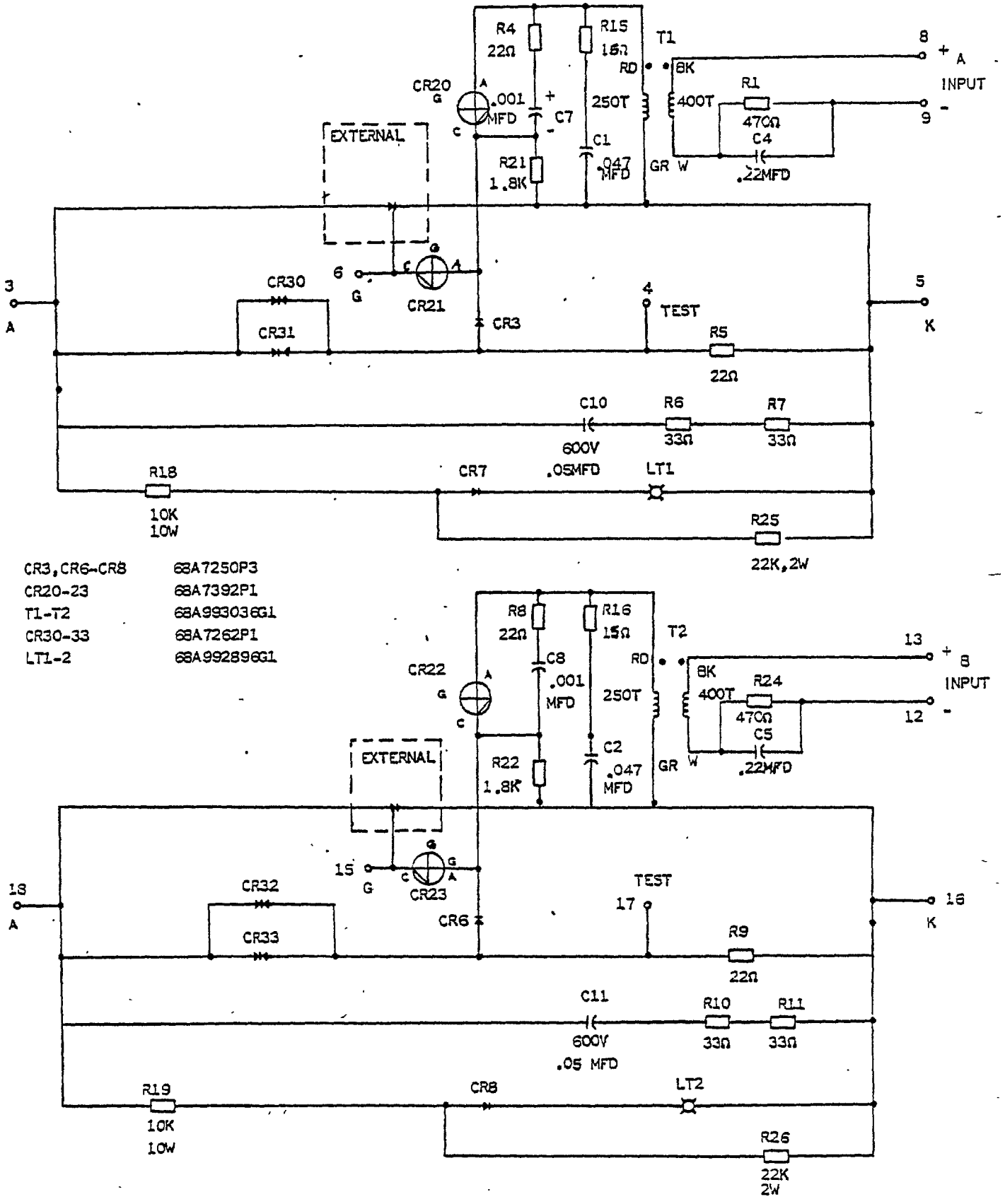
CURRENT LIMIT - 500 TO 600 MILLIAMPS

MOUNTING INFORMATION

THREE (3) FULL CARD SLOTS ARE REQUIRED FOR THIS POWER SUPPLY. STANDARD DOM II RECEPTACLE REQUIRED IN LEFT HAND SLOT (FRONT VIEW). RECEPTACLES MUST BE OMITTED IN ADJACENT RIGHT TWO SLOTS.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 CB 7/16/70	REV. 4	REV. 5	PRINTS TO 2N	APPROVALS	ELEMENTARY DIAGRAM ±50V, 0.4A REGULATOR
REV. 2 CEH 10-23-70	ISSUED Jan Eng 1/4/69 DL1 P6A		FIRST MADE FOR SILTROL	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600TPSJ1
REV. 3 ACW 7-22-71.	MADE BY RON MUCKENFUSS 1338		I.C. NO.		CONT. ON SH. FL. SH. NO. 3.2



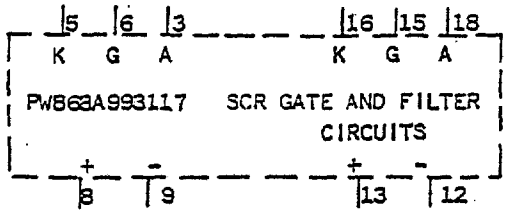
CR3, CR6-CR8	68A7250P3
CR20-23	68A7392P1
T1-T2	68A99306G1
CR30-33	68A7262P1
LT1-2	68A99289G1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 2 DL1	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SCR GATE AND FILTER 230V NRP C137	
ISSUED November 21-1966	PRINTS TO 1338	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	PWB68A993117	SH. NO. 3.0
MADE BY R.E.HANNAH	PRA	I.C. NO.		CONT ON SH. 3.1	

FORM IC 2219 (3-66)

SYMBOL



THIS GATE AND FILTER CIRCUIT CARD IS USED ON TYPE IC3605 SCR EXCITER PACKAGES. IT CONTAINS TWO IDENTICAL, BUT ELECTRICALLY INDEPENDENT GATE AND FILTER CIRCUITS.

IF 28V IS APPLIED TO INPUT TERMINALS 8 AND 9, THE VOLTAGE IS INITIALLY APPLIED TO T1 PRIMARY. C1 WILL CHARGE TO APPROXIMATELY 15V IN 2 TO 3 MICROSECONDS.

THIS IS THE NOMINAL BREAKDOWN VOLTAGE OF UNILATERAL SWITCHES CR20 AND 21 IN SERIES. THESE COLLAPSE AND T1 AND C1 DELIVER A .3 AMP PULSE TO THE GATE WITH A .2 MICROSECOND RISE TIME. AFTER 20 MICROSECONDS C4 STARTS TO REDUCE THE GATE CURRENT AND IT WILL BE MINIMIZED IN 100 MICROSECONDS. THE NOMINAL INPUT IS A 25 MICROSECOND PULSE WITH 50 MICROSECONDS OFF TIME, IN WHICH CASE C4 DOES NOT FULLY DISCHARGE AND THUS SUBSEQUENT PULSES HAVE SMALLER MAGNITUDES. ALSO, THE VOLTAGE ON C4 PROVIDES A RESET VOLTAGE FOR THE TRANSFORMER SO THAT A DIODE MAY BE USED ACROSS 8 AND 9 TO PREVENT VOLTAGE SPIKES.

IF THE PULSE INPUT VOLTAGE IS LOW DURING A LINE DIP, C7 WILL CHARGE UP AND FIRE CR20 FIRST AND THEN CR21 WILL FIRE PROVIDING HALF MANGITUDE GATE PULSES.

FILTER R6 AND R7 IN SERIES WITH C10 HAS BEEN SELECTED IN CONJUNCTION WITH EXTERNAL AIR CORE REACTORS IN THE SCR EXCITER TO HOLD VOLTAGE CHANGES TO UNDER 200 VOLTS PER MICROSECOND.

THYRECTORS CR30 AND 31 ARE ARRANGED TO ACCEPT APPROXIMATELY .4 AMP PULSES FROM AN INDUCTIVE LOAD. SOME ENERGY MUST BE ABSORBED AS SCR CURRENT FALLS BELOW ITS HOLDING CURRENT. ON HEAVY ABNORMAL CURRENTS THE THYRECTORS WILL FIRE THE SCR TO PROTECT THEMSELVES AND OTHER COMPONENTS.

THE LIGHTS ARE CONNECTED TO READ FORWARD CELL VOLTAGE. THUS THEY ARE BLACK AT 1/3 OUTPUT VOLTAGE AND ABOVE IN NORMAL OPERATION. HOWEVER, AN ABNORMAL UNBALANCE, OPEN OR SHORTED CELL WILL BE CLEARLY INDICATED BY UNBALANCED LIGHTS.

ISOLATION FROM THE SCR POWER SYSTEM IS PROVIDED BY THE PULSE TRANSFORMERS T_1 AND T_2.

APPLICATION DATA

GATE CIRCUIT PULSE TRANSFORMER INPUT (13+) (12-) OR GATE CIRCUIT INPUT (8+) (9-)

1. 24 TO 30 VOLT PULSES NOMINALLY 25 MICROSECONDS ON, 50 MICROSECONDS OFF.
2. WIDE (200 MICROSECOND) INPUT PULSES, CAUSED BY ADDING PULSES TO THE NORMAL PULSE TRAIN, ARE NOT HARMFUL. HOWEVER, OUTPUT GATE PULSE WILL BE 50 MICROSECONDS WIDE MAX.
3. PEAK CURRENT 250 MA.
4. DIODE CLAMP MAY BE USED ON INPUT TERMINALS IF DUTY CYCLE IS IN THE RANGE OF 25 MICROSECONDS ON AND 50 MICROSECONDS OFF.

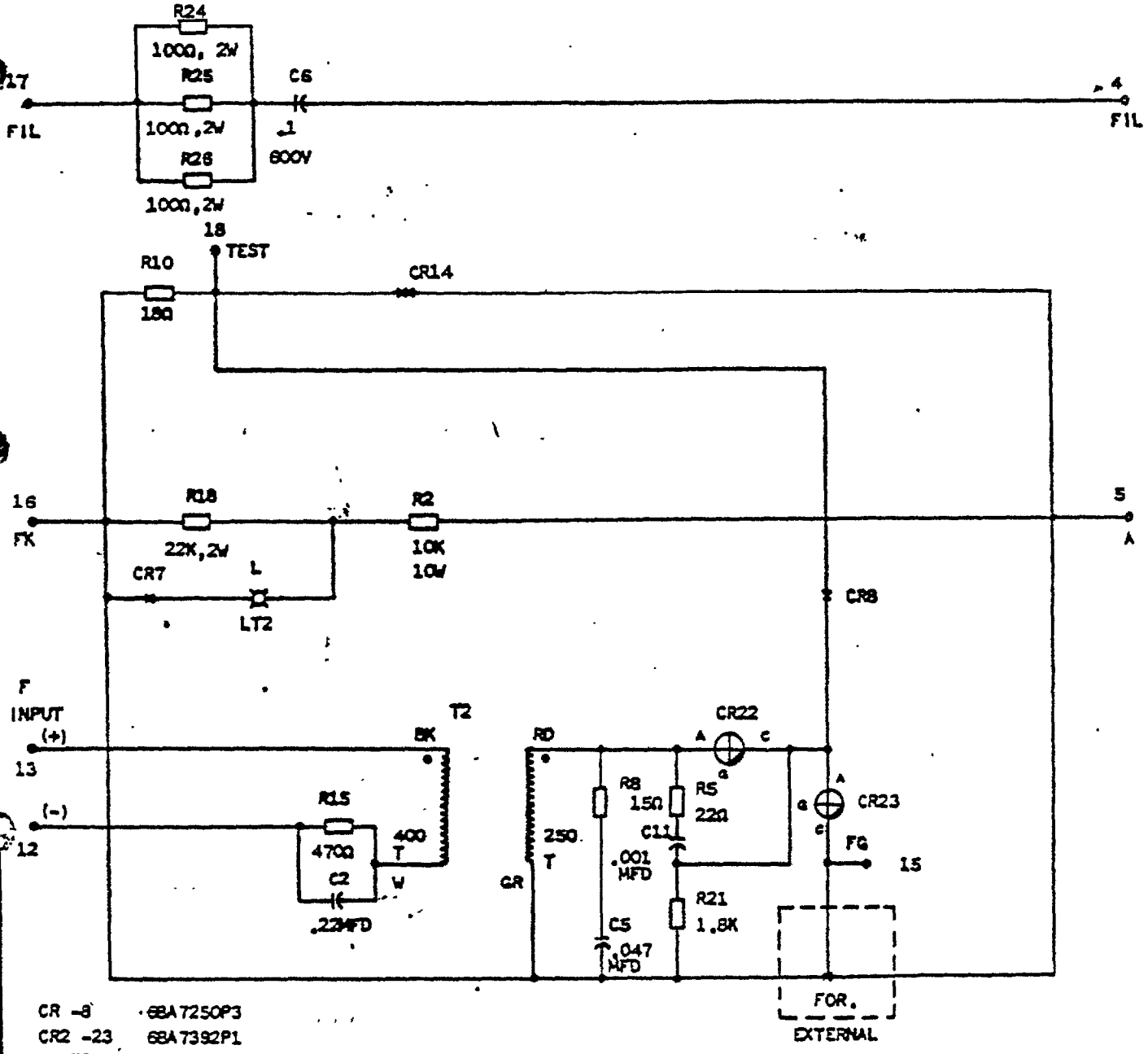
GATE OUTPUTS: (15) SCR #2; (6) SCR #1 (C137 TYPE)

1. FIRST PULSE 0.3 AMP FOR 20 MICROSECONDS WITH .2 MICROSECOND RISE.
2. CURRENT FALLS TO ZERO IN 50 MICROSECONDS WITH THE WIDE INPUT PULSES.
3. SUBSEQUENT PULSES HAVE NARROW 250 MA FRONT BUT BODY OF PULSE IS 100 - 150 MA.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 2 DL1	APPROVALS B.P. Chmura	GENERAL ELECTRIC	ELEMENTARY DIAGRAM GATE AND FILTER CKTS. (230V, NRP, C137)	
ISSUED November 21-1966	PRINTS TO 1338	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	PWB68A993117	SH. NO. 3.1
MADE BY R.E.HANNAH	P6A	I.C. NO.		CONT. ON SH. FL.	

FORM IC 2219 (3-66)



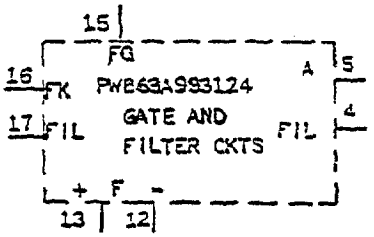
CR -8	8BA7250P3
CR2 -23	8BA7392P1
-T2	8BA993036G1
CR14	8BA7226P2
LT 2	8BA992896G1

REV 3 11-21-72

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 RCH 10-17-69	REV. 2 LCL DLL	APPROVALS B.P. Chausse	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SCR GATE AND FILTER CKTS (230V, NR, C380)	
ISSUED November 21-1966	PRINT NO. 1338	STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	PWB8A993124	3.0
DRAWN BY R.E. HANNAH	40A3			3.1	

SYMBOL



THIS GATE AND FILTER CIRCUIT IS USED ON TYPE IC3605 SCR EXCITER PACKAGE.

IF 28V IS APPLIED TO INPUT TERMINALS 13 AND 12, THE FULL VOLTAGE IS INITIALLY APPLIED TO T2 PRIMARY. C5 WILL CHARGE TO APPROXIMATELY 15V IN 2 TO 3 MICROSECONDS. THIS IS THE NOMINAL BREAKDOWN VOLTAGE OF UNILATERAL SWITCHES CR22 AND CR23 IN SERIES. THESE COLLAPSE AND T2 AND C5 DELIVER A .3 AMP PULSE TO THE GATE WITH A .2 MICROSECOND RISE TIME. AFTER 20 MICROSECONDS C2 STARTS TO REDUCE THE GATE CURRENT AND IT WILL BE ZERO IN 50 MICROSECONDS. THE NOMINAL INPUT IS A 25 MICROSECOND PULSE WITH 50 MICROSECONDS OFF TIME, IN WHICH CASE C2 DOES NOT FULLY DISCHARGE AND THUS SUBSEQUENT PULSES HAVE SMALLER MAGNITUDES. ALSO, THE VOLTAGE ON C2 PROVIDES A RESET VOLTAGE FOR THE TRANSFORMER SO THAT A DIODE MAY BE USED ACROSS 13 AND 12 TO PREVENT VOLTAGE SPIKES.

IF THE PULSE INPUT VOLTAGE IS LOW DURING A LINE DIP, C11 WILL CHARGE UP AND FIRE CR22 FIRST AND THEN CR23 WILL FIRE PROVIDING HALF MAGNITUDE GATE PULSES.

FILTER 4 TO 17 HAS BEEN SELECTED IN CONJUNCTION WITH EXTERNAL AIR CORE REACTORS IN THE SCR EXCITER TO HOLD VOLTAGE CHANGES TO UNDER 200 VOLTS PER MICROSECOND.

METAL OXIDE VARISTOR IS ARRANGED TO ACCEPT APPROXIMATELY .5 AMP PULSES FROM AN INDUCTIVE LOAD. SOME ENERGY MUST BE ABSORBED AS SCR CURRENT FALLS BELOW ITS HOLDING CURRENT. ON HEAVY ABNORMAL CURRENTS THE MOV WILL FIRE THE SCR TO PROTECT THEMSELVES AND OTHER COMPONENTS.

THE LIGHTS ARE CONNECTED TO READ FORWARD CELL VOLTAGE. THUS THEY ARE BLACK AT 1/3 OUTPUT VOLTAGE AND ABOVE IN NORMAL OPERATION. HOWEVER, AN ABNORMAL UNBALANCE OR OPEN OR SHORTED CELL WILL BE CLEARLY INDICATED BY UNBALANCED LIGHTS.

APPLICATION DATA

FOR GATE INPUT (13+) (12-)

1. 24 TO 30 VOLT PULSES NOMINALLY 25 MICROSECONDS ON, 50 MICROSECONDS OFF.
2. WIDE (200 MICROSECOND) INPUT PULSES, CAUSED BY ADDING PULSES TO THE NORMAL PULSE TRAIN, ARE NOT HARMFUL. HOWEVER, OUTPUT GATE PULSE WILL BE 50 MICROSECONDS WIDE MAX.
3. PEAK CURRENT 250 MA.
4. DIODE CLAMP MAY BE USED ON INPUT TERMINALS IF DUTY CYCLE IS IN THE RANGE OF 25 MICROSECONDS ON AND 50 MICROSECONDS OFF.

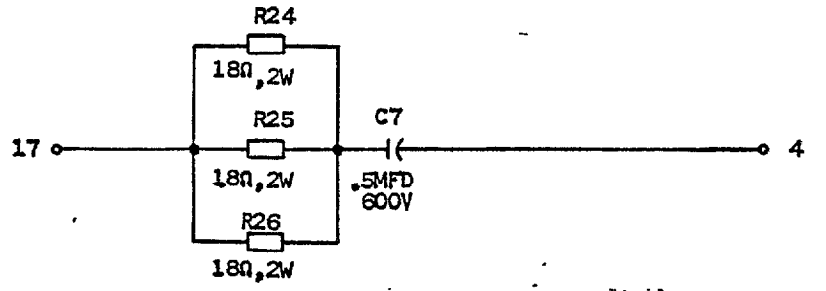
CONNECTIONS TO SCR CELLS

1. (16) TO FORWARD CELL CATHODE
2. (5) TO FORWARD CELL ANODE
3. (17) SAME AS (16), BUT PREFERABLY WITH SEPARATE WIRE.
4. (4) SAME AS (5) BUT PREFERABLY WITH SEPARATE WIRE.

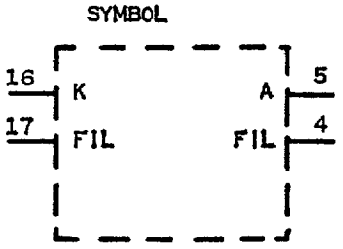
GATE OUTPUTS: (15) FOR CELL: (C350 OR C380 TYPE)

1. FIRST PULSE 0.3 AMP FOR 20 MICROSECONDS WITH .2 MICROSECOND RISE
2. CURRENT FALLS TO ZERO IN 50 MICROSECONDS WITH WIDE INPUT PULSES.
3. SUBSEQUENT PULSES HAVE NARROW 250 MA FRONT BUT BODY OF PULSE IS 100 - 150 MA.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY







THIS FILTER CIRCUIT IS USED ON THE IC3605 NX PACKAGES.

FILTER 4 TO 17 HAS BEEN SELECTED TO LIMIT REVERSE VOLTAGE TRANSIENTS ACROSS THE INDIVIDUAL RECTIFIER CELLS FOLLOWING COMMUTATION.

ADDITIONAL TRANSIENT PROTECTION IS PROVIDED BY METAL OXIDE VARISTOR CR14

APPLICATION DATA

CONNECTIONS TO RECTIFIER CELLS.

1. (16) TO CATHODE
2. (17) TO CATHODE
3. (4) TO ANODE
4. (5) TO ANODE
5. WIRES (16) AND (5) TO BE TWISTED TOGETHER FOR AT LEAST 4 INCHES.
6. WIRES (17) AND (4) TO BE TWISTED TOGETHER FOR AT LEAST 4 INCHES.

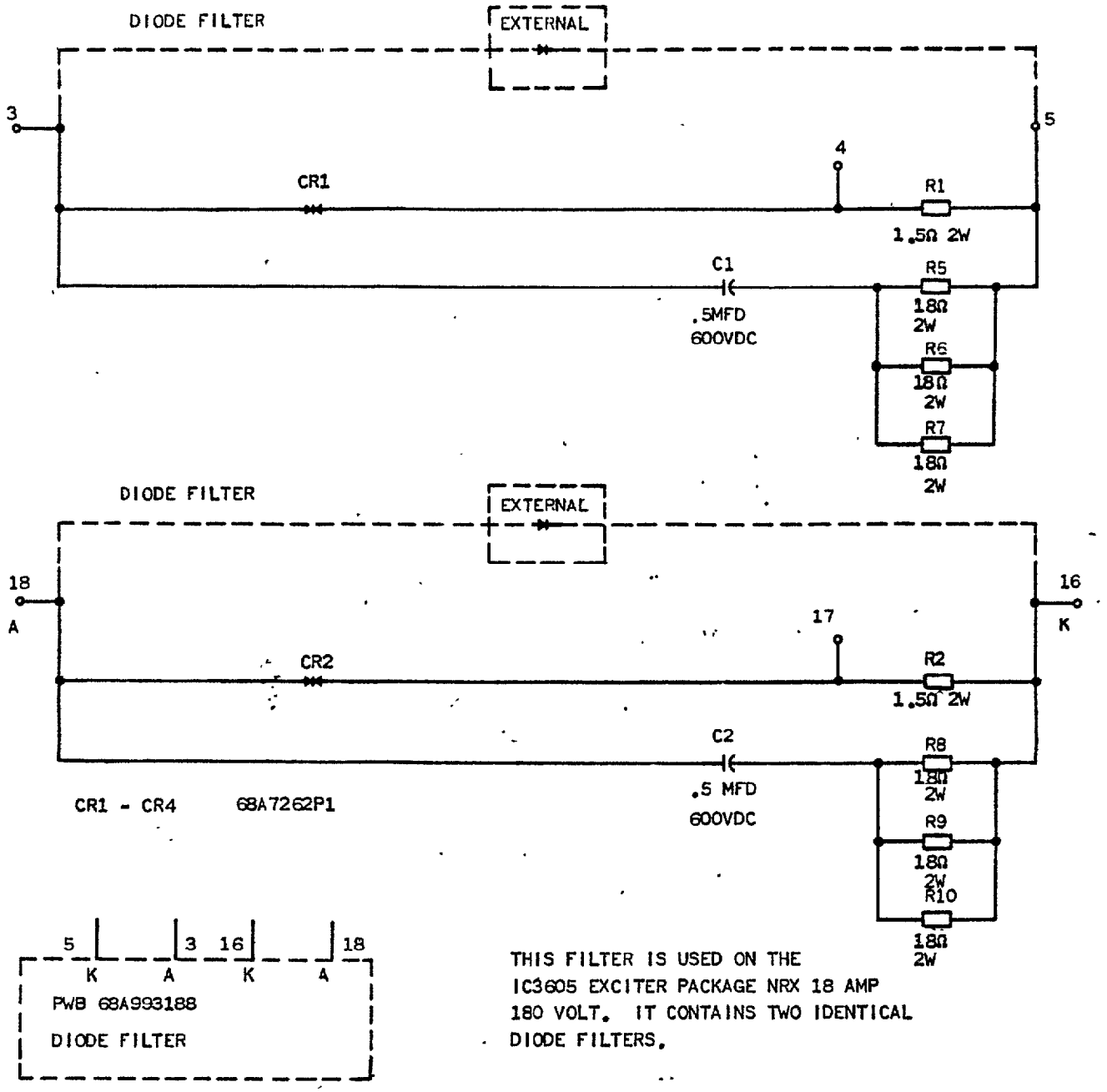
COMPONENTS

CR14- 68A7226P2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 11-21-72 B.D.	REV. 2 RDA 4/24/73	APPROVED	GENERAL ELECTRIC	ELEMENTARY DIAGRAM FILTER CIRCUIT 230V,NRX,A390	
ISSUED March 21-1967	PRINTS TO DL8	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	PWD68A993159 CONT. ON SH. FL.	SH. NO. 3.0
MADE BY J.R. INGRAM		T.C. NO			

FORM IC 2219



THIS FILTER IS USED ON THE IC3605 EXCITER PACKAGE NRX 18 AMP 180 VOLT. IT CONTAINS TWO IDENTICAL DIODE FILTERS.

EACH DIODE FILTER CONTAINS AN RC FILTER SELECTED TO LIMIT REVERSE VOLTAGE TRANSIENTS ACROSS INDIVIDUAL RECTIFIER CELLS FOLLOWING COMMUTATION. ADDITONAL PROTECTION AGAINST TRANSIENTS IS PROVIDED BY METAL OXIDE VARISTORS CR1, CR2

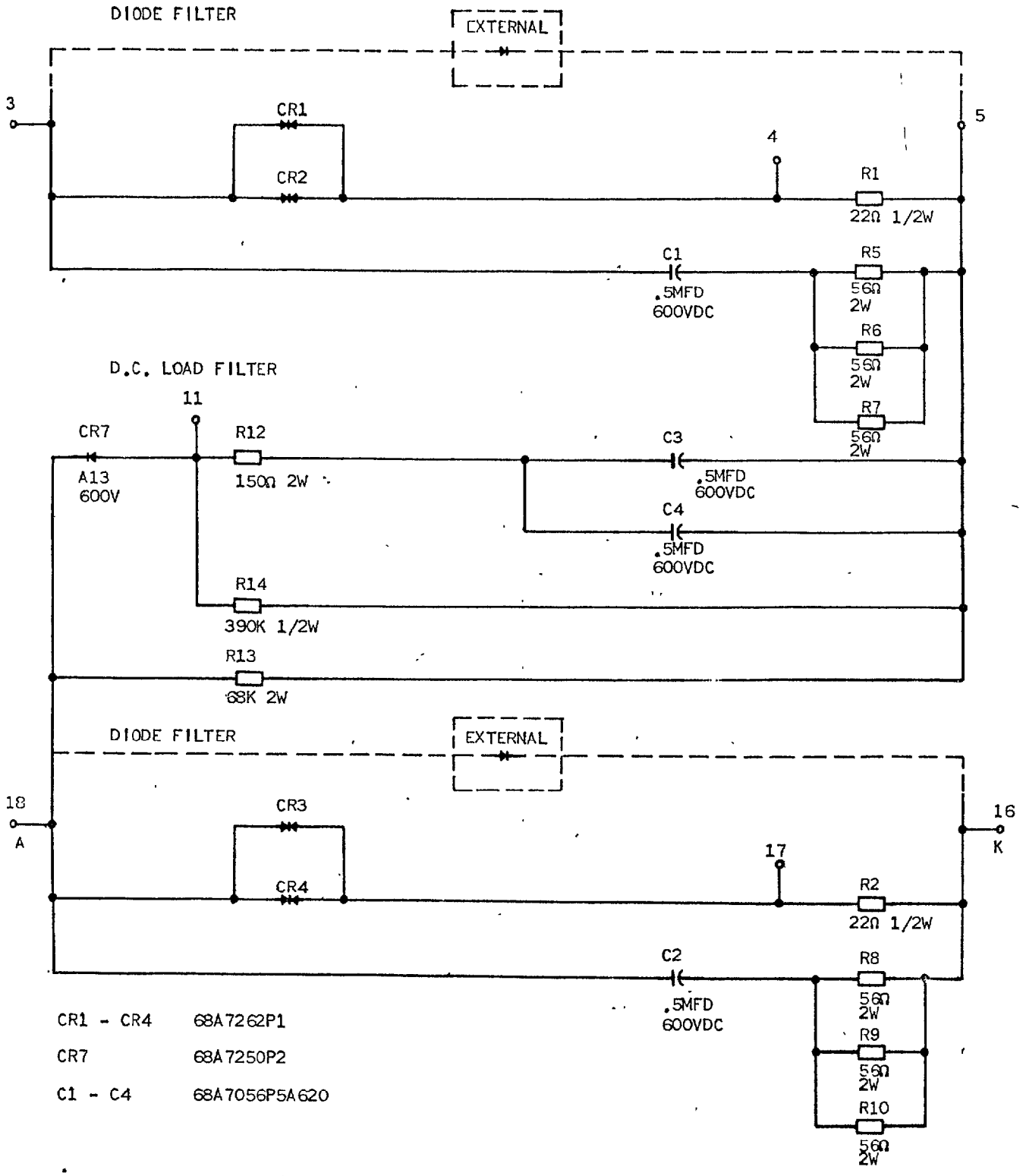
APPLICATION DATA

1. TERMINALS 3 AND 5 MUST BE USED AS A PAIR WITH ONE POWER DIODE.
2. TERMINALS 16 AND 18 MUST BE USED AS APAIR WITH ONE POWER DIODE.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 11-21-72 RB	REV. 2 RBA 4/24/73 DL1	APPROVALS R. A. Forden	GENERAL ELECTRIC	ELEMENTARY DIAGRAM DIODE FILTER 230V NRX IN2160	
ISSUED April 21-1967	PRINTS TO DL8	FIRST MADE FOR REQ.	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	PWB68A993188	SH. NO. 3.0
MADE BY J. R. INGRAM		I.C. NO.		CONT. ON SH. FL.	

FORM IC 2219



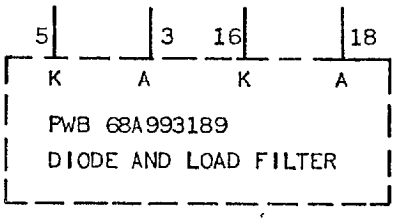
CR1 - CR4	68A7262P1
CR7	68A7250P2
C1 - C4	68A7056P5A620

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 2 DL1	APPROVALS R. A. Forden	GENERAL ELECTRIC	ELEMENTARY DIAGRAM DIODE AND LOAD FILTER 185V CPE 20A	
ISSUED April 21 1967	PRINTS TO P6A	FIRST MADE FOR REQ.	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	PWB68A993189	SH. NO. 3.0
MADE BY J. R. INGRAM	1338	IC NO.		CKT ON SH 3.1	

FORM IC 2219 (3-66)

SYMBOL



THIS FILTER IS USED ON THE IC3605 UNREGULATED EXCITER PACKAGE (250VDC C.P.E. - 20A) IT CONTAINS TWO IDENTICAL DIODE FILTERS AND ONE D.C. LOAD FILTERS

EACH DIODE FILTER CONTAIN AN R. C. FILTER SELECTED TO LIMIT REVERSE VOLTAGE TRANSIENTS ACROSS INDIVIDUAL RECTIFIER CELLS FOLLOWING COMMUTATION. ADDITIONAL PROTECTION AGAINST TRANSIENTS IS PROVIDED BY THYRECTOR CR1 AND CR2.

THE COMPONENTS BETWEEN TERMINAL 5 AND 18 MAKE UP THE D.C. LOAD FILTER. THE POSITIVE LOAD VOLTS ORIGINATE AT TERMINAL 5. VOLTAGAE TRANSIENTS APPEARING ON THIS POSITIVE LOAD BUS WILL PASS READILY THRU C3, C4, R12 AND CR7. THERE BY SHUNTING HIGH TRANSIENT VOLTAGES FROM THE POWER DIODES. CR7 IS TO PREVENT THE 360 CYCLE RIPPLE VOLTAGE FROM CIRCULATING CURRENT THRU THE C3, C4 AND R12. R14 IS USED TO DISCHARGE CAPACITOR C3 AND C4 WHEN A.C. POWER IS REMOVED. R13 WILL PROVIDE A CONSTANT LOAD MAKING THE DC OUTPUT VOLTMETER READ APPROXIMATELY THE SAME AT FULL LOAD AND NO LOAD CONDITION.

APPLICATION DATA

(1) TERMINAL 5 MUST CONNECT TO THE CATHODE OF THE POSITIVE POWER DIODES.

(2) TERMINAL 18 MUST CONNECT TO THE ANODE OF THE NEGATIVE CELLS.

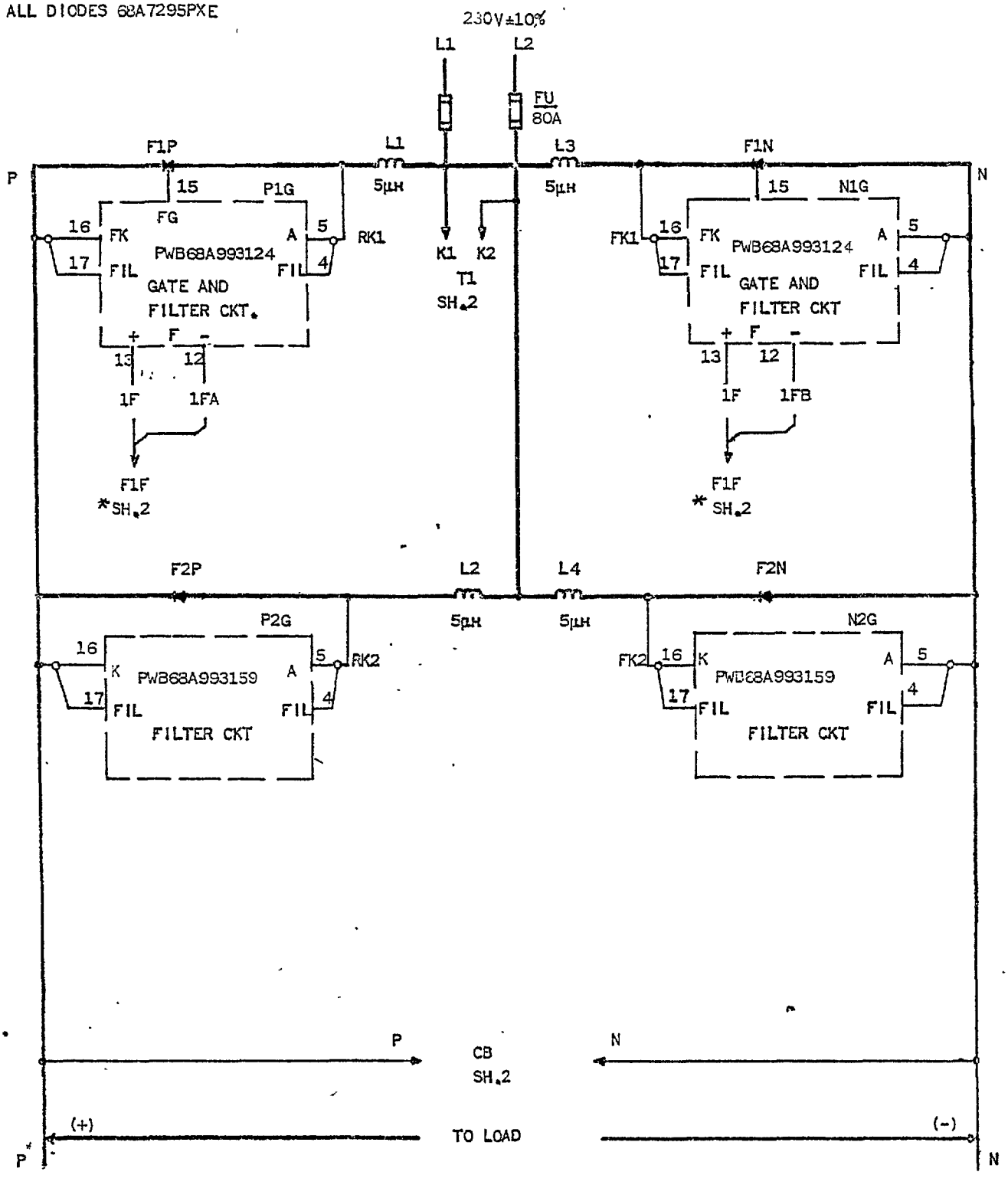
(3) THE COMPLETE EXCITER PACKAGE WILL CONTAIN THREE OF THESE FILTER CARDS. THEREFORE, THE TOTAL LINE FILTER CAPACITANCE WILL BE 3 MFD WITH 50 OHMS IN SERIES.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	EV. 2 DLI	APPROVALS R. C. Forden	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PWB 68A993189	
ISSUED April 21-1967	PRINTS TO P6A	FIRST MADE FOR REQ.	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.		
MADE BY J. R. INGRAM	1338	TC NO.		CONT ON SH FL.	SH. NO. 3.1

FORM IC 2219 (3-66)

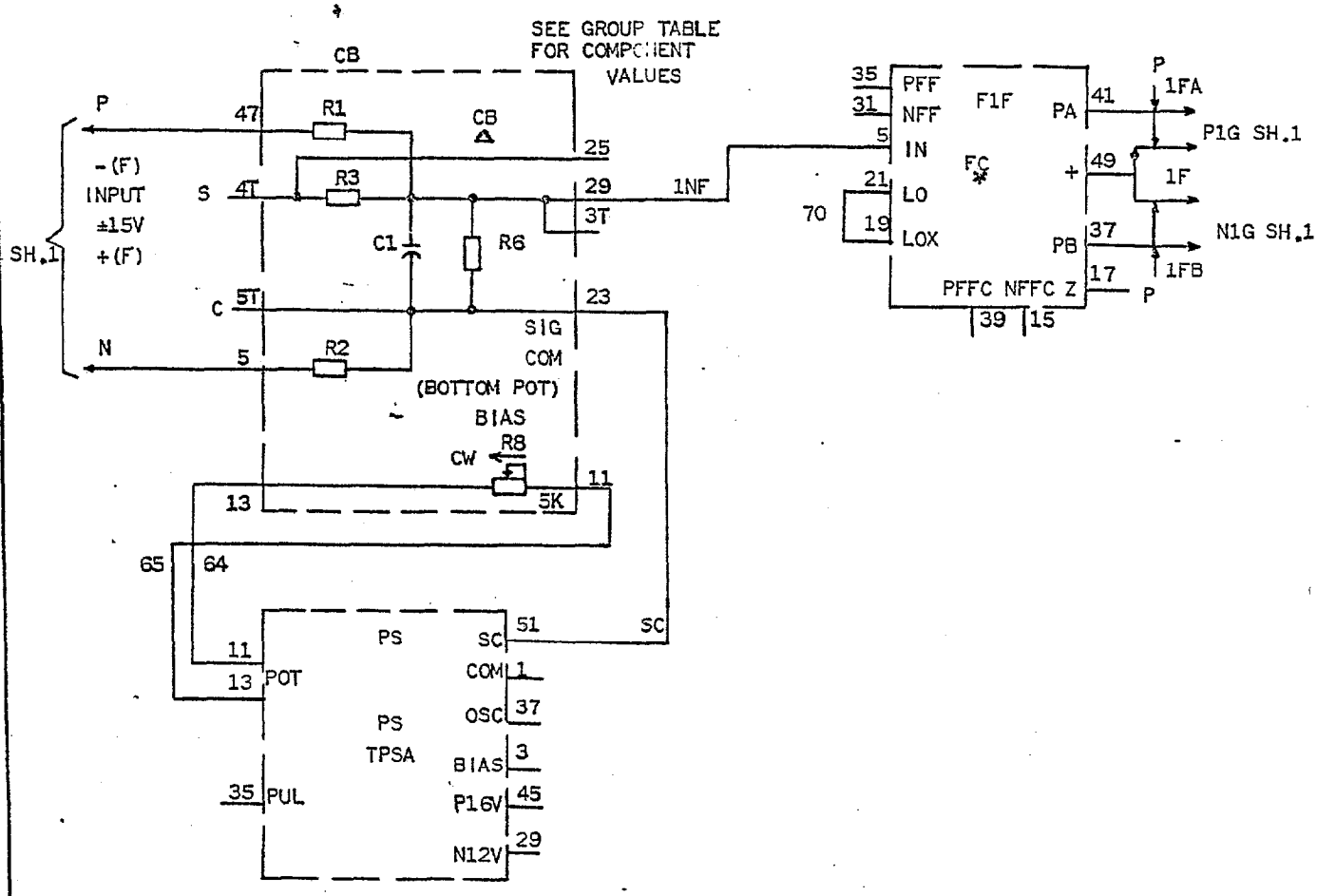
ALL SCR'S 68A7298PXE
C350 TYPE
ALL DIODES 68A7295PXE

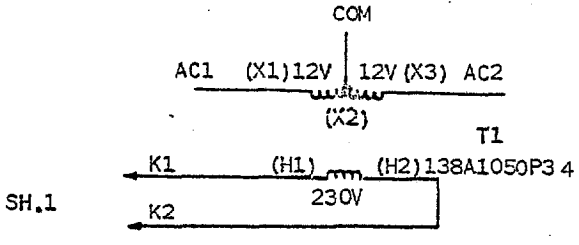


-* TWISTED PAIRS AS NOTED ON SH. 2

REV 1 MAY 67 B.H.O	REV 2 Oct. 31, 68 R. A. Wright	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	SCR EXCITER NRX 40 AMP 180 VOLT			
REV 3 12-5-69	REV						
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	68A9936711
ISSUED March 27-1967				67	2N		CONT ON SHEET 2 SH NO. 1

FORM IC 2143B (9-64)





WIRE TABLE FOR POWER SUPPLY AND REPETITIVE CONN.

WIRE NO. \ DEVICE	F1F	PS	T1
P16V	45	45	-
N12V	29	29	-
OSC	25	37	-
BIAS	13	3	-
COM	1-7	1	X2
AC1	33	47	X1
AC2	43	43	X3
PUL	27	35	-

GROUP TABLE

PACKAGE CAT. NO. IC3605	VOLTAGE FEEDBACK	FREQ. CPS	FC IC3600	△ CB PWB68A993614	R1 10 WATT	R2 10 WATT	R3 .5 WATT	R6 .5 WATT	C1 100V
E040NX180	NO	60	TFCU	G7	OMIT	OMIT	4.75K	825Ω	OMIT
F040NX180	NO	50	TFCV	G7	OMIT	OMIT	4.75K	825Ω	OMIT
C040NX180	YES	60	TFCU	G3	12K	12K	1.82K	OMIT	20MFD
D040NX180	YES	50	TFCV	G3	12K	12K	1.82K	OMIT	20MFD

FORM IC 2143B (9-64)

REV 2 1-31-68	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE SCR EXCITER NRX 40 AMP 180 VOLT			
REV 1 11-15-67	REV					
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338
ISSUED March 27 1967				6T	2N	

68A993671

CONT ON SHEET 3 SH NO. 2

INPUT:

A.C. POWER: 230V ±10% AMPS: 100% DC LOAD AMPS *4,6 TERMINALS: L1, L2
3,C *1,2,3

SIGNAL: 0→ -15V 8 MA FB
0→ -15V 3 MA NoFB (FOR MAX. OUTPUT)

OUTPUT:

D.C. 180V 40 AMPS 55°C AMBIENT AT PACKAGE P+, N-
180V 65 AMPS FAN COOLED WITH CFM

		OPEN LOOP	INTERNAL FEEDBACK
OUTPUT STABILITY:	WITH TEMPERATURE	- TYPICALLY 2.5% RATED OUTPUT	1.5% RATED OUTPUT
	WITH AC VOLTS	- VARIES DIRECTLY	10% OF AC CHANGE

TRANSFER FUNCTION (INCLUDES EFFECT OF TRANSPORT LAG)

OPEN LOOP *5

$$\frac{Vo}{V_i} = \frac{19}{1 + .001P}$$

INTERNAL FEEDBACK

$$\frac{Vo}{V_i} = \frac{12}{1 + .003P}$$

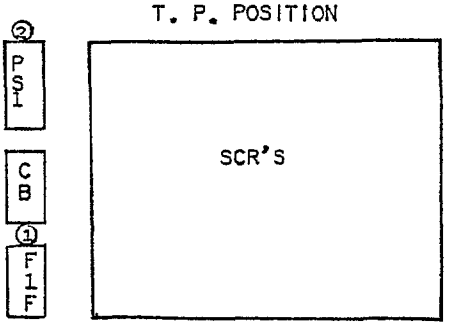
PACKAGE DIMENSIONS: 22"H x 24"W x 12"D (APPROX.)

OUTLINE: 157B6006
DULSEAL: 3851F

TEST POINTS

TP	① TFCU/TFCV	② TPSA
1	--	OSC
2	PA	GND
3	PB	--
4	AC1	+16V
5	--	BIAS
6	UNIJ.EMMITER	-12V
7	INPUT SIG	--

T. P. POSITION



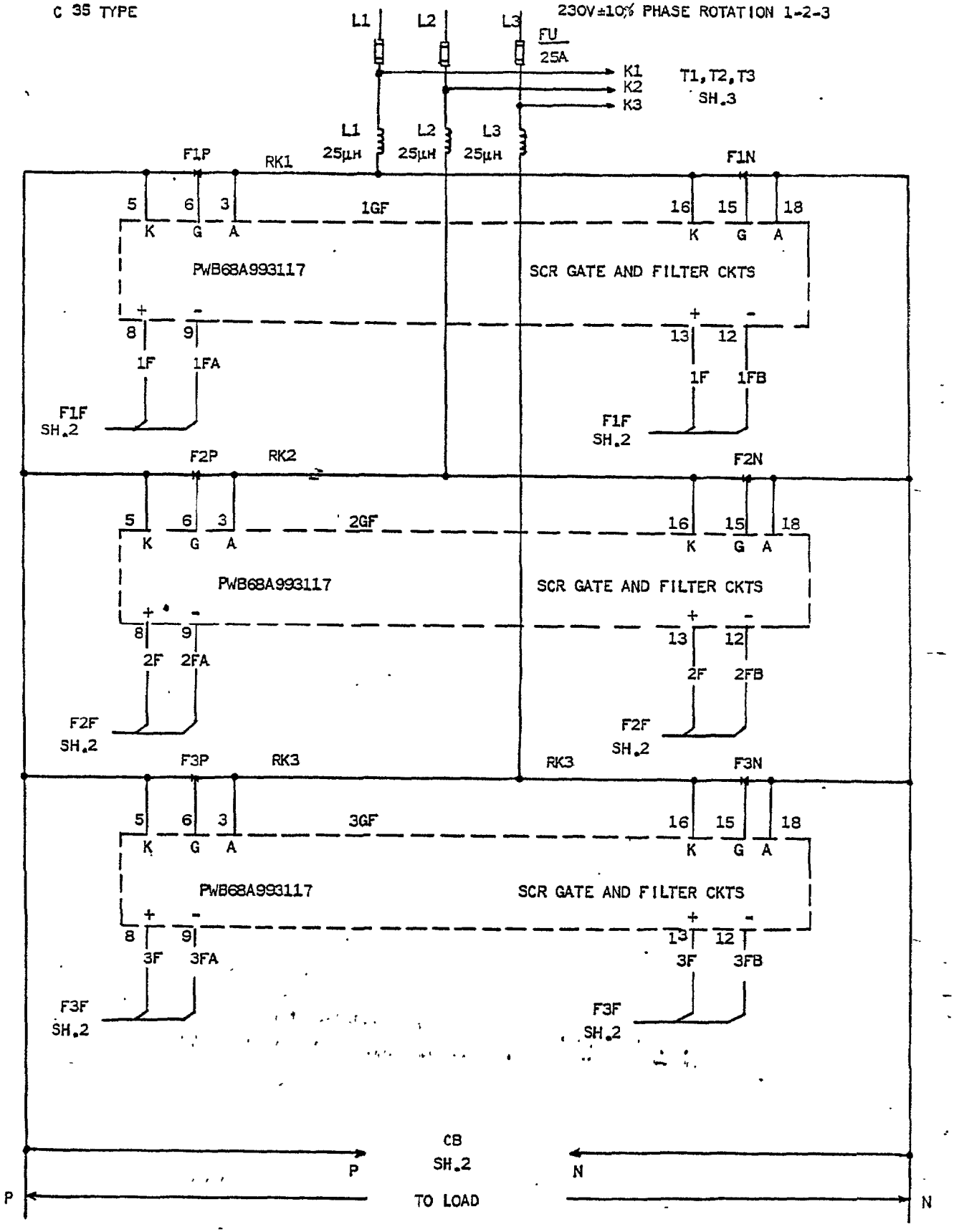
*NOTES: REFER TO GEK 8074 FOR DETAILED INFORMATION

1. DO NOT GROUND TERMINAL C WHEN INTERNAL FEEDBACK IS USED.
2. GROUND TERMINAL C OR USE AS COMMON WHEN NO FEEDBACK IS USED
3. POWER TERMINAL P IS POSITIVE WHEN INPUT S IS NEGATIVE WITH RESPECT TO C.
4. 100% D.C. LOAD AMPS WITH 180V HOT DROPS, LESS FOR LOW HOT DROPS.
5. AVOID INTERNAL 30, 45, AND 60 CYCLE RESONANT LOOPS 1Ø
6. FREQUENCY TOLERANCE - ±1/2 CPS
7. COMMON TERMINAL, C, HAS MICROSECOND SPIKES COUPLED FROM SCR PULSE TRANSFORMERS. IF SYSTEM USES FAST DIGITAL LOGIC, ADD 1K RESISTOR BETWEEN C AND REST OF SYSTEM.

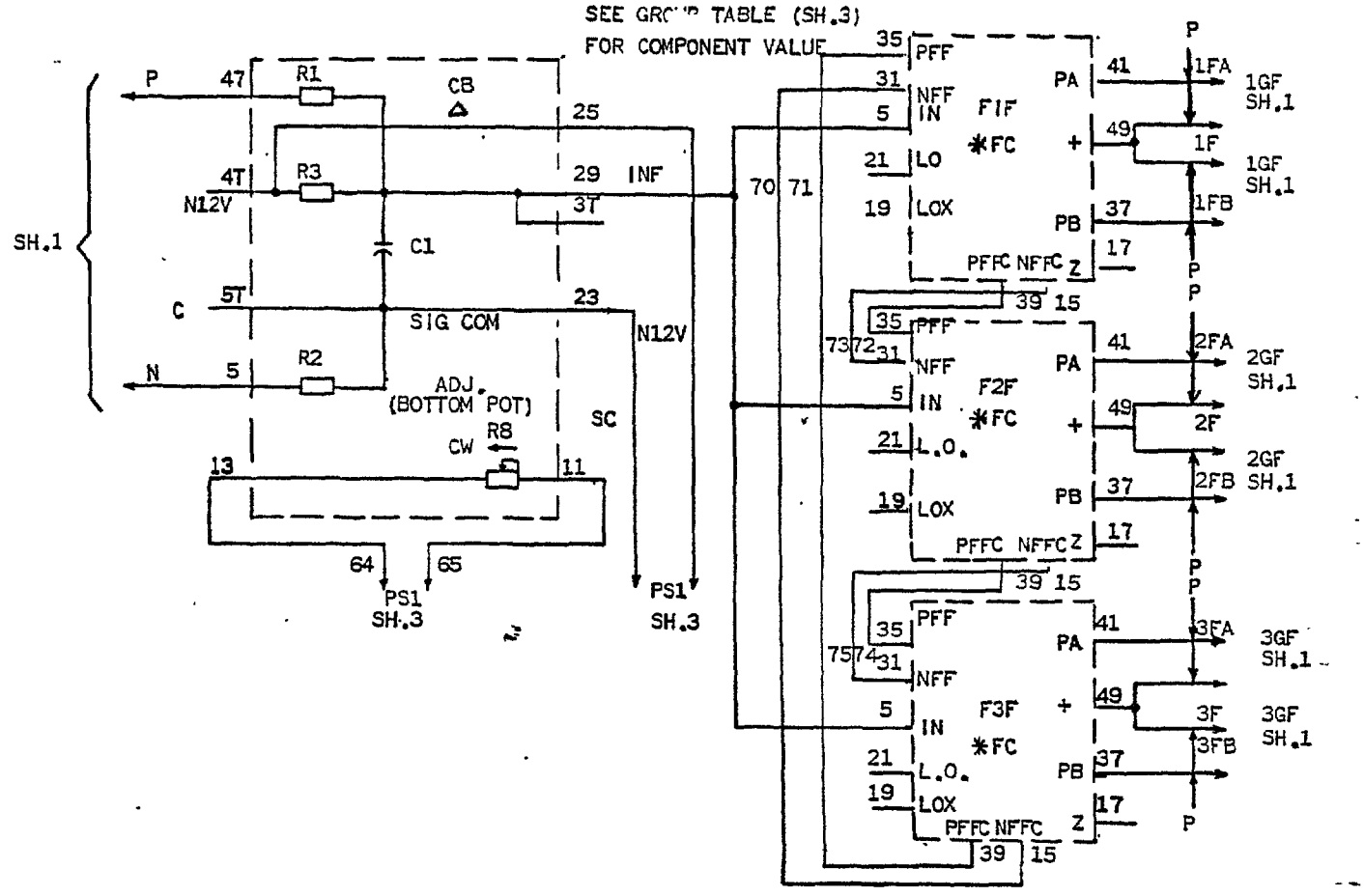
REV 2 12-5-67	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	SCR EXCITER NRX 40 AMP 180 VOLT			
REV 1 11-15-67	REV						
MADE BY B. H. OGDEN		APPROVALS D.J.M.	PRINTS TO	P6A	DL1	1338	68A993671
ISSUED MARCH 27-1967				6T	2N		CONT ON SHEET FL SH NO. 3

FORM IC 2143B (9-64)

ALL SCR'S 68A7297PXM
C 35 TYPE



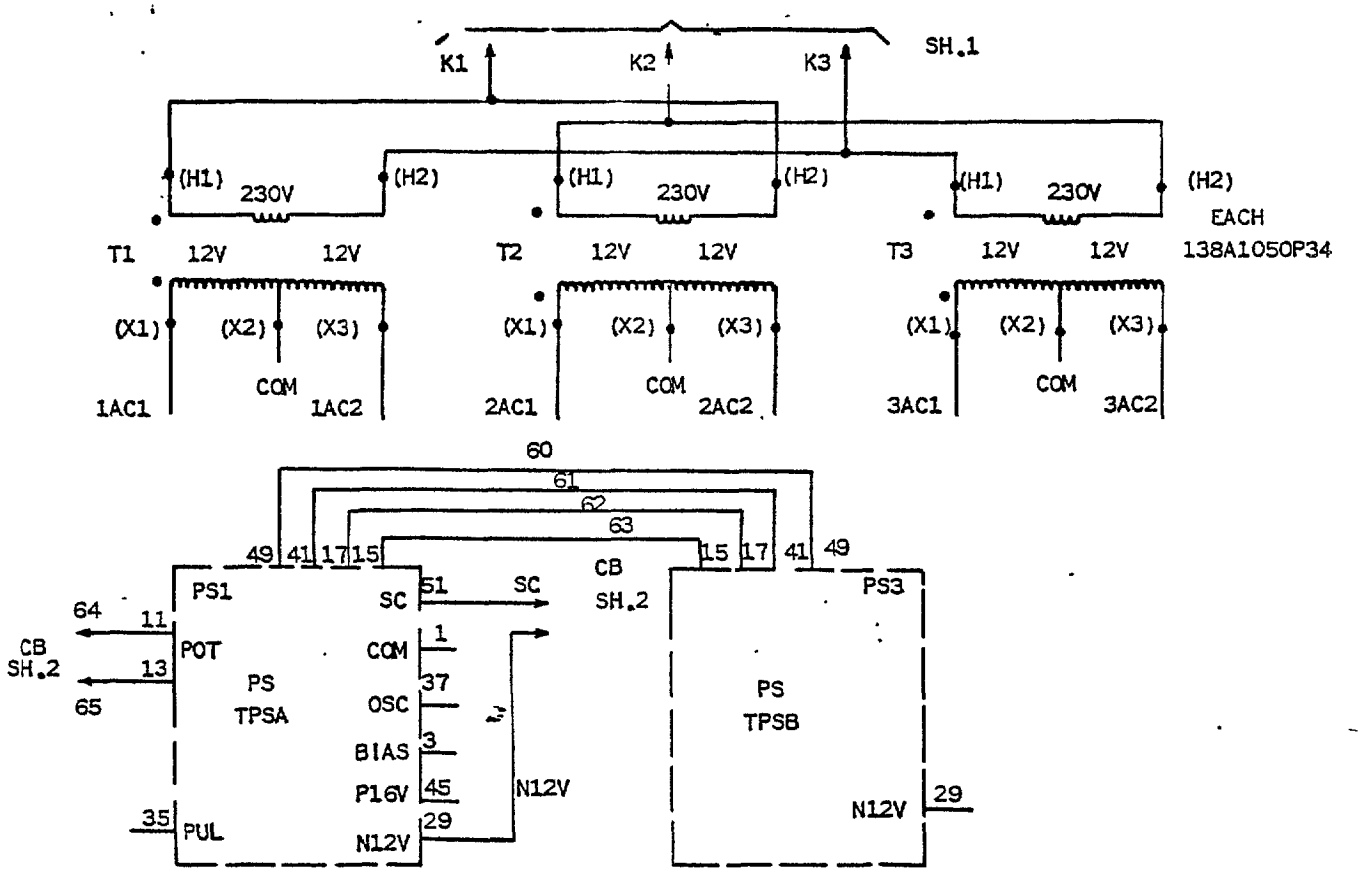
REV 1 MAY 67 2-68	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER			
REV	REV			CPE 22AMP 250 VOLT			
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	68A993690
ISSUED December 5-1966				6T	2N		CONT ON SHEET 2 SH NO. 1



*FC SEE GROUP TABLE (SH. 3)

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 22 AMP 250 VOLT			
REV	REV					
MADE BY B.H. OGDEN	APPROVALS	PRINTS TO	P6A	DL1	1338	68A993690
ISSUED December 5-1966			6T	2N		CONT ON SHEET 3 SH NO. 2

FORM IC 2143B (9-64)



WIRE TABLE FOR POWER AND REPETITIVE CONNECTIONS

WIRE NO \ DEVICE			F1F		F2F		F3F	PS1	PS3	T1	T2	T3
P16V			45		45		45	45	-	-	-	-
N12V			29		29		29	29	29	-	-	-
OSC			25		25		25	37	-	-	-	-
BIAS			13		13		13	3	-	-	-	-
COM			1_{51}		1_{51}		1_{51}	1_{51}	-	X2	X2	X2
1AC1			33		-			47	-	X1	-	-
1AC2			43		-		-	43	-	X3	-	-
2AC1			-		33		-	-	47	-	X1	-
2AC2			-		43		-	-	43	-	X3	-
3AC1			-		-		33	-	39	-	-	X1
3AC2			-		-		43	-	37	-	-	X3
PUL			27		27		27	35	-	-	-	-

GROUP TABLE

PACKAGE CAT. NO. IC3605	VOLTAGE FEEDBACK	FREQ. CPS	FC IC3600	CB PWB68A 993614	R1 10WATT	R2 10WATT	R3 .5WATT	C1 100V
C022CP250	YES	60	TFCU	G6	12K	12K	1.0K	200MFD
D022CP250	YES	50	TFCV	G6	12K	12K	1.0K	200MFD

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 22 AMP 250 VOLT				
REV 1 DHB 11-15-67	REV						
MADE BY B.H. OGDEN	APPROVALS	PRINTS TO	P6A	1338	DL1	68A9936900	
ISSUED December 5 1966			6T	2N		CONT ON SHEET 4	SH NO. 3

FORM IC 2143B (9-64)

INPUT:

A.C. POWER: 230V ±10% AMPS = 82% D.C. LOAD AMPS *6 TERMINALS: L1, L2, L3 *4,7

OUTPUT:

D.C. 250V 22 AMPS 55°C AMBIENT AT PACKAGES P+, N-

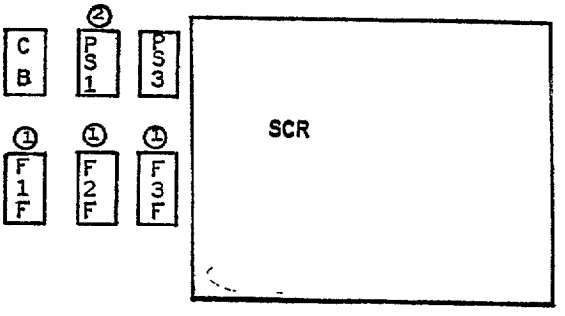
TRANSFER FUNCTION: (INCLUDES EFFECT OF TRANSPORT LAG)
RESPONSE TO 100% LOAD STEP .1 SEC.
REGULATION WITH LINE, LOAD, TEMP., ETC. ±1%

PACKAGE DIMENSIONS: 19"H x 24"W x 12"D (APPROX.)

OUTLINE: 157B6009
DULSEAL: 3851K
T.P. POSITION

TEST POINTS

TP	① TFCU/TFCV	② TPSA
1	---	OSC
2	PA	GND
3	PB	---
4	AC1	+16V
5	---	BIAS
6	UNIJ.EMITTER	-12V
7	INPUT SIG.	---

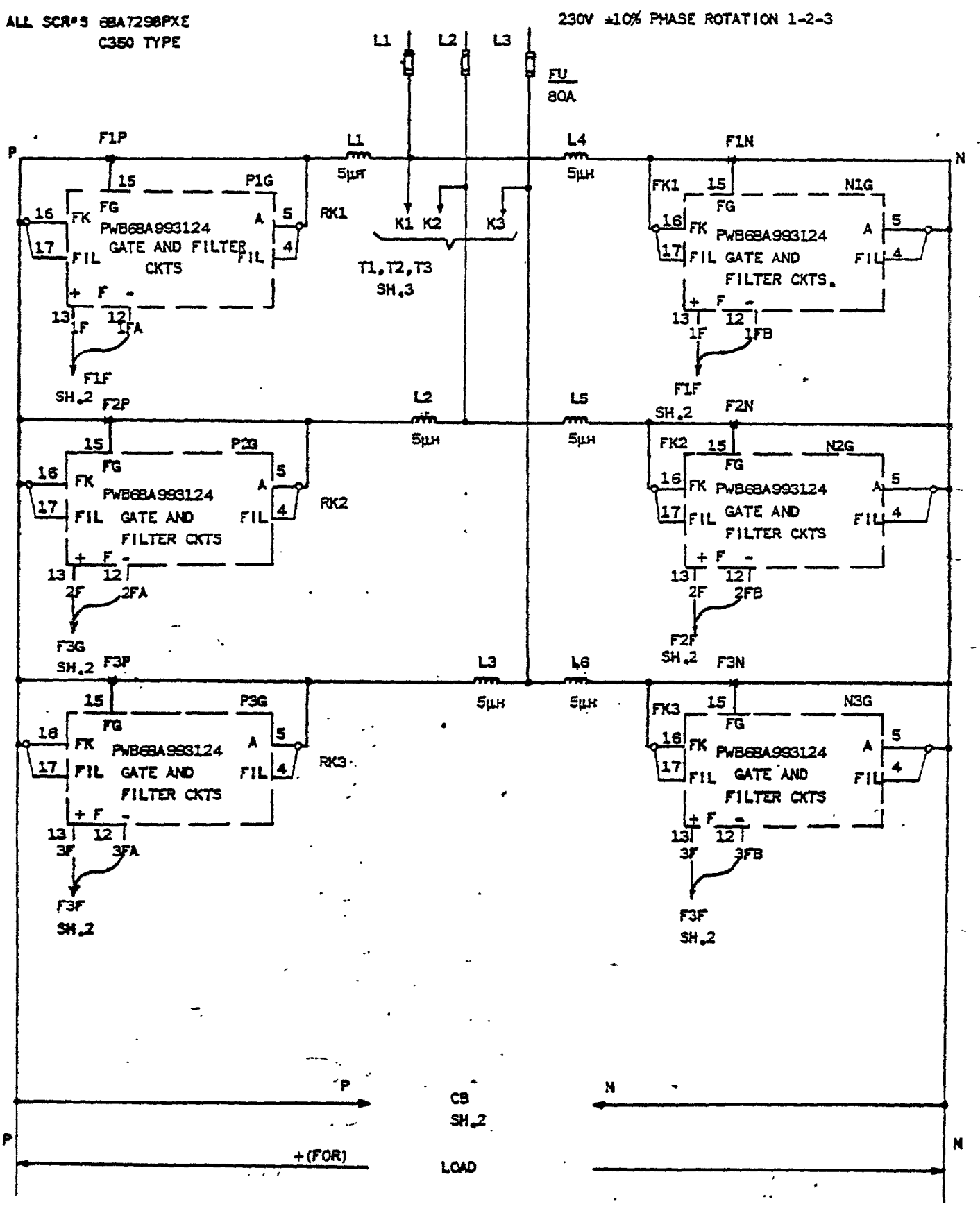


*NOTES: REFER TO GEK 8074 FOR DETAILED INFORMATION

1. DO NOT GROUND TERMINAL C WHEN INTERNAL FEEDBACK IS USED.
2. GROUND TERMINAL C OR USE AS COMMON WHEN NO FEEDBACK IS USED.
3. POWER TERMINAL P IS POSITIVE WHEN INPUT S IS NEGATIVE WITH RESPECT TO C.
4. PHASE ROTATION 1-2-3
5.
6. INDUCTIVE LOADS WITH .1 SEC. TC OR LONGER; RESISTIVE LOAD OR LESS THAN .1 SEC. TC - CURRENT MUST BE CALCULATED.
7. FREQ. TOLERANCE = ± 1/2 CPS
8. COMMON TERMINAL, C, HAS MICROSECOND SPIKES COUPLED FROM SCR PULSE TRANSFORMERS. IF SYSTEM USES FAST DIGITAL LOGIC, ADD 1K RESISTOR BETWEEN C AND REST OF SYSTEM.

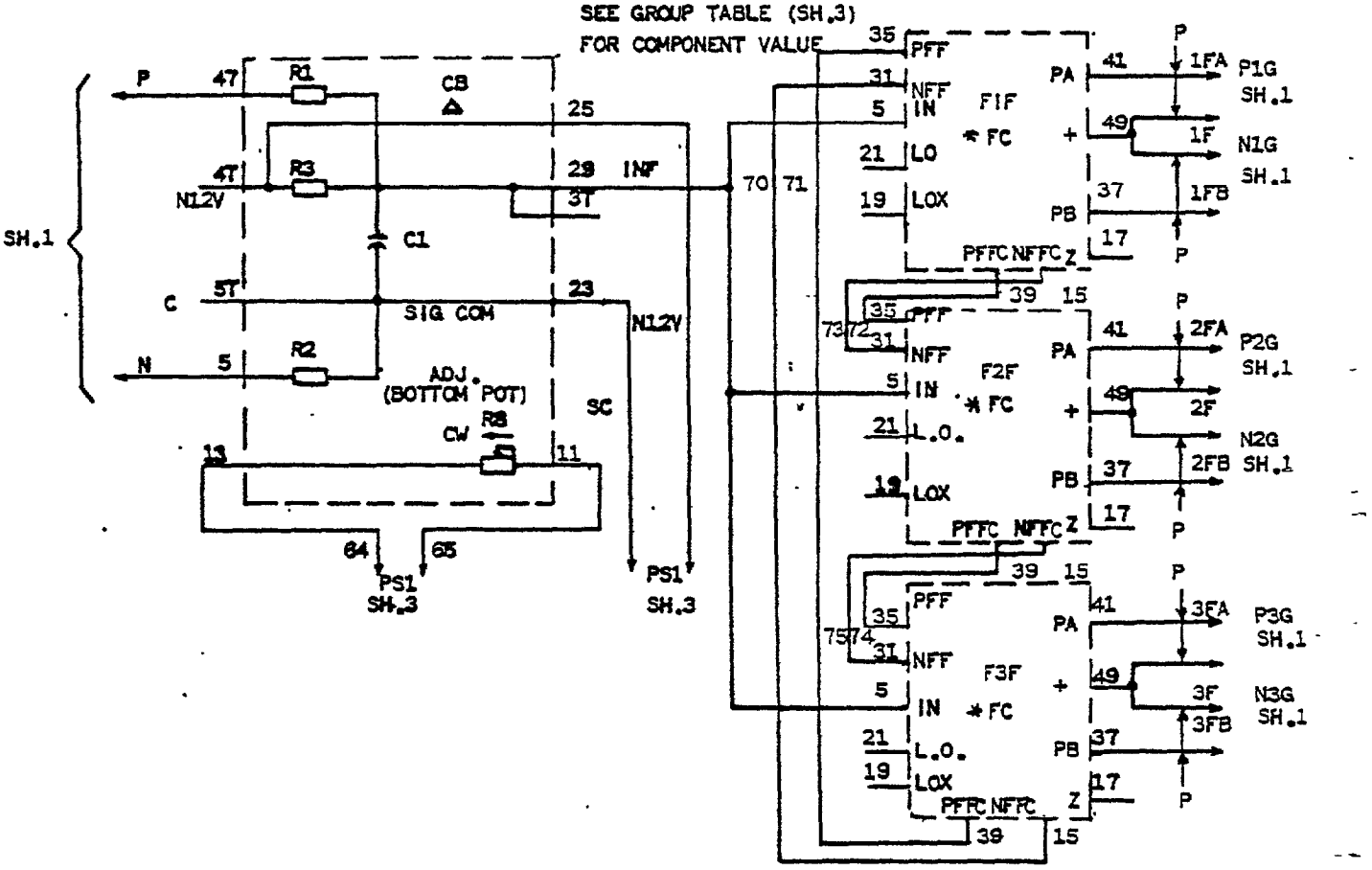
REV 11 11-15-67	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 22 AMPS 250 VOLTS				
REV 12 12-5-69	REV						
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	PGA	DL1	1338	GBA993690
ISSUED January 17 1967				6T	2N		CONT ON SHEET FL. SH NO. 1

FORM 10 4930 10-69



REV 1	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER			
REV 2 12-5-66	REV			CPE 50 AMP 250 VOLT			
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	6 8 A 9 9 3 6 9 1
ISSUED December 5 1966				6T	2N		CONT ON SHEET 2 SH NO. 1

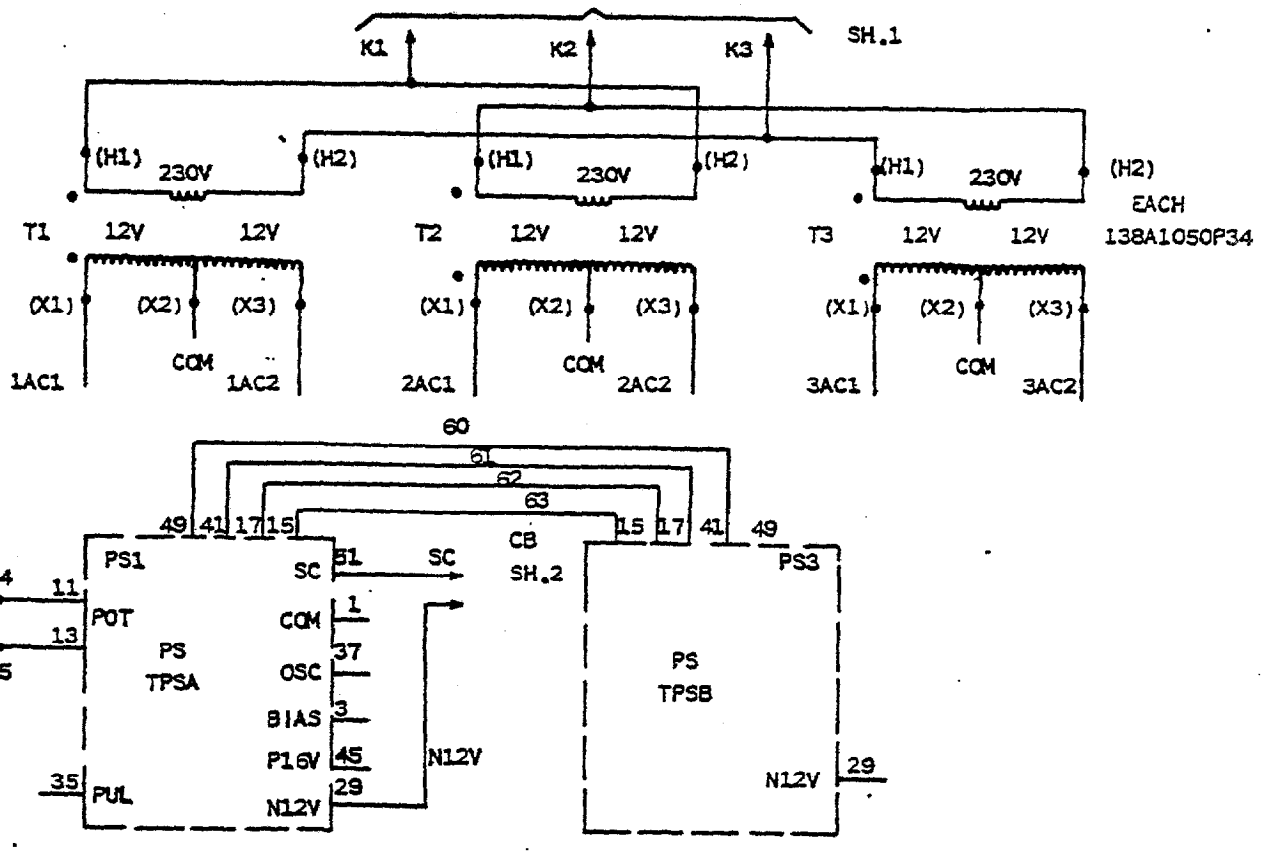
FORM IC 2143B R/C



* FC SEE GROUP TABLE (SH.3)

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 50 AMP 250 VOLT					
REV	REV							
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	6 8 A 9 9 3 6 9 1	
ISSUED December 5, 1966				6T	2N		CONT ON SHEET 3	SH NO. 2

FORM IC 2143B (9-67)



WIRE TABLE FOR POWER AND REPETITIVE CONNECTIONS

WIRE NO \ DEVICE			F1F	F2F		F3F	PS1	PS3	T1	T2	T3
P16V			45	45		45	45	-	-	-	-
N12V			29	29		29	29	29	-	-	-
OSC			25	25		25	37	-	-	-	-
BIAS			13	13		13	3	-	-	-	-
COM			$^{1}_{S1}$	$^{1}_{S1}$		$^{1}_{S1}$	$^{1}_{S2}$	-	X2	X2	X2
1AC1			33	-			47	-	X1	-	-
1AC2			43	-		-	43	-	X3	-	-
2AC1			-	33		-	-	47	-	X1	-
2AC2			-	43		-	-	43	-	X3	-
3AC1			-	-		33	-	39	-	-	X1
3AC2			-	-		43	-	37	-	-	X3
PUL			27	27		27	35	-	-	-	-

GROUP TABLE

PACKAGE CAT. NO. IC3605	VOLTAGE FEEDBACK	FREQ. CPS	*FC IC3600	△CB PWB68A 993614	R1 10WATT	R2 10WATT	R3 .5WATT	C1 100V
C050 CP250	YES	60	TFCU	G6	12K	12K	1.0K	200MFD
D050 CP250	YES	50	TFCV	G6	12K	12K	1.0K	200MFD

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 50 AMP 250 VOLT				
1 12-5-67							
REV	REV						
MADE BY B.H. OGDEN		APPROVALS DJM	PRINTS TO	PSA	1338	DL1	6 8 A 9 9 3 6 9 1
ISSUED December 5-1966				6T	2N		CONT ON SHEET 4 SH NO. 3

FORM IC2143B

INPUT:

A.C. POWER: 230V ±10% AMP = 82% D.C. LOAD AMPS *6, 7 TERMINALS: L1, L2, L3 *4

OUTPUT:

D.C. 250V 50 AMPS. 55°C AMBIENT AT PACKAGE P+, N-
250V 80 AMPS FAN COOLED WITH CFM
TRANSFER FUNCTION: (INCLUDES EFFECT OF TRANSPORT LAG)
RESPONSE TO 100% LOAD STEP .1 SEC.
REGULATION WITH LINE, LOAD, TEMP., ETC. ±1%

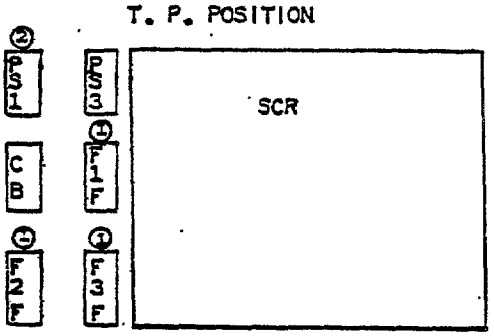
PACKAGE DIMENSIONS: 30"H x 24"W x 12"D (APPROX.)

OUTLINE: 157B6007P1
DULSEAL: 3851G

TEST POINTS

TP	① TFCU/TFCV:	② TPSA
1	--	OSC
2	PA	GND
3	PB	--
4	AC1	+16V
5	--	BIAS
6	UNIJ.EMITTER	-12V
7	INPUT SIG.	--

T. P. POSITION

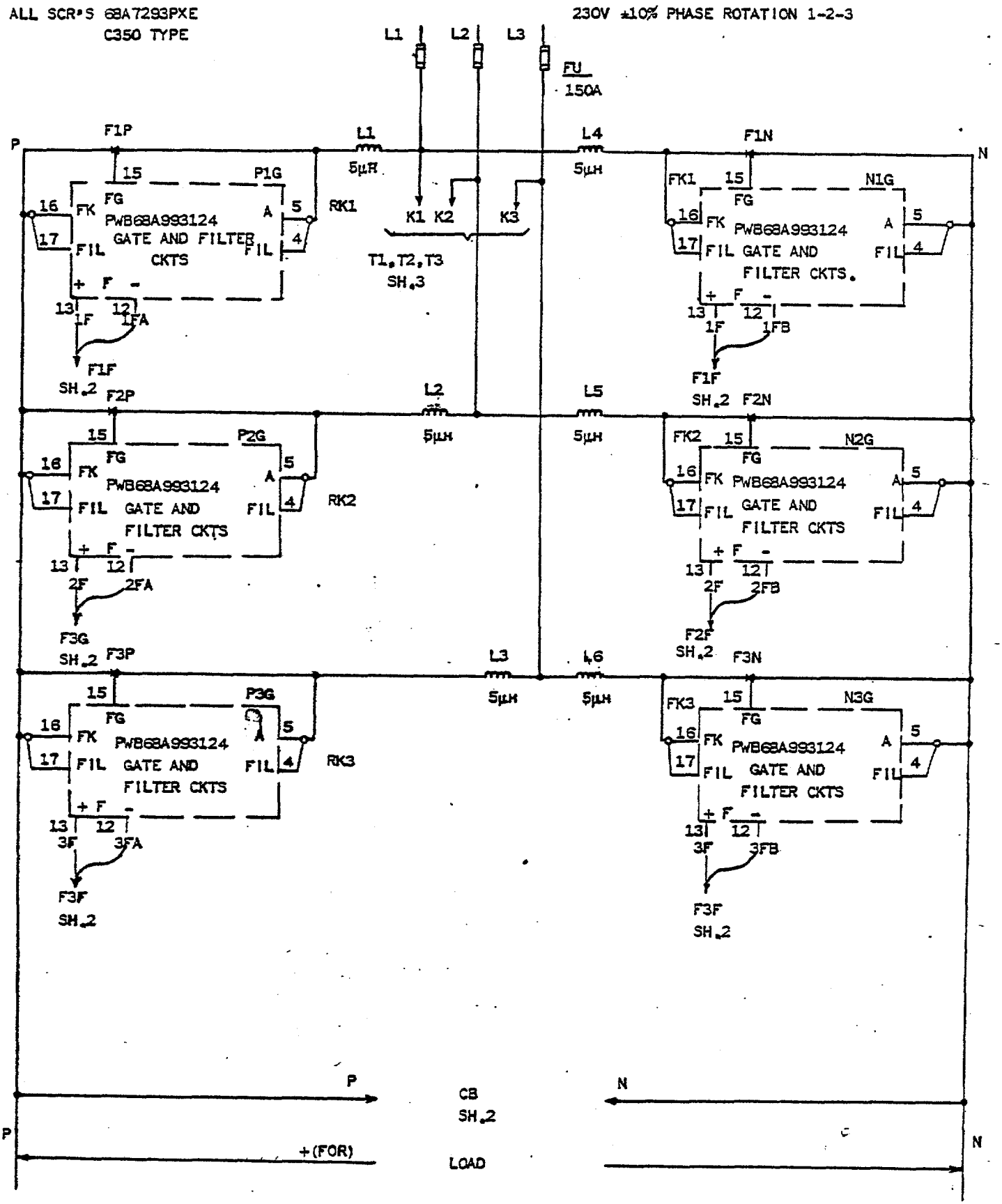


*NOTES: REFER TO GEK 8074 FOR DETAILED INFORMATION

1. DO NOT GROUND TERMINAL C WHEN INTERNAL FEEDBACK IS USED.
2. GROUND TERMINAL C OR USE AS COMMON WHEN NO FEEDBACK IS USED.
3. POWER TERMINAL P. IS POSITIVE WHEN INPUT S IS NEGATIVE WITH RESPECT TO C.
4. PHASE ROTATION 1-2-3
5.
6. INDUCTIVE LOADS WITH .1 SEC. TC OR LONGER; RESISTIVE LOAD OR LESS THAN .1 SEC. TC - CURRENT MUST BE CALCULATED.
7. FREQ. TOLERANCE - ± 1/2 CPS
8. COMMON TERMINAL, C, HAS MICROSECOND SPIKES COUPLED FROM SCR PULSE TRANSFORMERS. IF SYSTEM USES FAST DIGITAL LOGIC, ADD 1K RESISTOR BETWEEN C AND REST OF SYSTEM.

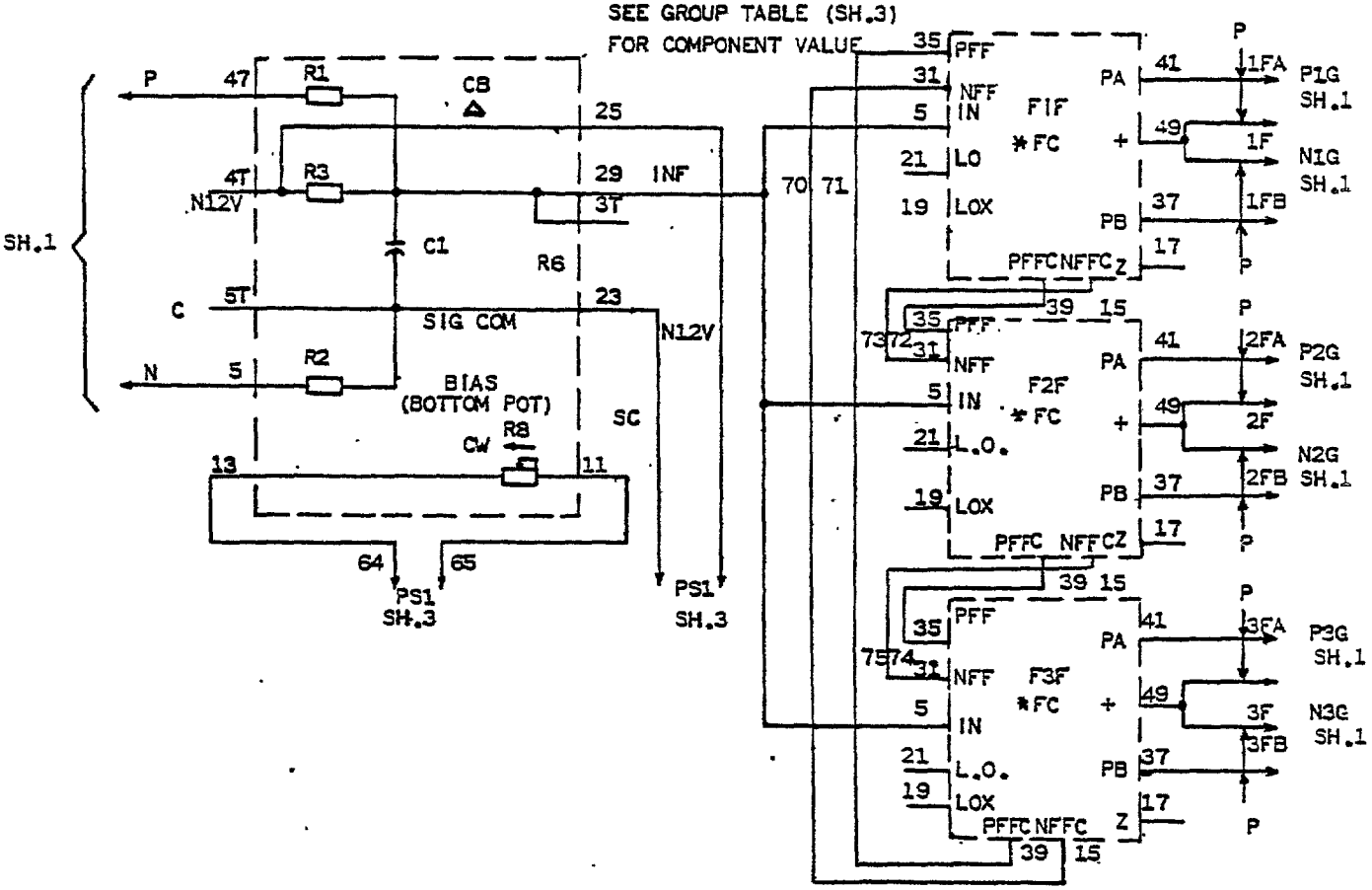
REV 1 11-15-67 REV-2 May 12, 1969	REV 3 12-5-69 REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 50 AMP 250 VOLT				
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	PSA	DLI	1338	6 8 A 9 9 3 6 9 1
ISSUED January 17-1969				ST	2N		CONT ON SHEET FL. SH NO. 4

FORM IC 2143B (9 6).



REV 12-5-69	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER CPE 100 AMP 250 VOLT			
REV	REV						
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	6 8 A 9 9 3 6 9 2
ISSUED December 5-1966				6T	2N		CONT ON SHEET 2 SH NO. 1

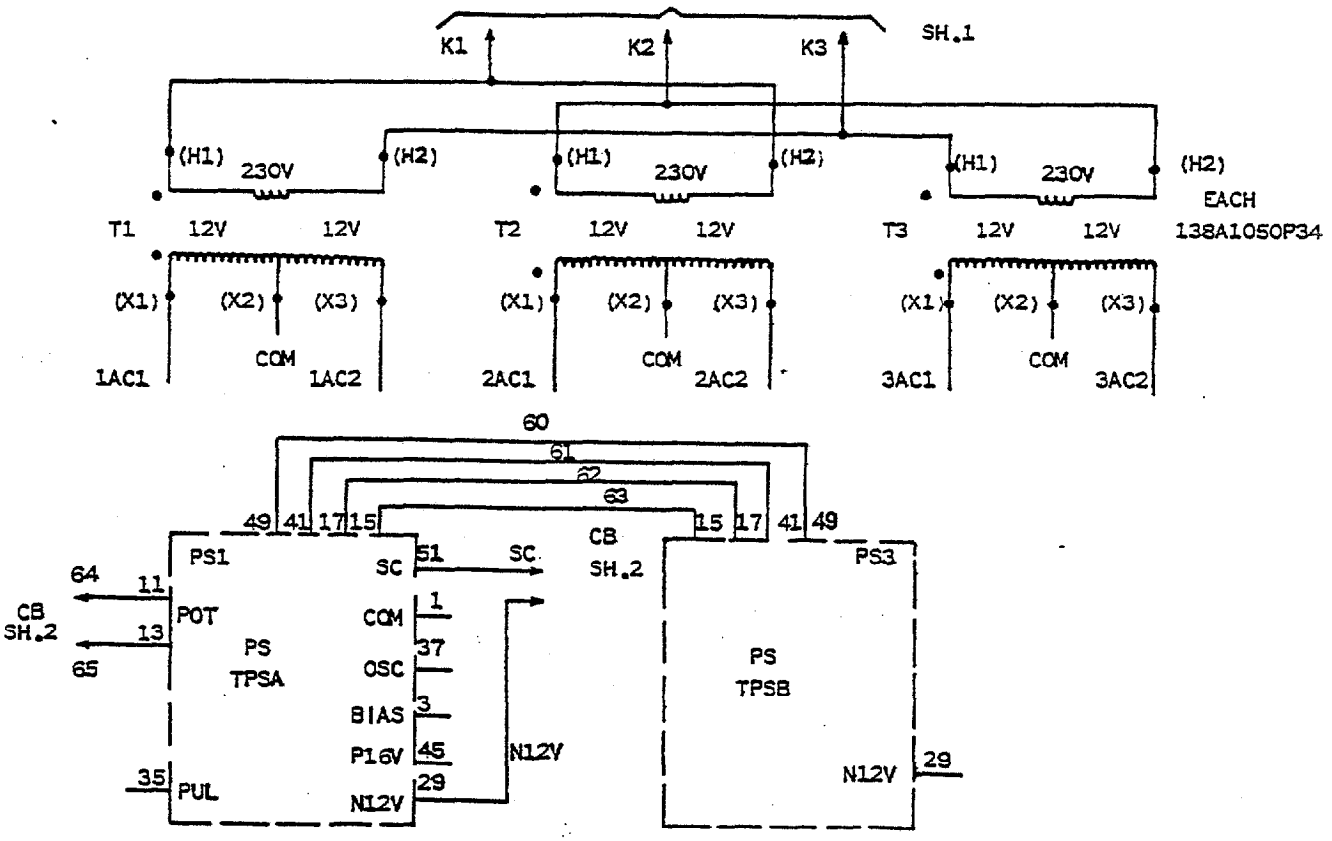
FORM IC 2143B (9-64)



* FC SEE GROUP TABLE (SH. 3)

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 100 AMP 250 VOLT				
REV	REV						
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	6 8 A 9 9 3 6 9 2
ISSUED December 5-1966				6T	2N		CONT ON SHEET 3 SH NO. 2

FORM IC 2143B (9 64)



WIRE TABLE FOR POWER AND REPETITIVE CONNECTIONS

WIRE NO \ DEVICE			F1F		F2F		F3F	PS1	PS3	T1	T2	T3
P16V			45		45		45	45	-	-	-	-
N12V			29		29		29	29	29	-	-	-
OSC			25		25		25	37	-	-	-	-
BIAS			13		13		13	3	-	-	-	-
COM			1_{51}		1_{51}		1_{51}	1_{51}	-	X2	X2	X2
1AC1			33		-			47	-	X1	-	-
1AC2			43		-		-	43	-	X3	-	-
2AC1			-		33		-	-	47	-	X1	-
2AC2			-		43		-	-	43	-	X3	-
3AC1			-		-		33	-	39	-	-	X1
3AC2			-		-		43	-	37	-	-	X3
PUL			27		27		27	35	-	-	-	-

GROUP TABLE

PACKAGE CAT. NO. IC3605	VOLTAGE FEEDBACK	FREQ. CPS	*FC IC3600	▲CB PWB68A 993614	R1 10WATT	R2 10WATT	R3 .5WATT	C1 100V
C100CP250	YES	60	TFCU	G6	12K	12K	1.0K	20MFD
D100CP250	YES	50	TFCV	G6	12K	12K	1.0K	20MFD

FORM IC2143B (9-64)

REV-1 2-28 May 12, 1969	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER CPE 100 AMP 250 VOLT	
REV	REV				
MADE BY B.H. OGDEN	APPROVALS D.J.M.	PRINTS TO	P6A 1338 DL1	6 8 A 9 9 3 6 9 2	
ISSUED Dec 10 5 1966			6T 2N	CONT ON SHEET 4	SH NO. 3

INPUT:

A.C. POWER: 230V ±10% AMP= 82% DC LOAD AMPS *6,7 TERMINALS: L1,L2,L3 *4

OUTPUT:

D.C. 250V 100 AMPS 55°C AMBIENT AT PACKAGE P+, N-
250V 150 AMPS FAN COOLED WITH 280 CFM

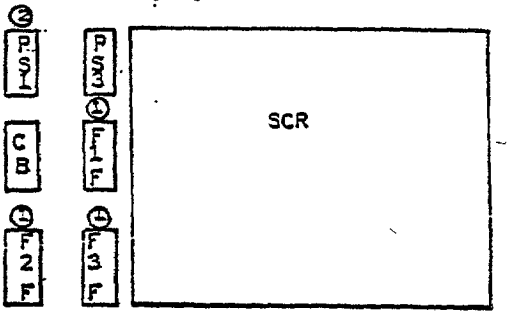
TRANSFER FUNCTION: (INCLUDES EFFECT OF TRANSPORT LAG)
RESPONSE TO 100% LOAD STEP .1 SEC.
REGULATION WITH LINE, LOAD, TEMP., ETC. ±1%

PACKAGE DIMENSIONS: 30"H x 24"W x 12"D (APPROX.) OUTLINE: 15786007P1
DULSEAL: 3851G

TEST POINTS

TP	① TFCU/TFCV	② TPSA
1	--	OSC
2	PA	GND
3	PB	--
4	AC1	+16V
5	--	BIAS
6	UNIJ.EMITTER	-12V
7	INPUT SIG.	--

T. P. POSITION

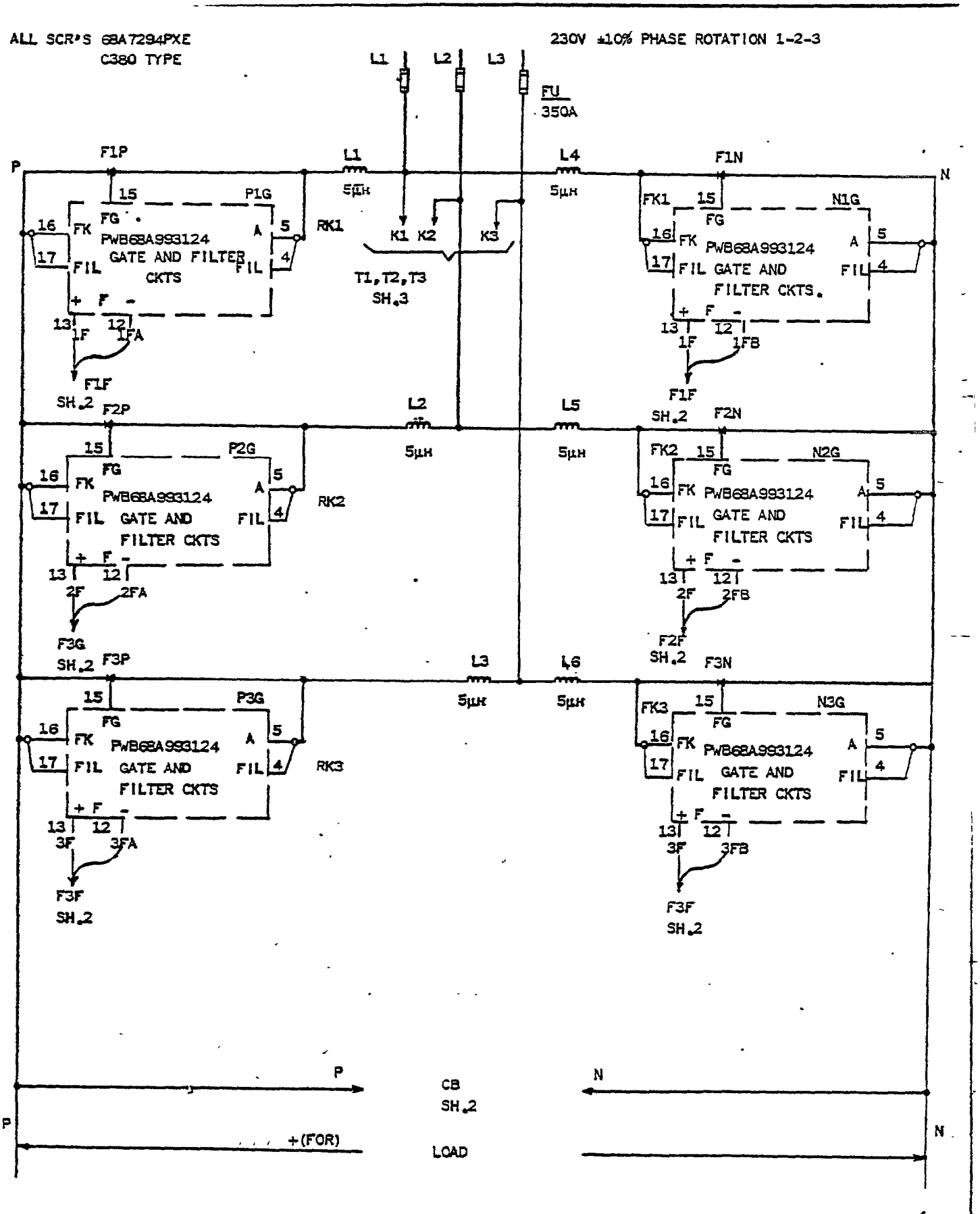


*NOTES: REFER TO GEK 8074 _ FOR DETAILED INFORMATION

1. DO NOT GROUND TERMINAL C WHEN INTERNAL FEEDBACK IS USED.
2. GROUND TERMINAL C OR USE AS COMMON WHEN NO FEEDBACK IS USED.
3. POWER TERMINAL P IS POSITIVE WHEN INPUT S IS NEGATIVE WITH RESPECT TO C.
4. PHASE ROTATION 1-2-3
5.
6. INDUCTIVE LOADS WITH .1 SEC. TC OR LONGER; RESISTIVE LOAD OR LESS THAN .1 SEC. TC - CURRENT MUST BE CALCULATED.
7. FREQ. TOLERANCE - ± 1/2 CPS.
8. COMMON TERMINAL, C, HAS MICROSECOND SPIKES COUPLED FROM SCR PULSE TRANSFORMERS. IF SYSTEM USES FAST DIGITAL LOGIC, ADD 1K RESISTOR BETWEEN C AND REST OF SYSTEM.

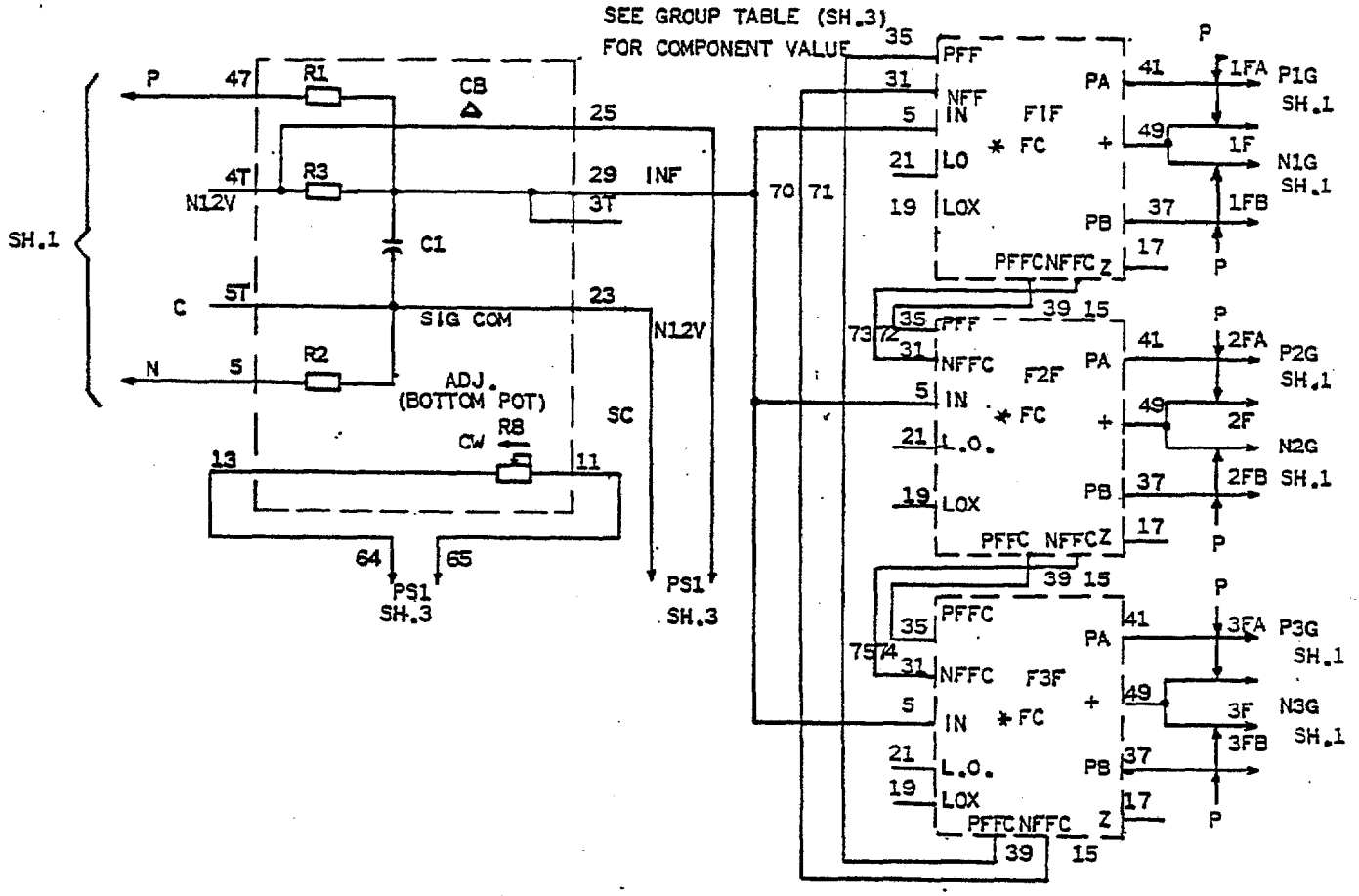
REV I LWO 1-22-67 D. Oddie	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENITIAL EXCITER CPE 100 AMPS 250 VOLTS				
REV J 12-5-68	REV						
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	6 8 A 9 9 3 6 9 2
ISSUED January 17 1969				6T	2N		CONT ON SHEET FL. SH NO. 4

FORM IC 2143B (9 64)



REV 12-5-69	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER			
REV	REV			CPE 150 AMP 250 VOLT			
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	6 8 A 9 9 3 6 9 3
ISSUED December 5-1966				6T	2N		CONT ON SHEET 2 SH NO. 1

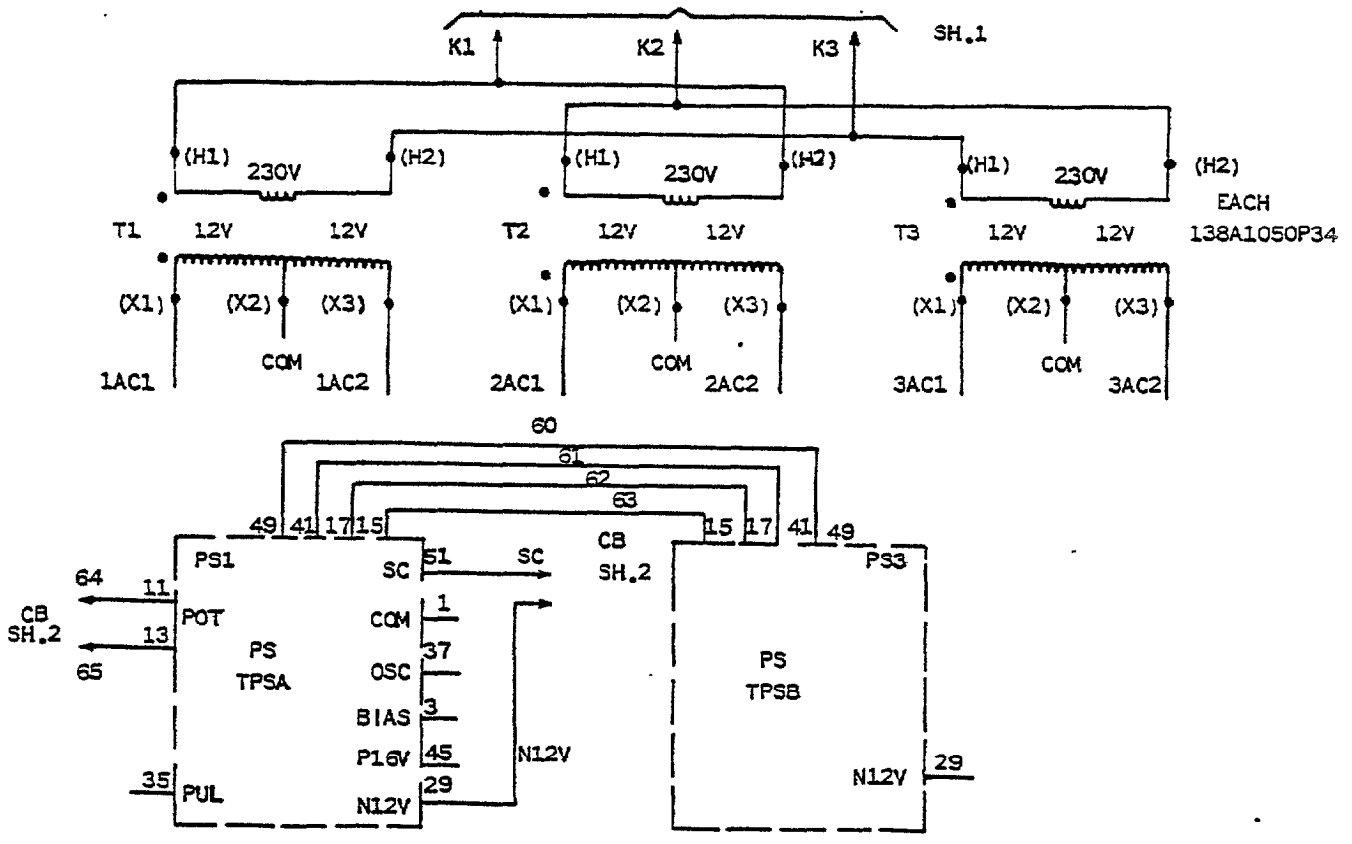
FORM IC21438 (9-66)



*FC SEE GROUP TABLE (SH.3)

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 150 AMP 250 VOLT					
REV	REV							
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P6A	DL1	1338	6 8 A 9 9 3 6 9 3	
ISSUED December 5 1966				6T	2N		CONT ON SHEET 3	SH NO. 2

FORM IC 2143B (9-64)



WIRE TABLE FOR POWER AND REPETITIVE CONNECTIONS

WIRE NO \ DEVICE			F1F		F2F		F3F	PS1	PS3	T1	T2	T3
P16V			45		45		45	45	-	-	-	-
N12V			29		29		29	29	29	-	-	-
OSC			25		25		25	37	-	-	-	-
BIAS			13		13		13	3	-	-	-	-
COM			$^{1}_{S1}$		$^{1}_{S1}$		$^{1}_{S1}$	$^{1}_{S1}$	-	X2	X2	X2
1AC1			33		-			47	-	X1	-	-
1AC2			43		-		-	43	-	X3	-	-
2AC1			-		33		-	-	47	-	X1	-
2AC2			-		43		-	-	43	-	X3	-
3AC1			-		-		33	-	39	-	-	X1
3AC2			-		-		43	-	37	-	-	X3
PUL			27		27		27	35	-	-	-	-

GROUP TABLE

PACKAGE CAT. NO. IC3605	VOLTAGE FEEDBACK	FREQ. CPS	*FC IC3600	ΔCB PWB68A 993614	R1 10WATT	R2 10WATT	R3 .5WATT	C1 100V
C150CP250	YES	60	TFCU	G6	12K	12K	1.0K	200MFD
D150CP250	YES	50	TFCV	G6	12K	12K	1.0K	200MFD

REV-1 RHB May 12, 1969	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 150 AMP 250 VOLT				
REV	REV						
MADE BY B.H. OGDEN	APPROVALS	PRINTS TO	P6A	1338	DL1	6 8 A 9 9 3 6 9 3	
ISSUED December 5, 1966			6T	2N		CONT ON SHEET 4	SH NO. 3

FORM IC2143B (9 64)

INPUT:

A.C. POWER: 230V ±10% AMPS = 82% D.C. LOAD AMPS *6, TERMINALS: L1, L2, L3 *4

OUTPUT:

D.C. 250V 150 AMPS 55°C AMBIENT AT PACKAGE
250V 275 AMPS FAN COOLED WITH 770 CFM P+, N-

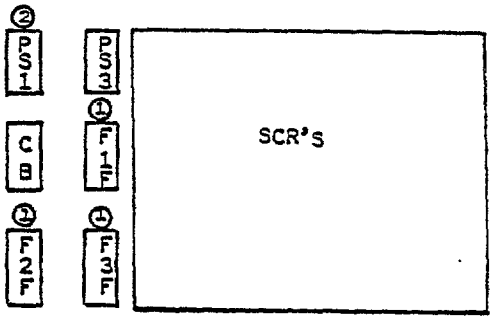
TRANSFER FUNCTION: (INCLUDES EFFECT OF TRANSPORT LAG)
RESPONSE TO 100% LOAD STEP .1 SEC.
REGULATION WITH LINE, LOAD, TEMP., ETC. ±1%

PACKAGE DIMENSIONS: 30"H x 24"W x 12"D (APPROX.)

OUTLINE: 157B6007P1
DULSEAL: 3851G
T.P. POSITION

TEST POINTS

TP	① TFCU/TFCV	② TPSA
1	--	OSC
2	PA	GND
3	PB	--
4	AC1	+16V
5	--	BIAS
6	UNIJ.EMITTER	-12V
7	INPUT SIG.	--



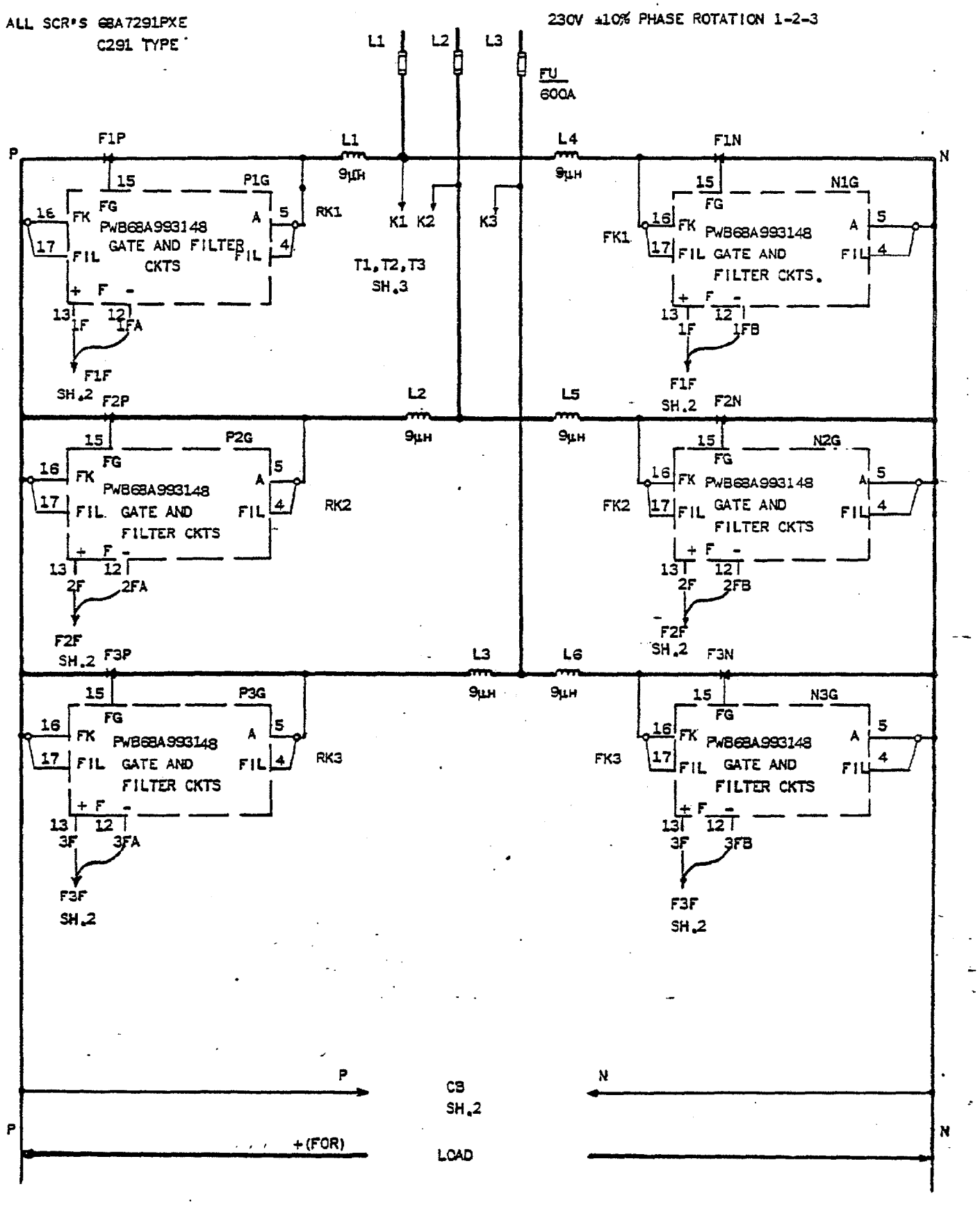
*NOTES: REFER TO GEK 8074 FOR DETAILED INFORMATION

1. DO NOT GROUND TERMINAL C WHEN INTERNAL FEEDBACK IS USED.
2. GROUND TERMINAL C OR USE AS COMMON WHEN NO FEEDBACK IS USED.
3. POWER TERMINAL P IS POSITIVE WHEN INPUT S IS NEGATIVE WITH RESPECT TO C.
4. PHASE ROTATION 1-2-3
5.
6. INDUCTIVE LOADS WITH .1 SEC. TC OR LONGER; RESISTIVE LOAD OR LESS THAN .1 SEC. TC - CURRENT MUST BE CALCULATED.
7. FREQ. TOLERANCE - ± 1/2 CPS
8. COMMON TERMINAL, C, HAS MICROSECOND SPIKES COUPLED FROM SCR PULSE TRANSFORMERS. IF SYSTEM USES FAST DIGITAL LOGIC, ADD 1K RESISTOR BETWEEN C AND REST OF SYSTEM.

REV 1	11-15-67	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER CPE 150 AMP 250 VOLT		
REV 2	12-5-69	REV					
MADE BY	B.H. OSDEN		APPROVALS	PRINTS TO	P6A	DL1	1338
ISSUED	January 17-1967				6T	2N	

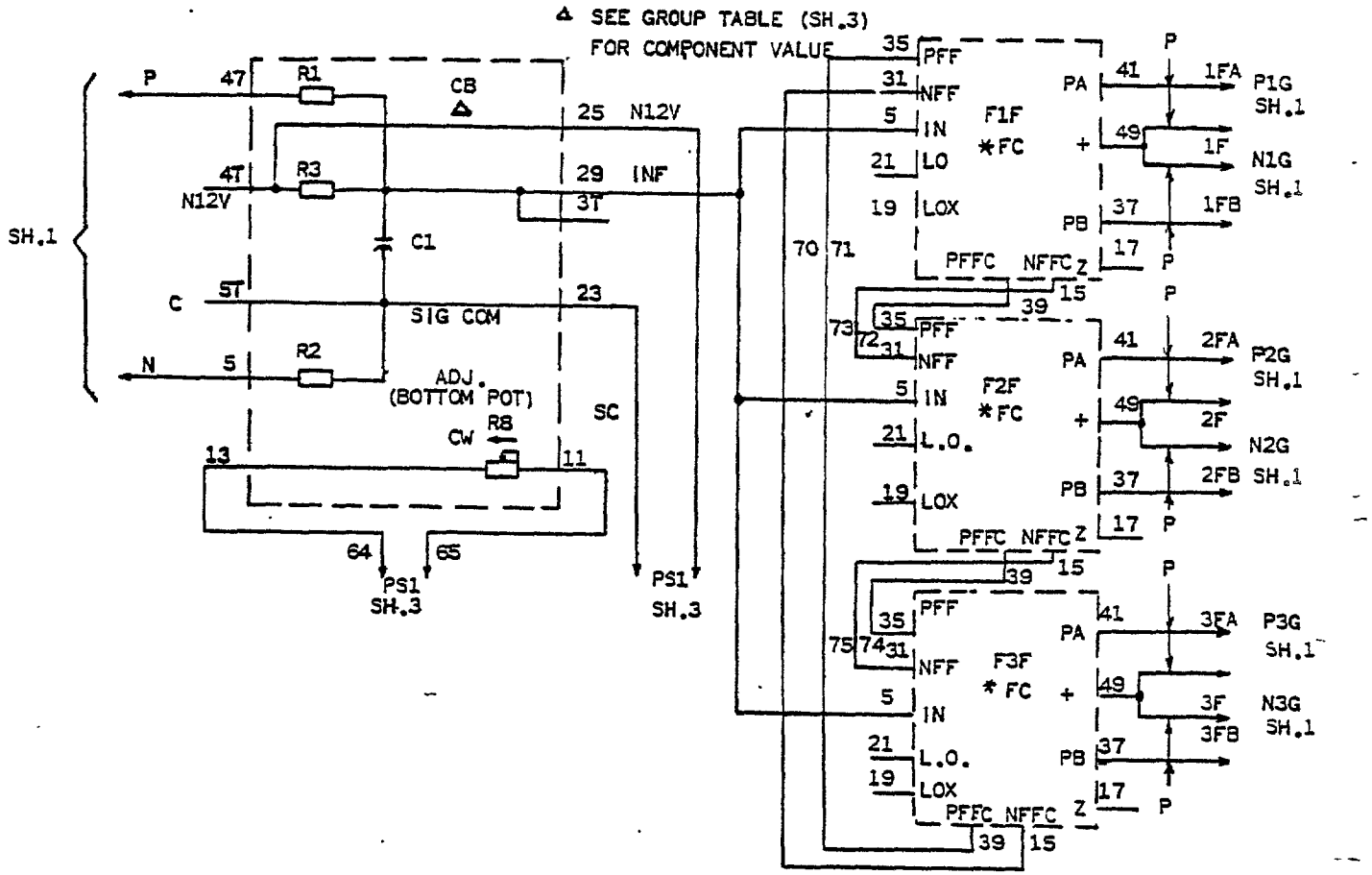
68A993693
CONT ON SHEET FL. SH NO. 4

FORM IC 2143B (9 64)



REV 12-5-69	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	SCR EXCITER CPE 300AMP 250VOLT			
REV	REV						
MADE BY B.H. OGDEN		APPROVALS	PRINTS TO	P&A	DL1	1338	68A993694
ISSUED January 30-1967				6T	2N		CONT ON SHEET 2 SH NO. 1

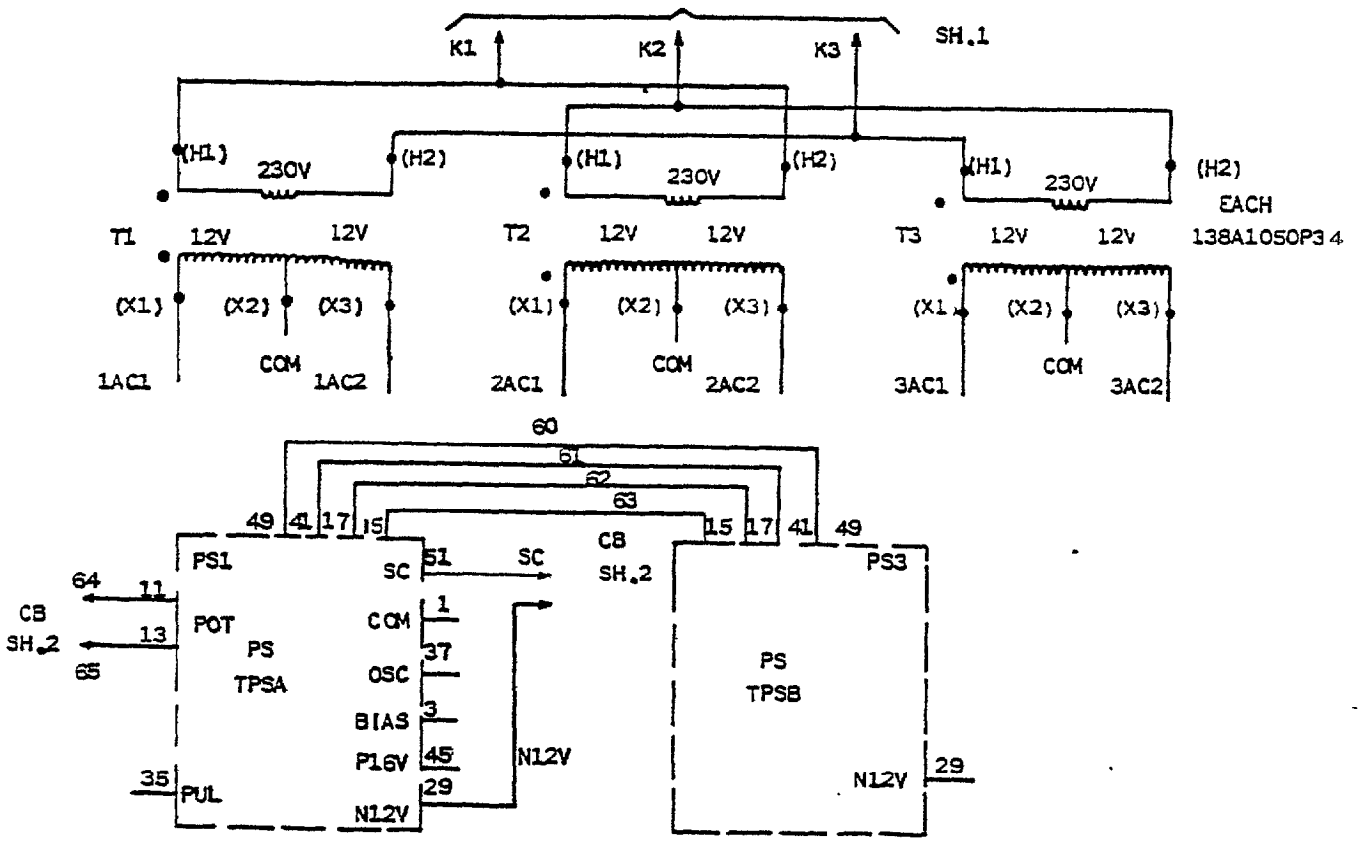
FORM IC 2143B (9-64)



*FC SEE GROUP TABLE (SH.3)

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE SCR EXCITER CPE 300 AMP 250 VOLT				
REV	REV						
MADE BY B.H. OGDEN	APPROVALS	PRINTS TO	P6A	DL1	1338	6 8 A 9 9 3 6 9 4	
ISSUED January 30 1967			6T	2N		CONT ON SHEET 3	SH NO. 2

FORM IC 2143B (9 64)



WIRE TABLE FOR POWER AND REPETITIVE CONNECTIONS

WIRE NO \ DEVICE			F1F		F2F		F3F	PS1	PS3	T1	T2	T3
P16V			45		45		45	45	-	-	-	-
N12V			29		29		29	29	29	-	-	-
OSC			25		25		25	37	-	-	-	-
BIAS			13		13		13	3	-	-	-	-
COM			1_{S1}		1_{S1}		1_{S1}	1_{S1}	-	X2	X2	X2
1AC1			33	-	-	-	-	47	-	X1	-	-
1AC2			43	-	-	-	-	43	-	X3	-	-
2AC1		-	-		33	-	-	-	47	-	X1	-
2AC2		-	-		43	-	-	-	43	-	X3	-
3AC1		-	-	-	-		33	-	39	-	-	X1
3AC2		-	-	-	-		43	-	37	-	-	X3
PUL	-		27		27		27	35	-	-	-	-

GROUP TABLE

PACKAGE CAT. NO. IC3605	VOLTAGE FEEDBACK	FREQ. CPS	*FC IC3600	CB	R1	R2	R3	C1
				PWB68A 993614	10WATT	10WATT	.5WATT	100V
C300CP250	YES	60	TFCU	G 8	12K	12K	1.0K	200MFD
D300CP250	YES	50	TFCV	G 8	12K	12K	1.0K	200MFD

REV	REV
REV	REV

GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.

TITLE SCR EXCITER
CPE 300AMP 250 VOLT

MADE BY B.H. OGDEN

ISSUED January 30-1967

APPROVALS

PRINTS TO

P6A	1338	DL1	6 8 A 9 9 3 6 9 4
6T	2N		CONT ON SHEET 4 SH NO. 3

FORM IC2143B (9 64)

INPUT:

A.C. POWER:- 230V ± 10% AMPS: 82% DC LOAD AMPS *6 TERMINALS: L1, L2, L3 *4

OUTPUT:

D.C. 250V 300 AMPS 55°C AMBIENT AT PACKAGE P+, N-
250V 600 AMPS FAN COOLED WITH 3600 CFM 55°C AIR
1420 50°C AIR
900 45°C AIR

TRANSFER FUNCTION RESPONSE TO 100% LOAD STEP .1 SEC.
REGULATION WITH LINE, LOAD, TEMP, ETC. ±1%

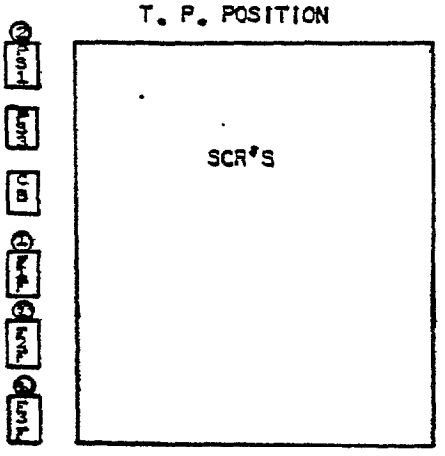
PACKAGE DIMENSIONS: 38"H X 36"W X 12"D (APPROX)

OUTLINE: 157B 6014
BULSEAL: 3851M

TEST POINTS

TP	① TFCU/TFCV	② TPSA
1	—	OSC
2	PA	GND
3	PB	—
4	AC1-	+16V
5	—	BIAS
6	UNIJ.EMMITER	-12V
7	INPUT SIG	—

T. P. POSITION

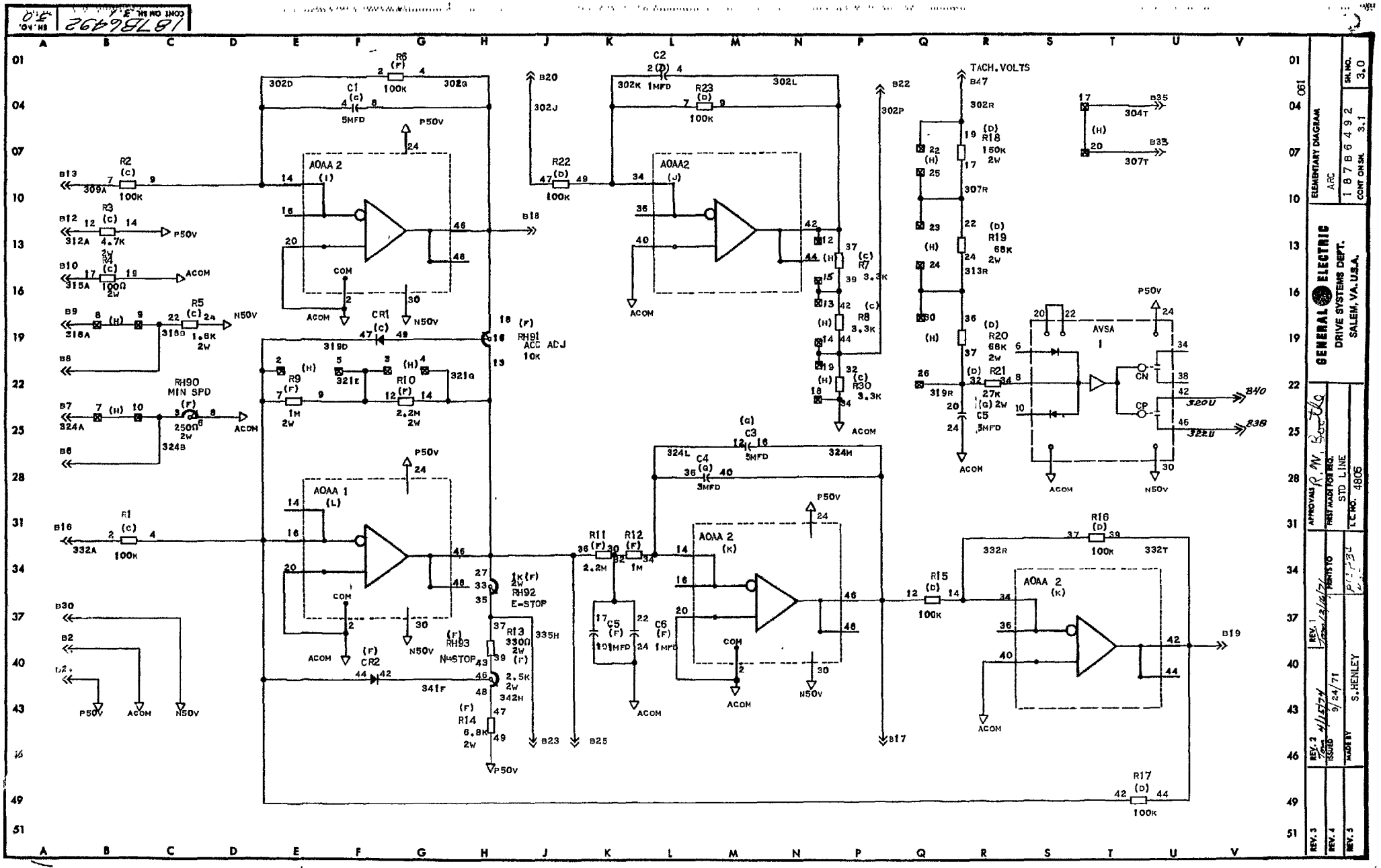


*NOTES: REFER TO GEK 8074 FOR DETAILED INFORMATION

1.
2.
3.
4. PHASE ROTATION 1-2-3
5.
6. INDUCTIVE LOADS WITH .1 SEC T. C. OR LARGER, RESISTIVE LOAD OR LESS THAN .1 SEC T.C. - CURRENT MUST BE CALCULATED.
7. COMMON TERMINAL, C, HAS MICROSECOND SPIKES COUPLED FROM SCR PULSE TRANSFORMERS. IF SYSTEM USES FAST DIGITAL LOGIC, ADD 1K RESISTOR BETWEEN C AND REST OF SYSTEM.

REV 12-5-69	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE SCR EXCITER CPE 300 AMP 250 VOLT				
REV	REV						
MADE BY B. H. OGDEN		APPROVALS D.J.M.	PRINTS TO	PGA 6T	DLI 2N	1338	68A993694
ISSUED June 2-1967						CONT ON SHEET	FL SH NO. 4

FORM IC 21438 (9-6...


GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
ARC
1 8 7 B 6 4 9 2
SH. NO. 3.0
CONT ON SH. 3.1
TACH. VOLTS
AOAA 2 (I)
AOAA2 (J)
AOAA 1 (L)
AOAA 2 (K)
AVSA
P50V
N50V
ACOM
COM
RH90 MIN SPD
RH91 ACC ADJ 10K
RH92 E-STOP
RH93 N-STOP
CR1
CR2
R1
R2
R3
R5
R6
C1
C2
C3
C4
R9
R10
R11
R12
R13
R14
R15
R16
R17
R18
R19
R20
R21
R22
R23
R30
C5
C6
CN
CP
STD LINE
T.C. NO. 4806
S. HENLEY
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5

GENERAL

THE PRIMARY PURPOSE OF THE MODULE IS TO PROVIDE A LINEAR RATE REFERENCE (WITH ROUNDING INTO AND OUT OF THE RATE) FOR A SPEED REGULATED DRIVE. AUXILIARY FUNCTIONS ALSO INCLUDED IN THIS MODULE ARE A ZERO SPEED RELAY AND A SPEED DROOP AMPLIFIER.

CIRCUIT OPERATION

ACCELERATION AND DECELERATION TIMES ARE ADJUSTABLE FROM A MINIMUM OF 30 SECONDS TO A MAXIMUM OF APPROXIMATELY 600 SECONDS. ACCELERATION AND DECELERATION RATES ARE INDEPENDENTLY ADJUSTABLE. THE DECELERATION RATE CAN BE RECALIBRATED (USING AN EXTERNAL INTERLOCK) TO GIVE A FASTER THAN NORMAL EMERGENCY STOP. PROVISION FOR EXTERNAL PROGRAMMING OF THE ACCELERATION RATE AS A FUNCTION OF ROLL DIAMETER IS PROVIDED IN ORDER TO ACCELERATE TO RUNNING SPEED IN THE SHORTEST POSSIBLE TIME. ROUNDING TIMES ARE INDEPENDENTLY ADJUSTABLE FOR ROUNDING INTO (1-2SECONDS) AND ROUNDING OUT OF (1-4 SECONDS) A RATE. A FUNCTIONAL REPRESENTATION OF THE CIRCUIT IS SHOWN BELOW.

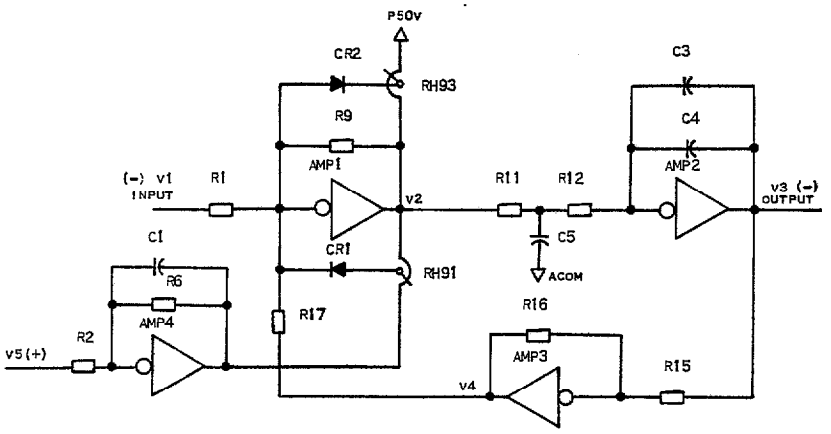


FIGURE 1

AMPLIFIER 2 IS AN INTEGRATING AMPLIFIER. AN INTEGRATOR HAS THE PROPERTY THAT A CONSTANT VOLTAGE INPUT CAUSES THE INTEGRATOR TO RAMP UP AT A LINEAR RATE. AMPLIFIER 2 IS USED AS THE MAIN RATE GENERATOR IN THIS CIRCUIT. THE SLOPE OF THIS RATE DETERMINES THE TOTAL ACCELERATION TIME. A SMALL SLOPE INDICATES A LONGER ACCEL TIME AND VICE VERSA. SLOPE IS EQUAL TO $\Delta V/\Delta T$ OR $V2/[(C3 + C4)(R11 + R12)]$ OR $V2/25.2$. THE ADJUSTABLE QUANTITY IN THIS EQUATION IS V2. TO FIND V2 REQUIRED FOR A GIVEN ACCELERATION TIME FROM 0 TO RATED VOLTAGE SOLVE THE FOLLOWING :

$$V2 = \frac{75\ 6.0}{\text{TIME REQ'D TO ACCELERATE}}$$

THE ABOVE EQTN ALSO APPLIES FOR DECELERATION EXCEPT THAT V2 IS OPPOSITE IN POLARITY. CAPACITOR, C5, ON THE INPUT OF AMPLIFIER 2 IS PRESENT TO PRODUCE ROUND-IN.

THE PURPOSE OF AMPLIFIER 1 IS TO PRODUCE THE CONSTANT VOLTAGE INPUT FOR AMPLIFIER 2 AND TO PRODUCE ROUND-OUT. AMPLIFIER 1 IS A

REV. 3	REV. 2	REV. 1	APPROVALS R. W. Beattie	GENERAL ELECTRIC	ASSEMBLY DIAGRAM
REV. 4	ISSUED 9/24/71	PRINTS TO	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	APPLICATION DATA
REV. 5	MADE BY S. HENLEY		T.C. NO. 4805		187B6492 CONT. ON SH. 3.2 — SH. NO. 3.1

DRAFTING MEDIA GRAPHIC CONTROLS CORPORATION Buffalo, New York Printed in U.S.A. ASTROFILM N685-4

CONSTANT GAIN SUMMING AMPLIFIER WITH DIODE CLAMPS TO LIMIT ITS OUTPUT SWING WITHIN ADJUSTABLE LIMITS IN BOTH THE FORWARD AND NEGATIVE DIRECTIONS. THE VOLTAGE AT WHICH AMP 1 CLAMPS IN THE NEGATIVE DIRECTION IS DETERMINED BY THE SETTING OF RH93. THE DIODE CLAMP IS ESSENTIAL TO CIRCUIT OPERATION SINCE A CONSTANT VOLTAGE INPUT TO AMP 2 IS NECESSARY FOR A LINEAR RATE. THE CLAMP WORKS AS FOLLOWS : DIODE,CR2,IS INITIALLY BACK-BIASED BY THE P50V BUS THROUGH RH93. WHEN THE NET VOLTAGE INTO AMP 1 POSITIVE, V2 INCREASES IN A NEGATIVE DIRECTION. THE SLIDER OF RH93 STARTS TO GO NEGATIVE AS THE CLAMPS VOLTAGE IS APPROACHED, HOWEVER,THIS FORWARD BIASES CR2 CAUSING IT TO CONDUCT CURRENT INTO THE AMPLIFIER SUMMING JUNCTION. THIS CURRENT IS IN A DIRECTION TO REDUCE THE AMPLIFIER VOLTAGE. NO FURTHER CHANGE IN OUTPUT VOLTAGE IS POSSIBLE (IN A NEGATIVE DIRECTION).

AMPLIFIER 4 IS USED AS A VOLTAGE SOURCE FOR AN AUTOMATIC RECALIBRATION OF THE ACCELERATION RATE. AMPLIFIER 4 IS SIMPLY A CONSTANT GAIN AMPLIFIER WITH ITS REFERENCE VOLTAGE OBTAINED FROM AN EXTERNAL VOLTAGE SOURCE. ITS OUTPUT VOLTAGE IS USED TO REVERSE BIAS DIODE CR1 AND THUS CONTROL THE POSITIVE CLAMP LEVEL TO AMPLIFIER 1.

AMPLIFIER 3 IS AN INVERTING AMPLIFIER. ITS OUTPUT VOLTAGE IS USED TO DRIVE AMPLIFIER 1 TO 0 VOLTAGE WHEN RATED REFERENCE IS REACHED.

FIGURE 2 BELOW SHOWS THE VOLTAGES FOR AN ACCELERATION FROM 0 TO RATED VOLTAGE.

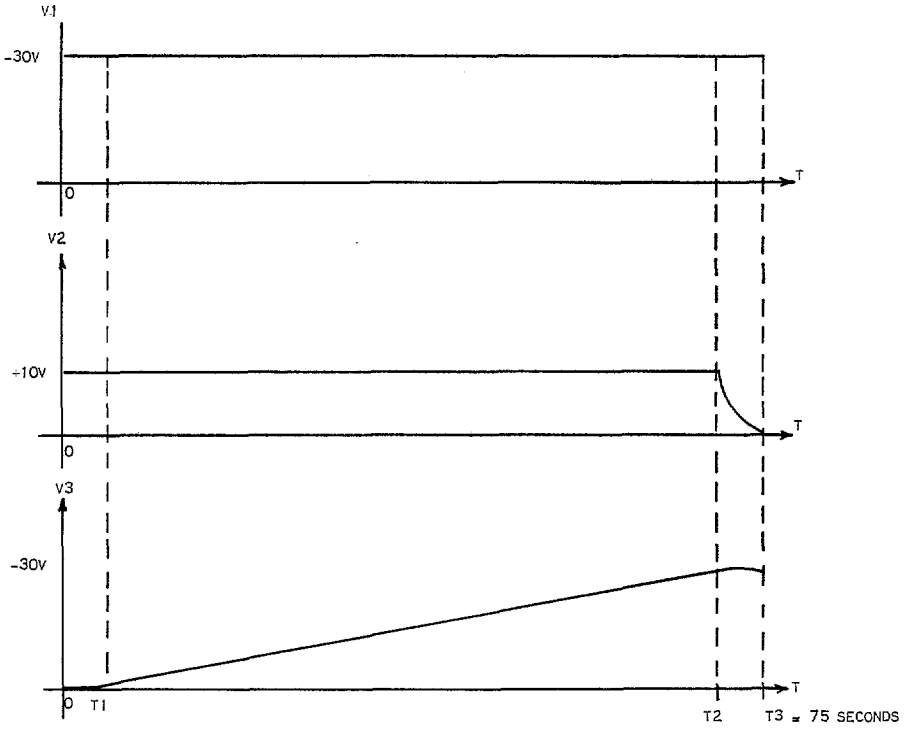


FIGURE 2

REFERRING TO FIGURE 2 ABOVE THE SEQUENCE IS AS FOLLOWS. THE INPUT REFERENCE OF (-) 30 VOLTS IS APPLIED AT T= 0.

AMPLIFIER 1

061

REV. 3	REV. 2	REV. 1	APPROVALS R W Rothe	GENERAL ELECTRIC	ASSEMBLY DIAGRAM	
REV. 4	ISSUED 9/24/71	PRINTS TO	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	APPLICATION DATA	
REV. 5	MADE BY S Z SLUSSER	9/24/71	I.C. NO. 4805		1 8 7 B 6 4 9 2 CONT. ON SH. 3,3	SH. NO. 3,2

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CLAMPS AT (+) 10 VOLTS INSTANTANEOUSLY. AMPLIFIER 3 STARTS TO INTEGRATE, HOWEVER, DURING THE TIME CAPACITOR C5 IS CHARGING, THE OUTPUT VOLTAGE V3 IS ROUNDING. AT TIME T1 CAPACITOR C1 IS FULLY CHARGED AND AMPLIFIER 2 INTEGRATES ON A LINEAR RATE. AMPLIFIER 3 FEEDS THE OUTPUT VOLTAGE V3 BACK TO THE INPUT. WHEN V4 REACHES A LEVEL SUCH THAT THE SUM OF V1 AND V2 MULTIPLIED BY THE GAIN OF AMP 1 IS LESS THAN THE CLAMP VOLTAGE THEN V3 STARTS TO ROUND TOWARD ITS FINAL VALUE. THIS OCCURS AT TIME T2. AT TIME T3 THE RATE IS COMPLETE AND THE OUTPUT VOLTAGE OF AMP 1 IS 0. A DISCUSSION OF DECELERATION WOULD BE SIMILAR, THEREFORE, IT WILL NOT BE INCLUDED.

IN SUMMARY, THE LENGTH OF TIME FOR ACCELERATION OR DECELERATION IS DETERMINED BY THE CLAMP VOLTAGE, V2. CAPACITOR C5 DETERMINES THE AMOUNT OF ROUND - IN. ROUND - OUT IS DETERMINED BY THE GAIN OF AMPLIFIER 1 (SINCE THIS GAIN DETERMINES THE DIFFERENCE BETWEEN THE OUTPUT VOLTAGE AND THE REFERENCE VOLTAGE WHICH BRINGS AMPLIFIER 1 OUT OF CLAMP).

AUXILIARY FUNCTIONS

SPEED DROOP, AN OPERATIONAL AMPLIFIER IS PROVIDED FOR STABILIZATION AND SCALING OF THE SPEED DROOP FUNCTION. IN NORMAL OPERATION A CURRENT FEEDBACK IS APPLIED TO B20. THE OUTPUT, B22, IS THEN CONNECTED TO THE SUMMING JUNCTION OF THE SPEED REGULATING MODULE.

ZERO SPEED RELAY. A VOLTAGE SENSITIVE RELAY IS PROVIDED FOR SENSING A SPEED CLOSE TO ZERO. AN INTERLOCK IS WIRED TO THE TERMINAL BOARD FOR USE IN THE DRIVE MAGNETICS.

TYPICAL VOLTAGE LEVELS

CONDITION	POINT	DESCRIPTION	LEVEL (VOLTS)
RATED REFERENCE	B16	REFERENCE INPUT	(-) 30
RATED REFERENCE	B25	AMPLIFIER 1 OUTPUT	0
RATED REFERENCE	B17	MODULE OUTPUT	(-) 30
RATED REFERENCE	B19	OUTPUT VOLTAGE FEEDBACK	(+) 30
RATED CURRENT	B20	ARMATURE CURRENT FEEDBACK	(-) 7.5
RATED CURRENT	B22	SPEED DROOP VOLTAGE (3o/o)	(+) 5.6V
RATED SPEED	B47	RATED TACH VOLTAGE	VARIABLE

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REV. 3	REV. 2	REV. 1	APPROVALS R. W. Boothe	GENERAL ELECTRIC	ASSEMBLY DIAGRAM APPLICATION DATA
REV. 4	ISSUED 9/24/71	PRINTS TO	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	1 8 7 8 6 4 9 2 CONT. ON SH. 3.4 / SH. NO. 3.3
REV. 5	MADE BY S Z SLUSSER	PIC, PSC, C/O	I.C. NO. 4805		

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ARC MODULE JUMPER ARRANGEMENT

JUMPER IN	JUMPER OUT	LOCATION	CONDITION	FUNCTION
X		H (22-25)	50 ≤ RATED TACH VOLTS < 150	ZERO SPEED RELAY TACH VOLTAGE CALIBRATION
	X	H (23-24)		
X		H (30-26)		
	X	H (22-25)	150 ≤ RATED TACH VOLTS < 200	
X		H (23-24)		
X		H (30-26)		
	X	H (22-25)	200 ≤ RATED TACH VOLTS < 250	
	X	H (23-24)		
X		H (30-26)		
	X	H (22-25)	250 ≤ RATED TACH VOLTS	
	X	H (23-24)		
	X	H (30-26)		
X		H (12-13)	3 o/o	SPEED DROOP
X		H (15-14)		
	X	H (19-18)		
X		H (12-13)	2 o/o	
	X	H (15-14)		
	X	H (19-18)		
	X	H (12-13)	1 o/o	
	X	H (15-14)		
	X	H (19-18)		
	X	H (12-13)	0 o/o	
	X	H (15-14)		
X		H (19-18)		
	X	F (17-19)	NO ROUND IN	ROUND IN (NOTE: CAPACITORS ARE MOUNTED IN THE LOCATIONS GIVEN)
	X	F (22-24)		
X		F (17-19)	1 SECOND	
	X	F (22-24)		
X		F (17-19)	2 SECONDS	
X		F (22-24)		
	X	H (2-5)	MAX ROUNDING TIME	ROUND OUT
X		H (3-4)		
X		H (2-5)	MED. ROUNDING TIME	
	X	H (3-4)		
	X	H (2-5)	MIN. ROUNDING TIME	
	X	H (3-4)		
X		H (8-9)	NO RECALIBRATION	SPEED REFERENCE RECALIBRATION
X		H (7 -10)		
	X	H (8-9)	FOR RECALIBRATION ADD APPROPRIATE RESISTORS	
	X	H (7-10)		
X		H (17-20)	NO RECALIBRATION	VOLTAGE REFERENCE RECALIBRATION
	X	H (17-20)	FOR RECALIBRATION ADD APPROPRIATE RESISTORS	

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REV. 3	REV. 2	REV. 1	APPROVALS R W Burke	GENERAL ELECTRIC	ASSEMBLY DIAGRAM APPLICATION DATA
REV. 4	ISSUED 9/25/71	PRINTS TO	FIRST MADE FOR REQ STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	1 8 7 8 6 4 9 2 CONT. ON SH. 3.5 SH. NO. 3.4
REV. 5	MADE BY S Z SLUSSER	PIC/PSC	I.C. NO. 4805		

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ADJUSTMENTS

COMPONENT	FUNCTION	RANGE
RH93	NORMAL STOP	50-600 SECONDS
RH92	EMERGENCY STOP	VARIABLE DEPENDING ON SETTING OF RH93
RH91	ACCELERATION	30 TO 600 SECONDS

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REV. 3	REV. 2	REV. 1	APPROVALS R.M. Bowtle	GENERAL ELECTRIC	ASSEMBLY DIAGRAM APPLICATION DATA
REV. 4	ISSUED 9/24/71	PRINTS TO	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	187B6492 SH. NO. 3.5
REV. 5	MADE BY S.HENLEY		I.C. NO. 4805		CONT. ON SH. 4.0

DRAFTING MEDIA GRAPHIC CONTROLS CORPORATION Buffalo, New York Printed in U.S.A. ASTROFILM N685-4

1.0 POWER REQUIREMENTS

1.1 ± 50V REGULATED POWER SUPPLY

2.0 EQUIPMENT

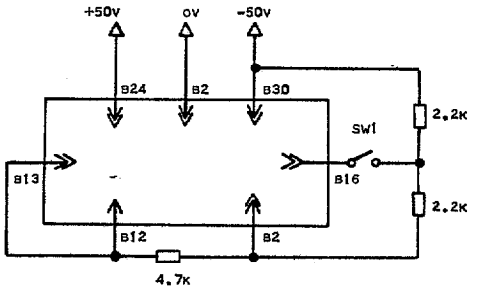
2.1 BRUSH RECORDER

2.2 SINGLE POLE SWITCH

2.3 (2) (2.2K) 2W RESISTORS

2.4 (1) (4.7K) 2W RESISTORS

3.0 WIRING



4. TEST PROCEDURE (AFTER WIRE CHECKING)

4.1 CHECK B18 FOR (-) 25 VOLTS.

4.2 REMOVE AMPLIFIER ROK. CLOSE SW1. SET RH91 FOR (+) 5 VOLTS AT B25. OPEN SW1 AND REPLACE AMPLIFIER ROK.

4.3 SET RH12 FULLY CCW. SET RH93 AT .2.

4.4 MONITOR B19 WITH A BRUSH RECORDER.

4.5 CLOSE SW1. THE TRACE OBTAINED SHOULD APPEAR AS SHOWN BELOW.

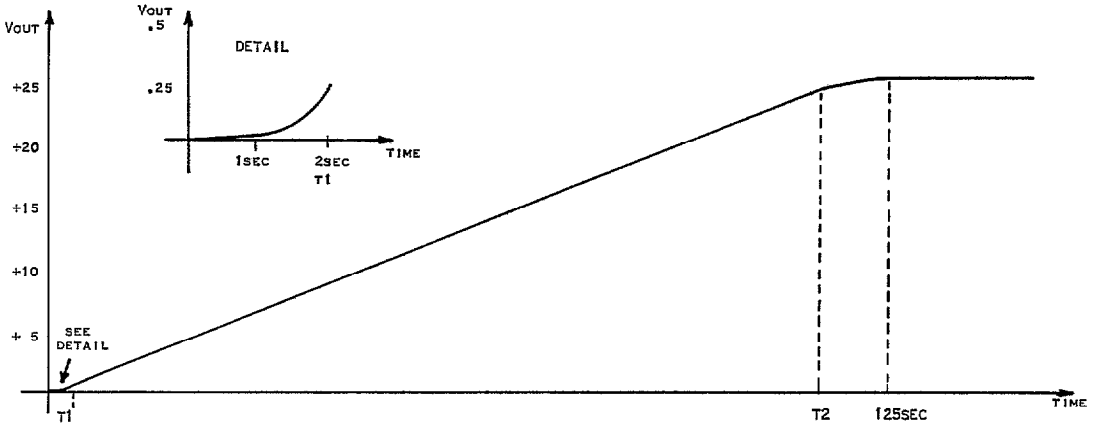


FIG. 1

4.6 OPEN SW1. VOLTAGE AT B19 SHOULD FOLLOW A CURVE SIMILAR TO FIGURE 1 FROM (+) 25 VOLTS TO 0 VOLTS IN 125 SECONDS.

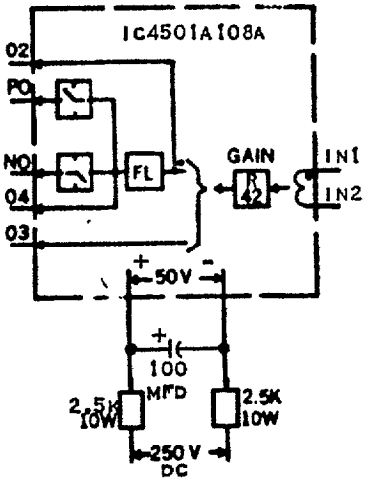
4.7 JUMPER B20 TO B12. CHECK B22 FOR (-) 8.3 VOLTS.

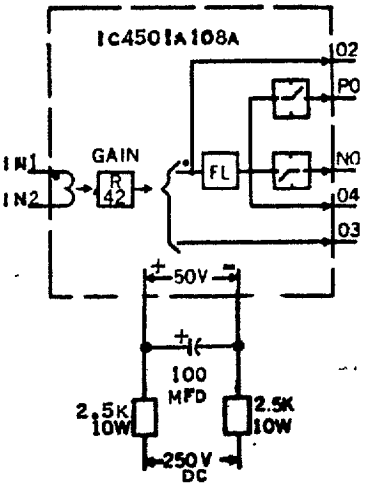
4.8 JUMPER B47 TO B24. CHECK B40 TO B38 WITH AN OHMMETER; CP SHOULD BE PICKED UP.

4.9 REMOVE ALL TEST WIRING AND TEST JUMPERS

REV. 3	REV. 2	REV. 1	APPROVALS R. W. Bourke	GENERAL ELECTRIC	ASSEMBLY DIAGRAM
REV. 4	ISSUED 9/24/71	PRINTS TO	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TEST SPECIFICATIONS
REV. 5	MADE BY S.HENLEY		T.C. NO. 4805	1B7B6492 CONT. ON SH. FL.	SH. NO. 4.0

DRAFTING MEDIA GRAPHIC CONTROLS CORPORATION Buffalo, New York Printed in U.S.A. ASTROFILM N665 4





SPECIFICATIONS

1. INPUT POWER - 50V ±7.5V, 50 MA MAX. 20 MA TYP.
2. OUTPUT - ±20V MAX., 10 ±8 MA.
3. GAIN ADJUSTMENT - POTENTIOMETER ON CARD PROVIDES 37 TO 225 VOLTS/VOLT.
4. ISOLATION - SIGNAL INPUT HAS 1000 VOLT INSULATION RATING TO OUTPUT AND TO POWER SUPPLY. OUTPUT HAS 250V INSULATION RATING TO POWER SUPPLY.
5. SIGNAL INPUT IMPEDANCE - 250K.
6. OFFSET - ADJUSTABLE TO ZERO WITH BAL. POT.
7. OFFSET STABILITY - IC3600AIAA. 500NV ± 10% /°C.
   IC3600AIAB. 50 V ± 10% /°C.
8. GAIN STABILITY - .01%/°C.
9. FREQUENCY RESPONSE - 3DB DOWN AT 400 CYCLES 10° PHASE SHIFT AT 50 CYCLES.
10. OUTPUT NOISE - LESS THAN 15 MV PEAK TO PEAK.

APPLICATION DATA

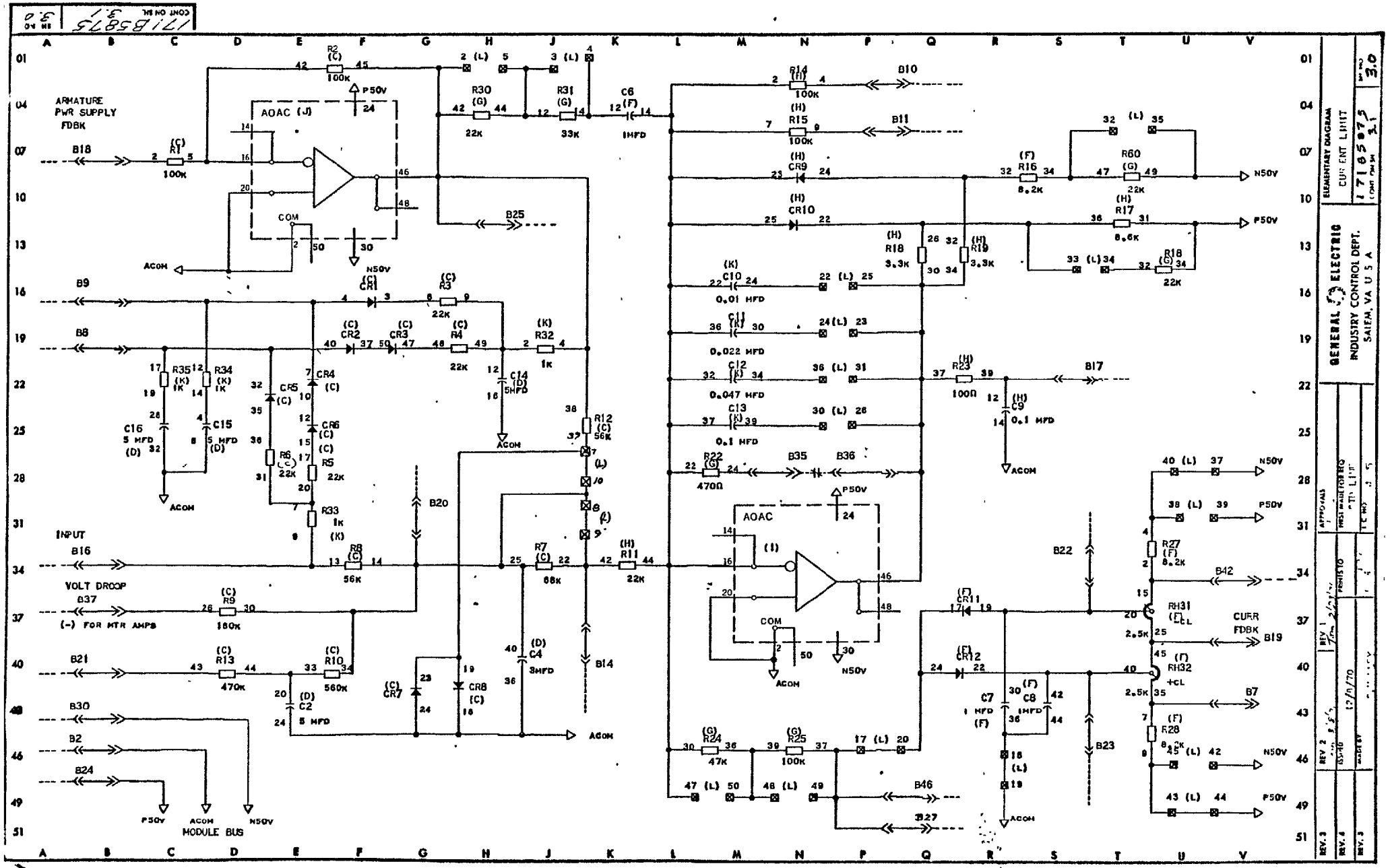
THIS PACKAGE CONSISTS OF CURRENT SENSOR AMPLIFIER IC3600AIAA OR B AND ISOLATION AMPLIFIER IC3600AVIB. THE SIGNAL INPUT TO THE ISOLATED CURRENT SENSOR WILL NORMALLY BE TAKEN FROM A 100 MILLIVOLT SHUNT. THE AMPLIFIER GAIN IS ADJUSTABLE AND SHOULD NORMALLY BE ADJUSTED TO GIVE 15 VOLTS OUTPUT AT THE MAXIMUM EXPECTED INPUT SIGNAL. EACH 1/10 REVOLUTION OF THE GAIN POT WILL GIVE A CHANGE OF APPROXIMATELY 36 MILLIVOLTS IN THE AMOUNT OF INPUT SIGNAL REQUIRED FOR 15 VOLTS OUTPUT.

THE ISOLATED CURRENT SENSOR PROTOTYPES PROVIDED MORE THAN 90DB REJECTION OF 60 CYCLE COMMON MODE VOLTAGE. SINCE THE AMPLIFIER HAS A HIGH GAIN THE INPUT WIRES MUST BE PROTECTED AGAINST NOISE PICK-UP. TWISTED INPUT LEADS WILL GIVE SATISFACTORY PROTECTION IF THE INPUT LEADS ARE 6 FEET OR LESS IN LENGTH. OUTPUT LINES SHOULD BE SHIELDED TWISTED AND SHOULD BE RUN IN A LEVEL 1 WIRE TRAY UNLESS THE ISOLATOR IS PAGE MOUNTED.

WHEN THE INPUT POWER IS TAKEN FROM A 250 VOLT EXCITER BUS, ADDITIONAL CIRCUITRY MUST BE USED AS SHOWN ON THE DULL SEAL SYMBOL.

REV. C3 REWORKED NOV 8 E.C. 68 ACM/WC TUCK	REV.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE ELEMENTARY ISOLATED CURRENT SENSOR (IC4501A108)				
REV.	REV.						
MADE BY W. C. TUCK		APPROVALS E.T.C	PRINTS TO	DL1	P6A	6T	6 8 A 9 9 3 7 4 6
ISSUED Nov. 26 1968							CONT ON SHEET FL. SH. NO. 1

IC 2143


ARMATURE PWR SUPPLY FDBK
AOAC (J)
AOAC (I)
INPUT
VOLT DROOP
(-) FOR MTR AMPS
CURR FDBK
MODULE BUS
P50V
N50V
ACOM
COM
GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA U S A
ELEMENTARY DIAGRAM
CURRENT LIMIT
1716587G1
SH NO 3.0

The CL module is used to provide two different types of current limit operation for speed regulators. One type is spillover current limit where the armature current is allowed to flow until a set value is exceeded. At this point, the armature current is limited to the set value. The other type provides a circuit where the armature current is regulated proportional to a reference. The limit occurs when the reference level is prevented from exceeding a set level, thereby, limiting the armature current. The spillover circuit is used for sectional drives, the external current regulator is used for winder speed regulators.

Figure 1 shows the spillover current limit circuit.

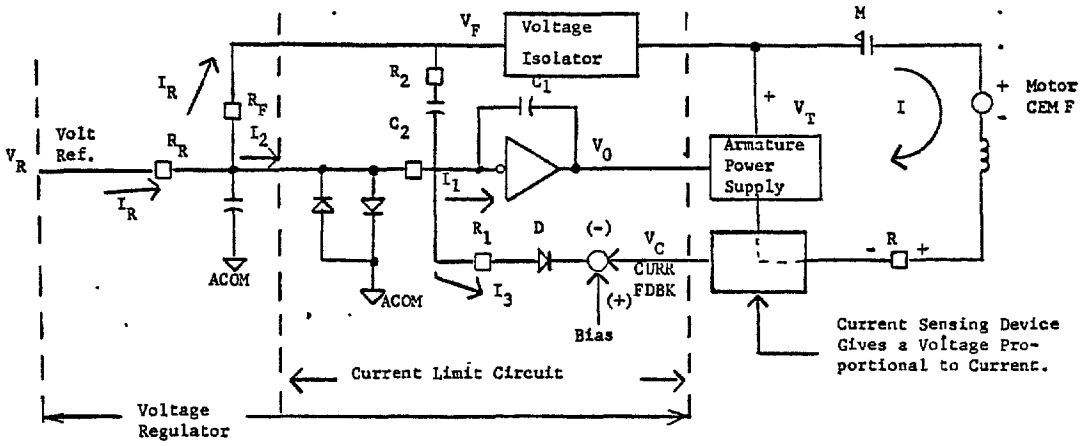


FIGURE 1 -- SPILLOVER CURRENT LIMIT

Figure 1 is split into two portions, the current limit and voltage regulators. The voltage regulator will be described first, then the effect of current limit will be discussed.

The current sensing device produces a voltage proportional to armature current. This signal (V_C) is compared to a bias. Diode D will be back biased as long as V_C is less that the bias level. With this condition, there will be no effect on the operational amplifier by the current feedback. The voltage isolator is used to give a voltage feedback signal proportional to the armature terminal volts (V_T). The amplifier configuration uses a capacitor (C_1) as the operational amplifier feedback. This means that at steady state, the input current (I_1) will be zero in the regulating mode. When the current feedback circuit has no effect on the amplifier, I_2 will also be zero. R_2 and C_2 are used for transient operation and the current through them will be zero at steady state (V_F a constant voltage). This means that $I_R=I_F$ at steady state. If any difference exists, then the current into the amplifier will change, causing the output (V0) to change. The amplifier output is used as the reference to the armature power supply and will cause V_T to change proportional to V_0. Since $R_R=R_F$, the amplifier will regulate $V_F= V_R$ and, consiquently, V_T proportional to V_R at steady state.

The current limit operation will be described for an acceleration sequence. As previously described, diode D will block current flow through R_1 as long as the bias voltage is greater than V_C. When V_C exceeds the bias diode, D, will conduct putting the regulator into current limit. If there is any appreciable inertia connected to the motor and its load, the CEMF cannot build up fast enough, following the terminal voltage, without circulating excessive current. Therefore, a current limit circuit is required. Upon initiation of a run, assume full reference is applied, this corresponds to rated terminal volts. Initially, I_2 will cause V_0 to change, increasing V_T. Since CEMF has not increased, significantly, due to the inertia, armature current will flow. When V_C exceeds the bias level, D conducts causing less current flow into the amplifier. This means that V_0 (and V_T) will be prevented from increasing any further. As CEMF increases (due to this acceleration current) the armature current would tend to decrease. However, I_2 is still tending to cause the amplifier (and V_T) to head towards rated and as CEMF increases V_0 will increase increasing V_T. As the speed reaches rated conditions,

V_T will near CEMF and the armature current will decrease. When V_C becomes less than the bias, I_3 becomes zero and the normal voltage regulator condition, as previously described, will exist, i.e., $I_F=I_R$; $I_2=I_1=0$.

Refering to the module elementary, the +CL and -CL rheostats adjust the bias level discussed above and, consequently, adjust the level of current limit.

Figure 2 shows the basic circuitry used for regulating armature current.

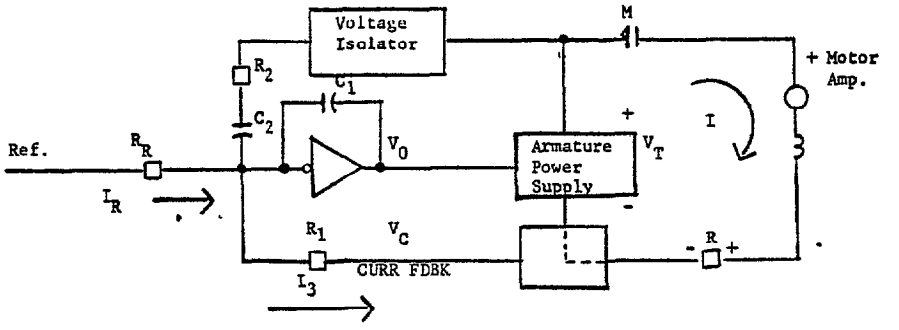


FIGURE 2

Note the simularity between Figures 1 and 2. In Figure 2, the reference (I_R) is compared to I_3 (the current feedback) in an integrating amplifier. At steady state, the amplifier will adjust the terminal volts (V_T) to cause $I_R=I_3$. Any unbalance between them will cause V_0 and V_T to change, changing the armature current so that the armature current will be held constant and proportional to the reference.

Again, refering to the module elementary, the +CL and -CL rheostats are used for current limit. The difference between this circuit and the spillover circuit is that these rheostats are connected (by jumpers in the module) to limit the reference. When the reference is limited, the armature current is limited.

G.E ITEM #	PANEL #	LAYER #	WIRE TABLE #	CAT #

REV 3	REV 2	REV 1	APPROVALS	GENERAL ELECTRIC	ASSEMBLY DIAGRAM CL
REV 4	ISSUED	PRINTED TO	FIRST MADE	INDUSTRY CONTROL DEPT SALEM VA USA	17185875 3.3 / 3.2
REV 5	MADE BY T. McLaughlin				

STABILITY

Current Limit

The stability of the current limit loop can be set by appropriate jumpers in the module. The two significant elements that need an approximate adjustment are the armature time constant and the IaRa drop in the loop. Referring to Figure 1, the product of RxC is used to offset the armature time constant (a lag) since the circuit used provides a compensating lead. The current feedback resistance R_1 is set to correspond to the IaRa drop

Addition Features

There are addition inputs and circuitry available to allow this module to be used in other applications.

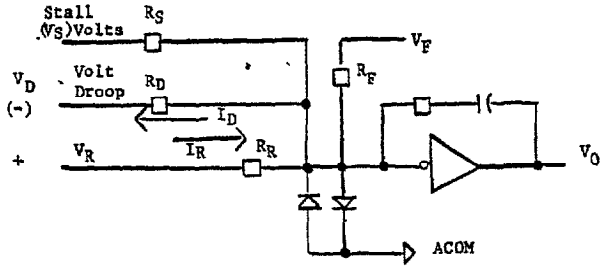


FIGURE 3

Voltage Droop

The voltage droop input is used to droop the terminal volts for increasing armature current. The polarity of the voltage droop signal (which is proportional to armature current, is opposite to the reference (V_R). Therefore, as the armature current increases, I_D increases, decreasing the net voltage reference (I_R-I_D). The terminal volts will decrease until V_F (which is proportional to V_T) satisfies the following equation.

$$\frac{V_F}{R_F} = \frac{V_R}{R_R} - \frac{V_D}{R_D}$$

Stall Current

In regenerative winders, the stall tension mode requires that motoring current be circulated in the windup motor to counteract the "pulling" effect of the sheet by the unwind stand. To accomplish this, the reference to the current regulator (Figure 2) is made proportional to the unwind motor current. So when stall tension current is circulated in the unwind stand, an equal current level is circulated in the wind up motor, thereby maintaining a taut sheet.

Typical Voltage Levels for Spillover Operation

Condition	Point	Description	Level (Volts)
Rated Arm. Volts	B16	Input	+15
Rated Arm. Volts	B17	Output	-15
Rated Arm. Volts	B18	Voltage Feedback	-15
Rated Arm. Volts	B19	Current Feedback	-7.5
Not in Current Limit	B20	Error	0

Typical Voltage Level for Current Regulator Operation

Condition	Point	Description	Level (Volts)
Rated Current	B16	Input	+7.5
Rated Arm. Volts	B17	Output	-15
Rated Arm. Volts	B18	Volt Feedback	-15
Rated Arm. Current	B19	Current Feedback	-7.5
Rated Arm. Current	B20	Error	+4.6

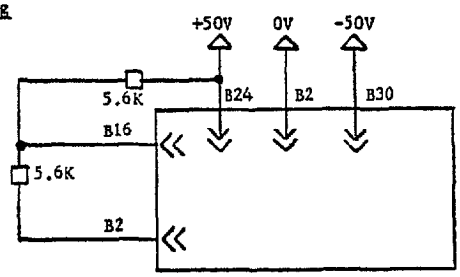
1.0 Power Requirements

1.1 ±50V Power Supply

2.0 Equipment

2.1 (2) (5.6K) 2W resistors

3.0 Wiring



4.0 Test Procedure (After Wire Checking)

4.1 Jumper B17 to B11.

4.2 Remove saddle point jumper (L) 7-10.

4.3 B17 should be -21V.

4.4 Jumper B22 to B46, B17 should go +7V when RH31 is rated full CW. and test resistors (5.6K) are removed.

4.5 Remove jumper B22 to B46.

4.6 Rotate RH32 full CW and jumper B23 to B46, B17 should go to -8V.

4.7 Remove jumper B22 to B46, and replace the two 5.6K test resistors.

4.8 Replace saddle point jumper L(7-10), and jumper B18 to B16, B17 should go to 0V.

4.9 Remove jumper B18 to B16.

G.E. ITEM #	PANEL #	LAYER #	WIRE TABLE #	CAT #

REV 3	REV 2	REV 1		GENERAL ELECTRIC	ASSEMBLY DIAGRAM
REV 4	ISSUED	PRINTS TO		INDUSTRY CONTROL DEPT SALEM VA USA	1716875 FL. 4.0
REV 5	MADE BY				

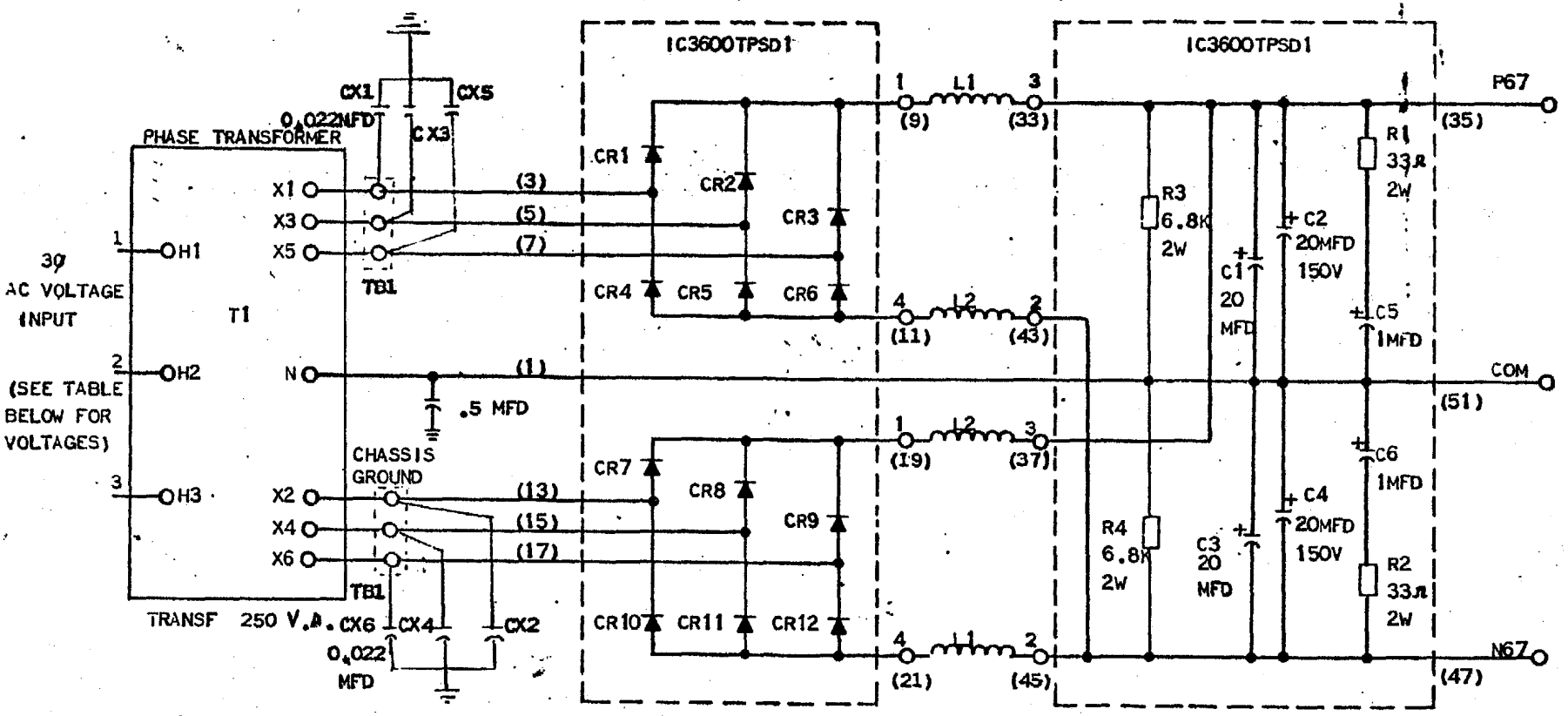
ISSUED 7-1-69 W. C. TUCK
REV. 1 11/26-69 MCW
REV. 2 3-6-72
REV. 3 7/19/79
APPROVALS
PRINTS TO
GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA. U.S.A.

TITLE

± 67V, 1 AMP UNREGULATED POWER SUPPLY

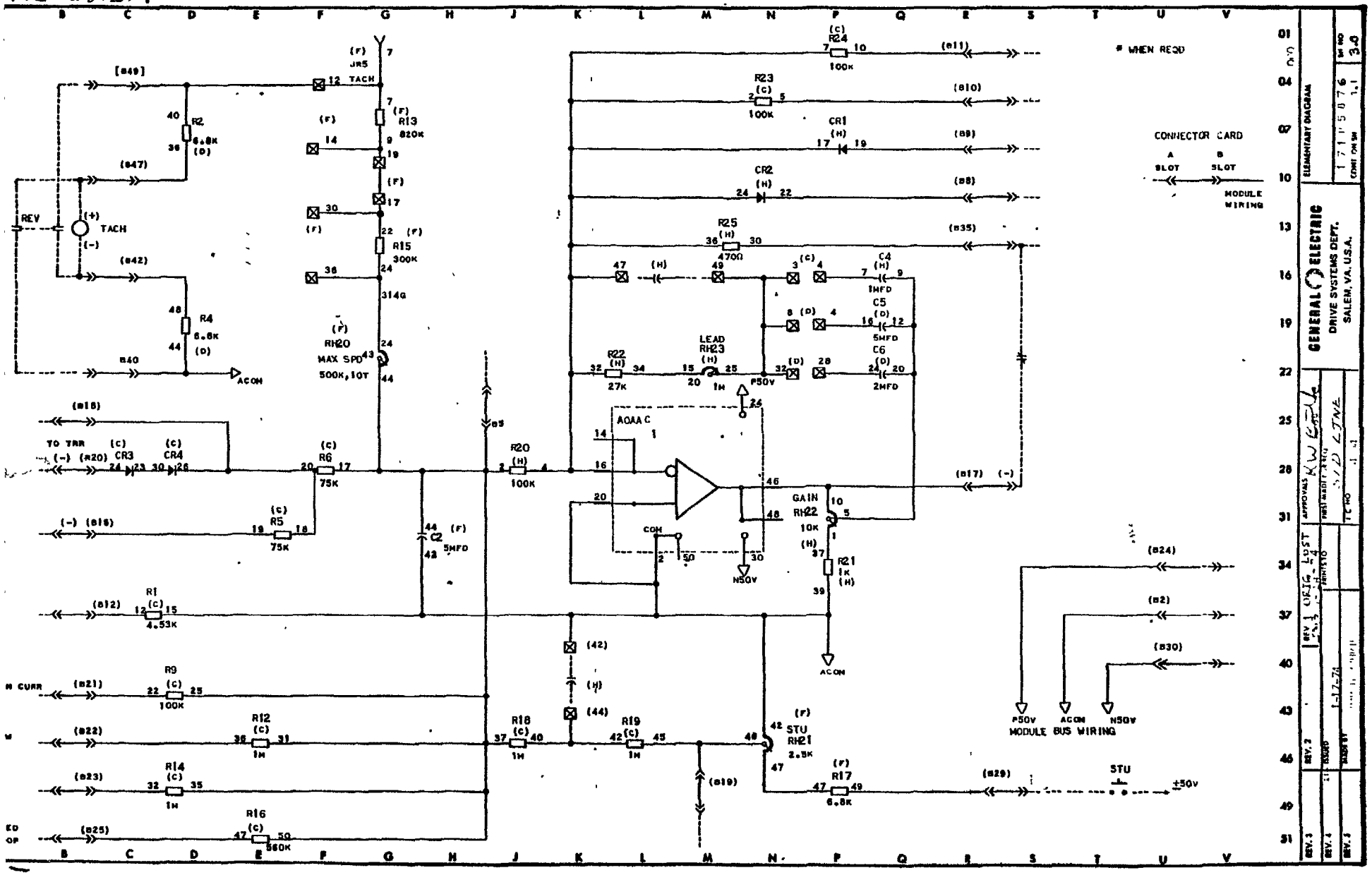
IC3601A133 3

OUTPUT RATING
P67 V.D.C. SUPPLY 1.2ADC
N67 V.D.C. SUPPLY 1.2ADS



NOTES:
1) NUMBERS IN PARENTHESIS ARE PIN NUMBERS ON PRINTED CIRCUIT IC3600TPSD1.
2) C1 THRU C6 ARE 0.022 MFD, 200VDC CAPACITORS MOUNTED ON TB1 NOT INCLUDED ON EARLY DESIGNS.
3) THERE ARE SEVERAL FORMS OF THIS NON-REGULATED ±67V.D.C. POWER SUPPLY. THE DIFFERENT FORMS ARE IDENTICAL WITH ONE ANOTHER EXCEPT FOR THE PHASING TRANSFORMER. THE PHASING TRANSFORMER CHANGES WITH THE CHANGE IN CUSTOMERS INCOMING A.C. VOLTAGE LEVELS.

+67V DC NON-REG POWER SUPPLY IC3601A133 ___ FORM	POWER TRANSF. PRIMARY VOLTAGE (NORMAL VOLTAGE RATING)	+67V DC NON-REG POWER SUPPLY IC3601A133 ___ FORM	POWER TRANSF. PRIM/SEC VOLTAGE (NORMAL VOLTAGE RATING)
A	220V A.C. 50/60 CYCLES	F	352V A.C. 50/60 CYCLES
B	460V A.C. 50/60 CYCLES	G	500/505V A.C. 50/60 CYCLES
C	380V A.C. 50/60 CYCLES	H	282V A.C. 50/60 CYCLES
D	575V A.C. 50/60 CYCLES	J	600VAC 50/60 CYCLES


GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
ELEMENTARY DIAGRAM
CONNECTOR CARD
A SLOT
B SLOT
MODULE WIRING
WHEN REQD
MODULE BUS WIRING
P50V
ACOM
N50V
STU
±50V
TACH
REV
ACOM
TO TRR
MAX SPD
500K,10T
LEAD
AOAA C
GAIN
CON
R1
R2
R4
R5
R6
R9
R12
R14
R16
R17
R18
R19
R20
R21
R22
R23
R24
R25
RH20
RH22
RH23
CR1
CR2
CR3
CR4
C2
C4
C5
C6
5MFD
1MFD
2MFD
1M
100K
75K
4.53K
6.8K
820K
300K
470Ω
27K
10K
1K
2.5K
560K
P50V
N50V

GENERAL
MAIN FUNCTION

THE SPEED MODULE IS USED FOR ACCURATE CONTROL OF SPEED FOR PAPER MILL DRIVES. FIGURE 1 SHOWS HOW THIS MODULE IS USED IN THE OVERALL SPEED REGULATING SYSTEM.

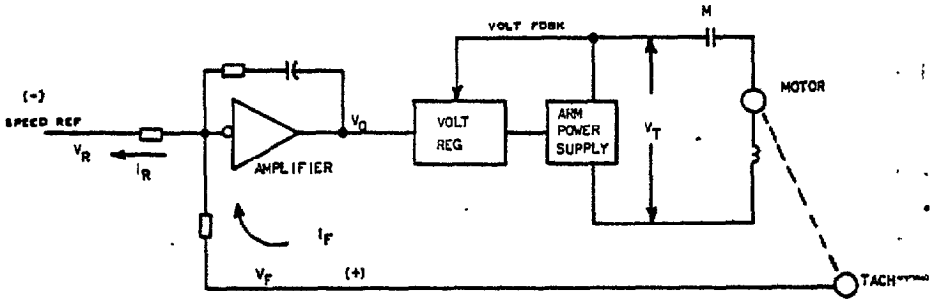


FIGURE 1 - SPEED REGULATOR

A FEW COMMENTS ON FIGURE ARE NECESSARY. THE ARMATURE POWER SUPPLY, SHOWN, CAN BE EITHER A SILTROL, SCM OR ROTATING GENERATOR. THE VOLTAGE REGULATOR IS EITHER A SEPARATE MODULE OR PART OF THE ARMATURE SUPPLY. WHENEVER AVAILABLE (SUCH AS THE SILTROL). IN ANY CASE, THE VOLTAGE REGULATOR WILL REGULATE THE TERMINAL VOLTS (VT) PROPORTIONAL TO THE INPUT VOLTAGE, VO. THE REASON FOR THE VOLTAGE REGULATOR IS TO PROVIDE A GOOD SUICIDE FEATURE AND FOR AIDING IN STABILITY.

SINCE THE AMPLIFIER IS CONNECTED IN AN INTEGRATING CIRCUIT, AT STEADY STATE $I_F = I_R$. IN OTHER WORDS, THE REGULATING ACTION WILL CAUSE THE FEEDBACK VOLTAGE (VF) TO BE PROPORTIONAL TO VR, THE SPEED REFERENCE. SINCE VF IS THE TACH VOLTAGE THE SPEED WILL BE MAINTAINED CONSTANT. IF THERE EXISTS AN UNBALANCE BETWEEN THE REFERENCE AND FEEDBACK, THE AMPLIFIER VOLTAGE WILL CHANGE. ASSUME THAT THERE IS AN INCREASE IN REFERENCE. VO WILL INCREASE, INCREASING THE TERMINAL VOLTS (VT), PROPORTIONALLY. VT WILL CAUSE AN INCREASE IN MOTORING ARMATURE CURRENT, ACCELERATING THE MOTOR. AT STEADY STATE, THE SPEED WILL BE REGULATED CONSTANT AND PROPORTIONAL TO THE REFERENCE.

ADDITIONAL FEATURES

FIGURE 2 SHOWS THE SPEED AMPLIFIER WITH ADDITIONAL INPUTS. SOME OF THESE INPUTS ARE CONTROLLED BY OTHER MODULES AND WILL BE MORE VIGOROUSLY DESCRIBED WITH THOSE MODULES.

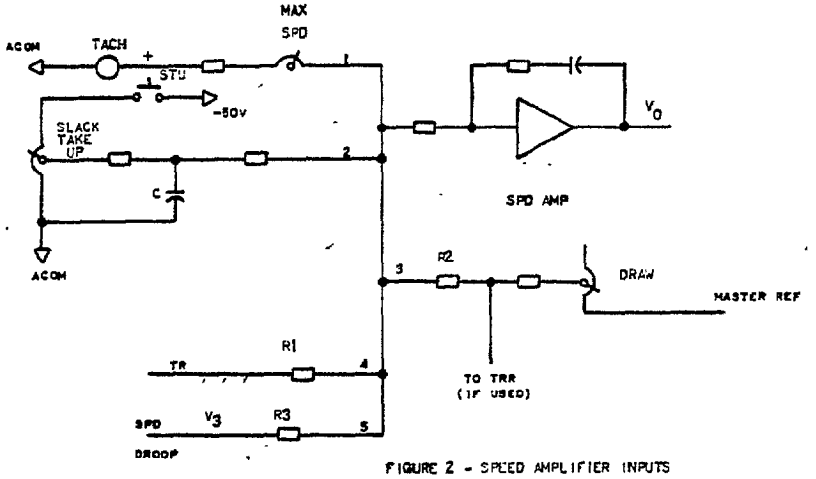


FIGURE 2 - SPEED AMPLIFIER INPUTS

REV 1	REV 1	REV LTRS. LOST	APPROVED	GENERAL ELECTRIC	ASSEMBLY DIAGRAM / APPLICATION DATA
REV 4	1-17 74			DRIVE SYS. PROD. DEPT.	1 7 1 B 5 8 7 6 3.2 3.1
REV 1				SALEM VA USA	

171B5876
3.1

INPUT 1 - TACH FDBK

THIS HAS BEEN PREVIOUSLY DESCRIBED. THE MAX SPEED RHEOSTAT IS USED TO ADJUST FOR DIFFERENT TACH VOLTAGES.

INPUT 2 - SLACK TAKE UP

TO ENABLE THE SECTIONS TO TAKE UP SLACK, THE CIRCUIT SHOWN IS USED. WHEN A SLACK TAKE UP IS INITIATED, ADDITION SPEED REFERENCE CURRENT IS INPUTTED INTO THE SPEED AMPLIFIER. THIS IS ADDED TO THE EXISTING MASTER REFERENCE AND THE SPEED IS INCREASED. CAPACITOR C MAY BE ADDED, AT STARTUP TO GIVE A LAG. SPACE HAS BEEN MADE AVAILABLE FOR THIS CAPACITOR.

INPUT 3 - REFERENCE

THIS SIGNAL NORMALLY COMES FROM THE DRAW RHEOSTAT THRU THE MR MODULE. THE MASTER REFERENCE IS MODIFIED BY THE DRAW RHEOSTAT. BEFORE THE NET SPEED REFERENCE IS INTRODUCED INTO THE SPEED MODULE, IT IS SOMETIMES MODIFIED BY THE TRR MODULE GIVING A RATE OF CHANGE OF SPEED REFERENCE SIGNAL.

INPUT 4 - TENSION REGULATOR (TR)

ANOTHER MODULE, THE TENSION REGULATOR, IS USED IN CONJUNCTION WITH THE SPEED MODULE. THE OUTPUT OF THIS MODULE IS DESIGNED TO CHANGE THE SPEED OF THE DRIVE IN ORDER TO CONTROL TENSION. (SEE THE DESCRIPTION FOR THE TR MODULE). WHEN THERE IS A SIGNAL ON INPUT 4, A CURRENT WILL FLOW THRU R, THIS WILL ADD TO OR SUBTRACT FROM THE SPEED REFERENCE SIGNAL (CURRENT THRU R2). THE SPEED OF THE SECTION WILL THEN CHANGE ACCORDINGLY.

INPUT 5 - SPEED DROOP

THE SPEED DROOP SIGNAL (WHEN SUPPLIED) COMES FROM A VOLTAGE PROPORTIONAL TO THE ARMATURE CURRENT. THE SPEED DROOP FUNCTION REQUIRES, THAT AS ARMATURE CURRENT INCREASES, THE SPEED IS TO DECREASE. THEREFORE, AS ARMATURE CURRENT INCREASES, V3 INCREASES, INCREASING THE CURRENT THRU R3. THE POLARITY OF THIS CURRENT IS SUCH THAT IT SUBTRACTS FROM THE CURRENT THRU R2 (THE SPEED REFERENCE SIGNAL) AND CONSEQUENTLY, THE SPEED OF THE MOTOR WILL DECREASE.

CLAMP CIRCUITS

LOCATED IN THE CL MODULE ARE COMPONENTS WHICH ARE USED IN CONJUNCTION WITH THE SPEED MODULE. THIS CIRCUIT INVOLVED IS TO KEEP THE SPEED AMPLIFIER IN RANGE WHENEVER THE DRIVE GOES INTO CURRENT LIMIT. FIGURE 3 SHOWS THE BASIC CIRCUITRY FOR ACCELERATING IN CURRENT LIMIT.

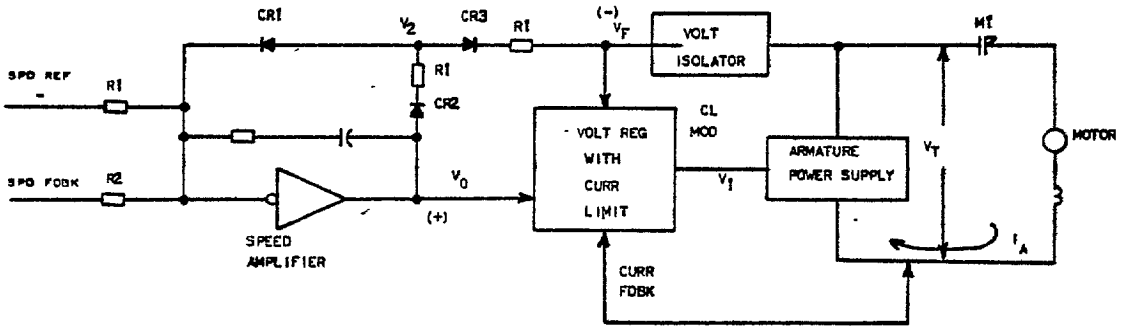


FIGURE 3 - CLAMP

REV 1	REV 2	REV 1 ORIG. LOST	APPROVED	GENERAL ELECTRIC	ASSEMBLY DIAGRAM
REV 4	1-17-74			DRIVE SYS. PROD. DEPT.	APPLICATION NOTES
REV 5				SALEM VA USA	171B5876 3.3 / 3.2

171B5876
3.2

CLAMP CIRCUIT-CONT

IF THE CLAMP CIRCUIT IS REMOVED, OPEN CIRCUIT CR1,CR2 AND CR3, AND AN ACCELERATION IN CURRENT LIMIT IS AMPLIFIER WOULD SATURATE. THE SPEED REFERENCE IS NEGATIVE AND FEEDBACK POSITIVE. THE REFERENCE IS FIXED AND WHEN AN ACCELERATION IS CAUSED, THE OUTPUT OF THE SPEED AMPLIFIER WILL, INITIALLY, GO POSITIVE, i.e., THIS WILL (AS DISCUSSED IN THE CL MODULE) WILL CAUSE AN ACCELERATING CURRENT. DUE TO THE CURRENT LIMITING ACTION, THE RATE OF CHANGE OF VI, WILL BE LIMITED, i.e., NOT UNDER CONTROL OF VO. WHEN THIS OCCURS, VO DOES NOT HAVE THE ABILITY TO CONTROL THE SYSTEM, AS IT WOULD NORMALLY DO, MAKING THE REFERENCE EQUAL THE FEEDBACK. THEREFORE, THERE WILL BE AN UNBALANCE IN THE REFERENCE AND FEEDBACK CURRENTS, WITH AN EXCESSIVE NEGATIVE CURRENT. THIS COULD CAUSE VO TO RAMP INTO POSITIVE SATURATING.

TO PREVENT THIS, THE CR1,CR2 AND CR3 CIRCUIT IS USED. NORMALLY THE SPEED AMPLIFIER OUTPUT (VO) IS EQUAL TO VF, THE VOLTAGE WHICH IS PROPORTIONAL TO VT. THE OBJECTIVE OF THIS CIRCUIT IS TO KEEP VO NEAR VF WHEN THE DRIVE IS IN CURRENT LIMIT. AS SOON AS THE DRIVE GOES INTO CURRENT LIMIT. VO INCREASES IN MAGNITUDE BEYOND VF. VO CONTINUES INCREASING UNTIL THE DIODES CONDUCT TO PREVENT ANY FURTHER INCREASE.

DIODE CR1 IS NORMALLY BACK BIASED BY THE VOLTAGE V2. WHEN VO EXCEEDS VF, BY THE BREAKOVER OF DIODES, V2 WILL BECOME POSITIVE CAUSING CR1 TO CONDUCT. WHEN THIS OCCURS CURRENT (POSITIVE) IS INTRODUCED AT THE AMPLIFIER TENDING TO MAKE VO LESS POSITIVE, PREVENTING VO FROM INCREASING.

AS THE MOTOR ACCELERATES VT AND VF WILL INCREASE ALLOWING VO TO ALSO INCREASE. WHEN THE DRIVE COMES OUT OF CURRENT LIMIT, CR1 WILL REVERT TO ITS BLOCKED CONDITION ALLOWING THE SPEED AMPLIFIER TO CONTROL SPEED.

STABILITY

FIGURE 4 REPRESENTS THE SPEED AMPLIFIER WITH ITS STABILITY COMPONENTS.

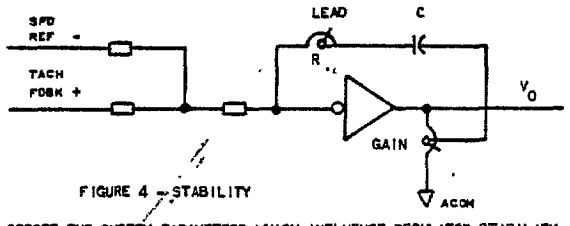


FIGURE 4 - STABILITY

TO OFFSET THE SYSTEM PARAMETERS WHICH INFLUENCE REGULATOR STABILITY, THE LEAD AND GAIN RHEOSTATS ARE USED. THE INERTIA OF THE MOTOR AND DRIVEN ROLL PRESENTS A DELAY TO THE REGULATING SYSTEM. WHEN A CHANGE IN SPEED REFERENCE IS MADE, THERE WILL BE A TIME LAG BEFORE THE SPEED CAN RESPOND TO THAT CHANGE. THIS TIME CONSTANT IS OFFSET BY THE LEAD ADJUST RHEOSTAT, THE SETTING OF THE RHEOSTAT ALONG WITH C, (i.e. RXC) GIVES A LEAD TO THE CIRCUIT TO OFFSET THE LAG OF THE INERTIA.

DEPENDING ON THE TYPE OF DRIVE DIFFERENT RESPONSE TIMES ARE REQUIRED. THE GAIN POT IS USED FOR THIS, THE SETTING OF THIS RHEOSTAT, ALONG WITH CAPACITOR C, DETERMINES THE RATE OF CHANGE OF THE OUTPUT VOLTAGE OF THE SPEED AMPLIFIER AND, CONSEQUENTLY, SPEED OF RESPONSE OF THE SPEED REGULATOR.

TO OFFSET MECHANICAL EFFECTS SUCH AS SHAFT TWIST AND BACKLASH, SPACE IS PROVIDED IN THE MODULE FOR ADDING A CAPACITOR IN PARALLEL WITH THE LEAD RHEOSTAT. THIS IS NOT REQUIRED, GENERALLY. WHEN NEEDED, A NOTE ON THE DRIVE ELEMENTARY WILL SPECIFY THE VALUE AFTER FIELD TUNE UP.

TYPICAL VOLTAGE LEVELS

CONDITION	POINT	DESCRIPTION	LEVEL (VOLTS)
"0" DRAW AT RATED SPD	B16	SPEED INPUT (REFERENCE)	-28.5
FOR SECTIONS W/O DRAW AT RATED	B16	SPEED INPUT	-30
500 TERMINAL VOLTS ON STATIC DRIVE AT NO LOAD	B17	SPEED AMPLIFIER OUTPUT	+15
AT STEADY STATE SPEED	B18	REF BRIDGE	HALF OF B16
FOR 2 o/o SLACK TAKE UP	B19	STU SETTING	-7.6 (FOR SECTIONS WITH DRAW)

RANGE OF ADJUSTMENTS

COMPONENT	FUNCTION	RANGE
RH20	MAX SPEED (ADJUSTS GEAR IN SPD)	TACH VOLTS. 60V - 160V WITH RS15 AND W/O RS13 140V - 240V WITH RS13 AND W/O RS15 200V - 300V WITH RS13 AND RS15
RH21	SLACK TAKE UP	0-3.5 o/o OF RATED SPEED
RH22	GAIN	ADJUSTS SPEED OF RESPONSE ADN STABILITY OF REGULATOR.
RH23	LEAD	

REV 3	REV 2	REV 1	APPROVED	GENERAL ELECTRIC	ASSEMBLY DIAGRAM
REV 4	1-17-74		FIRST MADE FOR MFG	DRIVE SYS. PROD. DEPT.	APPLICATION NOTES
REV 5	PAM TEMPLETON		4804	SALEM VA U.S.A.	171B5876 4.0 3.4

171B5876 3.4

1.0 POWER REQUIREMENTS

1.1 ± 50V POWER SUPPLY

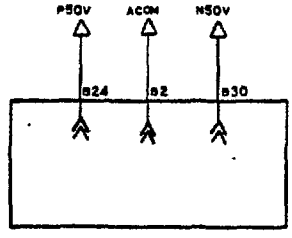
2.0 EQUIPMENT

2.1 NONE REQUIRED

3.0 WIRING

3.1 CHECK SLOT "B" PINS 43, 44

45, 46, 51 SHOULD BE REMOVED



4.0 TEST PROCEDURE (AFTER WIRE CHECKING)

4.1 SUICIDE THE AMPLIFIER BY JUMPERING B17 TO B35

4.2 ROTATE RH20, RH21, RH22, RH23 FOR 50 o/o ROTATION

4.3 MAKE SADDLE CLAMP JUMPER D (4-8)

4.4 CONNECT - 50V TO B49, AND REMOVE THE SUICIDE. B17 SHOULD RAMP TO SATURATION WITH A TIME TO +30V OF 2 SECONDS.

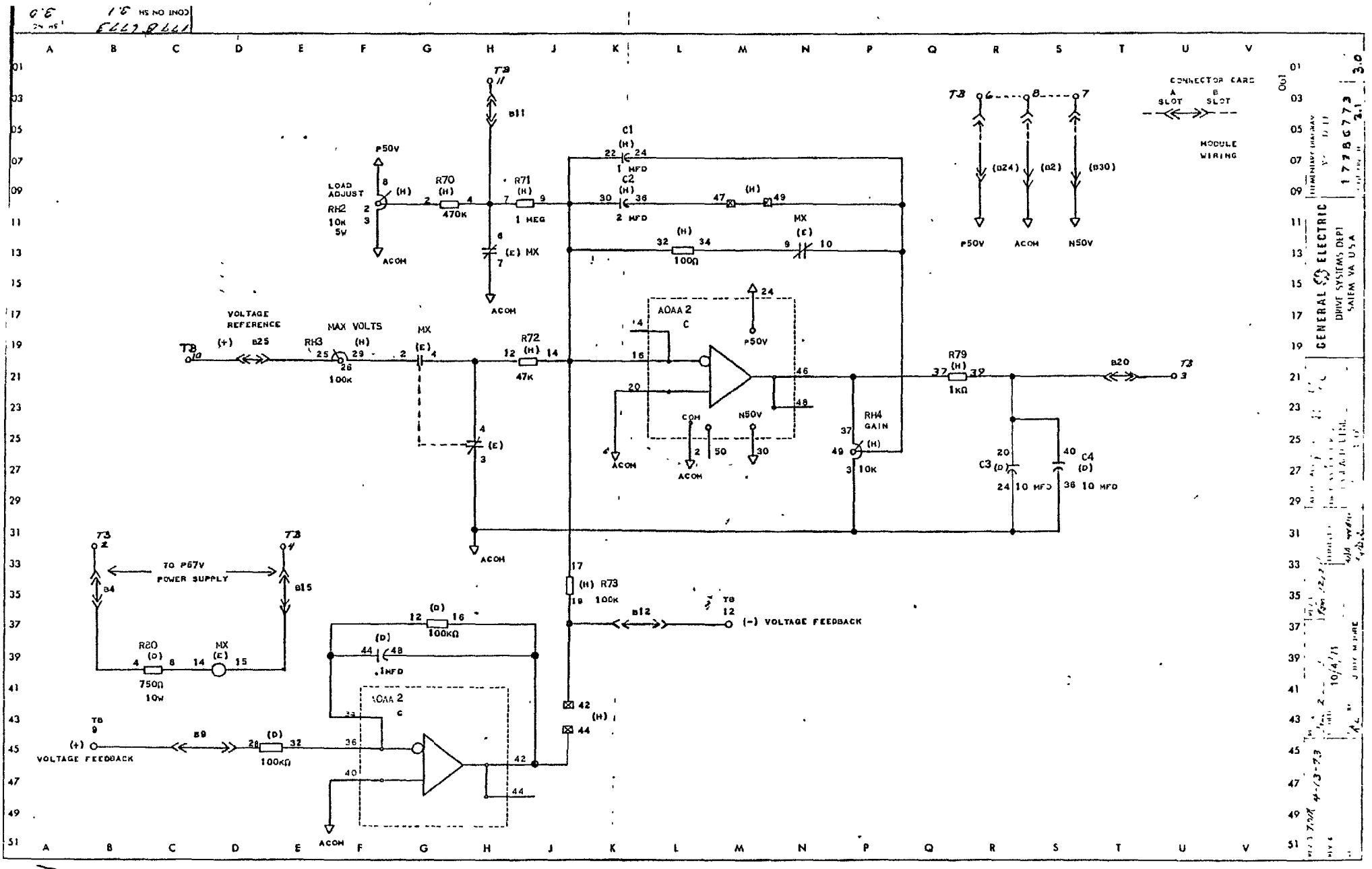
4.5 JUMPER C6 AND B17 SHOULD GO TO +18v. REMOVE -50V FROM B49.

4.6 JUMPER B29 TO -50V, B17 SHOULD GO TO +1.7V. REMOVE -50V FROM B29.

060

REV 3	REV 2	PLOT OFF-LOST		GENERAL ELECTRIC	ASSEMBLY DIAGRAM
REV 4		1-17-74		DRIVE SYS. PROD. DEPT.	TEST SPECIFICATIONS
REV 5		PWM TEST SPEC 5	4R04	SALEM VA U.S.A.	1718576 FL 4.0

1718576
4.0



GENERAL ELECTRIC
DRIVE SYSTEMS DEPT
SALEM VA U.S.A.

177B6773 3.0

GENERAL

THE VOLTAGE REGULATOR MODULE IS USED IN CONJUNCTION WITH A STATIC ARMATURE POWER SUPPLY FOR MOTOR ARMATURE VOLTAGE REGULATION. THE ARMATURE POWER SUPPLY MAY BE EITHER A SILTROL OR A MODIFIED CONSTANT POTENTIAL EXCITER. THE REFERENCE VOLTAGE MAY BE OBTAINED FROM EITHER A RHEOSTAT OR THE ARMATURE VOLTAGE OF ANOTHER DRIVE. FIGURE 1 BELOW SHOWS THE REGULATING LOOP IN WHICH THE DRIVE IS USED.

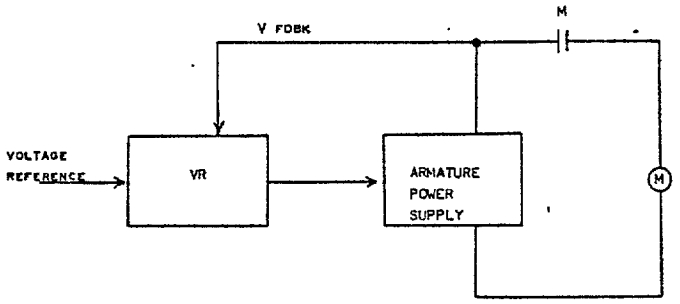


FIGURE 1

CIRCUIT OPERATION

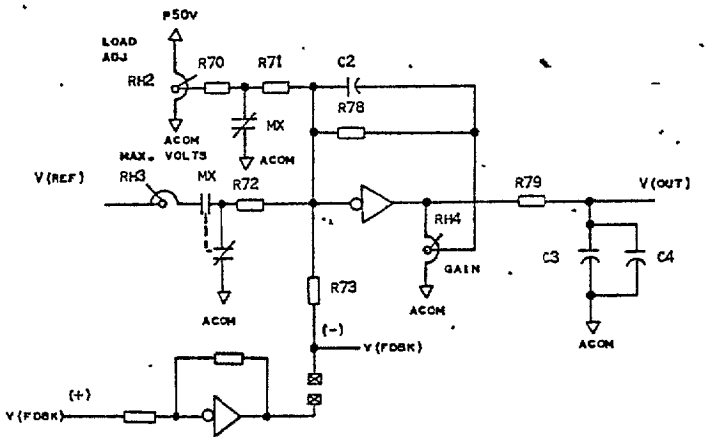


FIGURE 2

FIGURE 2 ABOVE SHOWS A FUNCTIONAL REPRESENTATION OF THE VR CIRUITRY. THE LAG TIME CONSTANT MADE UP OF R78 AND C2 ALONG WITH THE SETTING OF RH4 DETERMINES THE RESPONSE OF THE SYSTEM. R79 AND C3 + C4 MAKE UP A FILTER WHICH IS PRESENT TO ATTENUATE NOISE ON THE INPUT TO THE STATIC ARMATURE SUPPLY. POT RH2 IS A LOAD ADJUST. IF MOTOR SPEED OF THE VOLTAGE REGULATED MOTOR IS LESS THAN THAT OF THE SECTION SPEED REGULATED MOTOR DUE TO LOAD, ADDING A SECOND VOLTAGE REFERENCE WITH RH2 WILL COMPENSATE FOR THIS CONDITION. RELAY INTERLOCKS MX ARE USED TO SUICIDE THE VOLTAGE REGULATOR WHEN THE M CONTACTOR IS CLOSED.

061

REV 3	REV 2	REV 1 RG Humphreys 5-4-73	APPROV	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 4	DRWN	10/4/71		DRIVE SYSTEMS DEPT	APPLICATION DATA
REV 5	MADE BY	JUDY MOORE		SALEM, VA USA	1778 6773
					CONT ON SH 3.2 3.1

TYPICAL VOLTAGE LEVELS

CONDITION	POINT	DESCRIPTION	LEVEL (VOLTS)
RATED VOLTAGE	B20	MODULE OUTPOUT	(-) 15
RATED VOLTAGE	B25	REFERENCE	(+) 15
MAX LOAD COMP	B11	LOAD ADJUST REFERENCE	+ 32
RATED VOLTAGE	B12	VOLTAGE FEEDBACK	(-) 15
RATED VOLTAGE	B9	VOLTAGE FEEDBACK	(+) 15

ADJUSTMENTS

COMPONENT	FUNCTION	RANGE
RH2	LOAD COMPENSATION	0 - 100 o/o LOAD (DEPENDS ALSO ON I_R DROP)
RH3	MAX VOLTAGE	7.5 TO 22.5 VOLT REFERENCE SIGNAL
RH4	GAIN	10:1 RANGE

1.0 POWER REQUIREMENTS ± 50V POWER SUPPLY

2.0 EQUIPMENT. BRUSH RECORDER, 10K (2W) RESISTOR, 1K (2W) RESISTOR

3.0 WIRING NO SPECIAL CHECKS

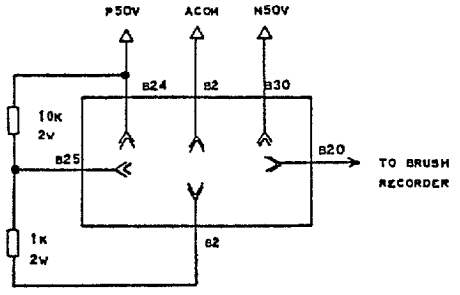


FIGURE 1

4.0 TEST PROCEDURE (AFTER WIRE CHECKING)

4.1 MAKE CONNECTIONS SHOWN IN FIGURE 1 ABOVE

4.2 JUMPER B4 TO B24. JUMPER B15 TO B2. RELAY MX SHOULD PICKUP.

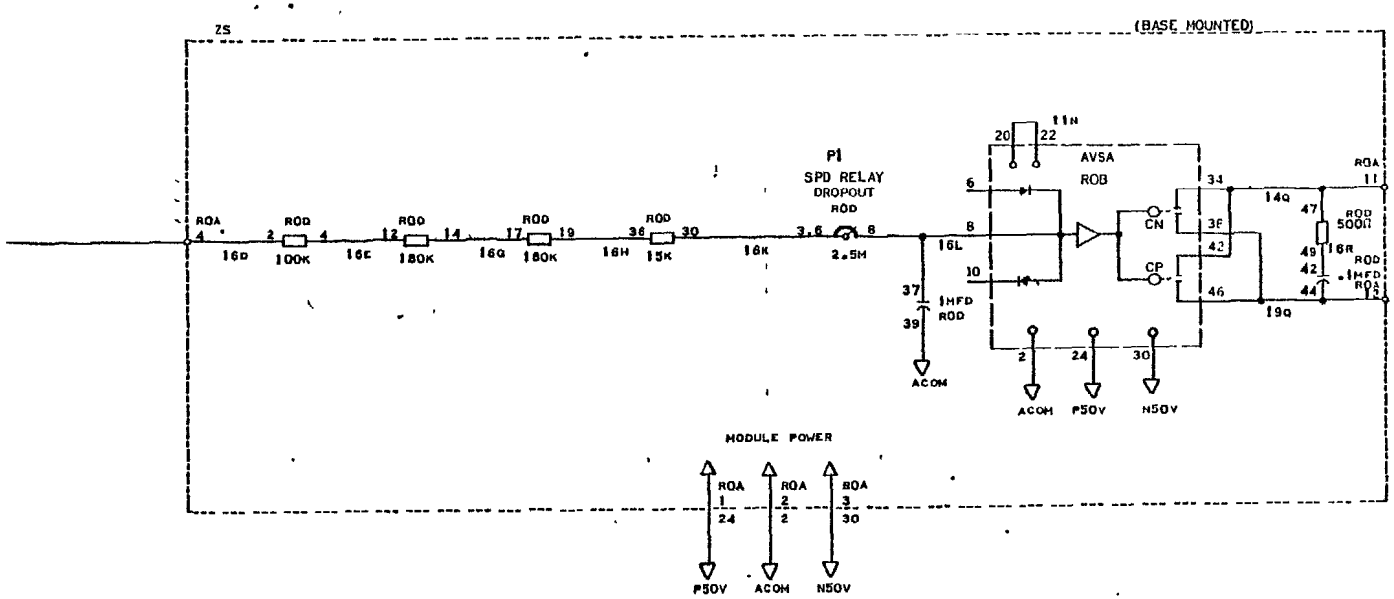
4.3 SET RH3 IN MID POSITION. SET RH4 FULLY CCW. SET RH2 FULLY CW. ADD JUMPER H(47 - 49)

4.4 DROPOUT MX. MONITOR B20 WITH A BRUSH RECORDER. STEP THE CIRCUIT BY PICKING UP MX. THE VOLTAGE AT B20 SHOULD RISE RAMP TO (-) 25.0 VOLTS WITH A TIME FROM 0 OUTPUT TO -25.0 VOLTS EQUAL 1.6 ± .2 SECONDS.

4.5 PLACE RH4 AT A SETTING OF 5. REPEAT STEP 4.4. THE RESPONSE SHOULD BE IDENTICAL WITH THE PREVIOUS ONE EXCEPT THAT THE FINAL VOLTAGE SHOULD BE .8 ± .2. RETURN RH4 TO FULL CCW POSITION.

4.6 REMOVE WIRE FROM COMMON POINT OF 10K AND 1K RESISTOR TO B25. DROPOUT MX. SET RH2 FULLY CCW. PICKUP MX. THE RESPONSE AT B20 SHOULD INCREASE FROM 0 TO -25.0 VOLTS IN 2.2 ±.2 SECONDS.

4.7 REMOVE ALL JUMPERS INSTALLED ABOVE.



GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA U.S.A.

PRELIMINARY DIAGRAM

19583479

REV 1
REV 2
REV 3
REV 4
REV 5

19583479
3.0

1.0 GENERAL

The Zero Speed module is used to indicate when section motor is either running (above a very low percentage of top speed) or stopped (below a very low percentage of top speed). The tachometer, voltage from the section's speed regulated motor, is the input to the module's voltage sensor (IC3600AVSA1) card. The Speed Relay Drop Out rheostat is for adjustment. The contacts from the AVSA work into the sectional magnetics.

2.0 SET UP

Set the ZS module to pick up at 3% of gear in speed. Drop out will be 85-95% of pick up. It is recommended that if an AVSA card is replaced that the pick up of the new card be checked.

TACH VOLTAGE					
@ Gear-in Speed	60 <	60 ≤160	160 ≤ 240	240 ≤ 320	320 <
@ 3%	1.8 <	1.8 ≤ 4.8	4.8 ≤ 7.2	7.2 ≤ 9.6	9.6 <

LOCATION FIXED RESISTANCE					
D (36-30)	15K	15K	15K	15K	15K
D (2-4)	Jumper	100K	100K	100K	100K
D (12-14)	Jumper	Jumper	180K	180K	180K
D (17-19)	Jumper	Jumper	Jumper	180K	Adjust as Necessary

3.0 POWER REQUIREMENTS

P50V (40MA) PIN 24

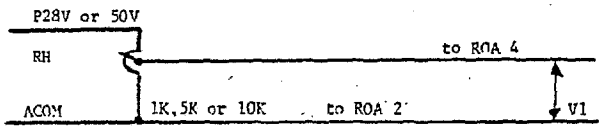
N50V (9MA) PIN 30

ACOM PIN 2

4.0 TEST PROCEDURE (After wire and component check)

4.1 Connect a multimeter (on ohms scale) to ROA11 and ROA12.

4.2 Connect a low voltage supply (probably P28V) as follows:



4.3 With all resistors in as above, adjust RH for V1 of 8.4 volts.

4.4 Then adjust P1 for AVSA contact closure, multimeter in 4.1 to zero ohms.

4.5 Adjust RH for V1 of .45 volts.

4.6 Install test jumper from. Slot D pin 2 to slot D pin 19.

4.7 Adjust P1 for AVSA contact closure, multimeter in 4.1 to zero ohms.

4.8 Remove test jumper in 4.6.

INSTRUCTIONS

IC 3605
THYRISTOR
FIELD EXCITERS

Information contained herein is Proprietary Information of the General Electric Company. The dissemination or use of this information for any other purposes than that for which it is provided is prohibited by the General Electric Company except by express permission.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

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THYRISTOR FIELD EXCITERS

I. INTRODUCTION

The following instructions refer to the theory, operation and maintenance of the General Electric AC to DC SCR power supply thyristor field exciters, Type IC3605.

The instructions do not purport to cover all details or variations in equipment, or to provide for every possible contingency to be met in connection with operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

The SCR exciters described in this manual are used as power amplifiers. They are fired by either a unijunction transistor firing circuit or an amplistat firing circuit. A control signal of a few milliamps will control outputs up to 300 amps convection cooled. The resulting power gain makes the SCR one of the most sensitive control devices. This extraordinary power gain makes possible control circuits using very low level signals. Low power control components, plus the fact that the device is static, make these SCR exciters free from many of the problems associated with rotating exciters. Exciter current ratings are established for operation in a 55 C ambient temperature.

II. SILICON CONTROLLED RECTIFIER NOTES

A. GENERAL DESCRIPTION

The silicon controlled rectifier, or SCR as it is more commonly called, is a three-terminal semiconductor device. It is much like an ordinary silicon rectifier which has been modified to block in the forward direction until a small positive current signal is applied to the gate load. After the gate signal is applied, the SCR will conduct in the forward direction with a forward characteristic very similar to that of an ordinary silicon rectifier, and will continue forward conduction even after the gate signal is removed. The SCR has characteristics similar to those of a gas thyratron, except that the forward drop is about 1/10th that of a thyratron (approximately 1.0 volt) and the turn-off time is much less than that of a thyratron (35-125 microseconds).

For a complete discussion of the theory of operation and all aspects of silicon controlled rectifiers, the reader is referred to the General Electric SCR Manual.

B. SCR AND DIODE TYPES UTILIZED IN EXCITERS

The following SCR's and diodes are utilized:

SCR C35	Stud Mount
SCR C137	Stud Mount
SCR C350	PRESS-PAK
SCR C380	PRESS-PAK
SCR C291	Clamp on Stud
Diode A390	PRESS-PAK
Diode IN2160	Stud Mount

See Table 1 for Cell Application.

TABLE 1. SCR AND DIODE APPLICATION

SCR THYRISTOR FIELD EXCITERS										
OUTPUT			CELL TYPE					POWER		FUSE
AMPS	VOLTS	KW	NP	NX*	FP	CP	RECT	VOLT	PHASE	AMP
018	090	1.62	C35	--	C137	--	---	230CT	1	25 C35 30 C137
018	180	3.24	C35	C35	C137	--	---	230	1	25 C35 30 C137
022	250	5.5	--	--	--	C35	---	230	3	25
040	090	3.6	C350	--	C350	--	---	230CT	1	80
040	180	7.2	C350	C350	C350	--	---	230	1	80
050	250	12.5	--	--	--	C350	---	230	3	80
040	360	14.4	C350	C350	C350	--	---	460	1	80
080	090	7.2	C350	--	C350	--	---	230CT	1	150
080	180	14.4	C350	C350	C350	--	---	230	1	150
100	250	25.0	--	--	--	C350	---	230	3	150
080	360	28.8	C350	C350	C350	--	---	460	1	150
100	560	56.0	C350	--	C350	--	---	460	3	150
120	180	21.6	C380	C380	C380	--	---	230	1	300
150	250	37.5	--	--	--	380	---	230	3	350
120	360	43.2	C380	C380	C380	--	---	460	1	300
150	560	84	C380	--	C380	--	---	460	3	350
300	250	75	--	--	--	C291	---	230	3	600
300	560	168	C291	--	C291	--	---	460	3	600
RECTIFIER FIELD EXCITERS										
020	250	5	--	--	--	--	IN2160	185	3	30
100	250	25	--	--	--	--	A70	185	3	100
300	250	75	--	--	--	--	A390	185	3	300

*The diode used with the SCR in the NX exciter is as follows:

C35 - IN2160
C350 - A390
C380 - A390

C. SCR AND DIODE CHARACTERISTICS

TABLE 2. SCR CHARACTERISTICS

Type	Repetitive Peak Rev. Voltage VROM V	Peak Forward Blocking Voltage VFOM V	Max. Avg. Fwd Current (induc. load) AMP	Maximum Holding Current I_{HO} ma	Peak Reverse and For. Blocking Current I_{ROM}, I_{FOM} (ma)	Max. Gate Trigger Current I_{GT} 25°C ma	Max. Gate Trigger Voltage V_{GT} V	Purchase Part Dwg. Number 68A72	Part No. After Potting 205A57	AC Power Voltage
C137 E	500	500	9 amp at 90°C case temp.	100	3.5	40	3	67PXE	--	230
C350 E	500	500	40 amp at 110°C case temp.	500	10	150	3	93PXE	93PXE	230
C350 P	1000	1000	40 amp at 110°C case temp.	500	7	150	3	93PXP	93PXP	460
C380 E	500	500	60 amp at 110°C case temp.	500	10	150	3	94PXE	94PXE	230
C380 P	1000	1000	60 amp at 110°C case temp.	500	7	150	3	94PXP	94PXP	460
C291 E	500	500	150 amp at 108°C case temp.	100	--	150	3.5	91PXE	--	230
C291 P	1000	1000	150 amp at 108°C case temp.	100	--	150	3.5	91PXP	--	460
C35 M	600	600	11.5 amp at 90°C case temp.	100	5	40	3	97PXM	--	230

TABLE 2. DIODE CHARACTERISTICS

Type	Repetitive Peak Rev. Voltage VRM	Maximum Avg. Fwd. Current Amp	Maximum Full Load Reverse Current ma	Purchase Part Dwg. No.	Part No. After Potting 205A57-	AC Power Voltage
A390 E	500	250 1ϕ, 130°C stud temp.	10	68A7295 PXE	95PXE	230
A390 P	1000	250 1ϕ, 130°C stud temp.	10	68A7295 PXP	95PXP	460
IN2160	600	25, 1ϕ 145°C stud temp.	2	167A8419 P15	---	230

D. GENERAL CONSIDERATIONS AND PRECAUTIONS

Gate Voltage

1. The gate should never be allowed to become more than five volts negative with respect to the cathode.

Gate Power

2. Peak gate power should be kept below 10 watts.

Recovery

3. If the anode-to-cathode current is reduced just to zero, the SCR will recover its ability to block forward voltage in approximately 100 microseconds. By applying a reverse voltage to the SCR from a low impedance source, reverse current will flow and the recovery time will be reduced.

Forward Breakover

4. The SCR may fire or turn on as a two-terminal device; if breakover action occurs below the rated repetitive forward blocking voltage, no permanent damage will be incurred.

Thermal Contact

5. Good thermal contact between the semiconductor and the heat sink requires adequate pressure between these two surfaces.

Ventilation

6. Be sure to keep the ventilation paths in the exciter open. Blocking ventilation paths in the exciter can cause it to overheat, causing erratic operation or damaged components.

   All exciters except those utilizing the C35 and C137 cells have forced ventilation ratings. The air flow requirements for individual exciters are stated on the elementary diagrams. In general, the forced ventilation rating will be just below twice the convection-cooled rating.

Fuse Protection

7. Fuses used in these exciters have been chosen for their fast-blow characteristic. A fast-blowing fuse is needed to protect the SCR cells, and an ordinary fuse of the same current rating will not protect the cells against short circuits.

SCR Removal and Replacement

8. The PRESS PAK SCR has been designed to operate with either the anode or the cathode toward the heat sink. If the SCR is removed from the heat sink, it is imperative that it be replaced with the same anode-cathode orientation. Instructions for removal and replacement of cells are printed on the back of the front cover. If doubt exists as to the proper orientation, the elementary diagram should be consulted. Eight hundred pounds pressure is required to compress the cell and allow current to flow through the thyristor. This pressure is provided by a special assembly. If the cell is not compressed, ohmmeter readings may not be possible.

Assembly of PRESS PAK SCR Cell in General Electric Mounting Kit

The PRESS PAK, when mounted as shown in Fig. 1, will have an external force of 800 lbs ± 10% properly applied.

1. Place the bolts (12) through the spring (11), the heat sink (10) as shown in the sketch.

2. Slip the insulating sleeves (4) over the bolts (12).

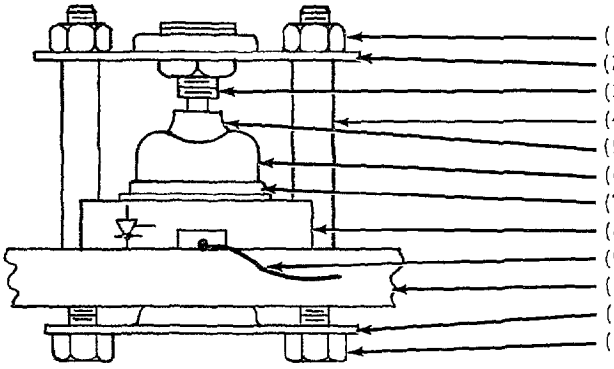
3. Place the spring with swivel (2) in position as shown, and add the nuts (1). Finger tighten the nuts just enough to support the spring with swivel and hold the bolts in an upright position.

4. Insert the PRESS PAK (8) in the locator, making sure that it drops into the centering hole of the locator. The PRESS PAK may be assembled in either of two ways depending upon the desired polarity of the heat sink.

5. For single-side cooling, center the copper terminal (7) and the insulation (6) over the PRESS PAK and under the swivel pad (5).

6. Finger tighten the set screw (3) just enough to hold all parts together.

7. Wrench tighten the set screw (3) 1-3/4 turns. This will place a load of 800 lbs ± 10% on the PRESS PAK unit.



Single-side cooling

Fig. 1. Assembly of PRESS PAK to heat sink

III. SCR THYRISTOR FIELD EXCITERS—UNIJUNCTION FIRED

A. MECHANICAL DESCRIPTION

Typical photographs of single-phase and three-phase exciters are shown in Figs. 2 and 3. All exciters, whether single or three phase, unijunction or amplistat, are a unitized assembly and approximately 12 inches deep. The width is 24 inches except those exciters which utilize the C291 SCR cell (36 inches wide). The height of the exciters is a function of the cell type, the number of power phases and the exciter function. See Table 3.

The printed wiring cards used with the exciters are located on the left side of the exciter and are mounted in individual Lexan® boxes. These cards can be spot checked in place by means of test points. If a card must be replaced, it is only necessary to remove two screws and pull the card from its box.

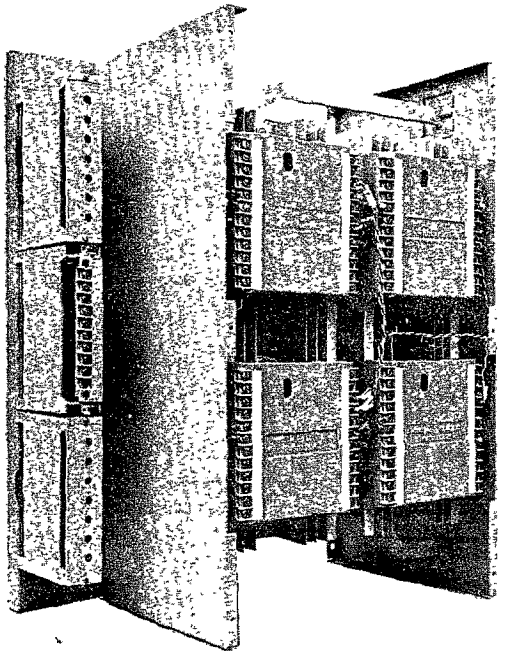


Fig. 2. One phase exciter

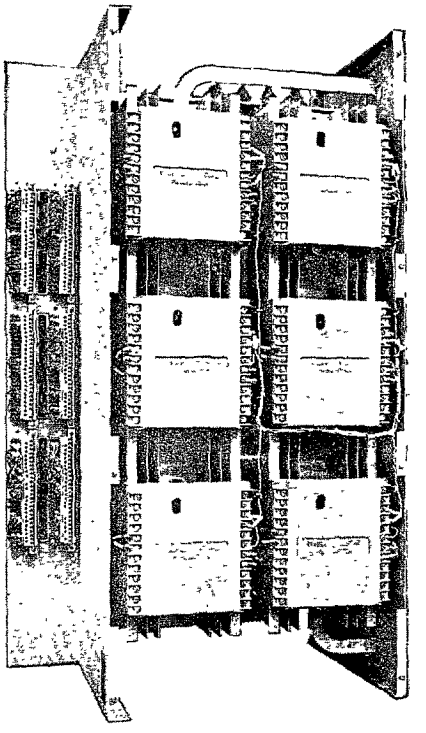


Fig. 3. Three phase exciter

TABLE 3. EXCITER HEIGHT

Number of Phases Exciter Function	SCR Cell			
	C137	C350	C380	C291
1ϕ NP, NX	11"	22"	22"	---
3ϕ NP, CPE	19"	30"	30"	38"
1ϕ FP	19"	22"	22"	---
3ϕ FP	---	31"	31"	62"

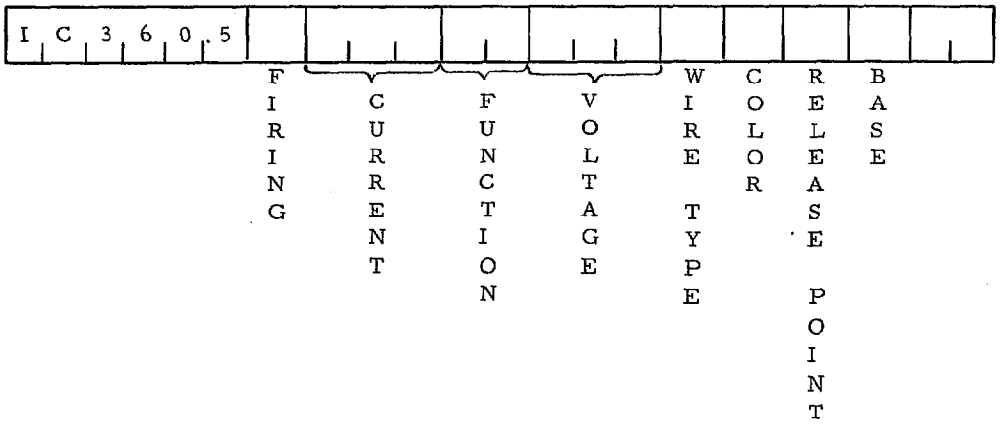
Subassemblies such as heat sinks are mounted on Lexan feet, which in turn are mounted on the main assembly by means of screws and captive nuts. Slotted or tear-shaped holes allow removal of some subassemblies by simply loosening the mounting screws.

These exciters are built using many interchangeable parts and subassemblies. This allows the customer whose system contains several different exciters to maintain a minimum spare parts inventory.

All the exciters are of rugged design, utilizing heavy gage steel construction. Electrical power connections are made on the side of exciters using PRESS PAKS; they are made on the top of exciters using C291 and C137 cells. The exciters are open on top and bottom to allow air flow.

1. CATALOG NUMBER CODE

1. Format –



2. Firing – A = Amplistat 60 Cycle
B = Amplistat 50 Cycle
C = Unijunction 60 Cycle Feedback
D = Unijunction 50 Cycle Feedback
E = Unijunction 60 Cycle No Feedback
F = Unijunction 50 Cycle No Feedback
X = Other

3. Current – 018 = 18 Amp.
022 = 22 Amp. (CPE only)
040 = 40 Amp.
050 = 50 Amp. (CPE only)
080 = 80 Amp.
100 = 100 Amp.
120 = 120 Amp.
150 = 150 Amp.
300 = 300 Amp.

4. Function – NP - Nonreversing, Plugging
NX - Nonreversing, Nonplugging
FP - Full Reversing, Plugging
CP - Constant Potential Exciter

5. Voltage – 090 = 90 Volt C.T. 1ϕ only
180 = 180 Volt 1ϕ only
250 = 250 Volt (CPE only)
360 = 360 Volt 1ϕ only
560 = 560 Volt 3ϕ only

6. Wire Type – F = Flamenol

7. Color – G = Gray (Standard)
B = Black (nonstandard)

8. Release Point – This applies to reversing exciters only.
A - NR, NX, CP, UR
B - 1ϕ FP
D - 3ϕ FP

9. Base – Base "Y"
No Base "N"

2. CATALOG NUMBER – ELEMENTARY DIAGRAM CROSS REFERENCE TABLE

CATALOG #			ELEMENTARY
IC3605	A or B	018NP090-	68A993640
	C, D, E, or F	018NP090-	68A993657
	C, D, E, or F	018FP090-	68A993677
	A or B	040NP090-	68A993642
	C, D, E, or F	040NP090-	68A993659
	C, D, E, or F	040FP090-	68A993679
	A or B	080NP090-	68A993645
	C, D, E, or F	080NP090-	68A993662
	C, D, E, or F	080FP090-	68A993682
	A or B	018NP180-	68A993641
	C, D, E, or F	018NP180-	68A993658
	A or B	018NX180-	68A993650
	C, D, E, or F	018NX180-	68A993670
	C, D, E, or F	018FP180-	68A993678
	A or B	040NP180-	68A993643
	C, D, E, or F	040NP180-	68A993660
	A or B	040NX180-	68A993651
	C, D, E, or F	040NX180-	68A993671
	C, D, E, or F	040FP180-	68A993680
	A or B	080NP180-	68A993646
	C, D, E, or F	080NP180-	68A993663
	A or B	080NX180-	68A993653
	C, D, E, or F	080NX180-	68A993673
	C, D, E, or F	080FP180-	68A993683
	A or B	120NP180-	68A993648
	C, D, E, or F	120NP180-	68A993665
	A or B	120NX180-	68A993655
	C, D, E, or F	120NX180-	68A993675
	C, D, E, or F	120FP180-	68A993685

CATALOG #			ELEMENTARY
IC3605	A or B	040NP360-	68A993644
	C, D, E, or F	040NP360-	68A993661
	A or B	040NX360-	68A993652
	C, D, E, or F	040NX360-	68A993672
	C, D, E, or F	040FP360-	68A993681
	A or B	080NP360-	68A993647
	C, D, E, or F	080NP360-	68A993664
	A or B	080NX360-	68A993654
	C, D, E, or F	080NX360-	68A993674
	C, D, E, or F	080FP360-	68A993684
	A or B	120NP360-	68A993649
	C, D, E, or F	120NP360-	68A993666
	A or B	120NX360-	68A993656
	C, D, E, or F	120NX360-	68A993676
	C, D, E, or F	120FP360-	68A993686
	C, D, E, or F	100NP560-	68A993667
	C, D, E, or F	150NP560-	68A993668
	C, D, E, or F	300NP560-	68A993669
	C, D, E, or F	100FP560-	68A993687
	C, D, E, or F	150FP560-	68A993688
	C, D, E, or F	300FP560-	68A993689
	C or D	022CP250-	68A993690
	C or D	050CP250-	68A993691
	C or D	100CP250-	68A993692
	C or D	150CP250-	68A993693
	C or D	300CP250-	68A993694
	X	020UR250-	68A993695
	X	100UR250	68A993697
	X	300UR250-	68A993699

B. COMPATIBILITY

Common subassemblies and cards are utilized as much as possible. See Table 4 for a listing of common parts vs exciter variables.

How to use Table 4: The columns list the component and description, then the firing, current, function, etc., as they appear in the catalog number. It is possible to determine what part is used in an exciter and if the part is common to other exciters. For instance: IC3605C080NP180FGAY

IC3605 - exciter series
C - unijunction fired, 60-cycle power, with internal exciter feedback
080 - 80 amps
NP - nonreversing plugging
180 - 180 volts rated output voltage
F - Flamenol® wire
G - gray
A - R5 = 00 ohm
Y - base provided

By matching the X's in the columns with the catalog number (X in unij; X in 60 cycle; X in 080; X in NP; X in 180), it can be seen that gate and filter circuit PWB68A993124 is used with the above exciter; the IC3600TFCU1 unijunction firing circuit is used, the IC3600TPSA1 single-phase power supply unijunction is used; etc. It can also be seen that many components are not used, such as the IC3600TRLA1 - Reversing Logic.

IC3605E150FP560FGAY catalog number uses a PWB68A989923 gate and filter circuit, an IC3600-TFCU1 unijunction firing circuit, an IC3600TPSA1 single-phase power supply unijunction, an IC3600-TPSB1 three-phase power supply unijunction and an IC3600TRLB1 reversing logic.

Some of these cards are common to both exciters.

C. PLUGGING

Plugging, or inversion, means pumping power back into the AC lines, assuming there is a source of power. A resistor is not a source of power, but the stored energy in an inductance, or a separate generator, would be a suitable source of power. This discussion will be limited to inductive loads.

TABLE 4. COMPATIBILITY CHART

COMPONENT	DESCRIPTION	UNIJ	AMPL	60 ~	50 ~	018 (22) AMP	040 (050) AMP	080 AMP	100 AMP	120 AMP	150 AMP	300 AMP	CP	NX	FP	NP	090 v	180 v	250 v	360 v	560 v
PWB68A989923	Gate & Filter CKT	X		X	X		X	X	X	X	X				X					X	X
PWB68A993117	Gate & Filter CKT	X	X	X	X	X							X	X		X	X	X	X		
PWB68A993122	Gate & Filter CKT	X		X	X		X	X		X					X		X	X			
PWB68A993123	Gate & Filter CKT	X	X	X	X		X	X	X	X	X			X		X				X	X
PWB68A993124	Gate & Filter CKT	X	X	X	X		X	X	X	X	X		X	X		X	X	X	X		
PWB68A993125	Gate & Filter CKT	X		X	X	X									X		X	X			
PWB68A993147	Gate & Filter CKT	X		X	X							X			X						X
PWB68A993148	Gate & Filter CKT	X		X	X							X	X						X		
PWB68A993149	Gate & Filter CKT	X		X	X							X				X					X
PWB68A993188	Filter CKT	X	X	X	X	X								X				X			
PWB68A993157	Filter CKT	X	X	X	X		X	X		X				X						X	
PWB68A993159	Filter CKT	X	X	X	X		X	X		X				X				X			
68A995018-G1	Amplistat		X	X		X	X	X		X				X		X	X	X		X	
68A995018-G2	Amplistat		X		X	X	X	X		X				X		X	X	X		X	
IC3600TFCA1	Amplistat F.C.		X	X	X	X	X	X		X				X		X	X	X		X	
IC3600TFCU1	Unijunction F.C.	X		X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
IC3600TFCV1	Unijunction F.C.	X			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
IC3600TPSA1	1ϕ P.S. Unijunction	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
IC3600TPSB1	3ϕ P.S. Unijunction	X		X	X	X	X		X		X	X	X		X	X			X		X
IC3600TPSC1	Amplistat P.S.		X	X	X	X	X	X		X				X		X	X	X		X	
IC3600TRLA1	Rev. Logic 1ϕ, 60~	X		X		X	X	X		X					X		X	X		X	
IC3600TRLB1	Rev. Logic 3ϕ, 60~	X		X					X		X	X			X						X
IC3600TRLC1	Rev. Logic 1ϕ, 50~	X			X	X	X	X		X					X		X	X		X	
IC3600TRLD1	Rev. Logic 3ϕ, 50~	X			X				X		X	X			X						X
IC3500P491AB103	Sat. Reactor	X		X	X	X	X	X	X	X	X	X			X		X	X		X	X
138A1050P34	Transformer	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
138A1050P35	Transformer	X	X	X	X		X	X	X	X	X	X		X	X	X				X	X
68A7267PXE	C137E-P1V500v	X		X	X	X									X		X	X			
68A7297PXM	C35M-P1V600v	X	X	X	X	X							X	X		X	X	X			
205A5793PXE	C350E-P1V500v	X	X	X	X		X	X	X				X	X	X	X	X	X	X		
205A5793PXP	C350P-P1V1000v	X	X	X	X		X	X	X					X	X	X				X	X
205A5794PXE	C380E-P1V500v	X	X	X	X					X	X		X	X	X	X		X	X		
205A5794PXP	C380P-P1V1000v	X	X	X	X					X	X			X	X	X				X	X
68A7291PXE	C291E-P1V500v	X		X	X							X	X						X		
68A7291PXP	C291P-P1V1000v	X		X	X							X			X	X					X
205A5795PXE	A390E-P1V500v	X	X	X	X		X	X		X				X				X			
205A5795PXP	A390P-P1V1000v	X	X	X	X		X	X		X				X						X	
167AB419P15	IN2160-P1V600v	X	X	X	X	X								X				X			

Figure 4 shows a typical circuit that will plug. It is a single-phase center tap circuit. The discussion will be based on this circuit. However, the wave shapes in a single-phase full wave bridge or a three-phase full wave bridge would be similar.

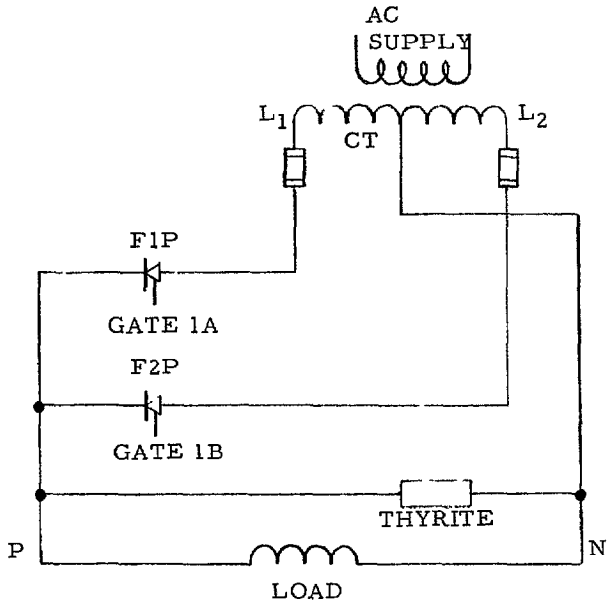


Fig. 4. Single phase center tap circuit

Figure 5 shows instantaneous wave shapes of the line voltage, load voltage, each SCR voltage and each SCR gate signal. These are shown for three different phase angles of the firing signals.

Since we assumed an inductive load, the load current is a straight line. This current is supplied half the time by F1P and half the time by F2P. The conducting time occurs when the SCR voltage is zero.

Figure 5 (a) shows the voltage for the "phased on" or full output condition, and these pictures occur either transiently or in steady state. Let us start at the time gate 1A signal occurs – this fires F1P, reducing its voltage to zero. This applies full phase voltage to the load, but note that it applies twice phase voltage to F2P in the reverse direction. This is true throughout Fig. 5. All parts of sinusoids on the load are at phase voltage, but all parts of sinusoids on the SCR's are at twice phase voltage.

At the first voltage zero, we have stored a half cycle of energy in the inductive load so the load will not let F1P stop conducting. The load voltage merely goes negative enough to overcome any voltage on phase 1 of the transformer. This negative load voltage, of course, adds to phase 2 of the transformer, putting double phase voltage on F2P in the forward direction.

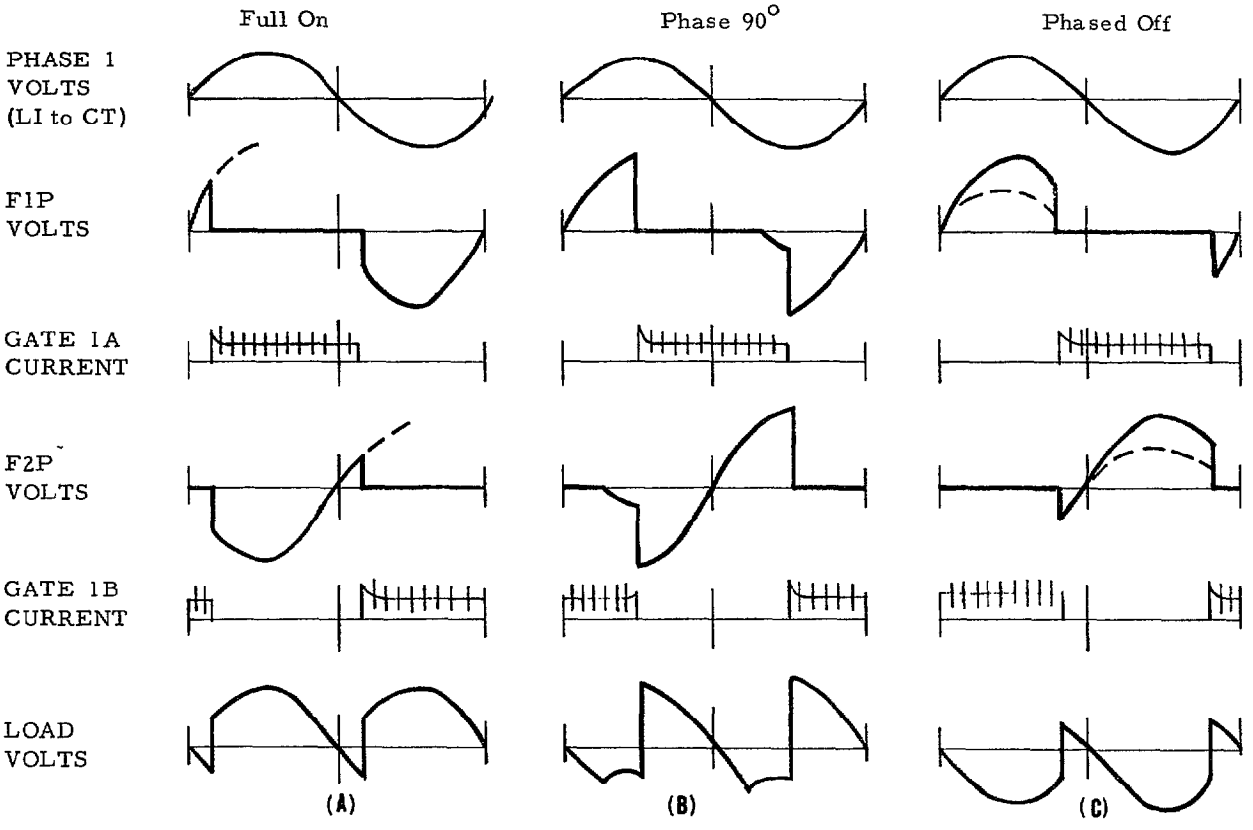


Fig. 5. Typical exciter waveshapes for center tap exciter with various output voltages

As gate 1B signal occurs, F2P goes to zero and starts conducting. This applies negative voltage to F1P, which stops the current flow in F1P. At this point, the load voltage is again positive. Since the circuit is symmetrical the same explanations apply to this half cycle.

Figure 5 (b) shows the gate signals firing at 90° and these curves are shown for the steady state condition. Starting at pulse 1, the explanation starts as before. However, we have stored only one quarter of a cycle of energy in the inductive load. Thus, by the time we are 80° into the next half cycle, virtually all the energy of the inductance has been returned to the line. At this point, the inductance usually finds an eddy current path where it can dissipate the remaining energy at a lower voltage than the line voltage. Thus, the negative wave of load voltage breaks toward zero and starts following an exponential. F1P goes out because the difference between line voltage and load voltage is now a small negative voltage on F1P. At the firing of pulse 1B, we continue as before.

A DC voltmeter across the load will read a few volts positive because the positive wave is larger than the negative. The amount of positive voltage depends on the circuit losses and is most dependent on the lamination thickness of the load or coupling into other fields (eddy current type losses).

Dynamically the curves are slightly different. If operation has been like Fig. 5 (a) and we suddenly phased back to Fig. 5 (b), there is no question that the inductive load has too much energy stored, temporarily, to dissipate it in eddy currents. Thus, the negative load voltage part of the wave will look like the positive part. In fact, it will be a couple of volts higher due to the drops in the SCR's. Thus, transiently a DC voltmeter would read minus two or three volts for the Fig. 5 (b) case.

Figure 5 (c) shows the firing pulse "phased off" and all the solid curves are for the transient case where operation has been like Fig. 5 (a) and is suddenly changed to Fig. 5 (c). The explanation is exactly like that for the dynamic case of Fig. 5 (b) – the only difference is the transfer points; there is no change in rules or explanation.

It is re-emphasized that the load must have stored energy so that it will keep current flowing. If the load causes current flow through an opposing transformer voltage we have plugged or inverted.

With the firing pulses 25° from the end of the half cycle (Fig. 5 (c)), transiently the shape of the curve is the inverse of Fig. 5 so a DC voltmeter will read -90 volts.

As the load current nears zero, the negative load voltage will start breaking off in exponentials as explained previously, until in the steady state it looks like the dotted load curve of Fig. 5 (c). In the steady state, the positive area is larger than the negative area so a voltmeter again reads very low positive voltage.

Now let us take the case where we are operating "phased on" (Fig. 5 (a)) and we suddenly remove all firing pulses. At the moment the pulses were removed let us assume F1P was conducting. With no firing pulse F2P will never conduct. (The following discussion shows that there are no abnormal voltages on F2P.)

If F2P will never conduct, the inductive load must keep F1P conducting continuously. In this circuit is phase 1 of the transformer whose DC voltage is zero. As the inductive load sees zero voltage in a discharge path it will decay on its natural time constant (there is no plugging or inverting) over one cycle. Actually, the load voltage is full positive one half cycle, receiving a block of power, then full negative one half cycle, plugging the same block of power back into the line. The 60-cycle voltage on the load applies double phase voltage to F2P, but F2P is designed for double phase voltage as shown in Fig. 5.

Thus, a plugging exciter must have "tail end" pulses – that is, fixed pulses that occur somewhere in the last 25° of the cycle which do not disappear regardless of what the control input calls for. This will cause the UJT to fire the SCR at the end of the cycle if it has not already fired due to an input signal. The function of the tail-end pulse is to dissipate the energy in the load by discharging the inductive energy during both positive and negative half cycles of the AC supply.

Figure 6 shows theoretical characteristic curves.

The dotted curve shows the characteristic for a nonplugging circuit. This curve is obtained with a resistive load or by adding a discharge rectifier around an inductive load. The solid curve represents the plugging characteristics. The negative portion of this wave is for a transient case.

D. SCR FIRING METHODS

There are several successful methods of firing SCR's. For this line of exciters we are using a unijunction transistor timing for SCR phase control.

The silicon unijunction transistor (UJT) is an ideal device for use in SCR firing circuits. It has

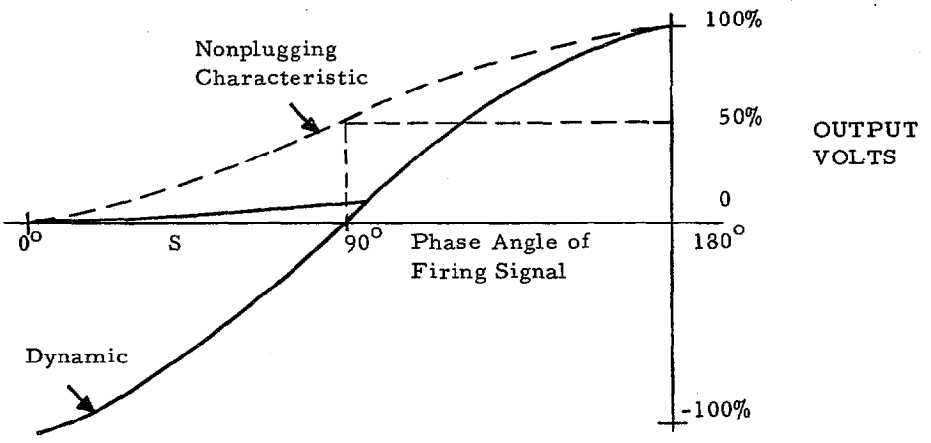


Fig. 6. Theoretical characteristic curves

the advantages of a stable firing voltage, a very low firing current, operation over temperature range of -55 C to +140 C, and a peak current rating of two amps. SCR firing circuits using the UJT are simple and compact with low power consumption and a high effective power gain in phase control circuits.

For a controlled DC output, the firing pulses must be synchronized with the power input voltage. In single phase exciters one firing circuit is used. In three phase exciters three firing circuits are used with a firing circuit synchronized to each phase.

The heart of the SCR firing circuit is the unijunction (UJT) transistor. Its symbol and basic circuit are shown in Fig. 7.

A DC power supply is connected from base 1 (B1) to base 2 (B2) with B2 positive. With no input signal on the emitter, very little current flows (microamperes). As the input voltage is raised (emitter positive), nothing happens until a critical voltage is reached. At this point, the UJT breaks into conduction and offers very low impedance in RB1. If the input is removed, the device recovers even when the power supply is kept on.

The UJT cannot be compared directly with either a common transistor or an SCR. In contrast to a transistor, the switching point is determined by voltage at the emitter – not current, and there is a definite, finite switching point – not proportional impedance. In contrast to an SCR, the control signal is a voltage level, instead of a pulse, and it recovers with no signal applied.

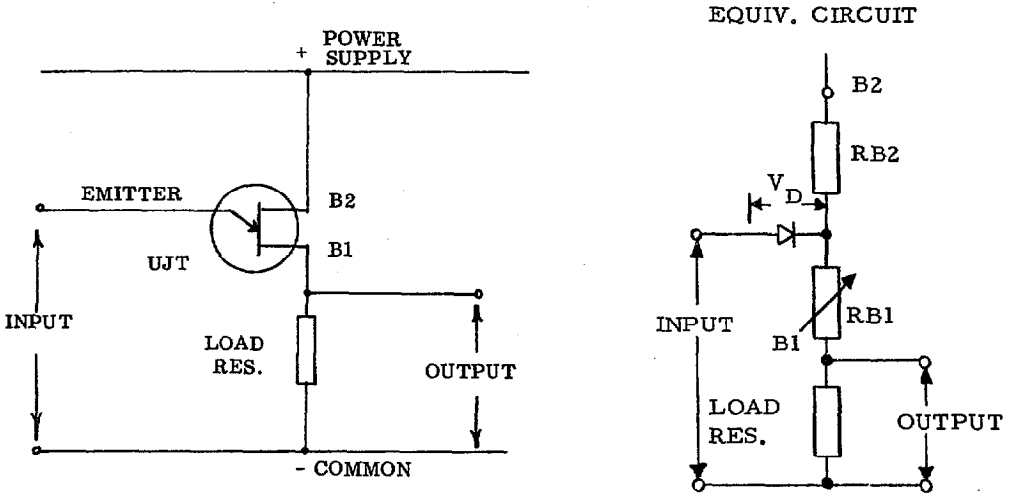


Fig. 7. Unijunction transistor

E. ELECTRICAL DESCRIPTION

1. BLOCK DIAGRAMS

See Figs. 8 and 9 for block diagrams of unijunction fired, single phase nonreversing and reversing thyristor exciters. The three phase exciters utilize the same general scheme; the number of components and firing sequence are different.

a. Nonreversing Exciter – Input signal is applied between S and C of TCBA. Negative voltage on terminal S is in a direction to cause the output voltage to increase in a positive direction. The input signal is scaled down in voltage and fed to the firing circuit card TFC. The firing circuit generates a pulse train to the SCR gates, the duration of which is approximately eight milliseconds long (60-cycle input power). This is true regardless of the magnitude of the input signal. The input signal shifts the timing of the pulse train so that when the input signal is zero volts the pulse train is applied to SCR gates whose anode to cathode voltage is reverse biased. As the input signal is increased the pulse train is shifted so that the gate pulses occur while the anode-cathode SCR voltage is forward biased. That is, the pulses are gradually shifted so that the SCR is partially on until the SCR's are on for about a half cycle at maximum input signal. The thyristor power circuit is the arrangement of SCR's which converts the AC input power to DC. The transformer converts AC input voltage to center tapped ± 12 volt RMS. This voltage is used in the power supply to generate the required + and - busses for the firing circuit. The ± 12 volt is also used to steer the firing circuit to deliver pulses to the proper SCR at the proper time.

b. Reversing Exciter – Input signal is applied between S and C of TCBA. Negative voltage on terminal S is in a direction to cause the output voltage to increase in a positive direction. Positive voltage on terminal S will result in a negative output voltage. The input voltage is scaled down on the CB card and fed to the forward firing circuit F1F and the reversing logic RL. If terminal S is negative the firing circuit F1F generates pulses and transfers them to the thyristor power circuit via the gate and filter circuits. If the terminal S is positive the input signal is inverted in the reversing logic and fed to the reverse firing circuit R1F. R1F generates pulses and transfers them to the thyristor power circuit via the gate and filter circuits. When one firing circuit is delivering pulses to the power circuit, the other firing circuit is disabled or locked out by the reversing logic. The function of the saturable reactor is to monitor the thyristors and determine whether they are conducting or not. In order to reverse the output voltage successfully the thyristors must not be conducting. When an exciter is delivering positive voltage to the load, F1F is firing the thyristors and R1F is locked out by the reversing logic. When the exciter is commanded to deliver negative voltage by means of a polarity reversal of the input signal, F1F is phased back or effectively turned off. R1F receives an input signal from the reversing logic of the proper polarity. However, it is still locked out. When the saturable reactor indicates all the thyristors are off, the reversing logic will unlock out R1F and lock out F1F, thus reversing the output voltage.

The transformer and power supply perform the same type of functions as they did in the nonreversing exciter.

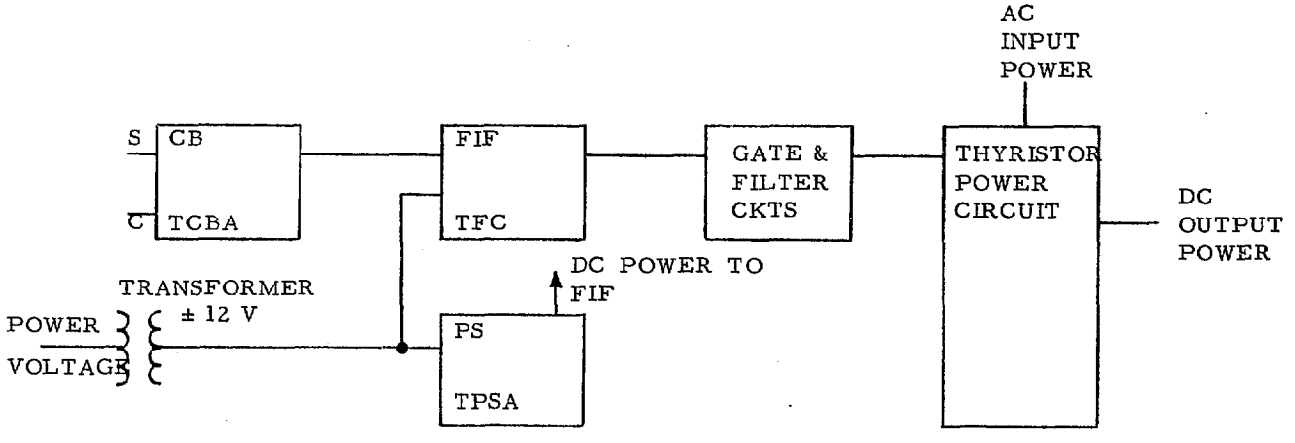


Fig. 8. Unijunction single phase nonreversing block diagram

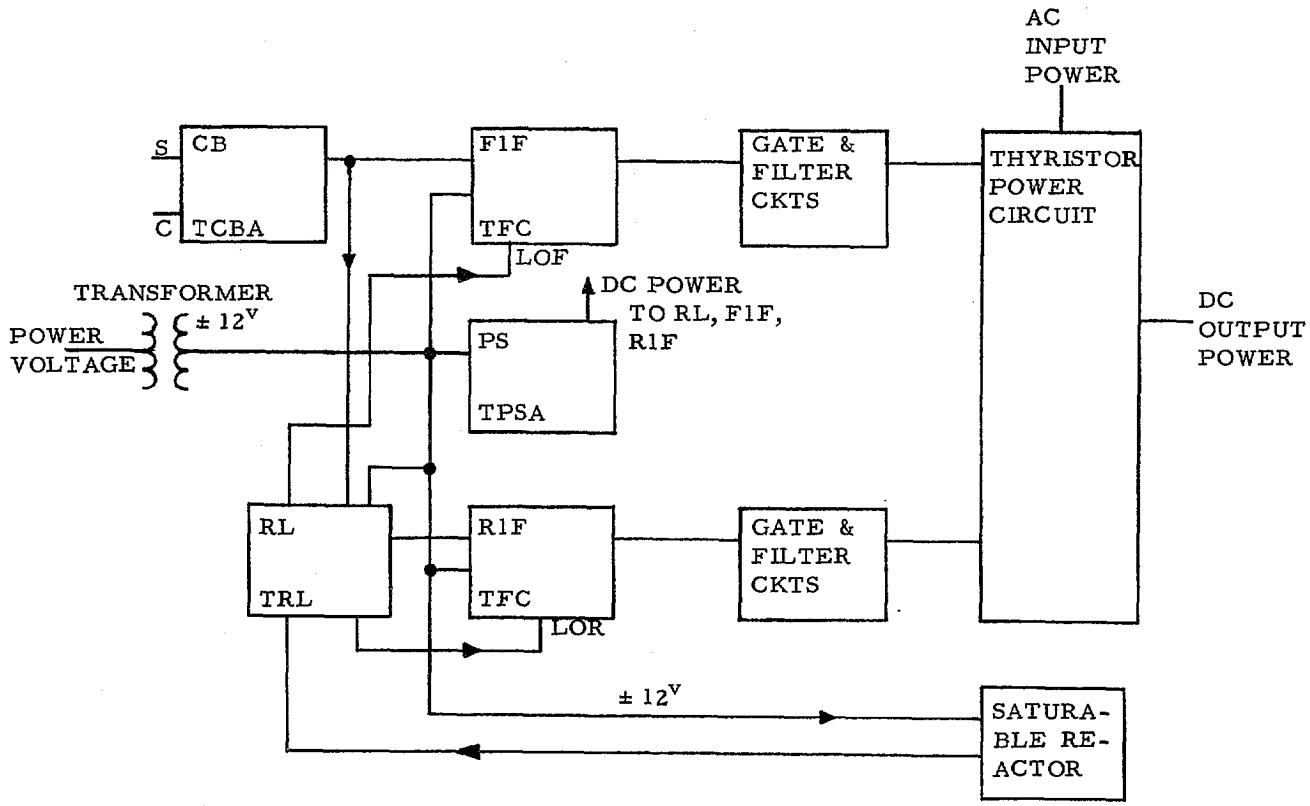


Fig. 9. Unijunction single phase reversing thyristor block diagram

2. BASIC POWER CIRCUITS

a. Thyristor Center Tap Exciters

IC3605-
A, B, C, D, E, F 018NP090
040NP090
080NP090

Basic Description:

This particular group of exciters uses a center tap power circuit as shown in Fig. 10. The output voltage is nominally 90 volts into any load and the circuit will plug with an inductive load applying minus 90 volts.

Circuit Description:

A thyristor is used in each leg of the power circuit.

A thyrite is used in this group of exciters to provide a load discharge path in the event both fuses should blow.

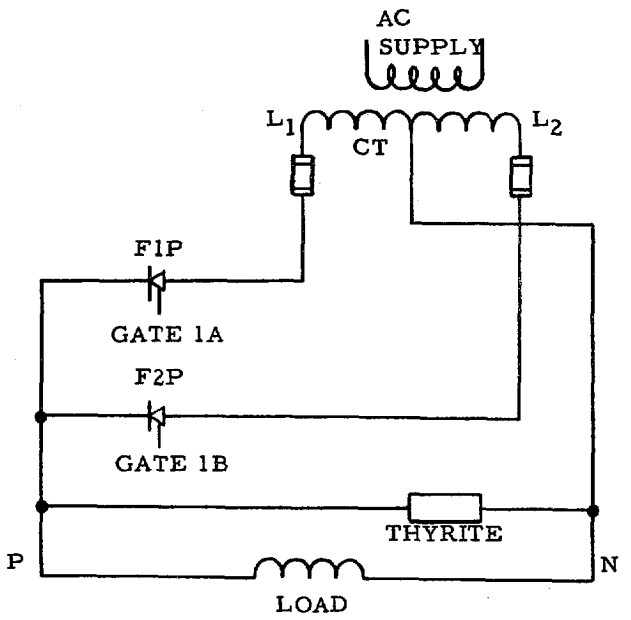


Fig. 10. Center tap power circuit

b. Single-phase Full-wave Bridge – Nonplugging

A, B, C, D, E, F	018NX180	A, B, C, D, E, F	040NX360
	040NX180		080NX360
	080NX180		120NX360
	120NX180		

Basic Description:

This group of exciters uses a single phase, full wave bridge power circuit as shown in Fig. 11. The output voltage for 230-volt supply is nominally 180 volts; the output voltage for 460-volt supply is nominally 360 volts.

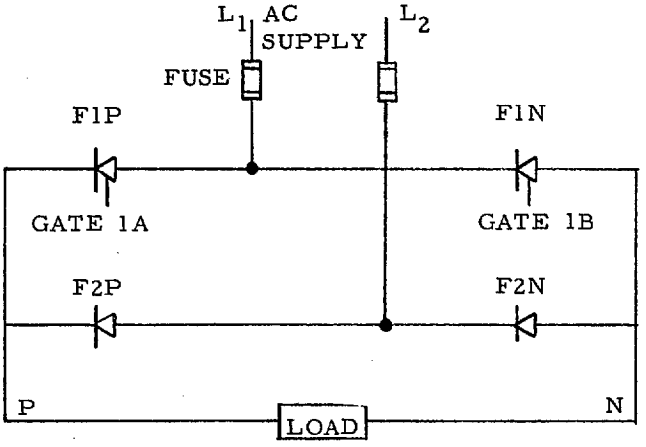


Fig. 11. Single phase, full wave bridge power circuit

Circuit Details:

A thyristor is used in two legs of the bridge. Conventional rectifiers are used in the other two legs.

This exciter will not plug an inductive load. F2P and F2N provide a discharge path so that the SCR's will always stop conducting at the end of a half cycle.

c. Single-phase Full-wave Bridge Exciters

A, B, C, D, E, F	018NP180	A, B, C, D, E, F	040NP360
	040NP180		080NP360
	080NP180		120NP360
	120NP180		

Basic Description:

This group of exciters uses a single phase, full wave bridge power circuit as shown in Fig. 12. The output voltage will be nominally 180 volts when the AC input power is 230 volts and 360 volts when the AC input power is 460 volts. It will plug (invert) applying minus voltage to an inductive load.

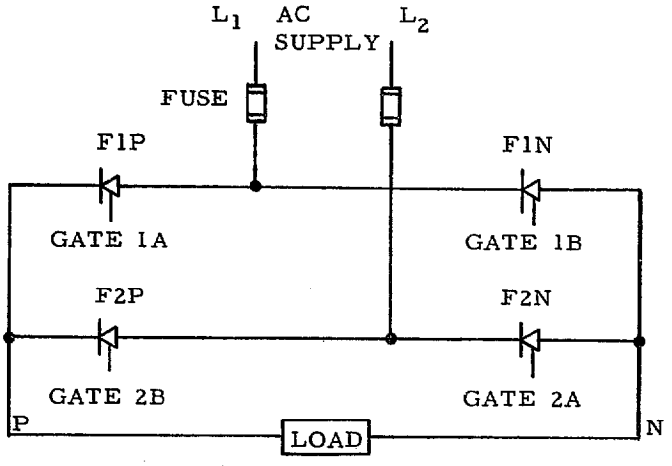


Fig. 12. Single phase, full wave bridge power circuit

Circuit Details:

A thyristor is used in each leg of the bridge.

The above circuit is for a nonreversing, plugging model NP.

d. Three-phase Full-wave Bridge Exciters

C, D, E, F	100NP560	C, D	022CP250
C, D, E, F	150NP560	C, D	050CP250
C, D, E, F	300NP560	C, D	100CP250
		C, D	150CP250
		C, D	300CP250

Basic Description:

This group of exciters uses a three-phase bridge power circuit as shown in Fig. 13. Nominal output varies with AC supply voltage and function.

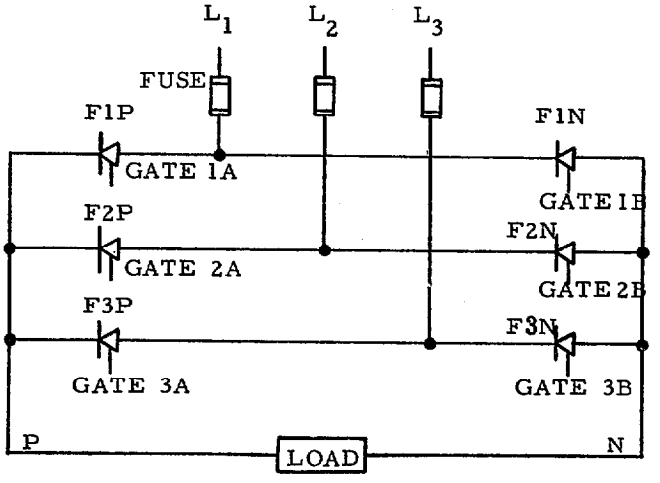


Fig. 13. Three phase bridge power circuit

AC Supply Voltage	Function	Nominal Output Voltage
230	CP	250
460	NP	560

It will plug with an inductive load, supplying negative output voltage.

Circuit Description:

A thyristor is used in each leg of the bridge.

The circuit (Fig. 13) is for a nonreversing, plugging exciter and a constant potential exciter.

Constant potential exciters will have waveshapes similar to the three-phase bridge NP exciters. The load voltage will be a steady 250V. This will be produced with a firing point of about 140°. The resultant load waveshape will have a large ripple content.

e. Thyristor Full-reversing Center-tap Exciters

C, D, E, F 018FP090
040FP090
080FP090

Basic Description:

This group of exciters uses a center tap power circuit as shown in Fig. 14. The output voltage is nominally plus or minus 90 volts into any load.

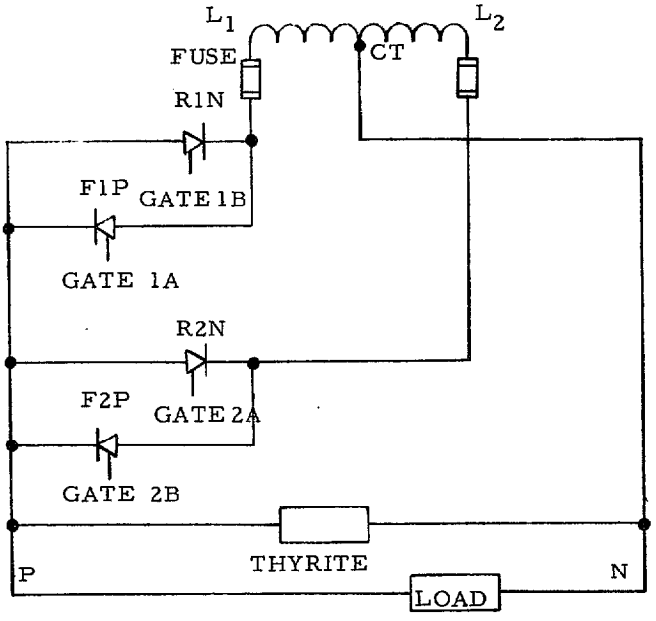


Fig. 14. Center tap power circuit

Circuit Description:

There are two thyristors in each leg of the power circuit; they are in parallel anode of one to cathode of the other. F1P and F2P provide positive voltage at terminal P while R1N and R2N provide negative voltage at terminal P.

The Thyrite® is used in this group to provide a load discharge path in the event both fuses should blow.

f. Thyristor Full-reversing Single-Phase Bridge

C, D, E, F 018FP180
040FP180
080FP180
120FP180
040FP360
080FP360
120FP360

Basic Description:

This group of exciters uses a single phase full wave bridge as shown in Fig. 15. It will deliver 180 or 360 volts to a load.

Circuit Details:

There are two thyristors in each leg of the power circuit; they are in parallel anode of one to cathode of the other. The terminal P will be positive or negative depending on whether the F or R cells are conducting.

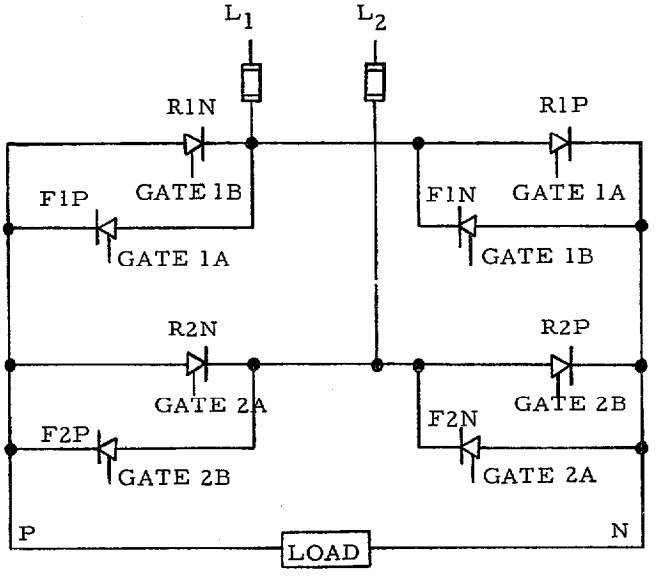


Fig. 15. Single phase, full wave bridge

g. Thyristor Full-reversing Three-Phase Bridge

C, D, E, F 100FP560
150FP560

Basic Description:

This group of exciters uses a three-phase bridge power circuit as shown in Fig. 16. Output is nominally 560 volts with an AC supply voltage of 460 volts.

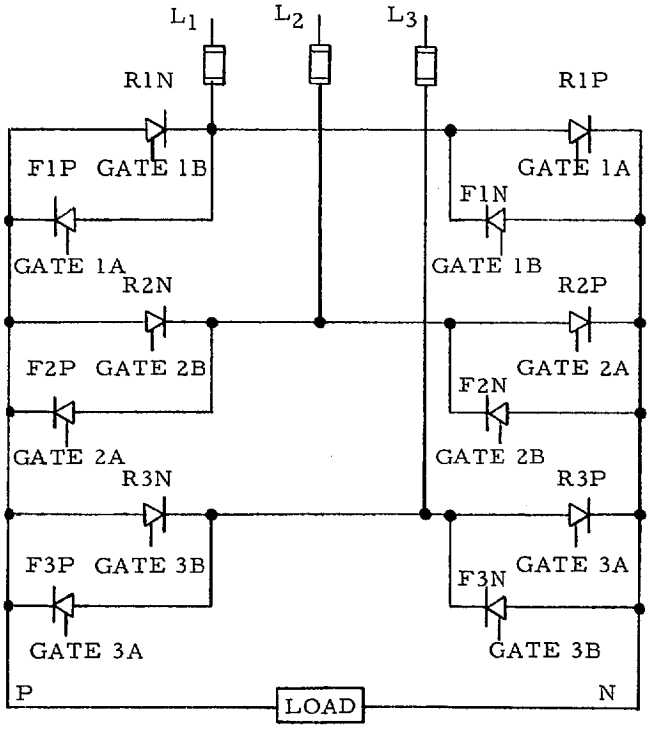


Fig. 16. Three phase bridge power circuit

Circuit Description:

There are two thyristors in each leg of the power circuit; they are in parallel anode of one to cathode of the other. Terminal P can be positive or negative depending on whether the R (reverse) or F (forward) cells are firing.

There is a 300-ampere full reversing exciter which is physically composed of two NP type exciters, one for positive and one for negative voltage. Electrically the exciter is similar to that described above.

h. Quarter-reversing Exciters (QP)

The following exciters may be converted to QP exciters:

A, B, C, D, E, F	018NP090
	040NP090
	080NP090
	018NP180
	040NP180
	080NP180
	120NP180
	040NP360
	080NP360
	120NP360
C, D, E, F	100NP560
	150NP560
	300NP560

3-phase

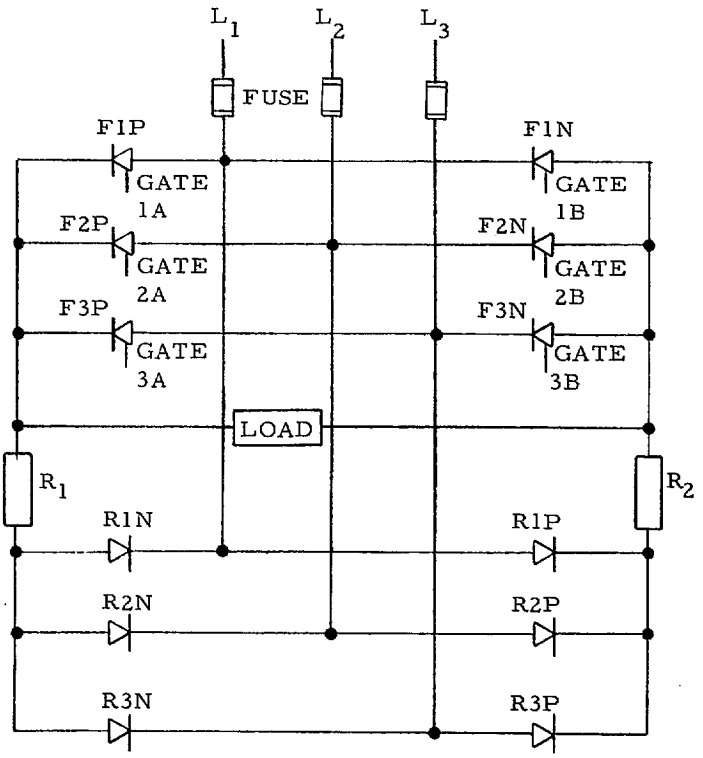


Fig. 17. Power circuit for three phase quarter reversing

Basic Description:

The circuit shown in Fig. 17 consists of a three-phase nonreversing plugging exciter with a three-phase full wave rectifier bridge. The rectifier bridge output opposes the SCR exciter and allows an output voltage reversal.

Circuit Description:

For most applications, the rectifier bridge draws its power through the same fuses as the SCR exciter as shown in Fig. 17.* The polarity of the rectifier bridge output is opposite to the SCR exciter; as a result, if none of the SCR's are conducting, the exciter output will be negative. When the SCR's do turn on they will overcome the rectifier bridge voltage and cause positive voltage across the load from P to N. R1 and R2 are chosen to limit reverse load current when the SCR bridge is not conducting.

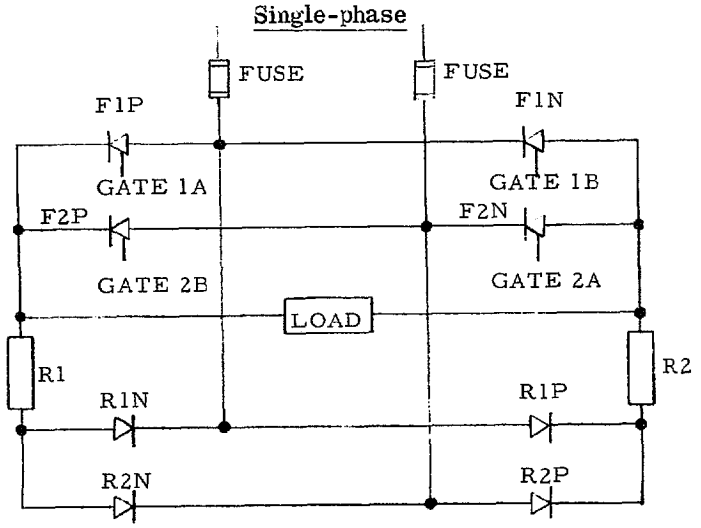


Fig. 18. Power circuit for single phase quarter reversing

Basic Description:

The circuit shown in Fig. 18 consists of a single phase nonreversing plugging exciter with a single phase full wave rectifier bridge. The rectifier bridge output opposes the SCR exciter and allows an output voltage reversal.

Circuit Description:

For most applications, the rectifier bridge draws its power through the same fuses as the SCR exciter as shown in Fig. 18.* The polarity of the rectifier bridge output is opposite to the SCR exciter; as a result, if none of the SCR's are conducting, the exciter output will be negative. When the SCR's do turn on they will overcome the rectifier bridge voltage and cause positive voltage across the load from P to N. R1 and R2 are chosen to limit reverse load current when the SCR bridge is not conducting.

*However, the elementary diagram should be checked to determine the actual circuitry, as it is possible to obtain power from separate fused supplies.

F. HOW TO PUT AN EXCITER IN SERVICE

1. Remove fuses.
2. Inspect for loose or damaged components.
3. Apply AC power. Check voltage at the top of fuse clips.

Input Voltage AC	Rated Output Voltage DC
230 CT ± 10% - 1 PH	90
230 ± 10% - 1 PH	180
230 ± 10% - 3 PH	250
460 ± 10% - 1 PH	360
460 ± 10% - 3 PH	560

If the input voltage is higher than rated voltage, change taps on the feeder transformer until the voltage is brought within limits.

4. If the exciter has a three-phase input, check phase sequence. Be sure the phase sequence is 1, 2, 3 when connected to L1, L2 and L3.
5. Remove AC power.
6. Replace fuses. The fuses used in these exciters are fast-acting current limiting fuses. Should one of these fuses be blown, it is important that it be replaced with the same type fuse. An ordinary fuse of the same current rating will not interrupt fast enough to protect the cells if a short circuit should occur in the load circuit of the exciter.
7. The exciter is now ready for service. It has been adjusted at the factory and should not require additional tune-up. If the exciter should appear to operate improperly, see the trouble-shooting notes to locate the trouble. DO NOT CHANGE internal adjustments unless the results of these adjustments are fully understood.

G. TROUBLE SHOOTING

1. SINGLE PHASE SCR EXCITERS

When a drive is down, the first concern is to get it going again. With this in mind, the following instructions will serve as a guide to locating the defective component. The exciters are so designed that most components can be changed easily, thus getting the drive back in operation in minimum time.

Most problems in the SCR exciters will be due to defective components, the majority of which can be located with a knowledge of exciter operation, the lights on the front of the exciter, and a volt-ohmmeter.

The lights on the exciter are very good indicators of the performance of the cells. These lights are connected directly across the SCR cells and are a

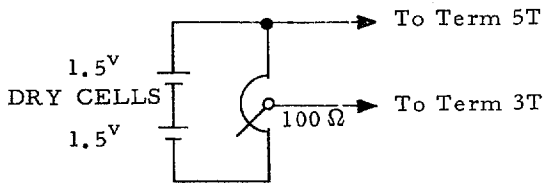
visual indication of the voltage across the cell. With the exciter phased off, the lights should burn relatively bright. As the exciter output increases, the intensity of the lights decreases until at half output the lights will go completely out. In reversing exciters, with no output in either direction, all lights will burn relatively bright. As the exciter is turned on in one direction, the lights corresponding to the cells for this direction will become dimmer while the lights for cells for the other direction will remain bright. Light condition will, of course, reverse when the exciter is turned on in the other direction. The lights will be brightest when the exciter is at full negative voltage (inverting).

Since these lights are such good indicators of exciter performance, it is important that good lamps be kept in the exciter at all times. If trouble should occur, a quick analysis of the light conditions will give a good idea where to start. Following is a list of possible light conditions, and the most common failure that could cause them.

LIGHT CONDITIONS	PROBABLE CAUSE	TROUBLE-SHOOTING PROCEDURE
1. All lights out and zero output.	This indicates loss of AC power across the SCR cells. For this condition there are two common causes:	
	1. Loss of AC input power to the exciter.	Check for voltage at the top of the fuses. This voltage should read 230 ± 10% volts or 460 ± 10% volts. If proper voltage is on the top of the fuses, check for voltage across each fuse. This will locate the blown fuse.
	2. One or more fuses blown.	Check all power semiconductors with an ohmmeter. Good SCR cells will have a resistance of 500 ohms or greater in both the forward and reverse direction. Any cell with less resistance than this is probably the cause of the blown fuses. Fuses can also be blown by shorts in the load circuit external to the exciter. To check this, measure the resistance across the load terminal. This resistance should be consistent with the value shown on the elementary diagram. As a power check for load shorts, disconnect one of the load terminals and put a high resistance across it, say 20KΩ10 watts. Also put small fuses (about 3 amps) in place of the blown fuse. Reapply AC power. If these fuses are blown, the trouble is inside the exciter and not in the external load circuit.
2. Light intensity becomes unbalanced - output decreases.	Bright lights indicate that a cell is not firing.	
	1. Trouble within the firing circuit card.	Interchange the firing circuit with spares or the other cards in the exciter. If the trouble follows the card, the card is not operating properly.
	2. Trouble is in the firing circuit outside the firing circuit cards.	Change the gate and filter cards with spares or other cards in the exciter. If the trouble follows the card, the card is not operating properly.

LIGHT CONDITIONS	PROBABLE CAUSE	TROUBLE-SHOOTING PROCEDURE
	3. Trouble is in the gate and filter cards.	Check the SCR gate-cathode junction with an ohmmeter. This can be done at the terminals on the gate and filter card. The resistance should be less than 250Ω. If it is greater, check wiring and SCR.
3. All lights bright - zero DC output.	This indicates that no cells are firing.	
	1. No input signal.	Check voltage across terminals C and S on the exciter terminal board. If voltage is present (C (+) with respect to S (-) greater than 0.5 volt or so) the exciter should have some DC output. (Reversing exciters will be turned on forward with this polarity input signal.) Check to see that this signal is reaching the firing circuit card. Test Point - 7 on TFCU-V. If the signal reaches the card, the trouble must be in the firing circuit.
	2. Firing circuit not functioning.	See firing circuit test point waveshapes.

Apparent trouble in the SCR is often not in the SCR at all, but in the control signal from the regulator. If you suspect that this may be the case, the following shortcut may prove helpful. Disconnect the control signal from the regulator terminals C and S on the exciter T.B. Wire in a source of DC signal, zero to three volts DC to card TCEA - Term (C) 5T and Term 3T. Vary this input signal from zero to about two volts and see if the exciter operates normally (two volts will turn the exciter full on). If the exciter operates normally under these conditions, the trouble is probably in the regulator instead of in the SCR. A simple DC signal source can be made from two 1.5 volt dry cells and a 100-ohm rheostat as shown in the sketch below.



Note: Exciters with internal voltage feedback can have 1/2 output voltage from input terminals to ground.

For additional assistance in locating trouble in your SCR exciters, oscilloscope patterns for properly operating exciters follow. These patterns are typical and any significant departure from them indicates trouble.

1. If exciter problems cannot be located with the trouble-shooting notes and the oscilloscope patterns, consult the General Electric Company.

2. When defective components are located, the following general procedure should be followed.

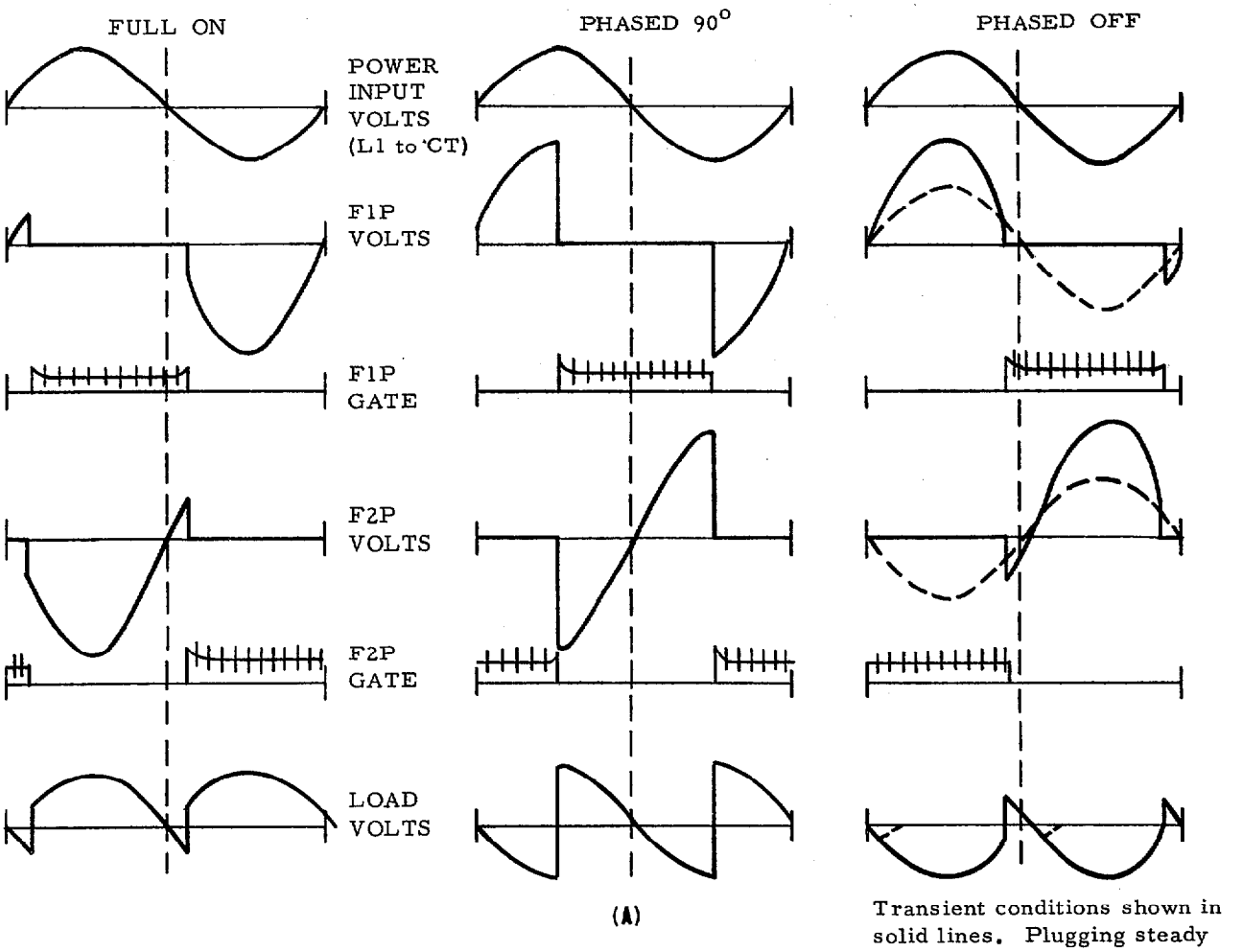
   a. Remove AC power from the exciter.

   b. Remove defective component. (Note wire numbers and location if wires have to be disconnected.)

   c. Replace with a component of the same catalog number. (Reconnect any wires that were disconnected; make sure cards are properly seated in their receptacles, all connections tight, etc.)

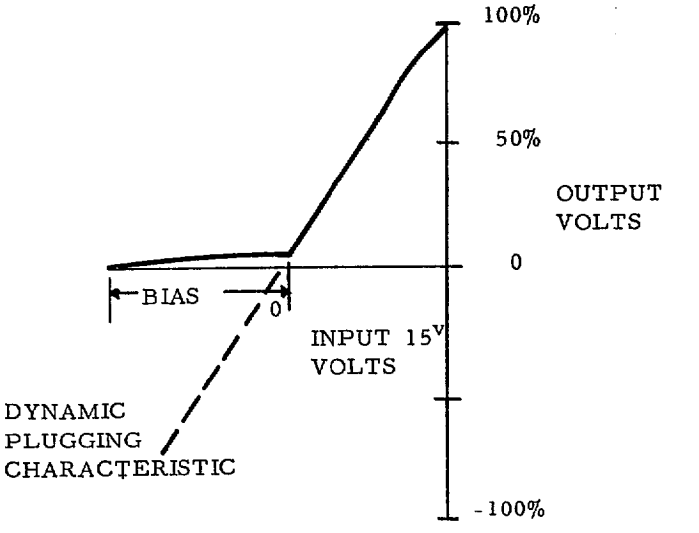
   d. Reapply AC power.

a. Single phase 90-volt Exciter (NP)
Center-tap Bridge



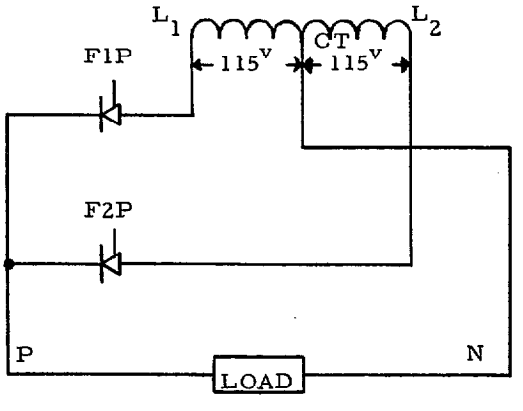
(A)

Transient conditions shown in solid lines. Plugging steady state condition shown in dotted lines.



NORMAL OPERATING CHARACTERISTIC

(B)



INDUCTIVE

POWER CIRCUIT

(C)

Fig. 19. Single-phase 90-volt exciter (NP) – center tap bridge

b. Single-phase Exciter Plugging – Full-wave Bridge (NP)

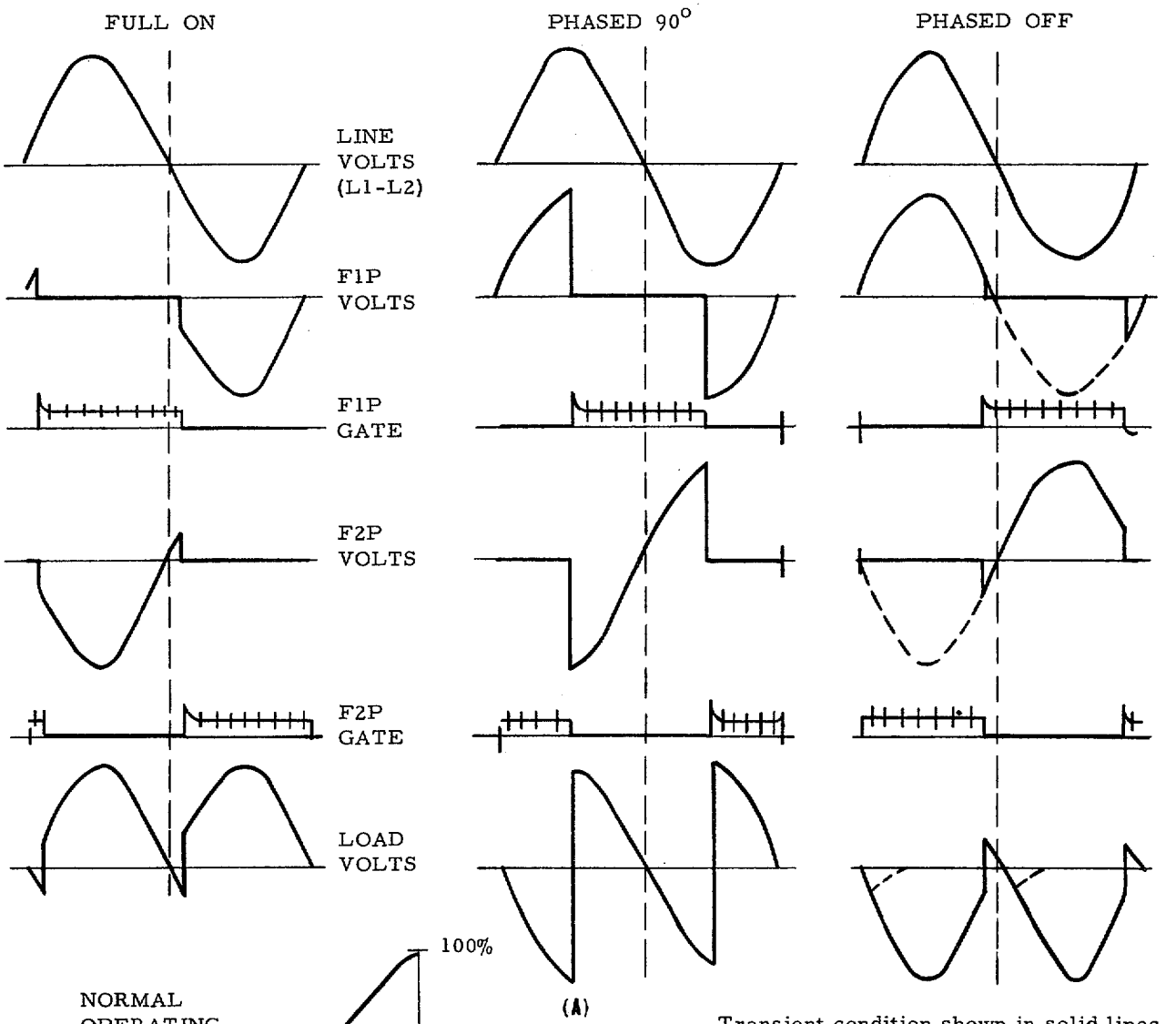


Fig. 20. Single phase exciter plugging – full wave bridge (NP)

c. Single-phase Exciter Nonplugging - Full-wave Bridge (NX)

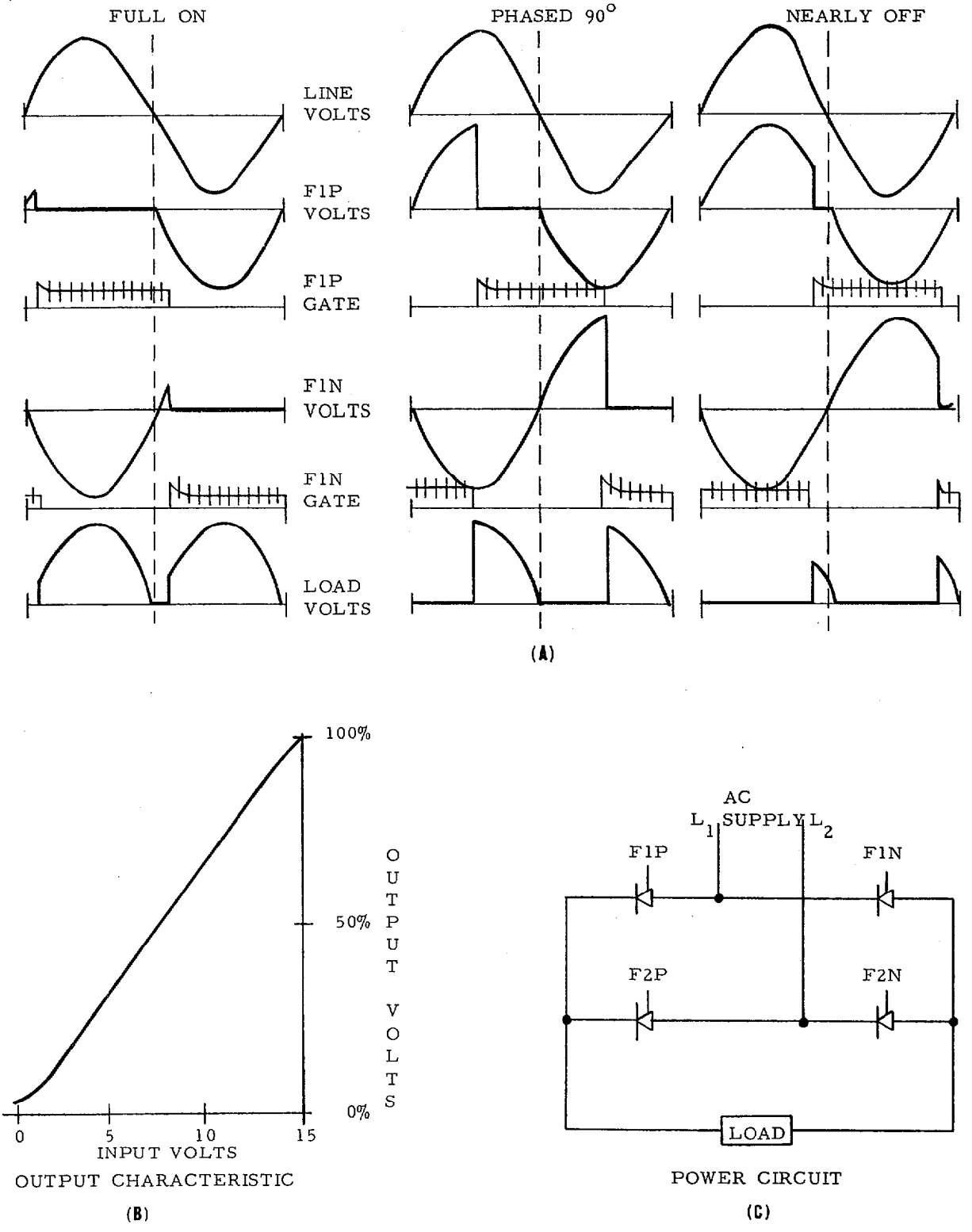
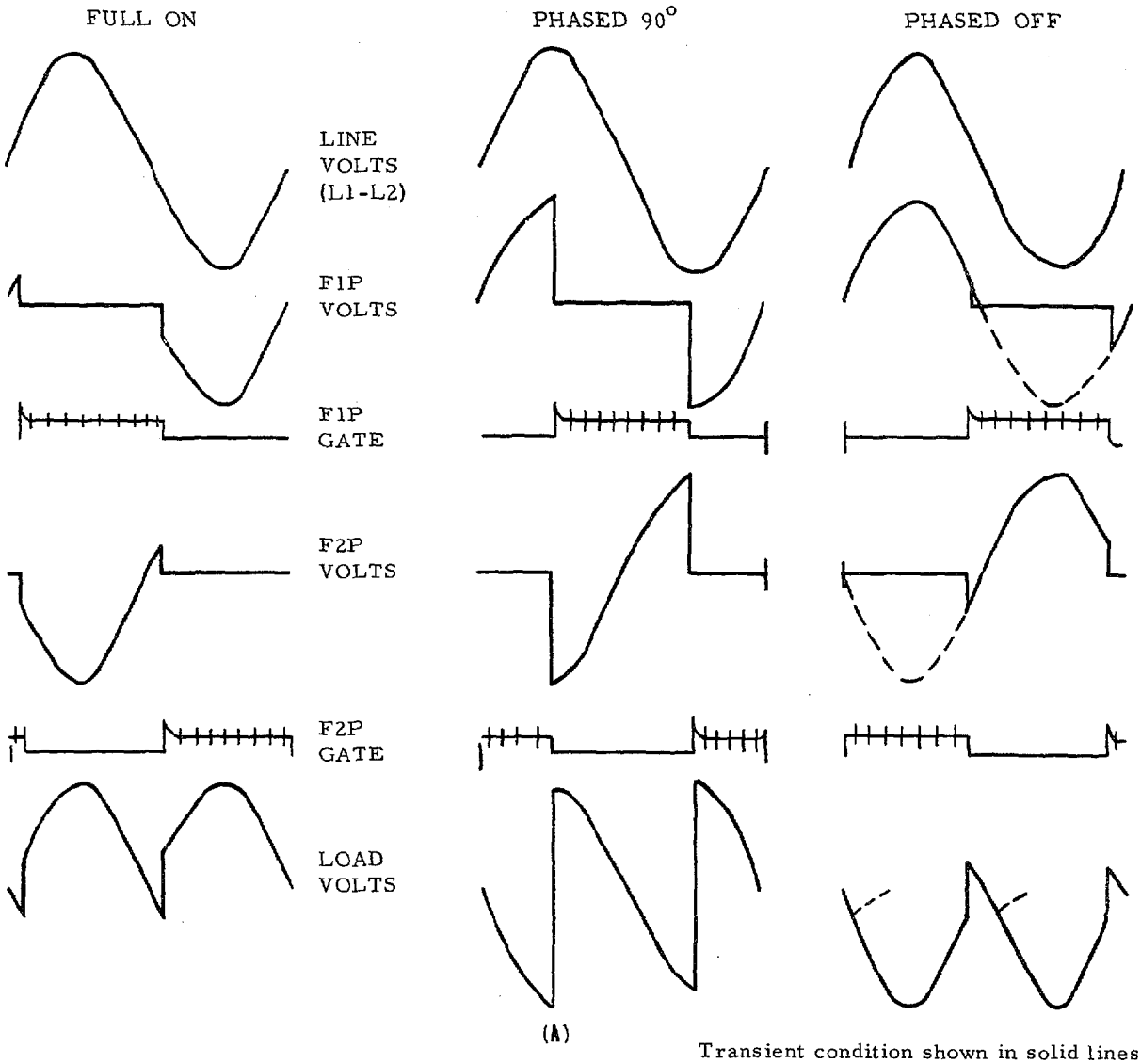


Fig. 21. Single phase exciter nonplugging - full wave bridge (NX)

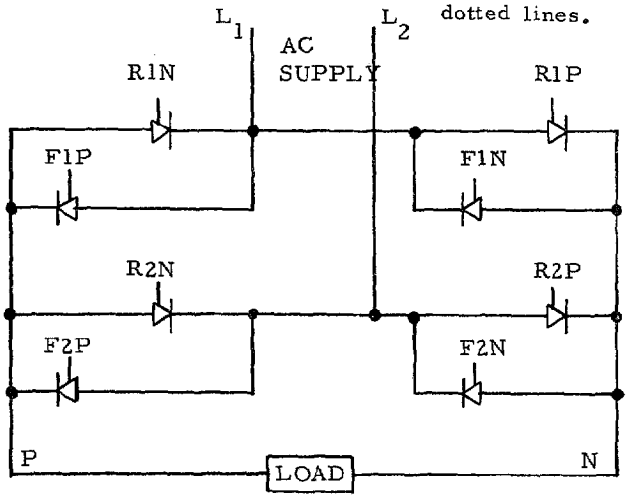
d. Single-phase Exciter, Full-reversing – Full-wave Bridge (FP)

P = +voltage



(A)

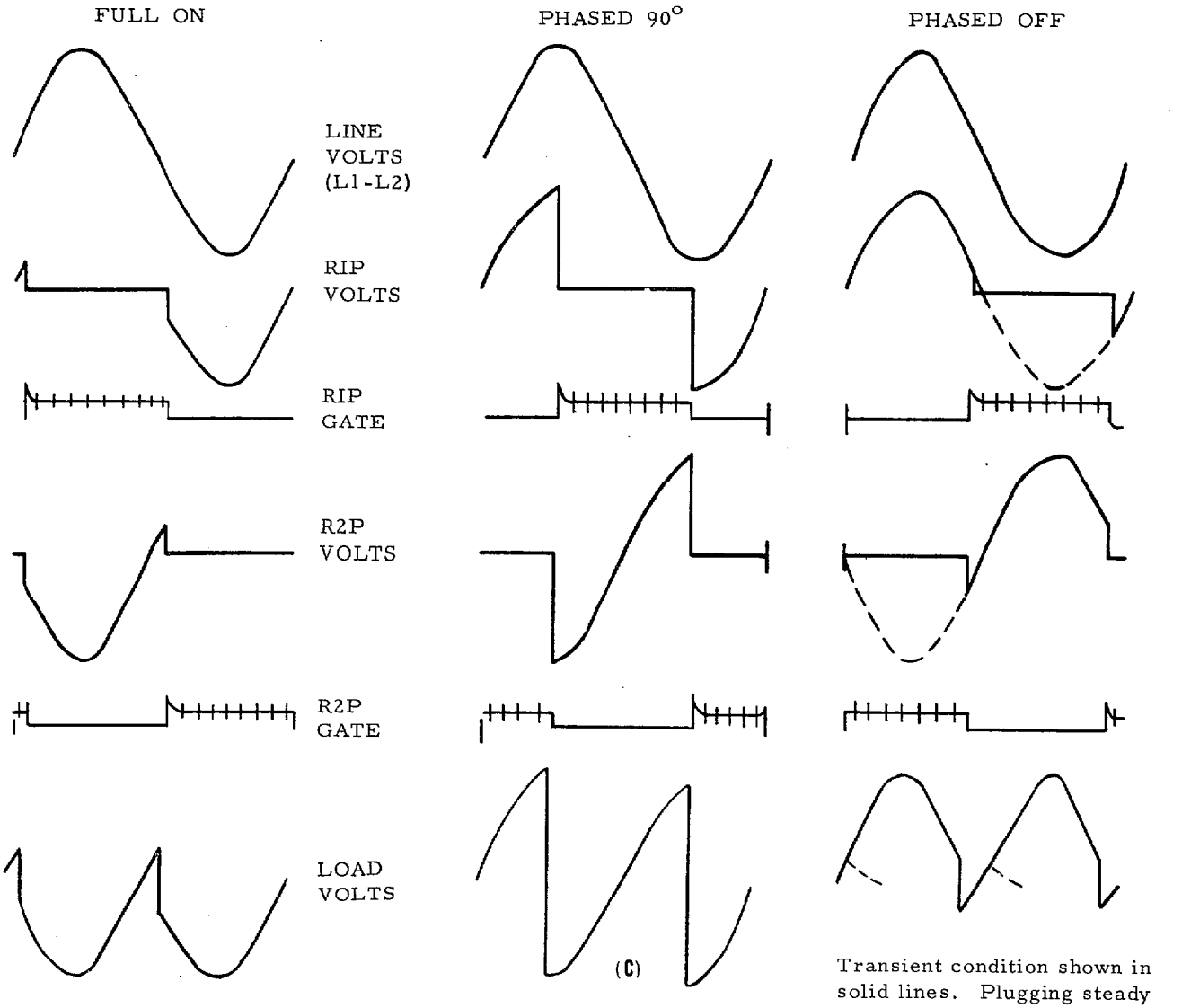
Transient condition shown in solid lines. Plugging steady state condition shown in dotted lines.



(B)

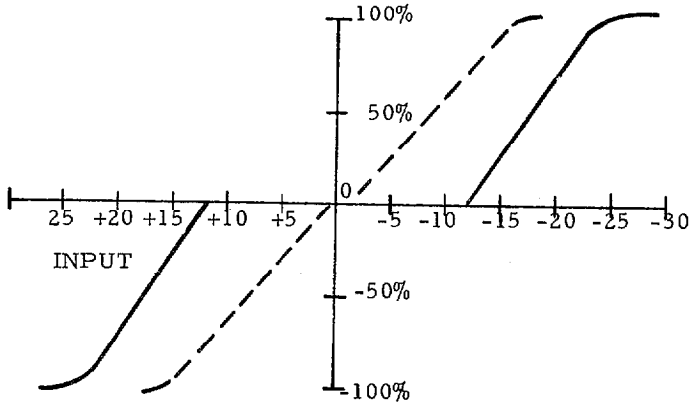
Fig. 22. Single phase exciter full reversing – full wave bridge

d. Single-phase Exciter, Full-reversing – Full-wave Bridge (FP) (cont.) P = -voltage



(C)

Transient condition shown in solid lines. Plugging steady state condition shown in dotted lines. Only + or - dotted lines will be present, i.e. either + or - will be locked out.



——— No internal feed back $\frac{\text{Input V}}{\text{Deadband}} = \pm 12^{V}$ with tailend pulse at 30°

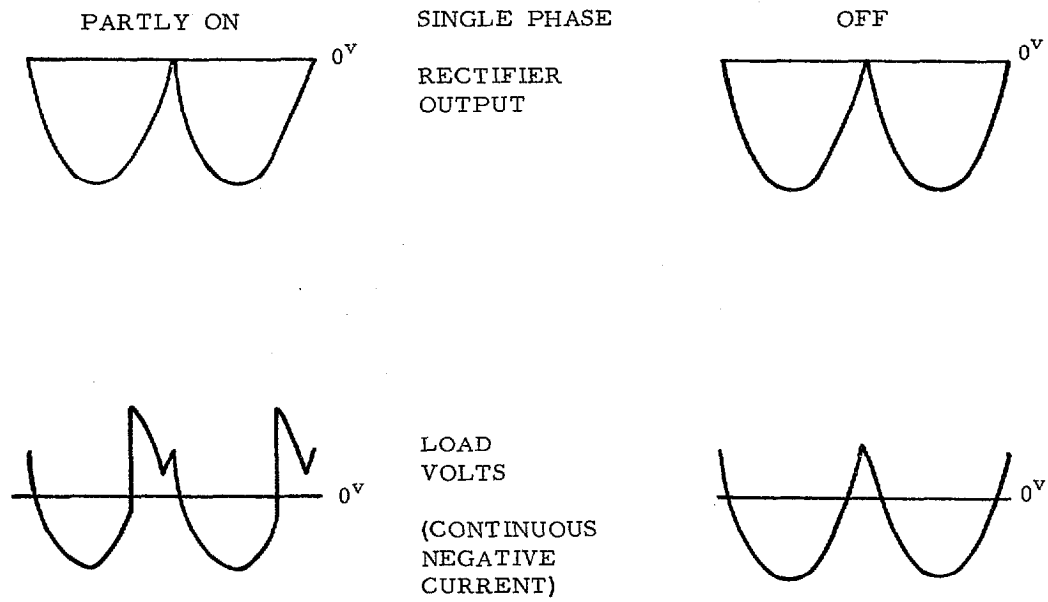
- - - Internal feedback $\frac{\text{Input V}}{\text{Deadband}} = \pm .5^{V}$

(D)

Fig. 22 (cont).

e. Single-phase Quarter-reversing Plugging Exciters

The waveshapes of the nonreversing nonplugging full wave bridge SCR exciters are similar to those covered previously. The load voltage will be different.



2. THREE PHASE SCR EXCITERS

When a drive is down, the first concern is to get it going again. With this in mind, the following instructions will serve as a guide to locating the defective component. The exciters are so designed that most components can be changed easily, thus getting the drive back in operation in minimum time.

Most problems in the SCR exciters will be due to defective components, the majority of which can be located with a knowledge of exciter operation, the lights on the front of the exciter, and a volt-ohmmeter.

The lights on the exciter are very good indicators of the performance of the cells. These lights are connected directly across the SCR cells and are a visual indication of the forward voltage across the cell. With the exciter turned off, the lights should burn relatively bright. As the exciter output increases, the intensity of the lights decreases until at half output the lights will go completely out. In reversing exciters, with no output in either direction, all lights will burn at medium brilliance. As the exciter is turned on in one direction, the lights corresponding to the cells for this direction will become dimmer, while the lights for cells for the other direction will become brighter. Light condition will, of course, reverse when the exciter is turned on in the other direction.

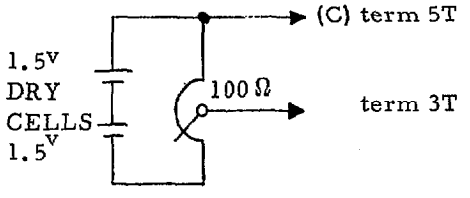
Since the lights are such good indicators of exciter performance, it is important that good lamps be kept in the exciter at all times. If trouble should occur, a quick analysis of the light conditions will give a good idea where to start. All three phase exciters will have six (6) indicating lights. In a properly operating exciter, all six (6) lights will glow with the same brilliance at all output levels. By observing light conditions a great deal of information is available if properly interpreted. The list of possible problems can be greatly reduced before the exciter is opened.

Following is a list of possible light conditions, and the most common failures that could cause them:

LIGHT CONDITIONS	PROBABLE CAUSE	TROUBLE-SHOOTING PROCEDURE
1. All lights out and zero output.	This indicates loss of AC power across the SCR cells. For this condition, there are two common causes:	
	1. Loss of AC input power to the exciter.	Check for voltage at the top of the fuses. This voltage should read line to line ± 10% volts. If proper voltage is on the top of the fuses, check for voltage across each fuse. This will locate the blown fuse.
	2. Two or more fuses blown.	Check all power semiconductors with an ohmmeter. Good SCR cells will have a resistance of 500 ohms or greater in both the forward and reverse direction. Any cell with less resistance than this is probably the cause of the blown fuses. Fuses can also be blown by shorts in the load circuit external to the exciter. To check this, measure the resistance across the load terminals. This resistance should be consistent with the value shown on the elementary. As a power check for load shorts, disconnect one of the load terminals and put a high resistance across it, say two 20K 10 watts resistors in series. Also, put small fuses (about 5 amps) in place of the blown fuse. Reapply AC power. If these fuses are blown, the trouble is inside the exciter and not in the external load circuit. All lights on the exciter should glow with the same intensity. Different light intensity within this group of six lights will indicate unbalance in the firing circuits.
2. Light intensity becomes unbalanced - output decreases.	Bright lights indicate that a cell is not firing.	
	1. Trouble within the firing circuit card.	Interchange the firing circuit with spares or the other cards in the exciter. If the trouble follows the card, the card is not operating properly.
	2. Trouble in the firing circuit outside the firing circuit cards.	Change the gate and filter cards with spares or other cards in the exciter. If the trouble follows the card, the card is not operating properly. If this is not the trouble, check the SCR gate-cathode junction with an ohmmeter. This can be done at the terminals on the gate and filter card. The resistance should be less than 250Ω. If it is greater, check wiring and SCR.
	3. One line fuse blown.	Positive and negative lights on the line with the blown fuse will be out. The remaining four (4) lights will behave normally. The maximum output voltage will be about 65% of the normal output and will have a large 120-cycle ripple.

LIGHT CONDITIONS	PROBABLE CAUSE	TROUBLE-SHOOTING PROCEDURE
	4. One SCR cell not firing.	One light will glow brightly at all times. The maximum output voltage will be about 60% of the normal output and will have a large 60-cycle ripple. When a cell is not firing, it could, of course, also mean that the cell is not receiving a firing signal. The best way of determining if a firing signal is present is to look at the SCR gate signal with a scope. If no scope is available, here are some things to try. 1. Check for loose connections on the heat sink card receptacle, firing circuit card receptacle, and the gate lead itself. 2. Interchange firing circuits. If the trouble follows the card, you have determined that the problem is in the firing circuit. 3. Interchange gate and filter cards. This will determine if the trouble is in the filter cards.
	5. One cell firing continuously.	One light will be out at all times. The maximum output voltage will increase very slightly. There will be a 60-cycle ripple increasing in magnitude as the output voltage is decreased. Continuous firing can be caused by noise on the input signal as well as by cell failure. A quick check to see whether the input signal is at fault is to remove the SCR gate lead from the filter circuit terminal. With the gate lead disconnected, a good cell will not fire at all, and the light that was out will not glow with the brilliance normally expected at zero output.
3. All lights bright - zero DC output.	This indicates that no cells are firing.	
	1. No input signal.	Check voltage across terminals C and S on the exciter terminal board. If voltage is present here (C (+) with respect to S (-) greater than 0.5 volt or so), the exciter should have some DC output. (Reversing exciters will be turned on forward with this polarity input signal.) Check to see that this signal is reaching the firing circuit card (Test point - 7 on TFCU or V). If the signal reaches the card, the trouble must be in the firing circuit.
	2. Firing circuit not functioning.	See firing circuit test point waveshapes.

Apparent trouble in the SCR is often not in the SCR at all, but in the control signal from the regulator. If you suspect that this may be the cause, the following shortcut may prove helpful. Disconnect the control signal from the regulator (terminals C and S on the exciter T. B.). Wire in a source of DC signal, zero to three volts DC to TCBA - term 5T and term 3T. Vary this input signal from zero to about two volts and see if the exciter operates normally (2 volts will turn the exciter full on). If the exciter operates normally under these conditions, the trouble is probably in the regulator instead of the SCR. A simple DC signal source can be made from two 1. 5 volt dry cells and a 100 ohm rheostat as shown in the sketch below.



For additional assistance in locating trouble in your SCR exciters, oscilloscope patterns for properly operating exciters are included. These patterns are typical and any significant departure from them indicates trouble.

1. If exciter troubles cannot be located with the trouble-shooting notes and the oscilloscope patterns, consult the General Electric Company.

2. When defective components are located, the following general procedure for changing them should be followed.

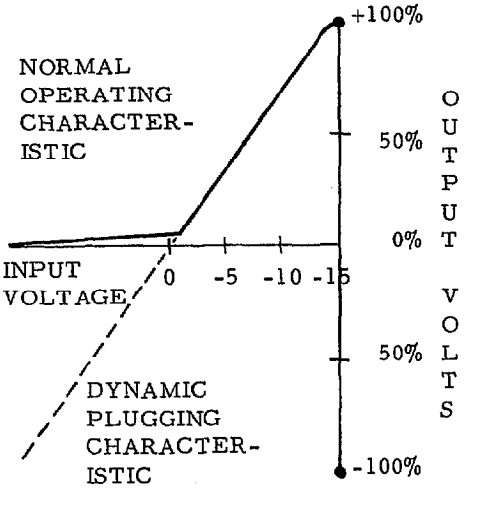
   a. Remove AC power from the exciter.

   b. Remove defective component. (Note wire numbers and location if wires have to be disconnected.)

   c. Replace with a component of the same catalog number. (Reconnect any wires that were disconnected; make sure cards are properly seated in their receptacles, all connections tight, etc.)

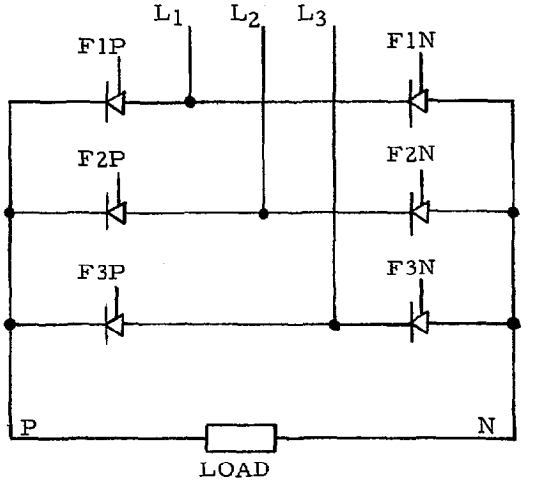
   d. Reapply AC power.

a. Three-phase Bridge (NP)



OUTPUT CHARACTERISTIC

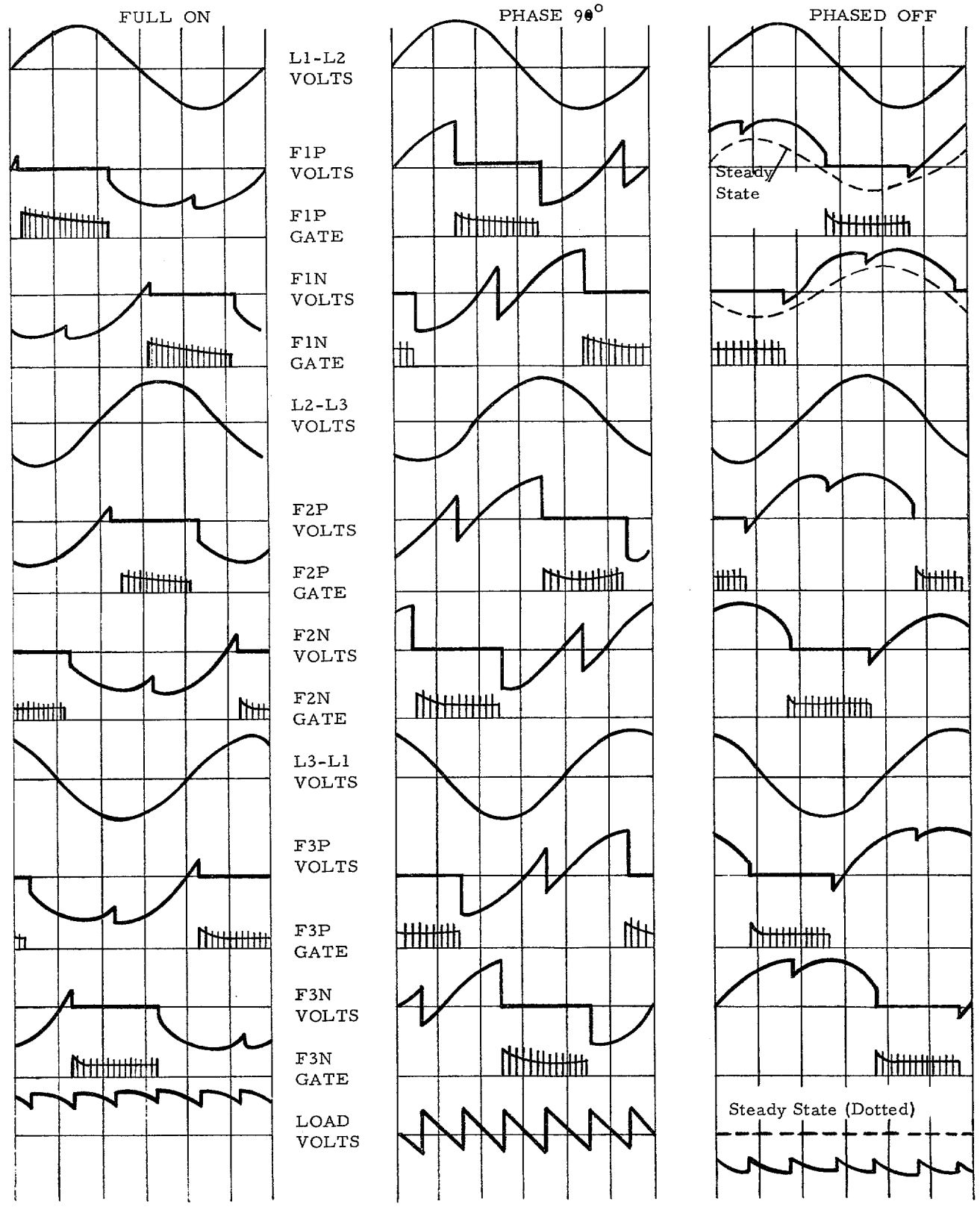
(A)



POWER CIRCUIT

(B)

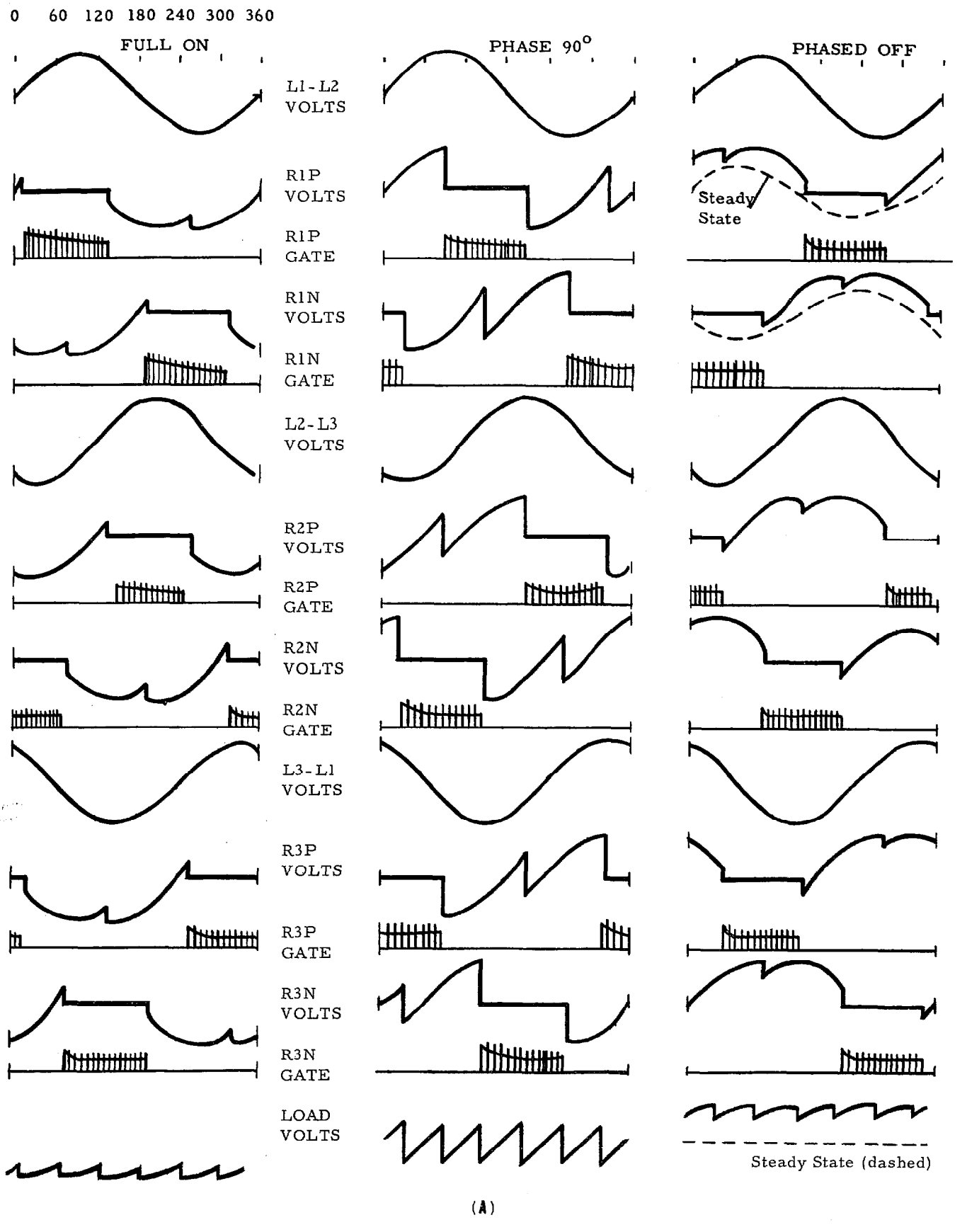
Fig. 23. Three-phase Bridge – NP



(C)

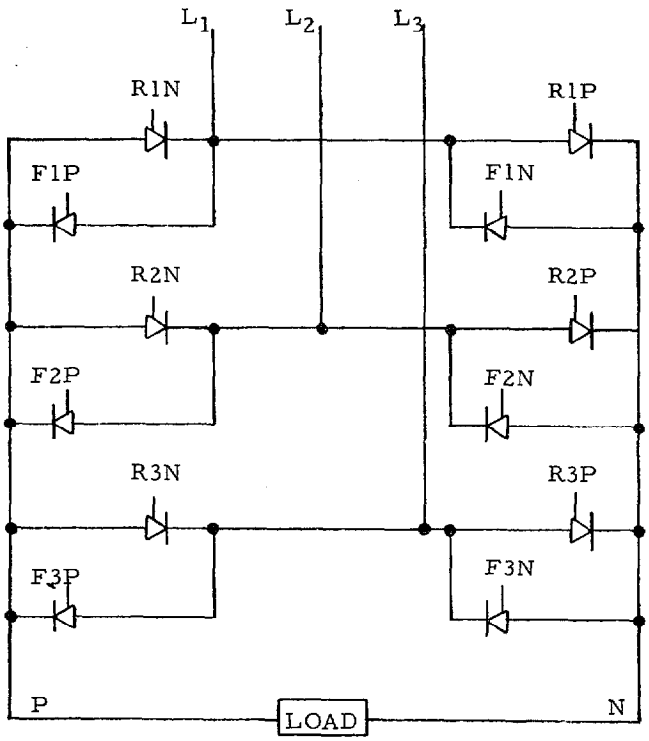
Fig. 23. Three-phase Bridge – NP (cont.)

b. Three-phase Bridge (FP)



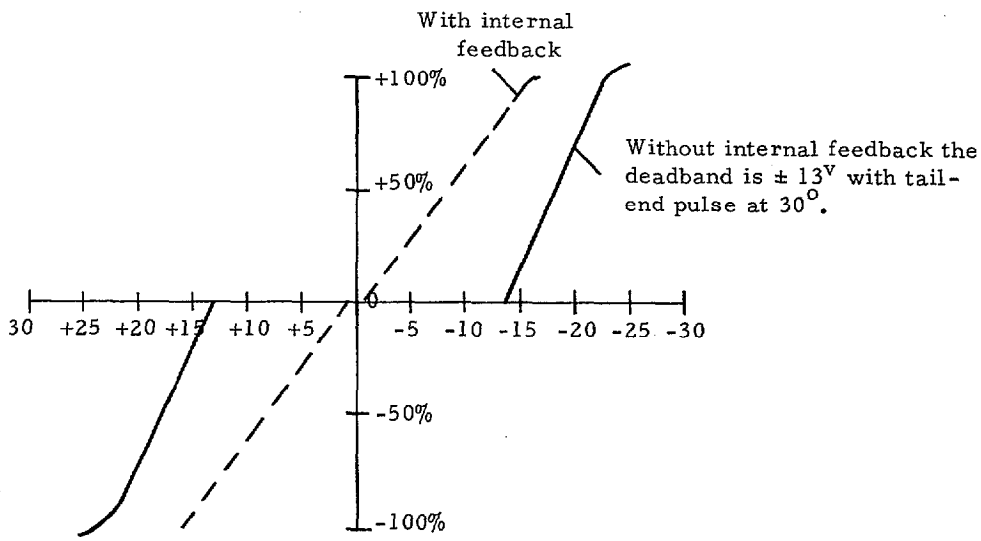
(A)

Fig. 24. Three-phase bridge – FP



POWER CIRCUIT

(B)



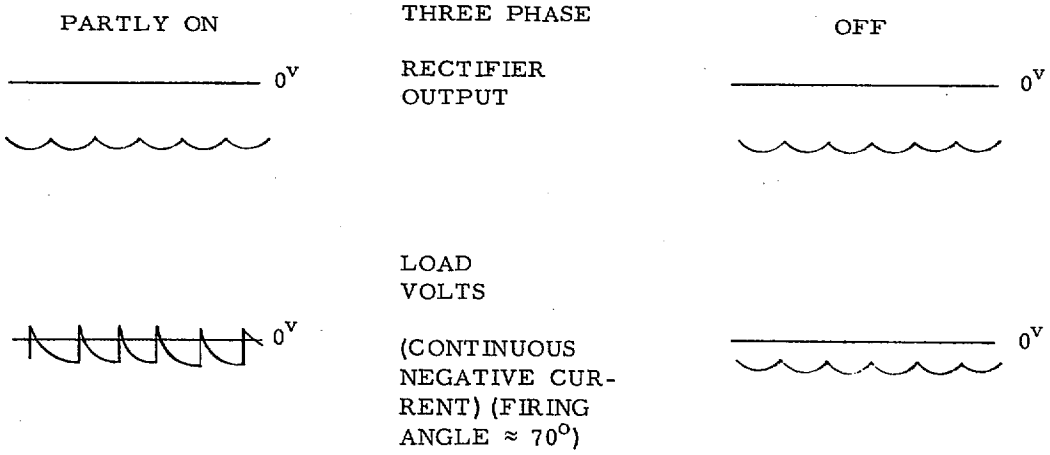
OUTPUT CHARACTERISTIC

(C)

Fig. 24. Three-phase bridge – FP (cont.)

c. Three-phase Quarter-reversing Plugging Exciters

The waveshapes of the nonreversing, nonplugging full wave bridge SCR exciters are similar to those covered previously. The load voltage will be different.



3. CARD TEST POINTS

Test Points are available on the cards to check for proper card operation.

a. Firing Circuit - Refer to schematic of IC3600TFCU or V, Fig. 26.

Test Point	Designation
1	Not Used
2	PA
3	PB
4	AC1
5	Not Used
6	Unijunction Emitter
7	Input Sig.

Explanation:

PA – Control signal which is connected to a pulse transformer on a gate and filter card. This is the source of the gate firing pulse for positive cells.

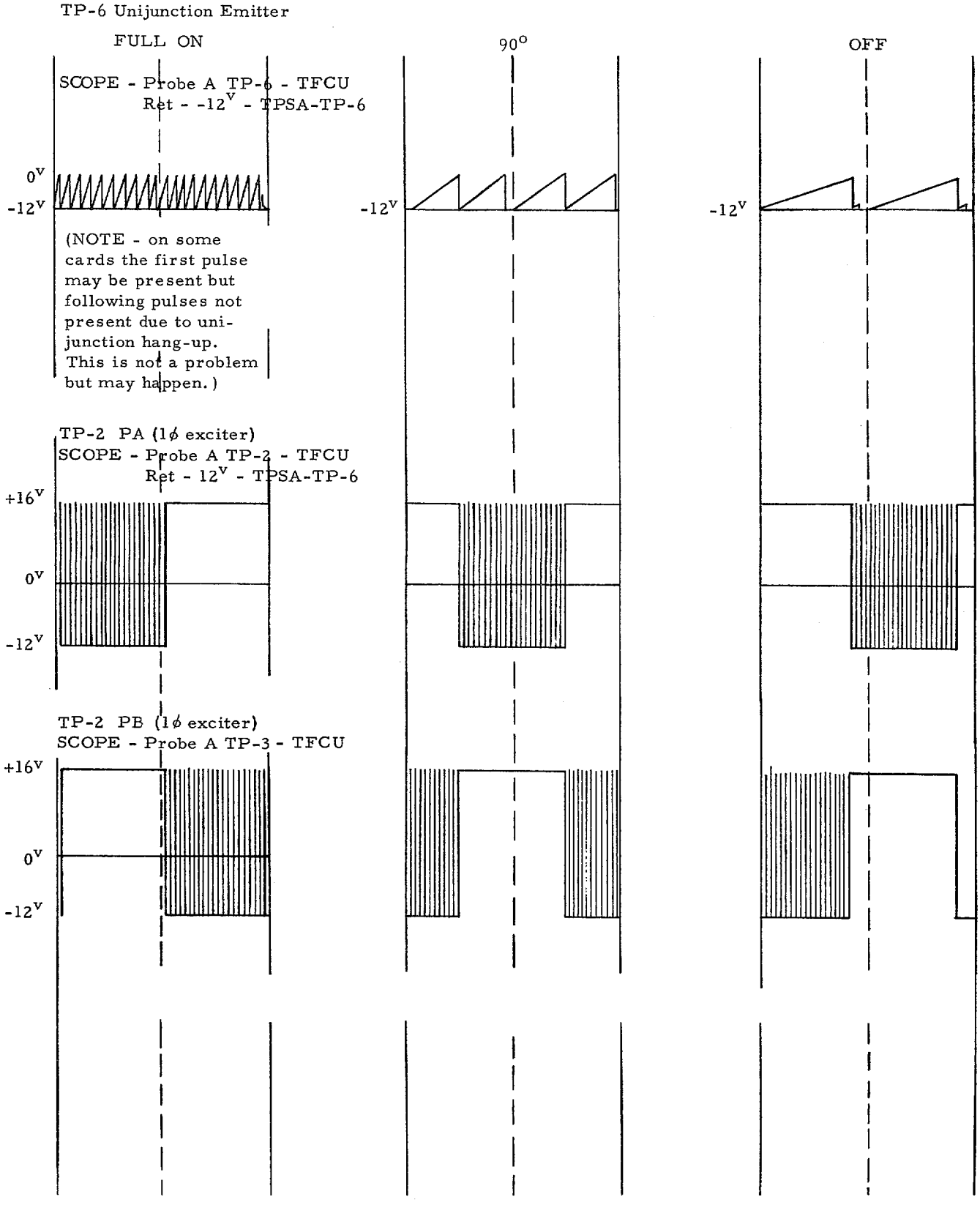
PB – Control signal which is connected to a pulse transformer on a gate and filter card. This signal is the source of the gate firing pulse for negative cells.

AC1 – 12V RMS reference voltage.

Unijunction Emitter – This point is in the timing circuit.

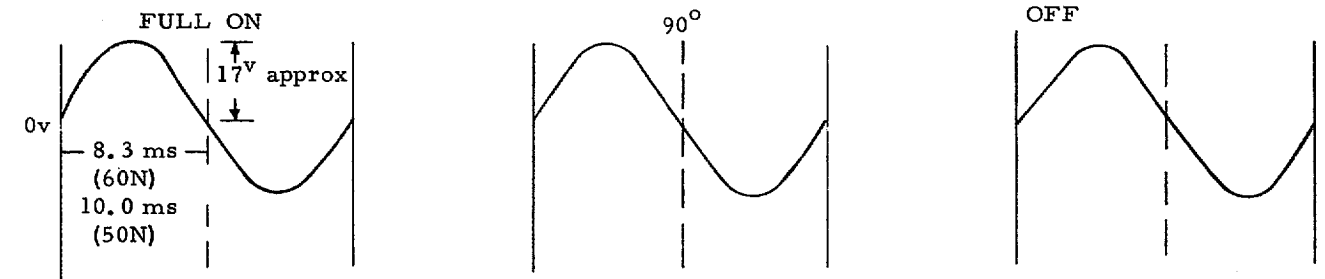
Input Signal – Error voltage which is fed into the firing circuit.

TFCU-TFCV Test Points



3 phase wave shapes would be similar but instead of 180 degrees long, they would be on (with pulses) for 120 degrees.

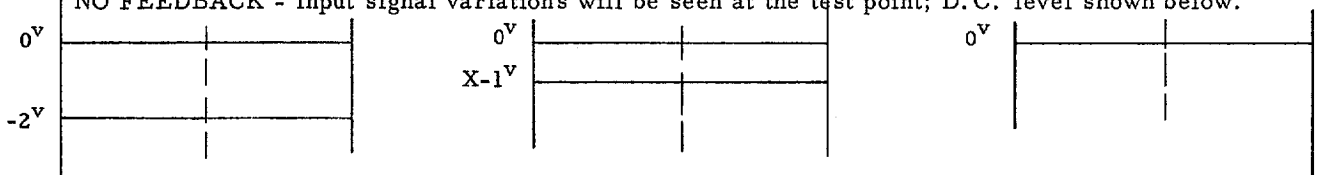
AC1-TP4
SCOPE PROBE A - AC1-TP4-TFCU or V
SCOPE RET. - COM. - TERM 5 of TCBA



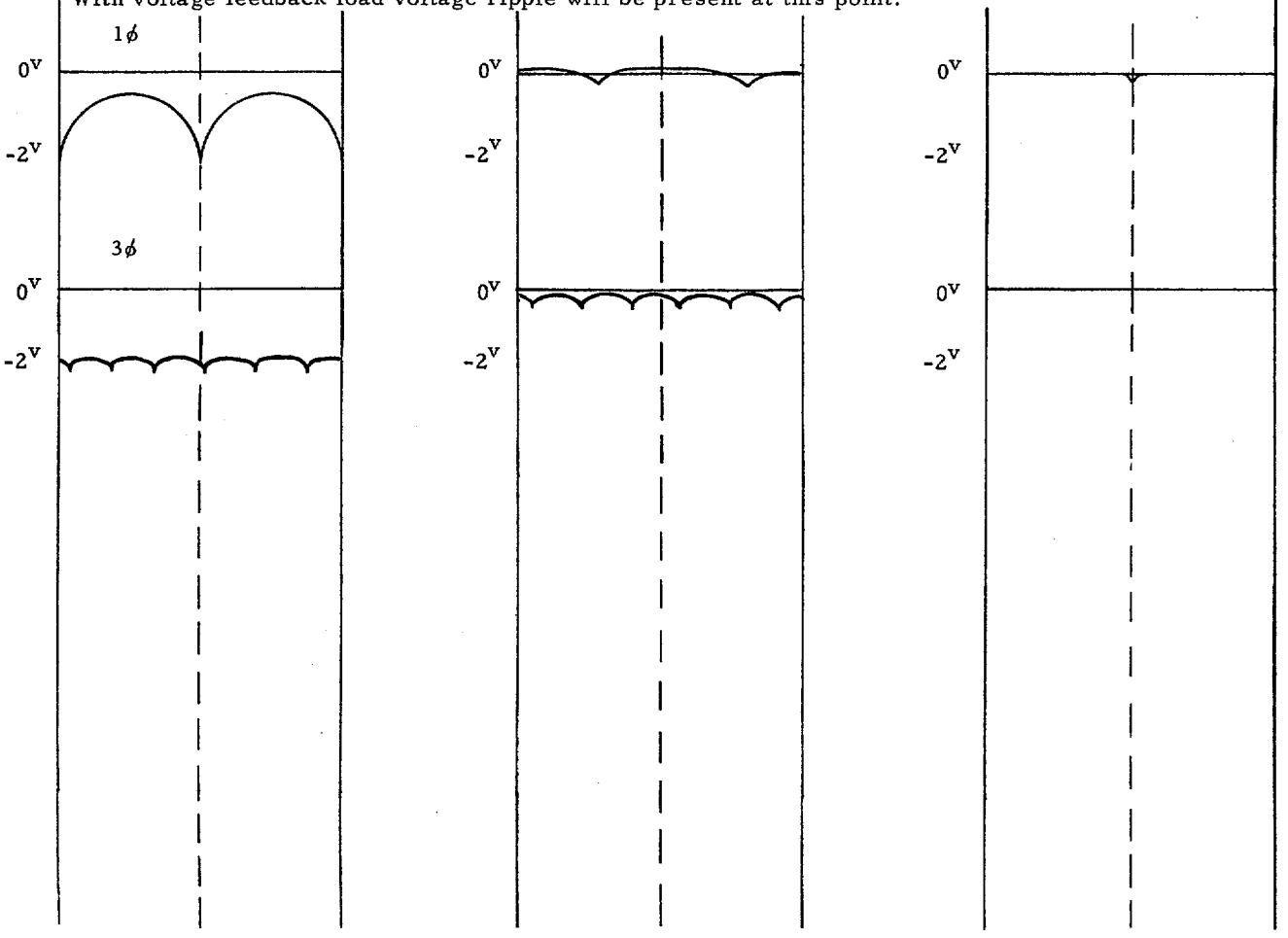
INPUT SIG= - TP-7
SCOPE PROBE A - INPUT SIGNAL - TP-7 - TFCU or V
SCOPE RET - COM. - TERM 5 of TCBA

(NOTE: Exciters with internal voltage feedback can have 1/2 output voltage from input terminals to ground.)

NO FEEDBACK - input signal variations will be seen at the test point; D.C. level shown below.



With voltage feedback load voltage ripple will be present at this point.



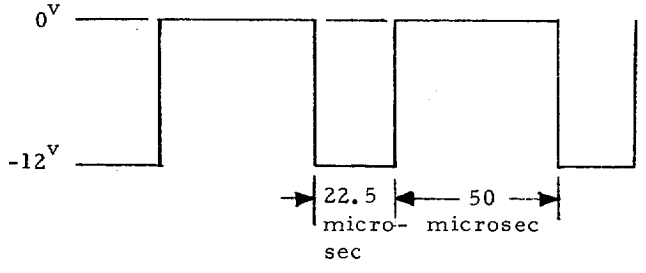
b. Power Supply - TPSA - Refer to schematic of IC3600TPSA, Fig. 28.

Test Point	Designation
1	OSC
2	GND
3	Not Used
4	+16V
5	Bias
6	-12V
7	Not Used

Explanation:

OSC - 20 KC multivibrator output
GND - 0V or common
+16V - unregulated DC bias
Bias - DC voltage used with firing circuit TFCU or V
-12V - regulated DC bias

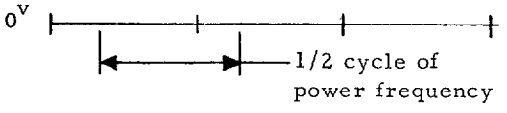
OSC - TP1-TPSA
SCOPE PROBE A - OSC - TP1-TPSA
SCOPE RET - TP2-TPSA



+16V - TP4-TPSA



NOTE - 3ϕ power supply will have 360 cycle ripple instead of the 120 cycle ripple shown here.



Bias and -12V are simply DC levels.

c. Reversing Logic - TRLA - B - C - D - Refer to schematic of IC3600TRLA, B, C, D, Fig. 30.

Test Point	Designation
1	S. R.
2	1 GO
3	Zero
4	LOF
5	T. D.
6	Read
7	LOR

Explanation:

S. R. - Saturable Reactor output which is fed into the Comparator circuit of the Reversing Logic.

1 GO - This is the Comparator circuit output.

Zero - This is zero pulses generated within the Reversing Logic from the zero pulses received from the Firing Circuit.

LOF - Lock out forward. This is a reversing logic output.

T. D. - Time Delay - Internal Reversing Logic Signal which controls flip-flop state.

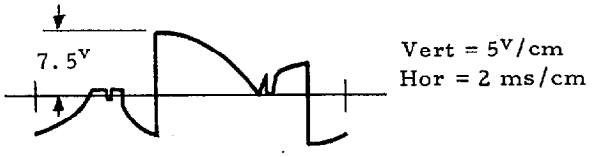
Read - Internal Reversing Logic signal which determines the time at which the comparator output is "looked at."

LOR - Lock out reverse. This is a reversing logic output.

S.R. - TP1 - TRLA (B, C, D)
SCOPE - Probe A - TP1 - TRLA
Ret - TP2 - TPSA

1ϕ Exciter (TRLA); (TRLC will be similar but will have a 10 msec. time base.)

FULL OUTPUT VOLTAGE

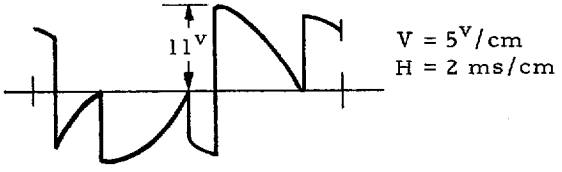


0V OUTPUT VOLTAGE

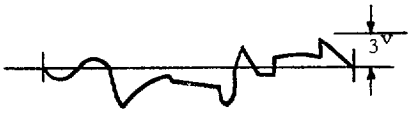


3ϕ Exciter (TRLB); (TRLD will be similar but will have a 10 msec. time base.)

FULL OUTPUT VOLTAGE

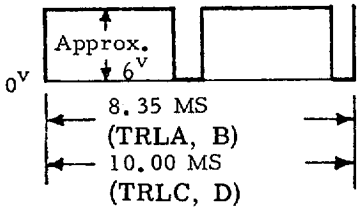


0V OUTPUT VOLTAGE

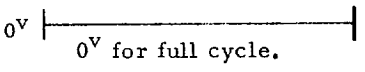


1GO - TP2 - TRLA (B, C, D)
SCOPE - Probe A - TP2 - TRLA
SCOPE - Ret - TP2 - TPSA

FULL OUTPUT VOLTAGE

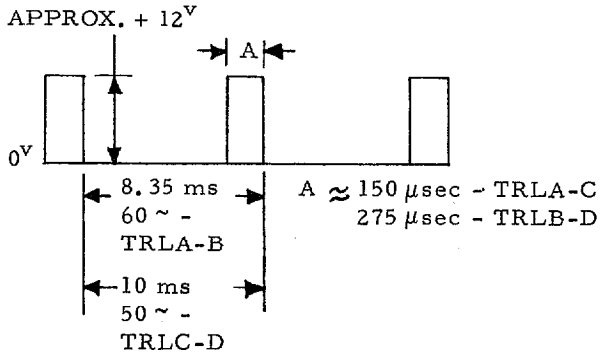


0V OUTPUT VOLTAGE



ZERO - TP3-TRLA(B, C, D)
SCOPE - Probe A - TP3-TRLA
Ret. - TP2-TPSA

FULL OUTPUT VOLTAGE



0V Output Voltage

same as full output voltage

LOF-TP4-TRLA(B, C, D)
D. C. Voltmeter Probe - TP4-TRLA
ret - TP2-TPSA

0V - indicates that the forward firing circuit is locked out.

-12V - indicates that the forward firing circuit is not locked out.

Note: It is possible to "lock out" the forward firing circuit by jumpering this TP4 to TP2-TPSA.

T. D. -TP5-TRLA(B, C, D)
Scope Probe A - TP5-TRLA
ret - TP2-TPSA

Full Output Voltage

0V 0V level

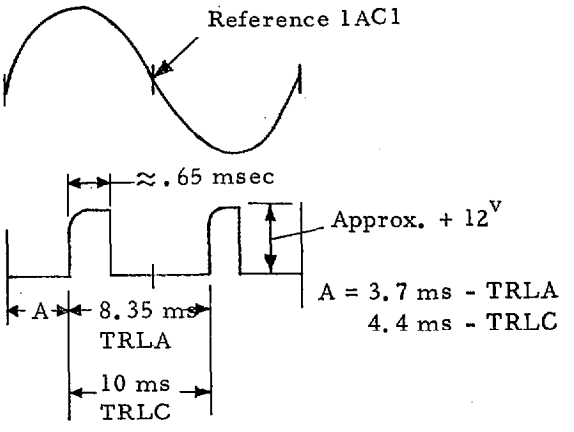
0V Output Voltage

+12V level

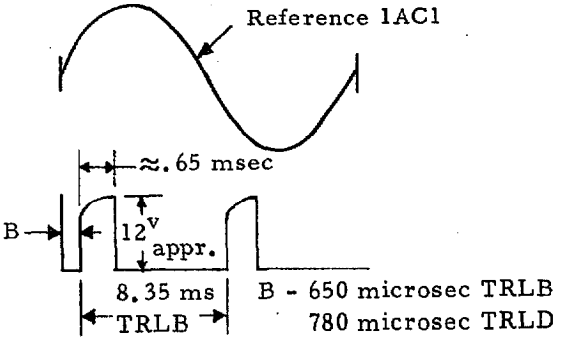
0V

Read - TP6 - TRLA (B, C, D)
Scope - Probe A - TP6 - TRLA
Ret - TP2 - TPSA

1ϕ package (TRLA, C)



3ϕ package (TRLB, D)



LOR-TP7-TRLA(B, C, D)
D. C. Voltmeter Probe - TP7-TRLA
ret - TP2-TPSA

0^V - indicates that the reverse firing circuit is locked out.

-12^V - indicates that the reverse firing circuit is not locked out.

Note: It is possible to "lock out" the reverse firing circuit by jumpering TP7-TRLA to TP2-TPSA.

4. EXCITER BIAS ADJUSTMENT

All unijunction SCR exciters include a bias or dead-band rheostat. This rheostat is used to provide a bias that advances the firing angle. Since a pure inductive load will not accept power until the firing angle advances beyond 90^{0}, a bias is needed to reduce the dead band.

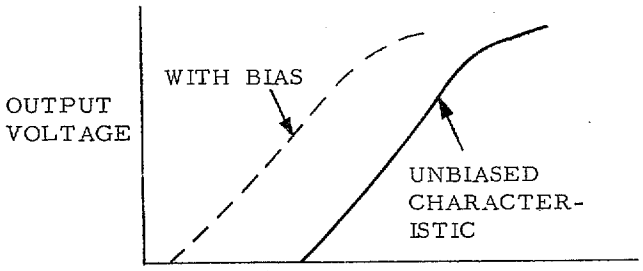


Fig. 25. Input signal

Figure 25 shows the effect of the bias on the characteristic of the exciter. The bias effectively shifts the characteristic curve to the left. The dead band can be reduced, the amount depending on the type of exciter involved. The bias voltage is generated within the power supply, TPSA.

NP (NONREVERSING; PLUGGING)

The bias adjustment (bias potentiometer on TCBA) is set just below that required to give some output voltage with the terminals (C and 3T) shorted. This can be checked by connecting a voltmeter to the exciter output terminals and observing the setting of the rheostat when voltage starts to increase. Then back the pot off one quarter turn.

QP (QUARTER REVERSING, PLUGGING)

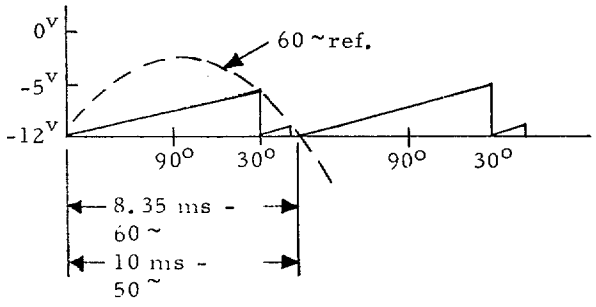
The bias adjustment in this exciter is set to just cancel the effect of the internal reverse current bridge. With the terminals (C and 3T) shorted, connect a voltmeter to the exciter output terminals. Adjust the rheostat until the voltmeter reads zero output volts.

FP (FULL REVERSING, PLUGGING)

The zero-input-voltage firing point is set at 30^{0} on these units. A rather wide dead band is maintained in exciters without internal feedback. If the exciter has internal feedback the dead band will be less. If a smaller dead band is mandatory, the bias pot may be used to reduce the dead band. It is recommended that the firing point not be advanced by the bias beyond 45^{0}. If instability does result, the bias must be backed off.

An oscilloscope is needed to set the dead band on reversing units. The following steps can be used to achieve the setting.

1. Jumper TCBA - term C to 3T.
2. Lock out the reverse firing circuits by jumpering TRLA (B, C, D) - TP7 to TPSA-TP2.
3. Connect scope from a forward TFCU (or V) - TP6 to TPSA-TP-6.
4. Adjust bias pot so that the unijunction firing point occurs at the desired point.



NX (NONREVERSING, NONPLUGGING)

The bias has been set on exciters with internal feedback by feeding - 15V from TCBA—term C to term 3T - 10V is fed to exciters without internal feedback; then adjusting the bias pot to achieve rated DC output voltage at the output terminals P to N.

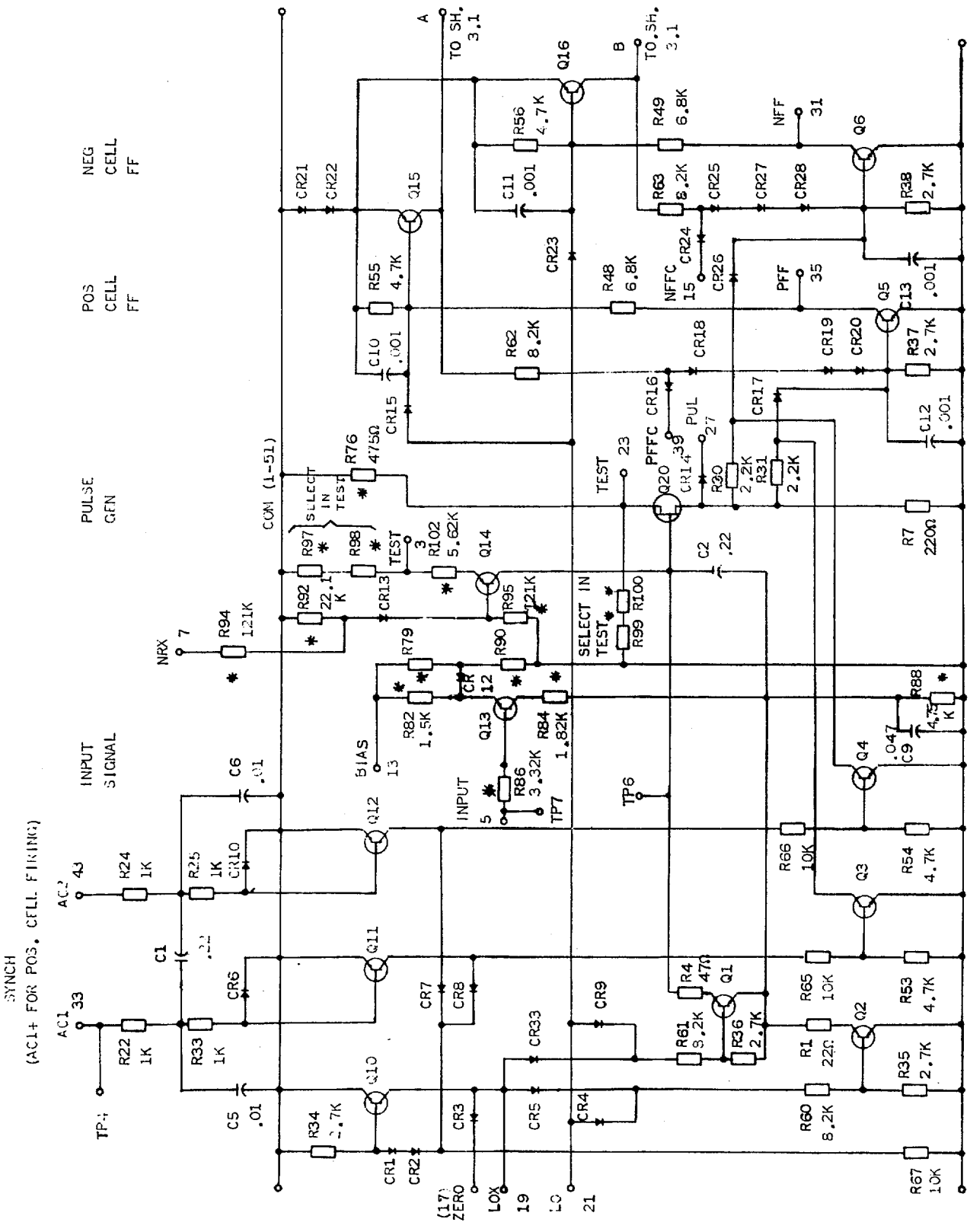


Fig. 26. SCR unijunction firing circuit. (Elementary diagram IC3600TFCU1)

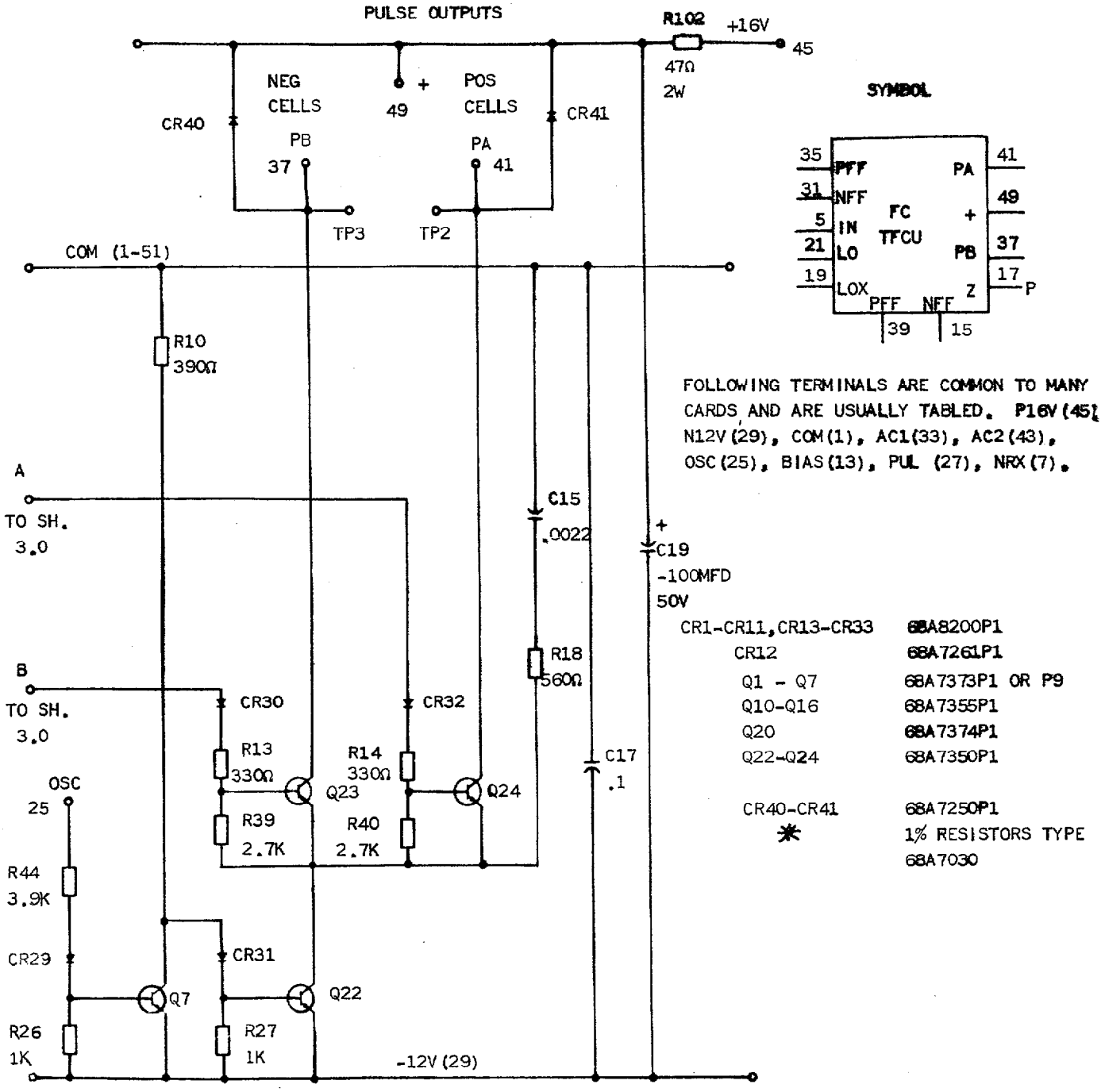

PULSE OUTPUTS
R102
+16V
45
47Ω
2W
NEG CELLS
49
POS CELLS
CR40
CR41
PB
37
PA
41
TP3
TP2
COM (1-51)
SYMBOL
35 PFF
31 NFF
5 IN
21 LO
19 LOX
FC TFCU
PA 41
+ 49
PB 37
Z 17
P
PFF 39
NFF 15
R10
390Ω
FOLLOWING TERMINALS ARE COMMON TO MANY CARDS AND ARE USUALLY TABLED. P16V (45), N12V (29), COM (1), AC1 (33), AC2 (43), OSC (25), BIAS (13), PUL (27), NRX (7).
A
TO SH. 3.0
C15
.0022
C19
-100MFD
50V
R18
560Ω
B
TO SH. 3.0
CR30
CR32
R13
330Ω
R14
330Ω
Q23
Q24
C17
.1
OSC
25
R39
2.7K
R40
2.7K
R44
3.9K
CR29
CR31
Q7
Q22
R26
1K
R27
1K
-12V (29)
CR1-CR11, CR13-CR33 68A8200P1
CR12 68A7261P1
Q1 - Q7 68A7373P1 OR P9
Q10-Q16 68A7355P1
Q20 68A7374P1
Q22-Q24 68A7350P1
CR40-CR41 68A7250P1
* 1% RESISTORS TYPE 68A7030

Fig. 26 (cont.)

H. PARTS

1. FIRING CIRCUIT - (IC3600TFCU-IC3600TFCV)

There are two types of unijunction firing circuits - one for 50-cycle operation and one for 60-cycle. The difference lies in the addition of capacitor C8 in parallel with C2 for the 50-cycle firing circuit.

The circuit may be broken down into several functional parts:

1. Synchronization Circuit
2. Input Signal
3. Pulse Generator
4. Positive Cell Flip-Flop
5. Negative Cell Flip-Flop
6. Pulse Outputs

Discussion: (See Elementary Diagram)

(1) Synchronization Circuit – 24-volt rms voltage transformer derived from the AC supply is fed into the circuit at AC1 (33) and AC2 (43). The zero crossovers of this sinusoidal voltage are used to generate "zero" pulses. These pulses are used both within this firing circuit and externally in the reversing logic. The zero pulse is used on the card to discharge the capacitors C2 and C9 by means of turning on transistors Q1 and Q2. The capacitors are discharged to insure the same initial conditions in the unijunction timing circuit (pulse generator) at the beginning of each half cycle of the AC voltage.

(2) Input Signal – Input signal is applied at pin 5. A negative voltage with respect to common will tend to turn on transistor Q13. Terminal 13 (Bias) is normally held at approximately ±.5 volts by the power supply card. Ideally a voltage at pin 5 will start Q13 to turn on at a slight change in the negative direction. Q13 acts as a current source which will charge capacitor C9. If Q13 manages to charge C9 so that the total charge on C9 and C2 in series reaches approximately 7 volts, unijunction Q20 will fire, discharging the capacitors and causing a voltage pulse across R7. The current generated by Q13 is proportional to the input signal. The proportionality is intentionally nonlinear to compensate for nonlinearities in the basic thyristor power circuit. R79 and CR12 act as a function generator to increase the current gain of Q13 at the high end of the input voltage range.

Since C9 is a small value of capacitance it is possible to charge it to 7 volts and fire the unijunction in a very short time. This results in a fast response firing circuit.

(3) Tail-end Pulse Current Generator – Q14 is a current generator whose output current is set in test to charge C2 to 7 volts and fire the unijunction at the end of each half cycle. If the input signal is 0V or positive the unijunction will fire each half cycle due to the action of Q14. These pulses are referred to as tail-end pulses.

It is possible to prevent unijunction firing by turning on transistors Q1 and Q2. These are turned on each half cycle by zero pulses to reset the timing capacitors C9 and C2. They can also be turned on by applying positive voltage to pin 21 (L.O.); this function is performed by reversing logic card in reversing operation.

(4) Unijunction Circuit – Q20 is the unijunction transistor. Each circuit is trimmed by resistor selection to insure that each firing circuit will track the other. This circuit-to-circuit balance is necessary, especially in three phase packages.

(5) Flip-Flop – Q15 and Q5 are the active transistors in the positive cell flip-flop; Q16 and Q6 are in the negative cell flip-flop. These flip-flops are externally connected so that if one is set the other is cleared. The FF's are steered by signals derived from the zero pulse generator. This is necessary to insure that when output voltage is desired gate pulses are sent to the thyristor with forward anode to cathode voltage.

The unijunction pulses generated across R7 will set either the positive or negative FF depending on the steering signal. The FF's can be reset or cleared by external signals on pin 39 (PFFC) or pin 15 (NFFC).

(6) Pulse Outputs – A square wave with a frequency of approximately 20KC is applied to pin 25 (OSC). (The oscillator is located in the power supply card.) This square wave will turn Q7 on and off at the same frequency. The positive and negative FF's control the states of Q23 and Q24. The FF's states are such that both Q23 and Q24 are never "on" simultaneously. Two external pulse transformers are connected in the collector circuits of Q23 and Q24. One transformer primary is connected

from 49 (+) to 37 (PB); the other transformer primary from 49 (+) to 41 (PA). Pulsating primary current occurs in the negative cell transformer (connected from 49 to 37) when Q23 is "on." Pulsating current occurs in positive cells (49 to 41) when Q24 is "on." Q22 is going on and off at a 20KC rate since it is driven by Q7.

2. SCR GATE AND FILTER CIRCUITS

Figure 27 shows the gate, filter and power circuits of PWB68A993123. This circuit is designed for use in nonreversing C380 SCR exciters powered with three phase or single phase 460V power.

The R-C filter from terminal 17 to terminal 4 is externally connected directly across the SCR from anode to cathode. Its purpose is to delay large voltage changes across the cell and thus reduce the maximum rate of change of voltage to which the cell is exposed $(\frac{dv}{dt})$.

The thyrectors CR14 and CR15 will protect the cell from high powered voltage spikes which are in the reverse direction of the SCR.

The light LT2 will be lit if the cell is off and there is forward voltage present on the cell. Diode CR7 prevents illumination of the light when reverse voltage is present.

Transformer T2 provides electrical isolation and a means of firing the SCR by means of a pulse train induced on the SCR gate.

This gate and filter circuit is generally representative of all the gate and filter cards. Specific differences are required because of different SCR cells, power voltage levels, and number of circuits on each card.

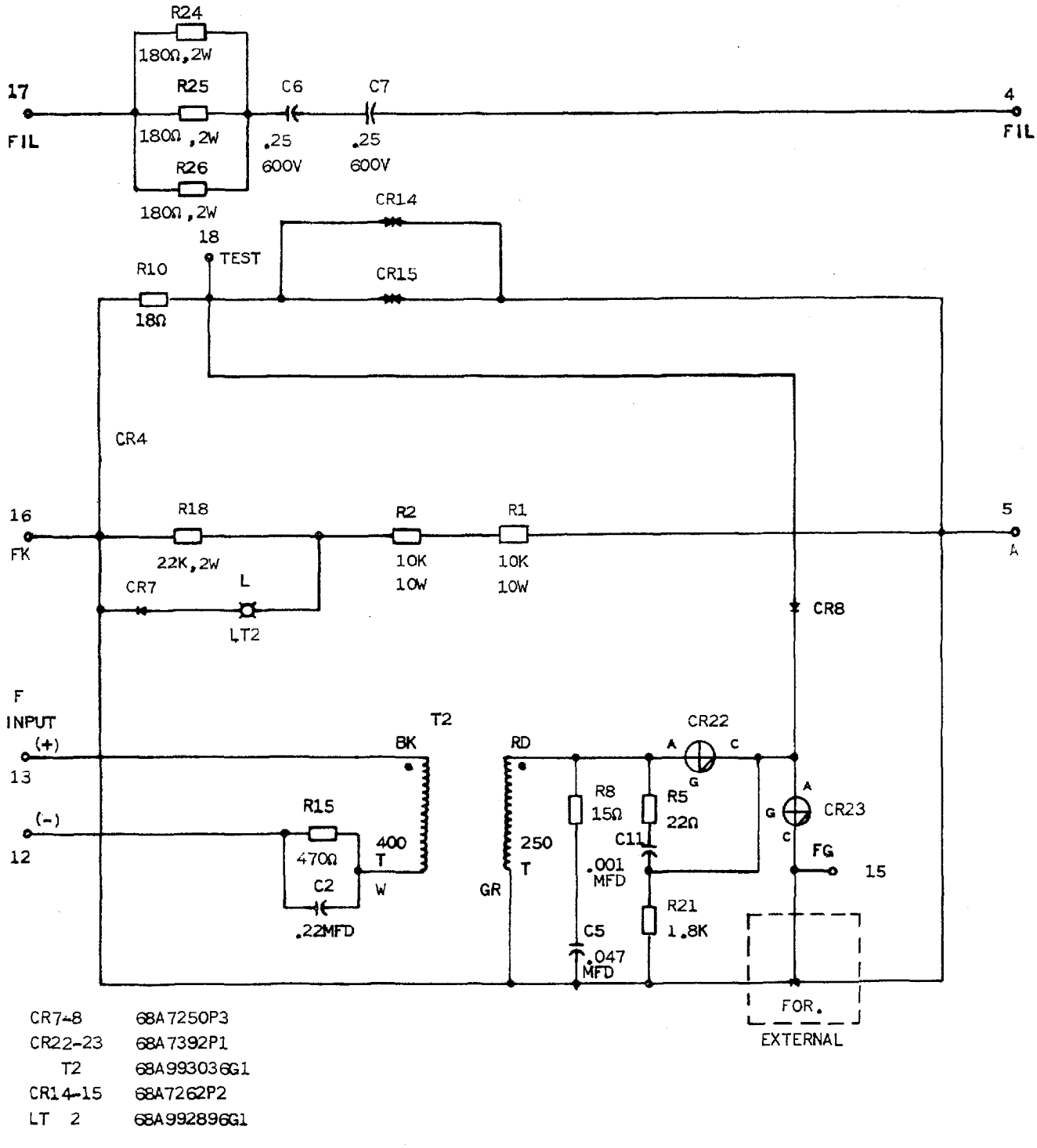
Diode Filter Circuits

These are similar to SCR gate and filter circuits except the components associated with the gate are absent. These would include the transformer T2 with all the components in its primary and secondary. The thyrectors are still present on the diode filters.

Table 5 is a compilation of filter cards which indicates the different cards for use with different type SCR exciters, input power levels and cells.

TABLE 5. SCR GATE AND FILTER CARDS

DIODE FILTER CARDS			
EXCITER FUNCTION	INPUT POWER VOLTAGE	CELL	CARD NUMBER
FR	480	C380P-C350P	PWB68A989923
FR	230	C380E-C350E	PWB68A993122
FR	230	C137E	PWB68A993125
NR	480	C380P-C350P	PWB68A993123
NR	230	C380E-C350E	PWB68A993124
NR	230	C35M	PWB68A993117
1/2R	460	C291P	PWB68A993147
CPE	230	C291E	PWB68A993148
NR	460	C291P	PWB68A993149
NX	460	C390P	PWB68A993157
NX	230	C390E	PWB68A993159
NX	230	IN2160E	PWB68A993188



CR7-8	68A7250P3
CR22-23	68A7392P1
T2	68A993036G1
CR14-15	68A7262P2
LT 2	68A992896G1

Fig. 27. SCR gate and filter circuit – 460 V, NR, C380. (Elementary diagram PWB68A993123)

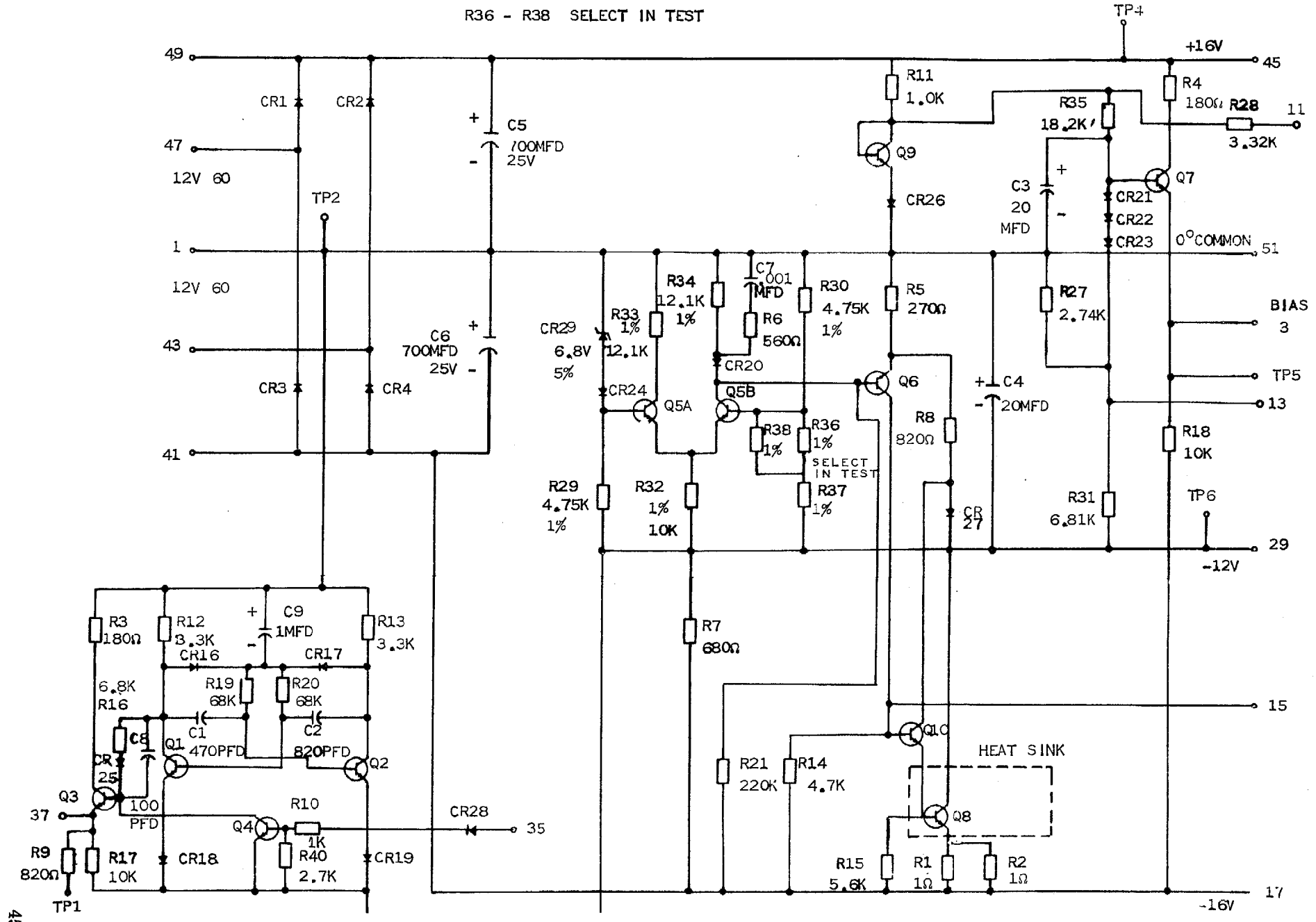
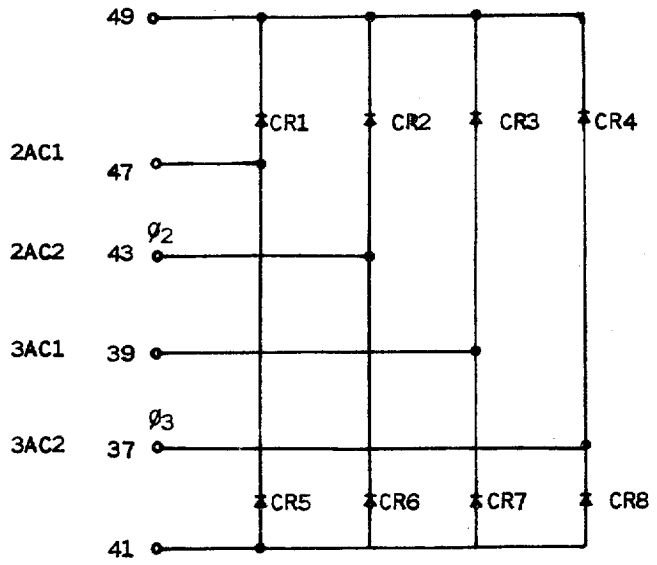


Fig. 28. Single phase power supply. (Elementary diagram IC3600TPSA1)

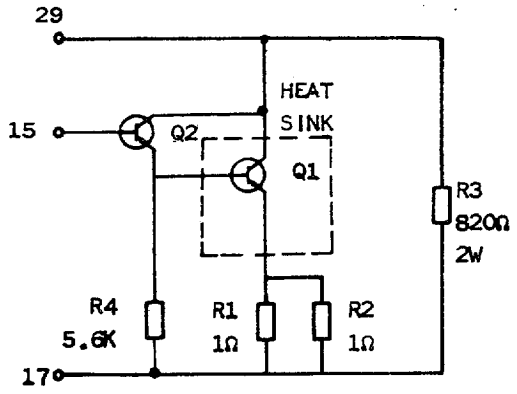


Q1 - 68A7356P1
WITH HEAT SINK
SIMILAR TO 68A997024P1

CR1-6 68A7250P1

Q2 68A7373

NOTE NEW HEAT SINK MUST FIT ON TPSA PRINTED CIRCUIT



Application Data

This card provides two single-phase full-wave rectifier bridges. These are normally used with the single-phase bridge on card IC3600TPSA to provide a six-phase rectifier.

Also a Darlington power amplifier stage is provided which can be connected in parallel with the power output stage of IC3600TPSA. The combined cards will provide up to 600 ma of regulated output.

Fig. 29. Three phase power supply. (Elementary diagram IC3600TPSB1)

3. POWER SUPPLY (C3600TPSA AND IC3600TPSB)

The function of the power supplies TPSA and TPSB is to provide:

a. +16^{V} DC unregulated

b. Regulated -12^{V} DC

c. 20 KC square wave for use in the firing circuits

d. Bias voltage for use in the firing circuits.

Single phase exciters use a TPSA card. Three phase exciters use a TPSA and a TPSB card.

The power source for the single phase power supply is the center tapped 24V AC transformer derived from the AC power source. The three phase supply utilizes a three phase center tapped 24V derived from the power source.

See Fig. 28 and Fig. 29 for circuit details.

4. COMPONENT BOARD (IC3600TCBA1)

This card TCBA contains the components which are peculiar to the different type exciters. These include input scaling resistors, feedback resistors, stabilizing capacitors, release level resistors and a bias pot.

All components except the potentiometer make electrical connection by means of screw terminals.

The signal input is fed in through a screw type terminal board attached to the top of this card.

5. REVERSING LOGIC (IC3600TRLA, - TRLB, - TRLC, - TRLD, - TRLH, - TRLJ)

This card is used only in exciters which are full-reversing.

Card IC3600TRLH1 single-phase (IC3600TRLJ1 three-phase) has a permanent memory logic feature to maintain proper operation in the event AC power momentarily is lost or drops below normal operating specifications. This card can be substituted for IC3600TRLA1 or IC3600TRLC1 neither of which has permanent memory logic. Card IC3600TRLJ1 likewise can be substituted for IC3600TRLB1 or IC3600TRLD1.

For description of operation of a typical card (IC3600TRLB1) without permanent memory logic, see pages 48-50. For operation of a typical card (IC3600TRLJ1) with permanent memory logic see pages 51-55.

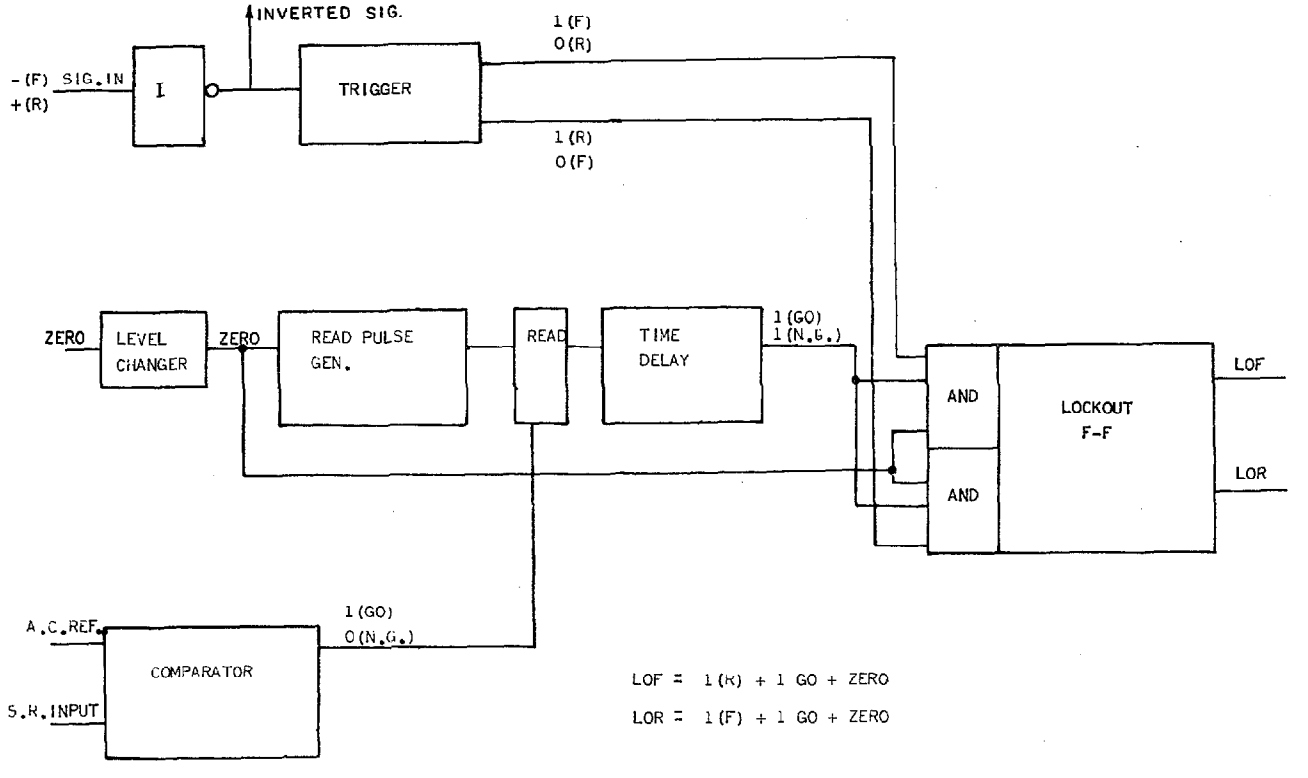


Fig. 30A. Reversing logic block diagram (IC3600TRLB1)

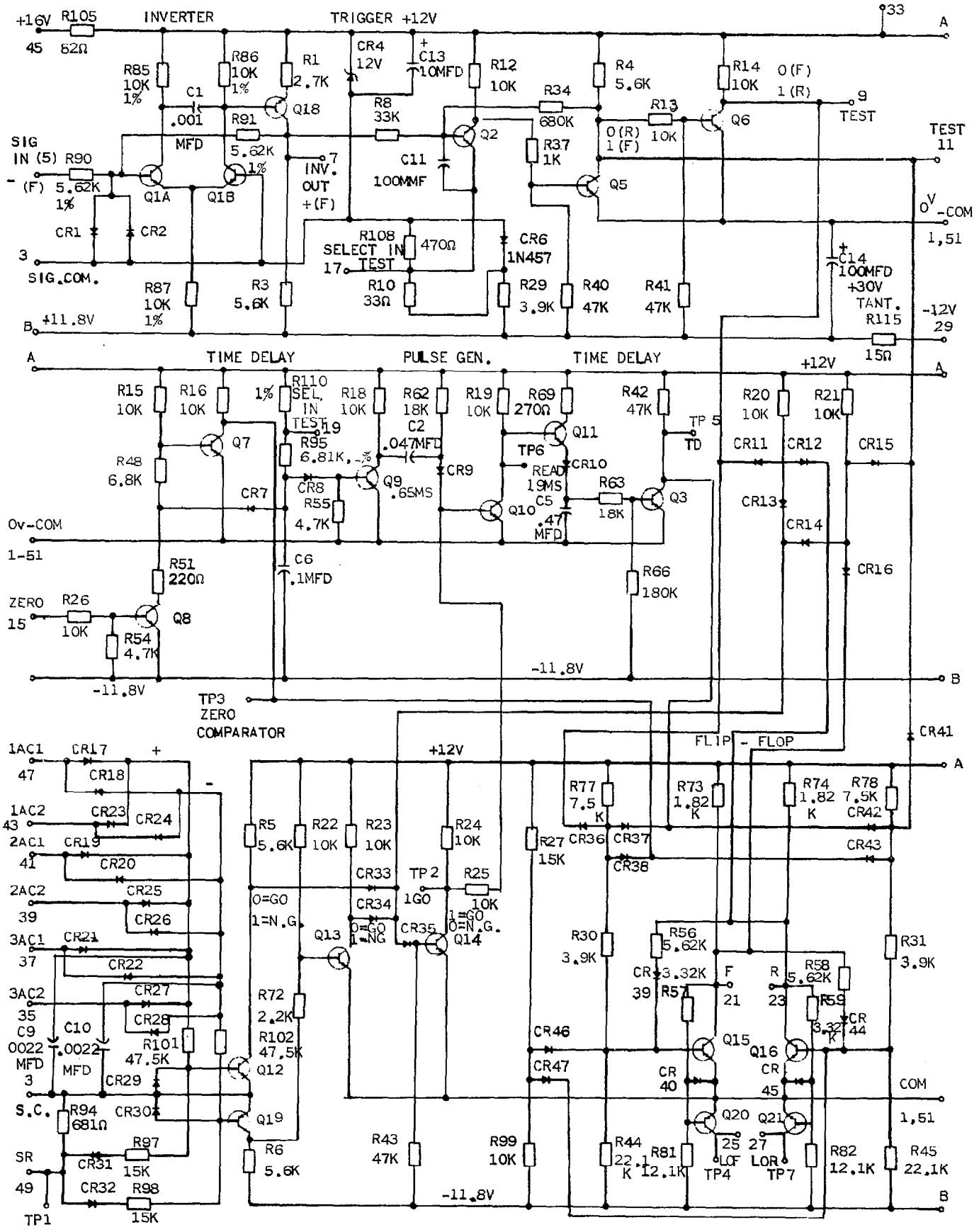
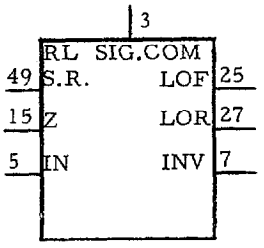


Fig. 30B. Reversing logic. (Elementary diagram IC3600TRLB1)

SYMBOL



TRLA - 1ϕ-60∿
TRLB - 1ϕ-50∿
TRLC - 3ϕ-60∿
TRLD - 3ϕ-50∿

Function: The reversing logic card is used in full reversing SCR exciters. Its function is to allow polarity reversal when conditions are correct for a safe reversal. All reverse cells must be extinguished before a forward cell is turned on and all forward cells must be extinguished before a reverse cell is turned on. The reversing logic controls the forward and reverse firing circuits by means of lockout signals. It determines input command by monitoring the input signal, cell current conditions by means of cell voltage and a saturable reactor. It makes use of 60∿ power voltage zero crossovers as a timing reference.

Inputs

Pin 5 to 3	Signal input
Pin 15 to 1	Zero input from firing circuit card
Pin 49 to 3	Saturable reactor input
Pin 47, 43, 41, 39, 37, 35	12V rms reference voltage with respect to 0V
Pin 25	Forward lockout (LOF)
Pin 27	Reverse Lockout (LOR)
Pin 45	+16V 100 ma max
Pin 29	-12V 20 ma max

Parts (See Fig. 31 for block diagram)

1. Inverter
2. Trigger
3. Comparator
4. Read Pulse Generator and Time Delay
5. Lockout-Flip-Flop

(1) Inverter – The inverter accepts signal voltage and inverts its polarity in order to provide the proper polarity signal to drive the reverse firing circuits. It also provides an input for the trigger circuit.

The input signal is applied between pin 5 and 3. The input signal is applied to the base of Q1A, half of a differential pair Q1A-Q1B, the base of Q1B is tied directly to ground. The output is taken from the collector of Q1B and fed directly to the base of Q18. Q18 provides voltage amplification and polarity inversion; the output is taken from its collector. Feedback is provided by means of R91. Overvoltage protection at the input is provided by CR1-CR2. These diodes limit the base voltage of Q1A to ± 0.5V.

(2) Trigger – The trigger monitors the inverted input signal. If the input signal goes from 0V to approximately -200 mv the trigger will change the conduction states of two transistors which indicate the input signal polarity to the reversing logic. Once the trigger has operated the circuit is such that the input signal must go to approximately +200 mv to change the transistor states. This overlap avoids rapid transistor state change when a signal is in the vicinity of the trip point.

The inverter output is applied to the base of Q2 through R8. The emitter is biased in the negative direction to allow turn on of transistor Q2 at a voltage below the 0.5V base-emitter drop. As the input voltage goes negative the inverter output goes positive turning on the transistor Q2. The collector voltage of Q2 approaches zero volts causing the base of Q5 to go negative, thus turning Q5 off. Q5 collector voltage goes positive, thus turning on Q6 and providing current through R34 to the base of Q2, thus "locking" it on. The situation is that a signal input less than a -200 mv causes the collector of Q5 to go positive and the collector of Q6 to go to zero volt. A signal input of greater than +200 mv overcomes the positive feedback through R34 and changes the collector states to Q5-zero volts and Q6 to positive voltage. These voltages are used as 2 inputs to the lockout flip-flop.

(3) Comparator – This circuit monitors the output of a saturable reactor which indicates the conduction state of the cells. If the voltage is zero the cells are off; if it is present (approximately 12V rms) the cells are conducting. The circuit compares the S. R. input against a reference derived from the AC power source. The S. R. input level will determine the state of transistor Q10 which affects the operation of the time delay circuit. The AC power supply is selected as the prime source of the reference voltage to attain some cancellation of noise which is present in the power source.

The reference voltage is a full wave rectifier bridge output utilizing CR17, 18, 23 and 24 for 1Ø exciters and CR17, 18, 23, 24, 19, 20, 25, 26, 21, 22, 27 and 28 for 3Ø exciters. The S. R. input is an AC voltage. The comparison is made during each half cycle; therefore, the necessity of both a + and - bus. If the S. R. input is 0V transistors Q12 and Q19 will be turned on. The collector of Q12 is 0V, thus back biasing CR33. The collector of Q19 is 0V, thus causing the base of Q13 to be positive and turning the transistor on. The collector of Q13 will be 0V back biasing CR34. Transistor Q14 would be biased off by R43 if CR13 and 14 were also back biased (CR13 and 14 will be discussed later). This turns Q14 off and its collector rises to approximately +6V. As the S. R. input rises it will exceed a level determined by R44 and an external resistor. At this level Q12 and Q19 will turn off, resulting in Q14 turn-on and therefore Q14 collector at 0V.

(4) Read Pulse Generator and Time Delay – This circuit accepts zero pulses derived from the power voltage zero crossovers. The pulses are changed in polarity for use in the comparator flip-flop and also delayed an interval of time for use as "read" pulses. These "read" pulses are the sampling times at which the comparator output is looked at. If the comparator says "go" for approximately 18 ms a final time delayed "go" will be fed to the lockout flip-flop. If the comparator says "no go" the go signal is not sent to the flip-flop.

(5) Lockout Flip-Flop – This circuit is the logic output device. It provides the lockout signals to the forward or reverse firing circuits. The circuit is controlled by two "and" circuits. The trigger inputs are "mirror images" of one another. That is, if one "and" receives a forward input signal as a + voltage the other will receive a forward input signal as a minus voltage. To change the state of the flip-flop the following are required:

a. A positive voltage forward or reverse input signal indication from the trigger.

b. A "go" signal from the comparator which has been "read" and maintained for the time delay period (approx. 18 msec.).

c. A "zero" signal.

A firing circuit will be locked out when 0V is present at LOF or LOR.

A diode bridge made up of CR11 through CR16 eliminates unwanted command conditions by freezing the flip-flop in its present state when these commands occur. It freezes the state by making the comparator output look like a no go whether this is the case or not. The comparator output will be allowed to act only when a forward input signal is indicated and a FORWARD lockout is present or when a reverse input signal is indicated and a REVERSE lockout is present. This is a safety device to help eliminate unwanted effects of noise and other transients.

The resistors R27 and R99 form a divider whose output voltage is fed via diodes CR46 and 47 to the bases of the flip-flop transistors Q15 and Q16. Upon turn-on of the exciter this circuit functions to lock out both forward and reverse firing circuits until the supply voltages stabilize.

The flip-flop has been designed so that Q15 will turn on before Q20 and Q16 before Q21. This avoids the situation of the lockouts operating before the flip-flop has and a possible condition of neither lockout energized.

(6) Differences between Reversing Logic Card Types – The differences between card types appear in the comparator and read pulse generator. The comparator full wave rectifier circuit is different for 3Ø and 1Ø exciters. R94 is different for 3Ø and 1Ø exciters to compensate for the difference in read pulse times with respect to the zero pulses.

The read pulse generator requires different delay times to allow for 1Ø, 3Ø, 50~, and 60~ power inputs. This is done by changes in R110 and C6.

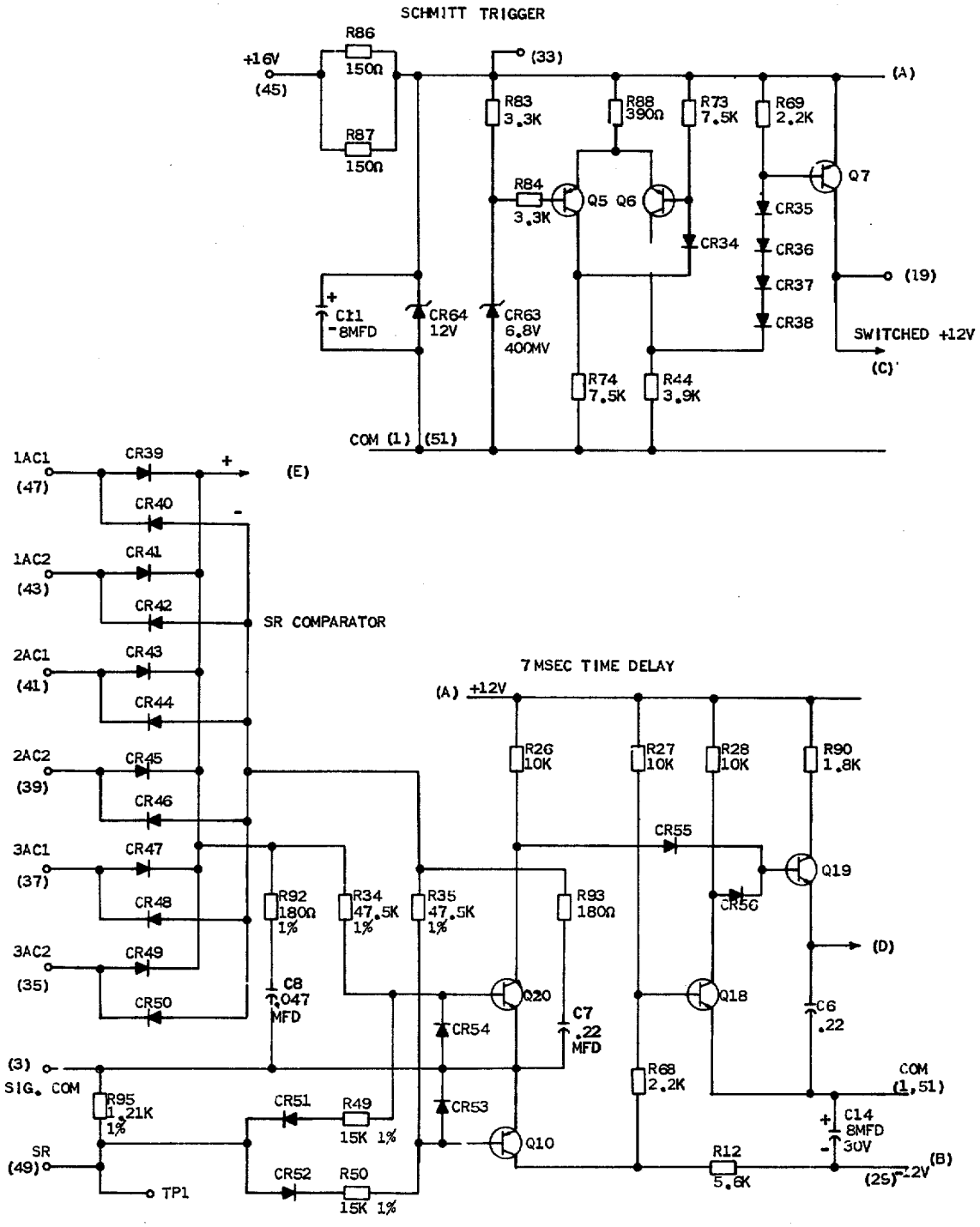


Fig. 31A Reversing logic (IC3600TRLJ1)
(Cont'd page 52)

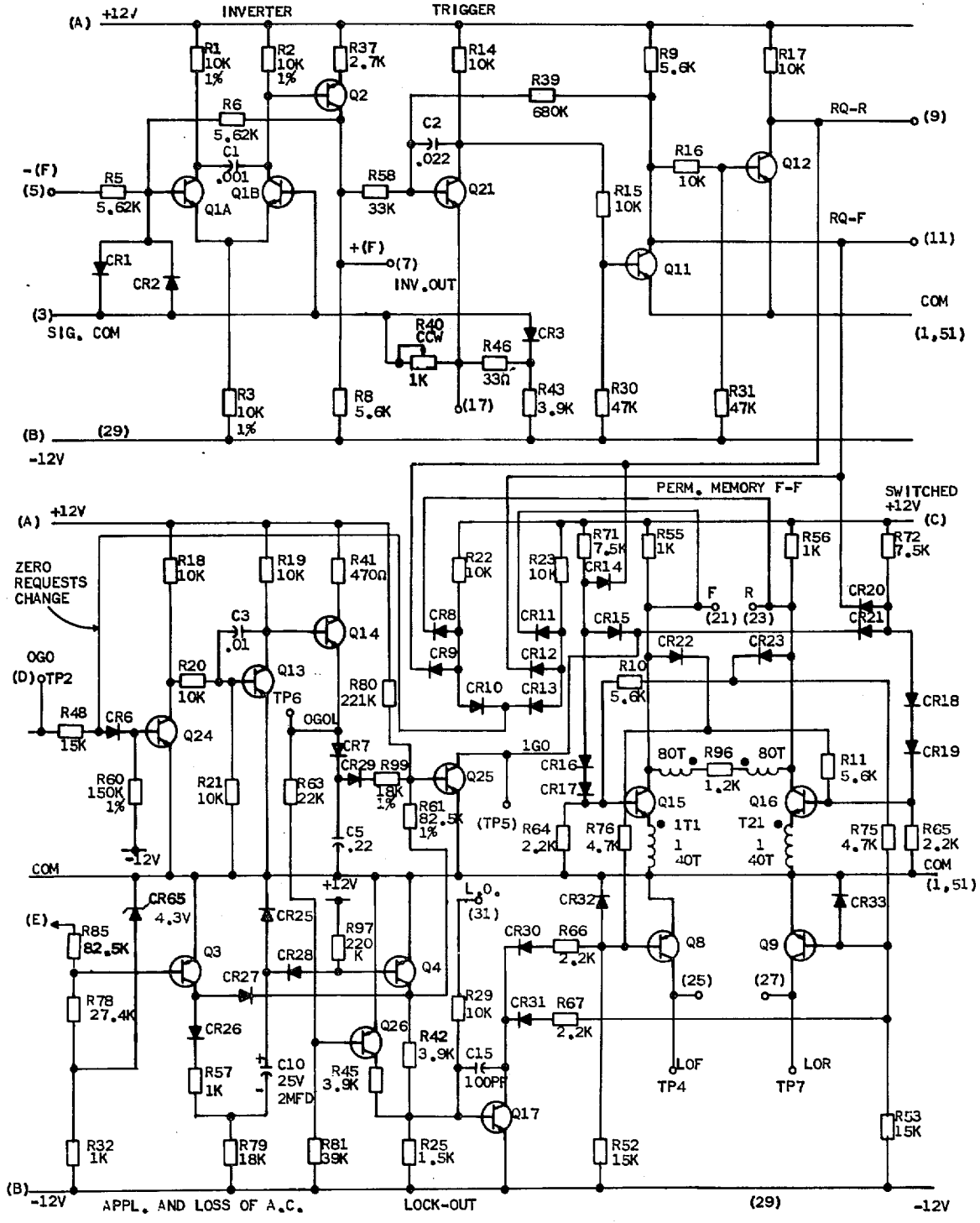


Fig. 31A Reversing logic (IC3600TRLJ1)
(Cont'd from page 51)

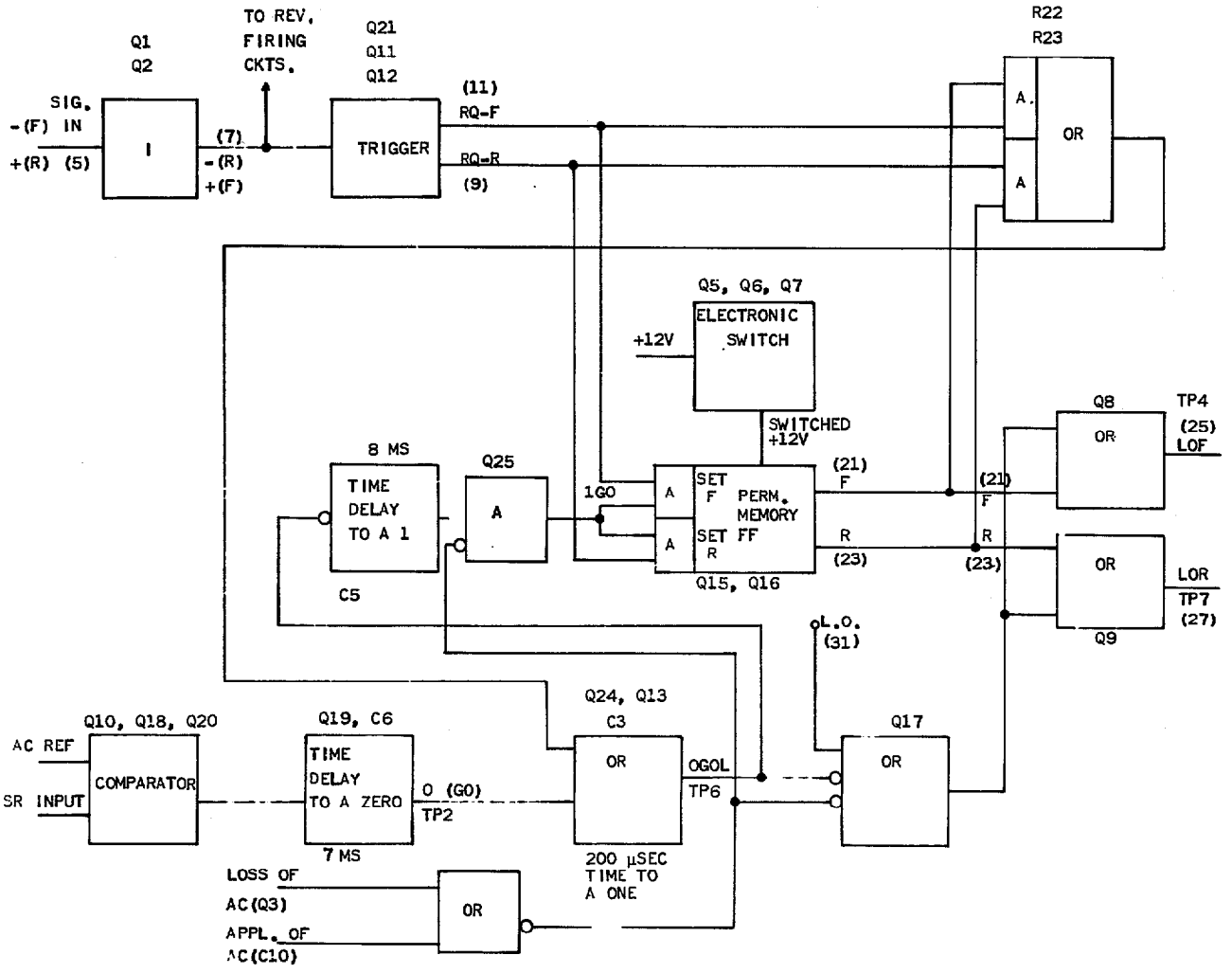
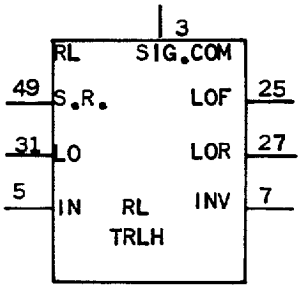


Fig. 31B. Block diagram—Reversing logic with permanent memory (IC3600TRLJ1)

SYMBOL



Function: The reversing logic card is used in full reversing SCR exciters. Its function is to allow polarity reversal when conditions are correct for a safe reversal. All reverse cells must be extinguished before a forward cell is turned on and all forward cells must be extinguished before a reverse cell is turned on. The reversing logic controls the forward and reverse firing circuits by means of lockout signals. It determines input command by monitoring the input signal and cell current conditions by means of cell voltage with a saturable reactor. During a normal transfer both lockouts are energized together for 8 milliseconds. Application or loss of AC voltage locks out both sides for a short time.

Inputs

Pin 5 to 3	Signal input ±2V, 5K load
Pin 31	External Lockout, tie to com for lo.
Pin 49 to 3	Saturable reactor input ±18V max, 1.2K load
Pin 47, 43	12V RMS reference voltages with respect to COM, 20K Load

Pin 25 Forward Lockout (LOF) can drive 12MA.
Pin 27 Reverse Lockout (LOR) can drive 12 MA.
Pin 45 +16V 70 MA max; 55 MA. NOM
Pin 29 -12V 40 MA. max (during lockout), 25 MA NOM

Parts (See Fig. 1 for block diagram)

1. Inverter
2. Trigger
3. Comparator
4. Electronic Switch
5. Lockout-Flip-Flop
6. AC Detection Circuits

(1) Inverter — The inverter accepts signal voltage and inverts its polarity in order to provide the proper polority signal to drive the reverse firing circuits. It also provides an input for the trigger circuit.

The input signal is applied between pin 5 and 3. The input signal is applied to the base of Q1A, half of a differential pair Q1A-Q1B, the base of Q1B is tied directly to ground. The output is taken from the collector of Q1B and fed directly to the base of Q2. Q2 provides voltage amplification and polarity inversion; the output is taken from its collector. Feedback is provided by means of R6. Overvoltage protection at the input is provided by CR1-CR2. These Diodes limit the base voltage of Q1A to ±.5V.

(2) Trigger — The trigger monitors the inverted input signal. If the input signal goes from OV to approx. -200 MV. the trigger will change the conduction states of two transistors which indidicate the input signal polarity to the reversing logic. Once the trigger has operated the circuit is such that the input signal must go to approximately +200 MV. to change the transistor states. This overlap avoids rapid transistor state change when a signal is in the vicinity of the trip point.

The inverter output is applied to the base of Q21 thru R58. The emitter is biased in the negative direction to allow turn on of transistor Q21 at a voltage below the .5V base-emitter drop. As the input voltage goes negative the inverter output goes positive turning on the transistor Q21. The collector voltage of Q21 approaches zero volts causing the base of Q11 to go negative thus turning Q11 off. Q11 collector voltage goes postive thus turning on Q12 and providing current thru R39 to the base of Q21 thus "locking" it on. The situation is that a signal input less than a -200 MV. causes the collector of Q11 to go postive and the collector of Q12 to go to zero volt. A signal input of greater than +200 MV. overcomes the positive feedback thru R39 and changes the collector states of Q11 to zero volts and Q12 to postive voltage. These voltages are used as 2 inputs to the lockout flip-flop.

(3) Comparator — This circuit monitors the output of a saturable reactor which indicates the conduction state of the cells. If the voltage is zero the cells are off, if it is present (approximately 6V RMS) the cells are conducting. The circuit compares the S. R. input against a reference derived from the A. C. power source. The S. R. input level will determine the state of transistor Q19 which affects the operation of the time delay circuit. The A. C. power supply is selected as the prime source of the reference voltage to attain cancellation of variations present in the power source.

The reference voltage is a full wave rectifier bridge output utilizing CR39, CR40, CR41, and CR42 for 1Ø exciters. The S. R. input is an A.C. voltage. The comparison is made during each half cycle, therefore, the necessity of both A + and -BUS.

If the saturable reactor voltage at TP1 (TERM 49) is over 5 volts negative, then Q20 is off and OGO is a one thru CR55 and Q19. If positive over 5V, then Q10 and Q18 are off and OGO is also a one thru CR56 and Q19. As SR is an AC signal it will be less than ±5V each half cycle. The nominal setting of SR is ±8V peak and the time between ±5 volts is 3.5 to 4 MS which is killed by the 7 MS time delay in C6. Only if the SCR cells are off will SR be below 5V for 7 MS.

For the standard single phase SCR packages, the SR load resistor R95 may be externally parallel by 4.7K or higher to set ±8V (or higher). With the 4.7K resistor the lockouts will release at 50% rated package voltage. With no resistor, release is at 25% voltage.

If the request signal from the trigger and the flip flop state agree, then the output of the and-or circuits R22 and R23 is a one holding Q24 saturated. When the request signal changes, this signal releases but Q24 remains saturated until the saturable reactor (thru OGO) detects the SCRS are safe. OGOL (TP6) then becomes a zero which immediately locks out both forward and reverse thru Q4, Q17, Q8 and Q9. OGOL also starts C5 timing and after 8 MS the flip-flop will change to the requested state. As the flip-flop and request signals again agree, Q24 saturates to remove the lockout signal thru Q17 and the opposite SCRS are now fired.

(4) Flip-Flop — This circuit has two square loop core reactors added to a standard flip-flop circuit. The collector voltages of Q15 and Q16 always drive one core into positive saturation and the other into negative saturation. Thus when AC power is lost and reapplied the reactors add voltages in series with the emitters to force the flip-flop to return to the same state it had before the loss of power.

AC power may have been lost with a large current flowing in the inductive load which requires many seconds to reduce to zero. If the flip-flop returns to the correct state, control of the load may be regained safely as soon as AC power returns.

The reactors require about 40 microseconds to change state as the flip-flop changes state but several times this is provided by the time delay of C3. As the delay in the cores is small, power to the flip-flop must be applied and removed quickly which is the function of the electronic switch described later.

This flip-flop always energizes one of the lockouts, so forward and reverse SCRS cannot both be fired. Other circuits can energize both lockouts as described elsewhere.

(5) Electronic Switch — This circuit uses a zener to detect when the +12V BUS is near the 8 volt level. Slightly above or below this point operates the schmitt trigger (Q5, Q6) which controls Q7 to apply or remove power to the flip-flop circuitry. Rise and fall of Q7 output is under 10 microseconds.

The electronic switch opens when the AC voltage drops to 60 percent of nominal.

(6) AC Detection Circuits — As AC voltage is applied the -12V BUS starts increasing and acts thru C10 to lockout both sides thru Q4, Q17, Q8, Q9. These circuits are active when the BUS reaches 4 volts which is before any SCR can be fired. C10 nominally holds both lock outs on for .1 to .2 seconds.

If the AC voltage drops to 60 per cent voltage, Q3 will turn on and also lockout both sides. Q3 also discharges C10 so it is ready to act on reapplication of power.

If the AC voltage is suddenly lost, the saturable reactor can no longer detect that load current may be flowing. Thus Q3 and Q4 bring 1GO (Q25) to zero within 10 milliseconds so the flip flop cannot change state. If current were flowing, FF cannot change in less than 15 milliseconds (C5 and C6 delays).

On re-application of power the 0.1 second delay due to C10 keeps the 1GO signal zero, which is sufficient time for the saturable reactor to determine whether load current is still flowing.

6. SATURABLE REACTOR (IC3500A491AB103)

The saturable reactor is the device which determines the conduction state of the thyristors in a reversing exciter (see Fig. 32).

If no cell is conducting, the voltage at P and Y will be approximately zero. This results in no current flow in the P-Y winding of the saturable reactor. The 12V RMS will see maximum impedance in the 1 to 3 windings and little current will flow through the resistor R. This causes a low voltage to be developed across R.

If one of the cells is firing (FIP, for instance) P assumes L1 line voltage and Y will be at approximately half line voltage. This will cause current to flow in P-Y. This results in saturation of one core and a resultant increase of current. The output will look like a current source and the value of R will determine the voltage at A. This voltage is compared to a reference in the reversing logic; if it falls below a specific level the thyristors are definitely off and the exciter is ready to refire.

7. LINE FILTERS

An RC filter is used to attenuate high frequency line spikes and other fast rise time voltage transients.

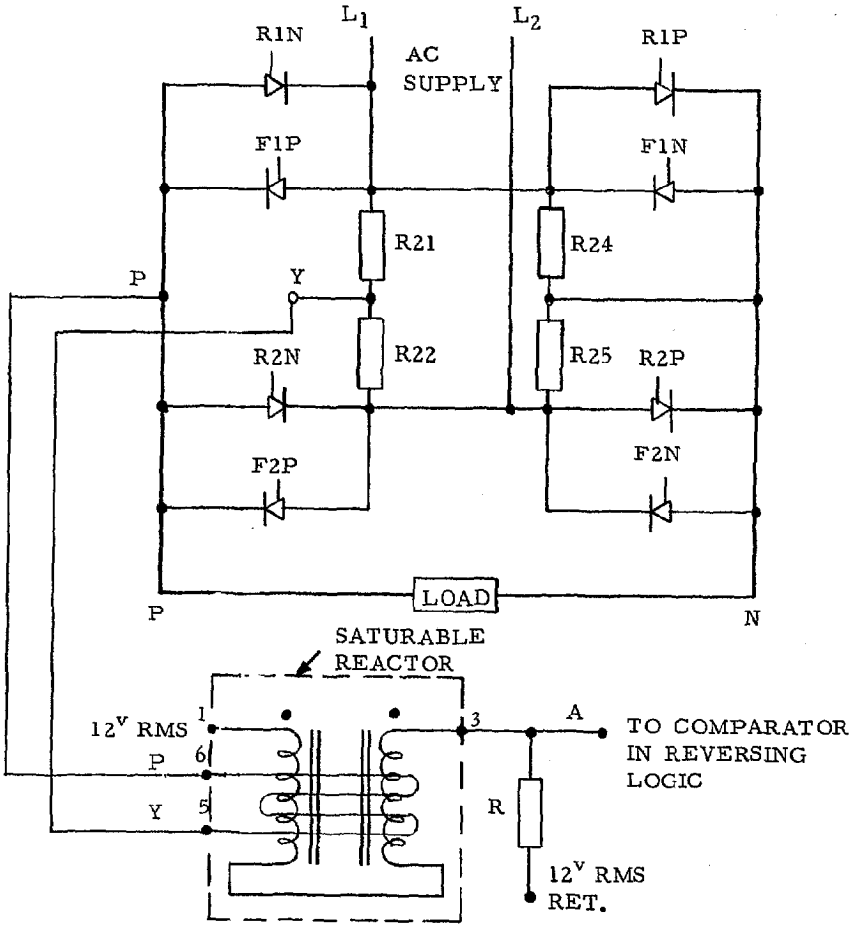


Fig. 32. Schematic diagram for determining state of SCR conduction

IV. SCR THYRISTOR FIELD EXCITERS—AMPLISTAT FIRED

A. MECHANICAL DESCRIPTION

The amplistat exciters utilize the same structures as the unijunction exciters for the same output ratings.

B. COMPATIBILITY

See IIIB.

C. HOW TO PUT AN EXCITER IN SERVICE

1. STEPS

See IIIC. In addition to IIIC insure that the DC bias voltage necessary for normal amplistat operation is present.

2. EXCITER BIAS ADJUSTMENT

All amplistat exciters utilize an external DC bias voltage to adjust the firing angle. This bias is applied to one of the seven control windings.

Tail-end pulses are set at 30° firing angle in the factory. This adjustment is made with potentiometer R40 on the power supply card TPSC1. Normally it will not be necessary to adjust this setting.

The external bias voltage should be set as follows:

a. NP Exciters – Nonreversing; Plugging

The output should be monitored with a DC voltmeter while connected to the load. All amplistat windings except F1-F2 and the bias winding should be disconnected. Adjust bias so that the voltmeter reads 0^{v}. This is attained by adjusting the bias to read a voltage, then back off on bias until the voltmeter just reads 0^{v}. Replace windings.

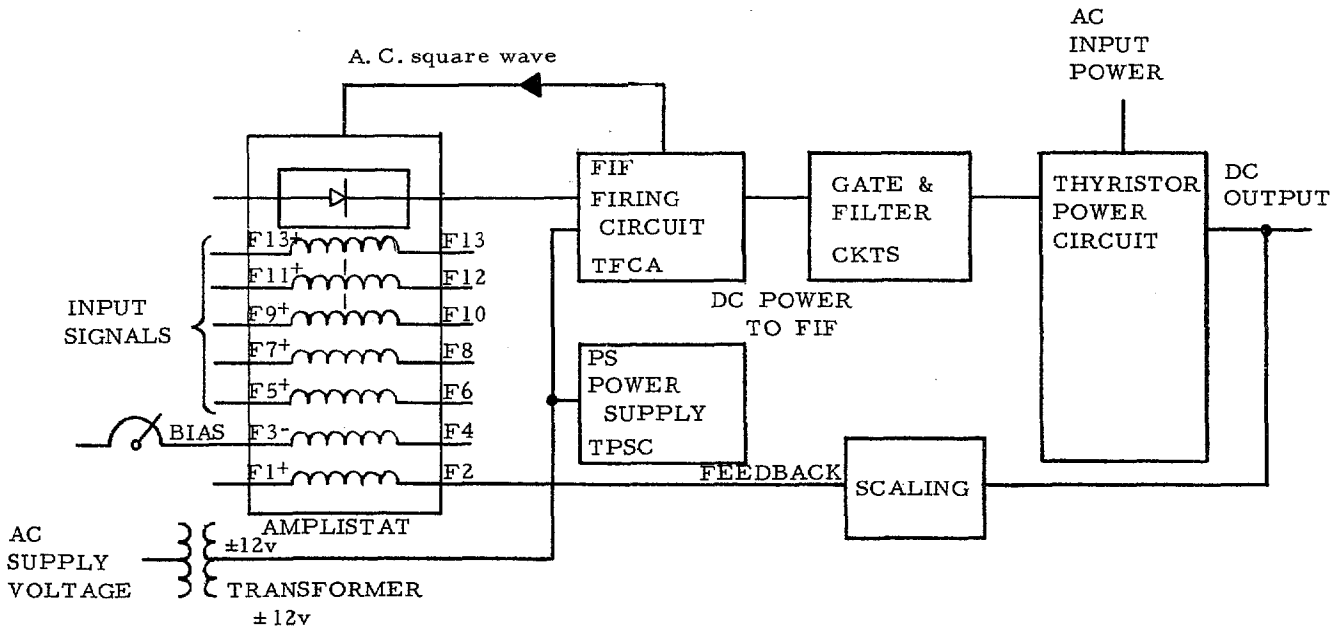


Fig. 33. Amplistat single phase nonreversing thyristor block diagram

b. NX Exciters – Nonreversing; Nonplugging

The output should be monitored with a DC voltmeter while connected to load. All amplistat windings except F1-F2 and the bias winding should be disconnected. Adjust bias so that the voltmeter reads 0^V. This is attained by adjusting the bias to read a voltage, then back off on bias until the voltmeter just reads 0^V. Replace windings.

D. ELECTRICAL DESCRIPTION

1. BLOCK DIAGRAM

1. The amplistat NP thyristor block diagram is shown in Fig. 33. The input signals may be fed in on five separate amplistat windings. These windings allow input summing of several signals with ground isolation. A winding is required for setting the bias and another for internal exciter feedback. The amplistat output is fed to the firing circuit. This output is essentially a variable width pulse whose timing is derived from the stepped down AC supply voltage.

The firing circuit converts this amplistat output to a pulse train which is fed to the thyristor power circuit via the gate and filter circuits.

±12 volts is applied to the power supply and the firing circuit. The power supply rectifies this voltage and generates the plus and minus voltages utilized in the firing circuit. The firing circuit is ultimately steered by the ± 12 volts.

Internal feedback is utilized to insure identical exciter gain from exciter to exciter.

2. BASIC POWER CIRCUITS

Basic power circuits are identical to the unijunction power circuits.

E. PLUGGING

Refer to unijunction plugging.

F. TROUBLE SHOOTING

The general procedures are the same as those for unijunction exciters.

However, when the exciter is to be disconnected from the regulator and energized to insure its proper operation, a separate DC signal input is not required. Remove all wires from the amplistat windings except F1-F2 and the bias input. By varying the bias it is possible to vary the output from 0^V to full output voltage.

1. and 2. CT AND NP WAVESHAPES

The waveshapes for center-tapped and nonplugging exciters are the same as for unijunction fired exciters.

3. CARD TEST POINTS

a. Firing Circuit - TFCA

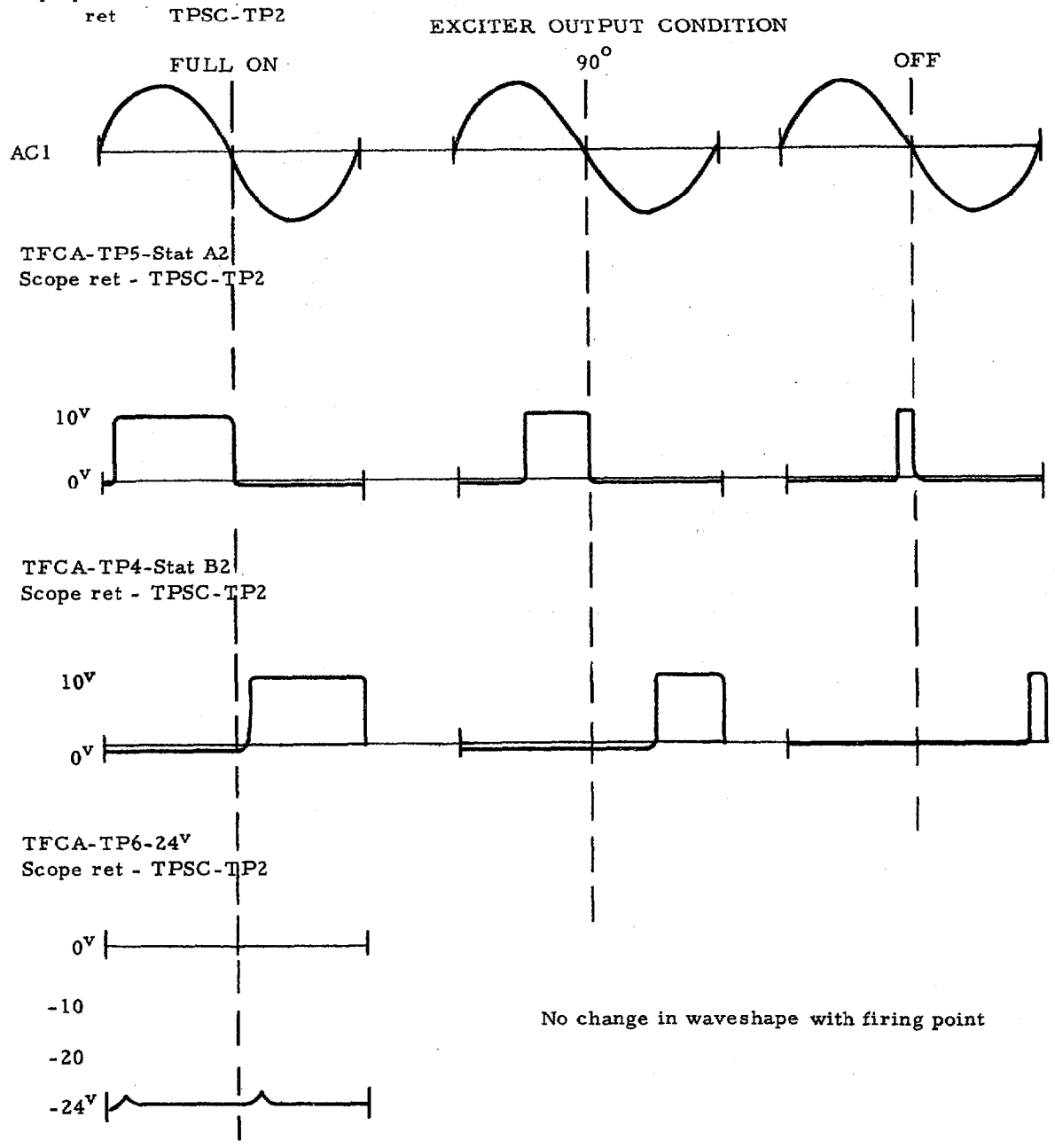
The following test points are present on the card.

Test Point	Function
1	Not Used
2	PA
3	PB
4	Stat. B2
5	Stat. A2
6	-24^V
7	AC1

Explanation of Test Points:

2 and 3 – Same as that for the TFCU card PA and PB
4 – Stat B2 – Amplistat output voltage
5 – Stat A2 – Amplistat output voltage
6 – -24^V – Bias voltage generated within the firing circuit
7 – AC1 – $+12^V$ reference sine voltage with respect to ground

C - Card Test Point (Cont'd)
1 - Firing Ckt (cont'd)
TFCA-TP7-AC1
Scope probe A-TP7
ret TPSC-TP2



No change in waveshape with firing point

G. PARTS

The parts peculiar to the amplistat fired exciters are:

1. Amplistat Cores
2. Firing Circuit - TFCA
3. Power Supply - TPSC

1 and 2. Amplistat Cores and Firing Circuit – TFCA

The amplistat cores and firing circuit are an entity and are separated only due to packaging requirements. Since this is the case they shall be discussed together. (Refer TFCA elementary diagram.)

± 12 volts is applied to AC1 and AC2. Since these voltages are sine waves referenced to common, Q1 will be turned on when AC1 is negative; Q2 will be "off." When AC2 is negative, Q2 will be "on" and Q1 "off." When Q1 is "on," Pt. 5 is connected to common and current will flow through B1-B2, CR3-CR4, R45 and ultimately turn Q7 on. However, the impedance offered by B1-B2 is a function of the state of its magnetic circuit. If the circuit is saturated Q7 will be turned on immediately. If it is not saturated B1-B2 offers high impedance and Q7 will not turn on.

B1-B2 can saturate at some time during the period that Q1 is on. The time of saturation is a function of the sum of the ampere turns in all the control windings. When Q2 is on, Pt. 3 is connected to common, current can flow through A1-A2, CR1-CR2, R42 and turn Q6 on. Again the impedance offered by A1-A2 is a function of the input signals applied to the input windings.

When Q6 turns on the Q6-Q3 flip-flop is set and turns on Q12. This allows a pulsating current to flow through a transformer, connected from pins 41 to 49. This transformer is on a gate and filter card and provides SCR gate current to the power circuit. The Q6-Q3 and Q7-Q4 flip-flops are externally connected so that when one flip-flop is set the other is cleared. This results in Q11 and Q12 never being on simultaneously and gating only positive or negative cells simultaneously. When Q7 turns on the Q7-Q4 flip-flop is set, the Q6-Q3 flip-flop cleared and Q11 turned on, allowing pulsating current to flow through the gate transformer connected between 49 and 37.

An oscillator voltage is applied to pin 25; it drives Q8 and Q10 so that it is the control for the pulsating currents driven through Q11 or Q12.

An amplistat output will turn on slightly when driven hard in the negative A. T. direction. To prevent this CR7, CR8, R14 and R15 provide a current bias which permits negative control drives of over 10 times normal.

3. Power Supply-TPSC (See Elementary Diagram)

This power supply is basically similar to TPSA except there is no separate bias output. The -12 volt DC output level can be controlled by potentiometer R40. This is utilized in factory adjustment of the tail-end pulse.

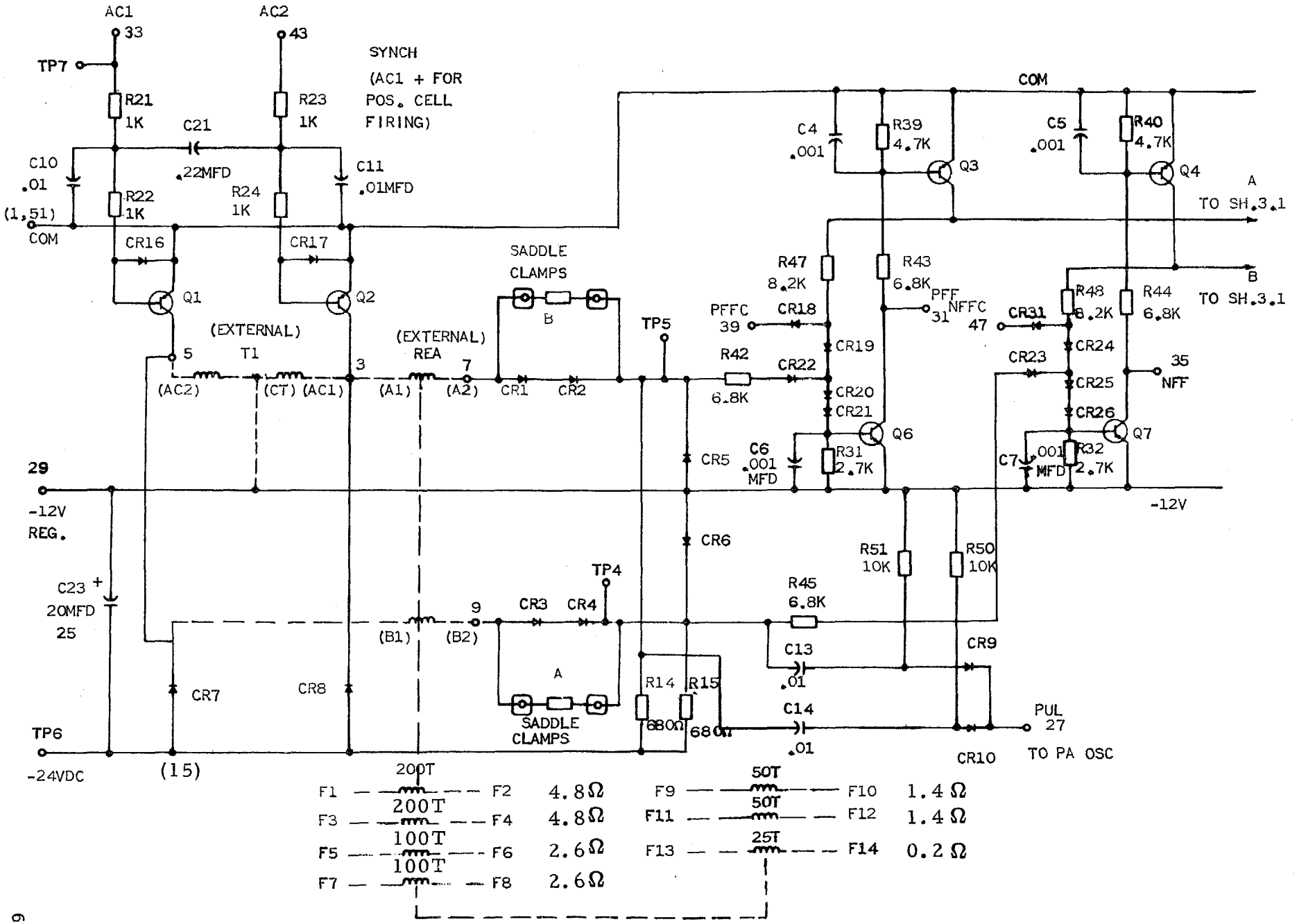
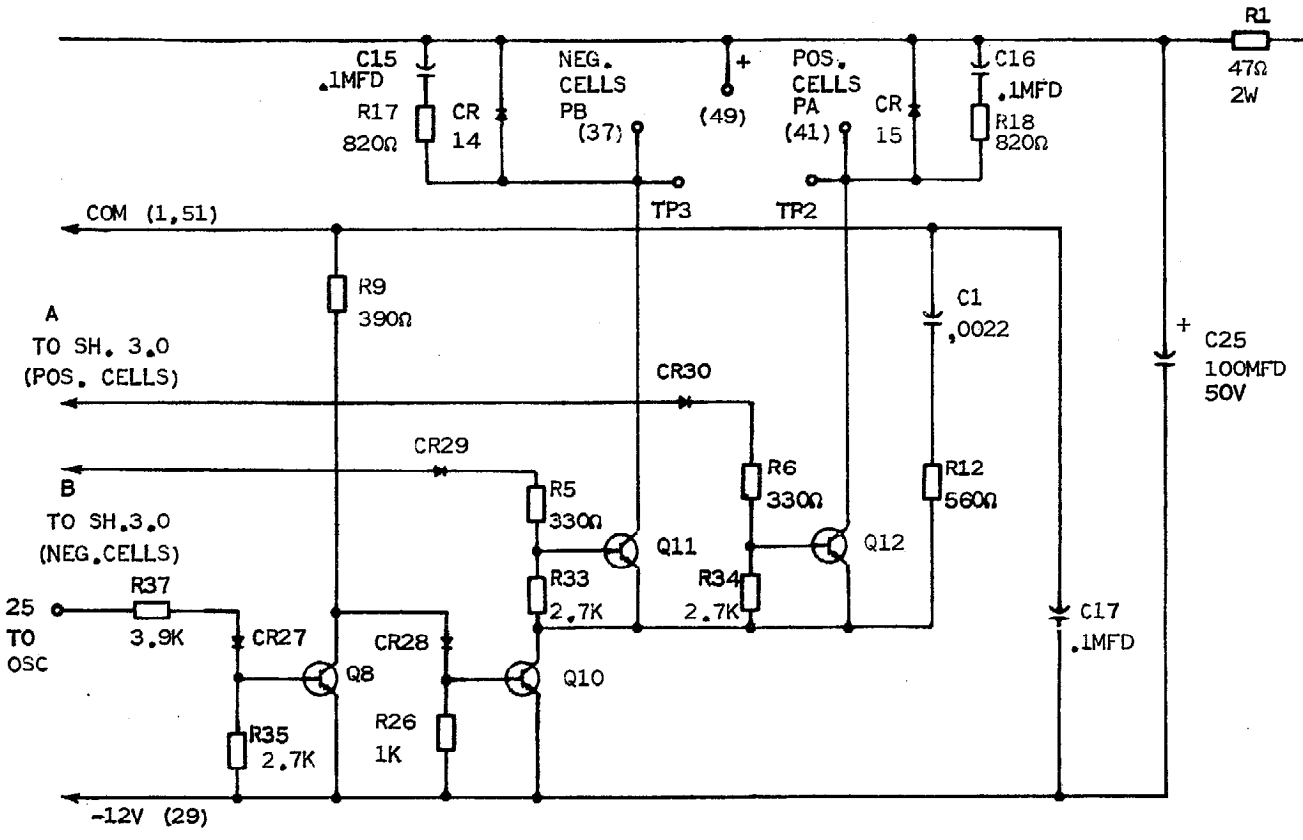
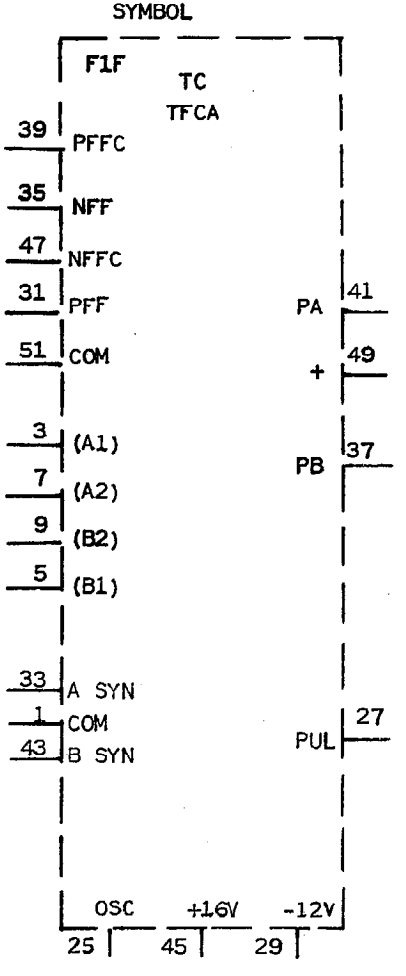


Fig. 34. Firing circuit amplistat. (Elementary diagram IC3600TFCA1)





JUMPERS

NRP PACKAGE	35 to 39	31 to 47
NRX PACKAGE	29 to 39	29 to 47

Q1 TO Q4 68A7355P1
Q6 TO Q8 68A7373P1
Q10 TO Q12 68A7350P1

Fig. 34 (cont.)

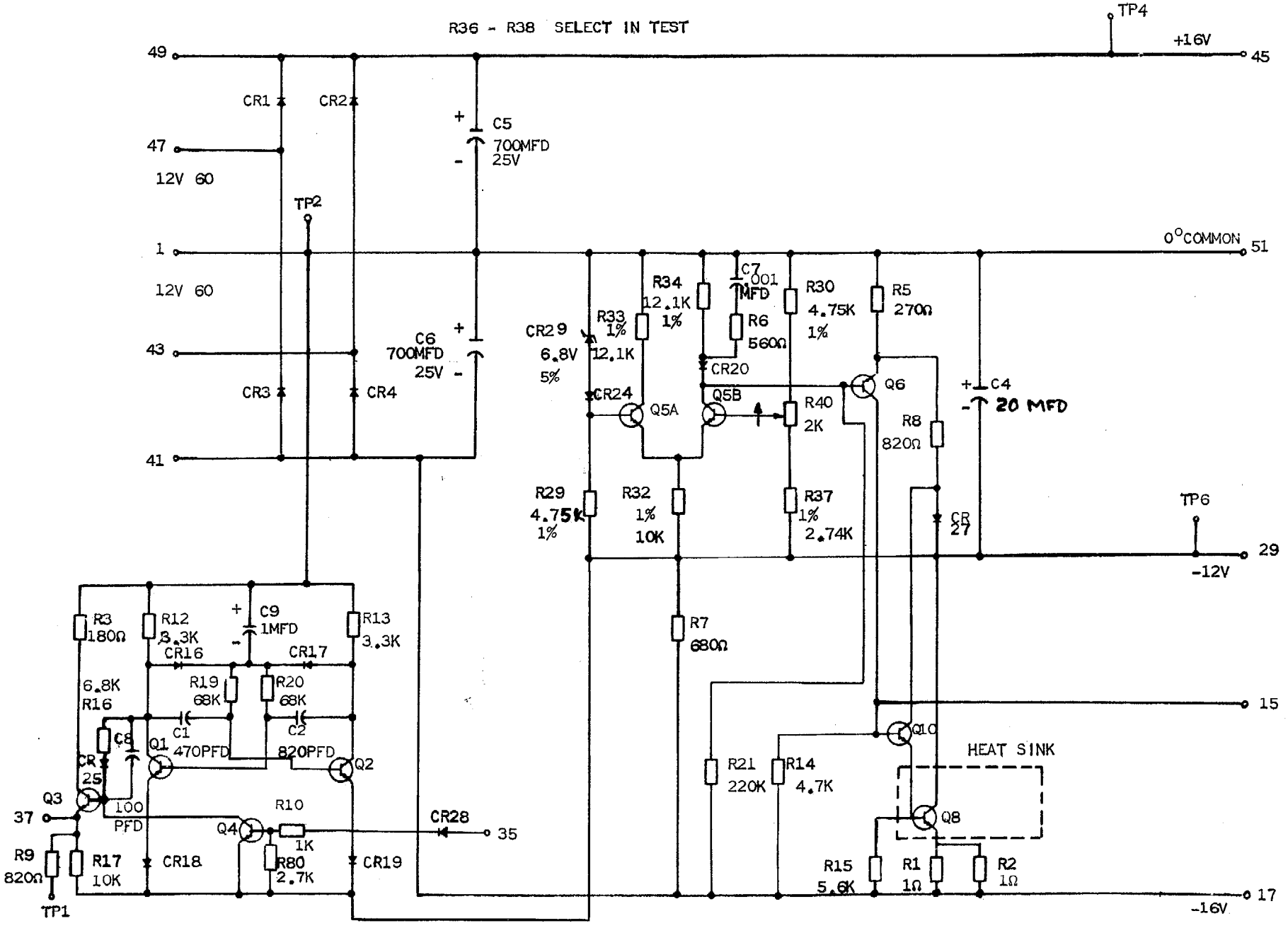
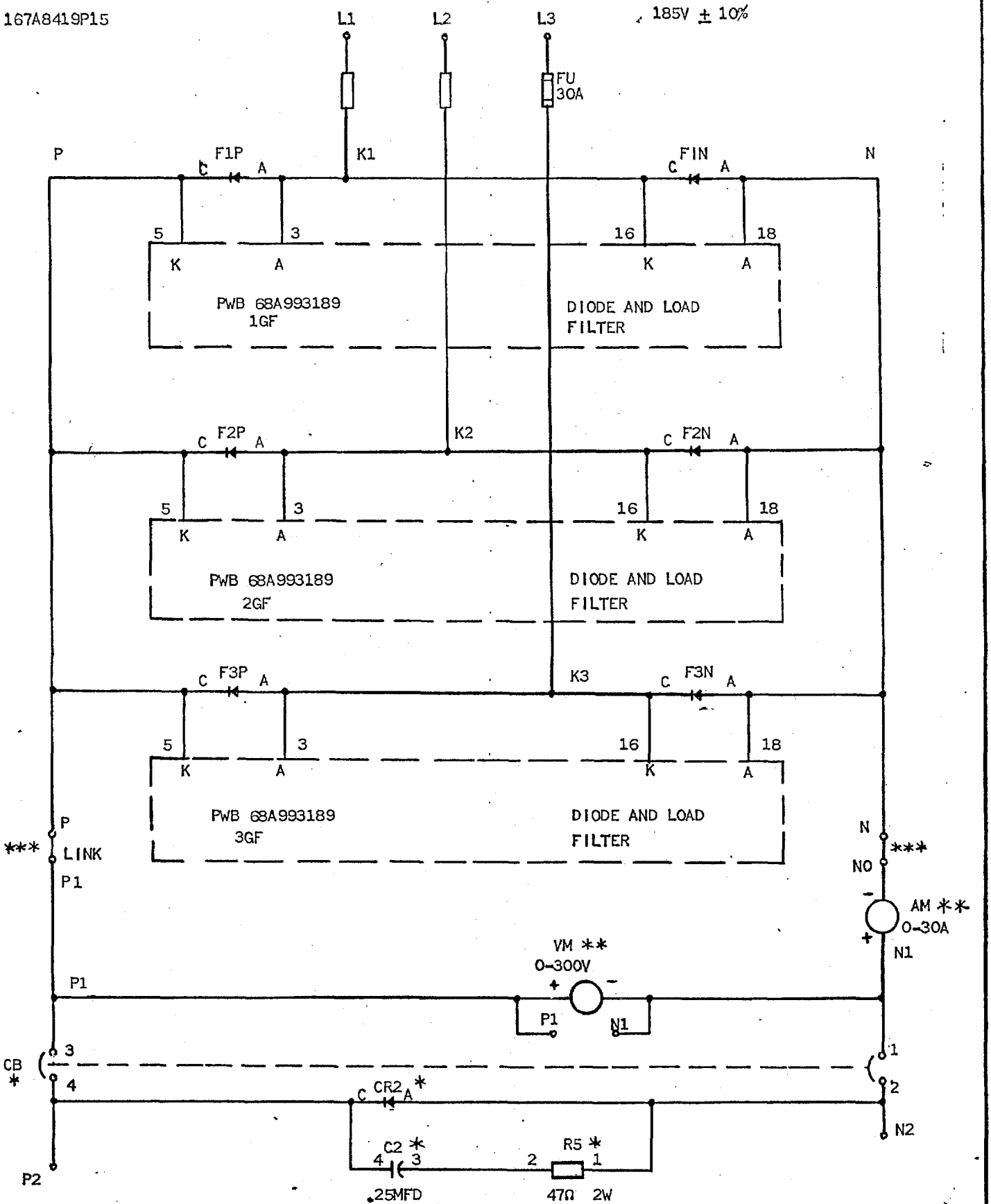


Fig. 35. Single phase power supply. (Elementary diagram IC3600TPSC1)

DRIVE SYSTEMS PRODUCT DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

ALL DIODES
ARE 167A8419P15



* - BREAKER AND DISCHARGE NETWORK-OMIT ON IC3602A125ANL1 AND 0, IC3602A125ANN1 AND 0.
** - METERS - OMIT ON IC3602A125ABL0, ABNO, ANL0 AND ANNO.
*** - LINKS - OMIT ON IC3602A125ABN1 AND 0, IC3602A125ANN1 AND 0.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 Deleted 7-10-72	REV. 2 6T	APPROVALS R. Jarvinen	GENERAL ELECTRIC	ELEMENTARY DIAGRAM UNREGULATED EXCITER 250V CPE 30A
ISSUED 7-2-29-1968	PRINTS TO P6A	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3602A125 CONT. ON SH. FL. SH. NO. 3.0
MADE BY L. W. DINKEL	1338	I.C. NO.		

FORM IC 2219 (3-66)

GENERAL ELECTRIC

219A3404

CONT ON SHEET 1 SH NO. 0

REV NO.	TITLE
219A3404 CONT ON SHEET 1 SH NO. 0	SCR Exciter 22 Amp 280 Volts FIRST MADE FOR

REVISIONS

Take one IC3605C022CP250FGAN Package and remove the wire from terminal 25 on CB to terminal 29 on TPSA.

Remove card PWB68A993614G6 and replace with card PWB219A3403G1.

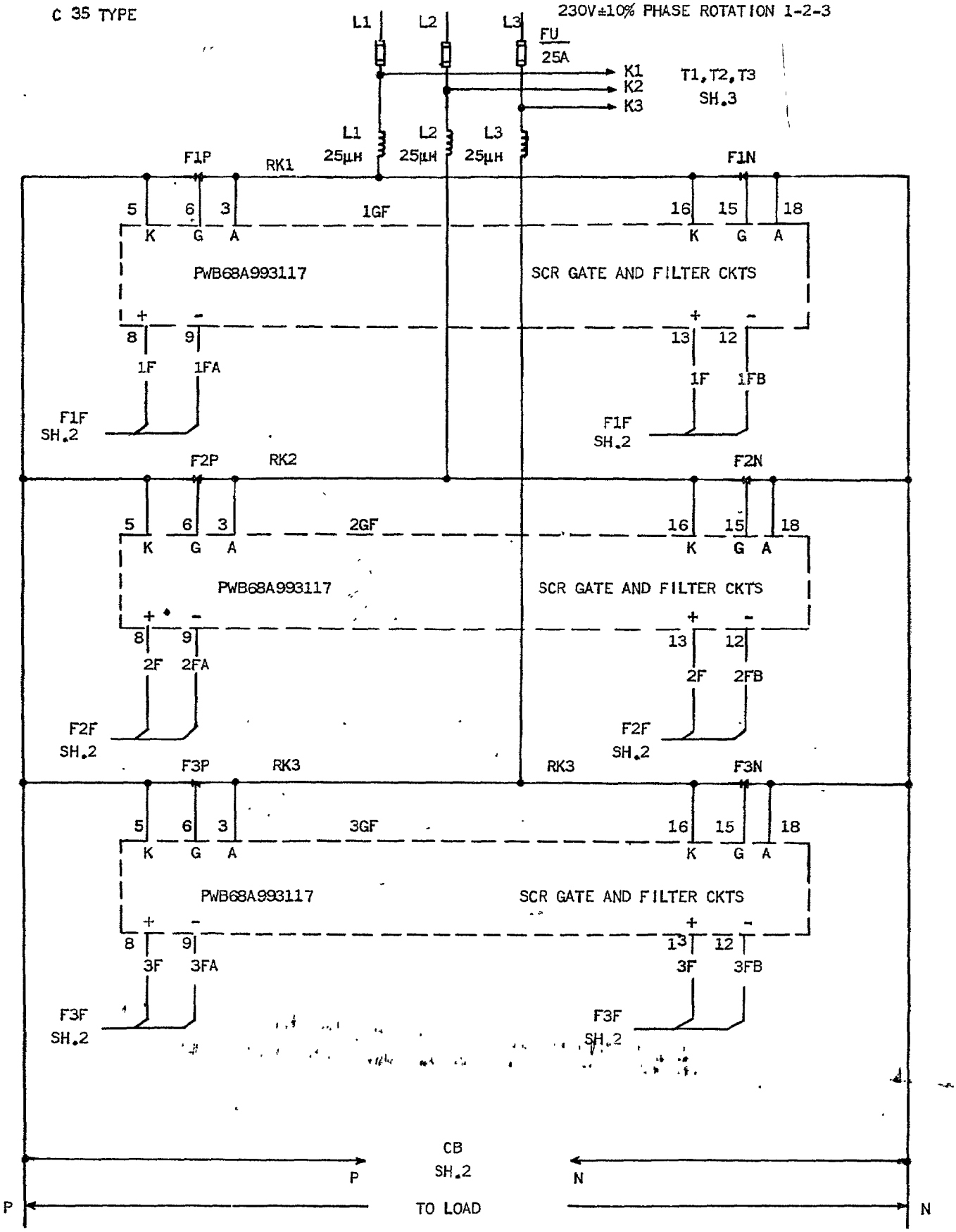
The elementary will then be sheets 1-4 of this drawing.

Remove existing NP and add a 151A4925 NP stamped "219A3404".

PRINTS TO

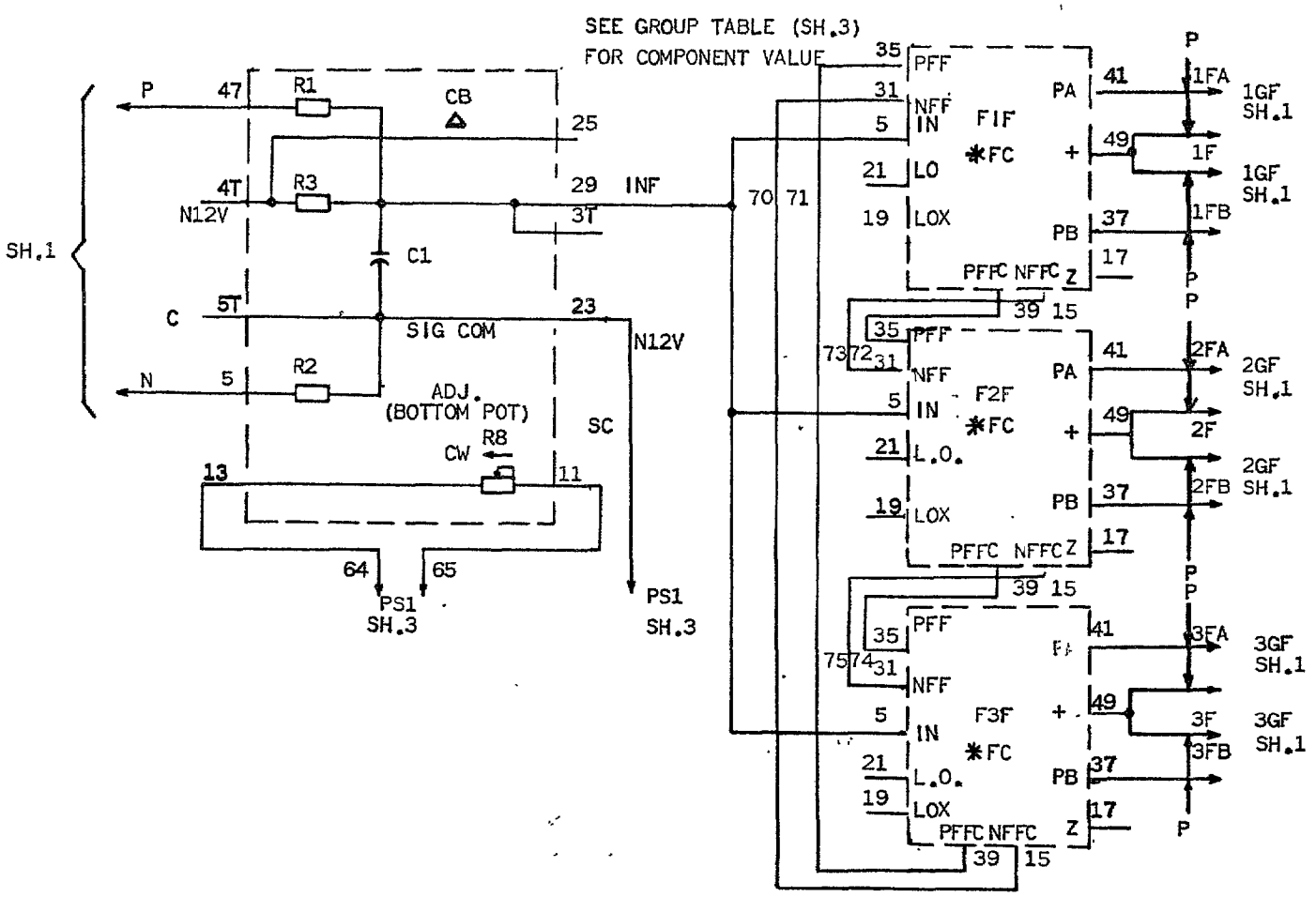
MADE BY	APPROVALS	DIV OR DEPT.	
J.A. Finch		Industry Control	219A3404
ISSUED 6/25/66		LOCATION Salem, Virginia	CONT ON SHEET 1 SH NO. 0

ALL SCR'S 68A7297PXM
C 35 TYPE



REV A	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER				
REV	REV			CPE 22AMP 250 VOLT				
MADE BY J. A. FINCH		APPROVALS	PRINTS TO				219A3404	
ISSUED 6/25/69							CONT ON SHEET 2	SH NO. 1

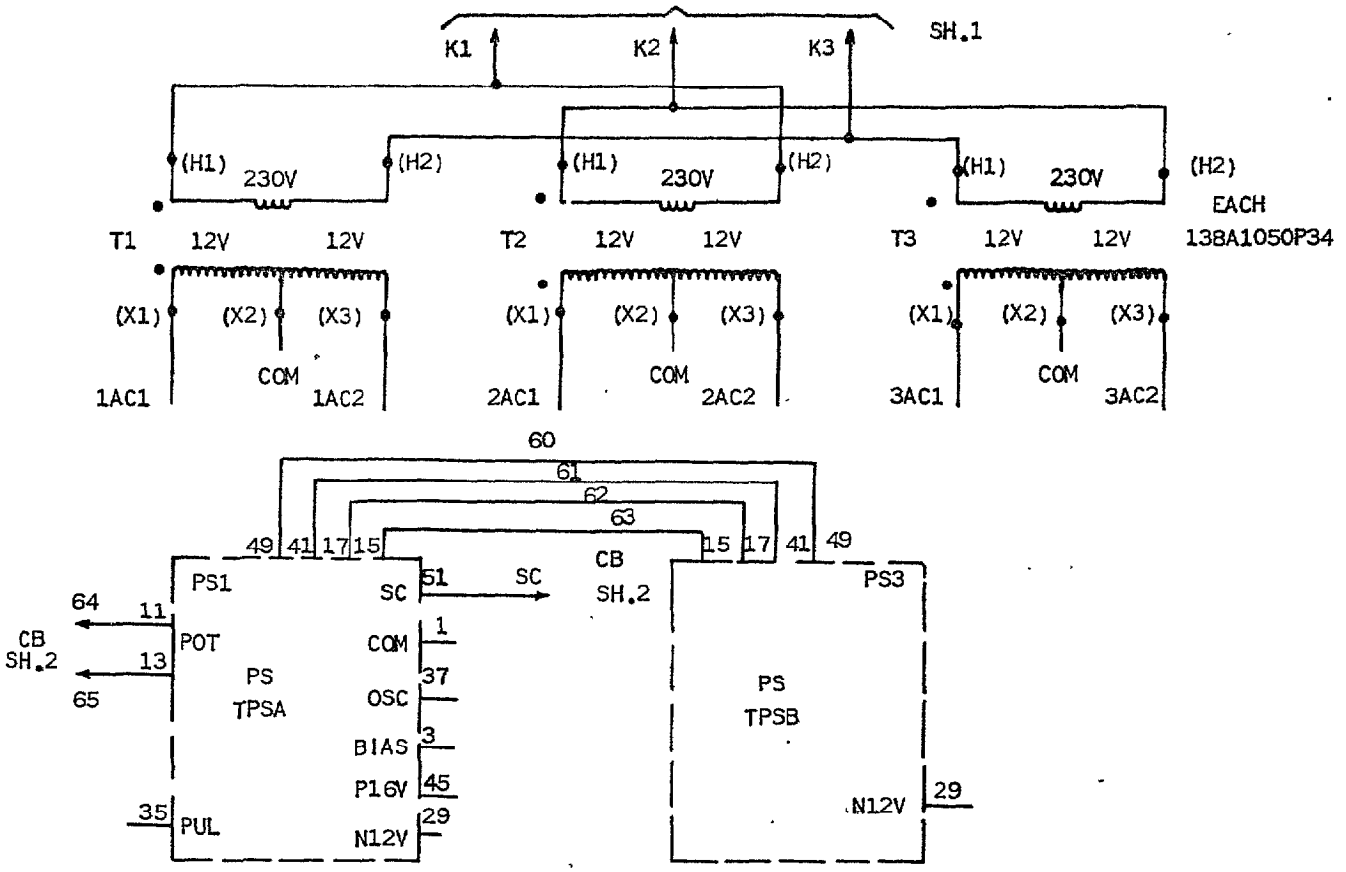
FORM IC 2143B (9-64)



*FC SEE GROUP TABLE (SH. 3)

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER CPE 22 AMP 250 VOLT				
REV	REV							
MADE BY J.A.FINCK		APPROVALS	PRINTS TO				219A3404	
ISSUED 6/15/69							CONT ON SHEET 3	SH NO. 2

FORM IC 2143B (9-64)



WIRE TABLE FOR POWER AND REPETITIVE CONNECTIONS

WIRE NO \ DEVICE			F1F		F2F		F3F	PS1	PS3	T1	T2	T3
P16V			45		45		45	45	-	-	-	-
N12V			29		29		29	29	29	-	-	-
OSC			25		25		25	37	-	-	-	-
BIAS			13		13		13	3	-	-	-	-
COM			$^{1}_{51}$		$^{1}_{51}$		$^{1}_{51}$	$^{1}_{51}$	-	X2	X2	X2
1AC1			33		-			47	-	X1	-	-
1AC2			43		-		-	43	-	X3	-	-
2AC1			-		33		-	-	47	-	X1	-
2AC2			-		43		-	-	43	-	X3	-
3AC1			-		-		33	-	39	-	-	X1
3AC2			-		-		43	-	37	-	-	X3
PUL			27		27		27	35	-	-	-	-

GROUP TABLE

VOLTAGE FEEDBACK	FREQ. CPS	FC	ΔCB	R1	R2	R3	C1
		IC3600	PWB219A 3403	10WATT	10WATT	.5WATT	50V
YES	60	TFCU	G1	12K	12K	1.0K	20 MFD
YES	50	TFCV	G1	12K	12K	1.0K	20 MFD

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 22 AMP 250 VOLT
REV	REV		
MADE BY J.A. FINCH	APPROVALS	PRINTS TO	219A3404
ISSUED 6/25/69			CONT ON SHEET 4 SH NO. 3

FORM IC 2143B (9-64)

INPUT:

A.C. POWER: 230V ±10% AMPS = 82% D.C. LOAD AMPS *6 TERMINALS: L1, L2, L3 *4,7

OUTPUT:

D.C. 250V 22 AMPS 55°C AMBIENT AT PACKAGES P+, N-

TRANSFER FUNCTION: (INCLUDES EFFECT OF TRANSPORT LAG)
RESPONSE TO 100% LOAD STEP .1 SEC.
REGULATION WITH LINE, LOAD, TEMP., ETC. ±1%

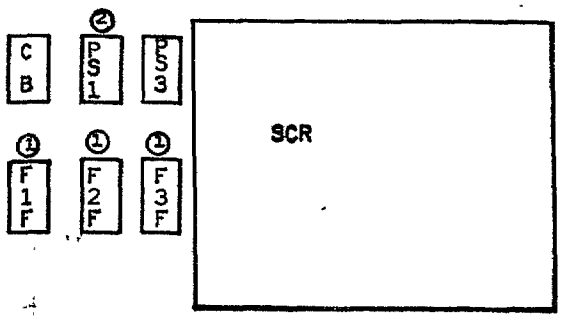
PACKAGE DIMENSIONS: 19"H x 24"W x 12"D (APPROX.)

OUTLINE: 157B6009
DULSEAL: 3851K

TEST POINTS

TP	① TFCU/TFCV	② TPSA
1	---	OSC
2	PA	GND
3	PB	---
4	AC1	+16V
5	---	BIAS
6	UNIJ.EMITTER	-12V
7	INPUT SIG.	---

T.P. POSITION



*NOTES: REFER TO GEK 8074 FOR DETAILED INFORMATION

1. DO NOT GROUND TERMINAL C WHEN INTERNAL FEEDBACK IS USED.
2. GROUND TERMINAL C OR USE AS COMMON WHEN NO FEEDBACK IS USED.
3. POWER TERMINAL P IS POSITIVE WHEN INPUT S IS NEGATIVE WITH RESPECT TO C.
4. PHASE ROTATION 1-2-3
5.
6. INDUCTIVE LOADS WITH .1 SEC. TC OR LONGER; RESISTIVE LOAD OR LESS THAN .1 SEC. TC - CURRENT MUST BE CALCULATED.
7. FREQ. TOLERANCE = ± 1/2 CPS

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 22 AMPS 250 VOLTS			
REV	REV					
MADE BY J.A.FINCK		APPROVALS	PRINTS TO			219A3404
ISSUED 6/25/69						CONT ON SHEET FL. SH NO. 4

FORM IC 2143B (9-64)

GENERAL ELECTRIC

219A3405
CONT ON SHEET 1 SH NO. 0

REV NO.	TITLE
219A3405 CONT ON SHEET 1 SH NO. 0	SCR Exciter 50 Amp 280 Volts FIRST MADE FOR

REVISIONS

Take one IC3605C050CP250FGAN Package and remove the wire from terminal 25 on CB to terminal 29 on TPSA.

Remove card PWB68A993614G6 and replace with card PWB219A3403G1.

The elementary will then be sheets 1-4 of this drawing.

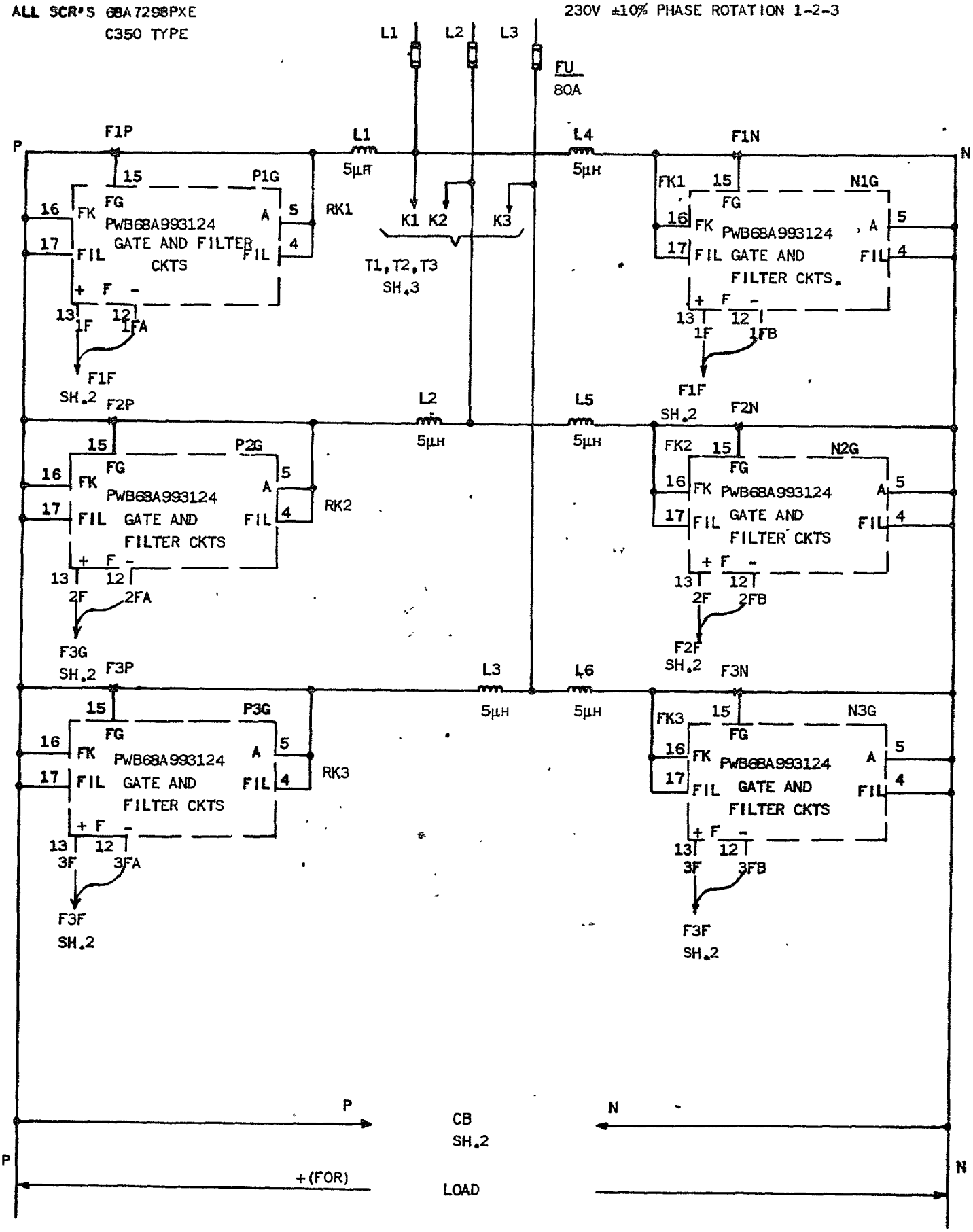
Remove existing NP and add a 151A4925 NP stamped "219A3405".

PRINTS TO

MADE BY	APPROVALS	DIV OR DEPT.	
J.A. Finch		Industry Control	219A3405
ISSUED 6/15/		Salem, Virginia LOCATION	CONT ON SHEET 1 SH NO 0

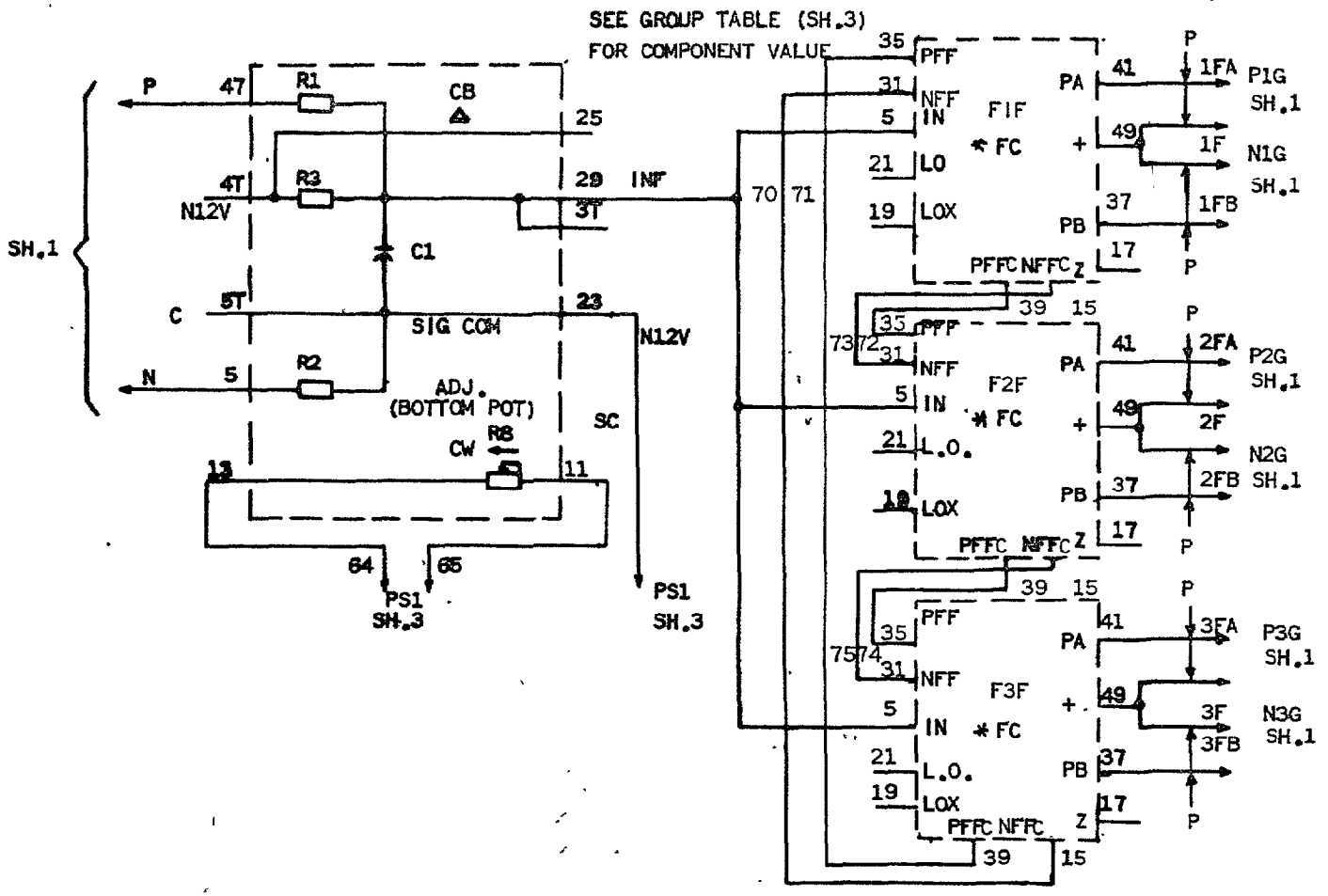
FF-803-WA (5-68)
PRINTED IN U S.A.

CODE IDENT



REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	CONSTANT POTENTIAL EXCITER
REV	REV			CPE 50 AMP 250 VOLT
MADE BY J.A. Finch		APPROVALS	PRINTS TO	219A3405
ISSUED 6/25/65				CONT ON SHEET 2 SH NO. 1

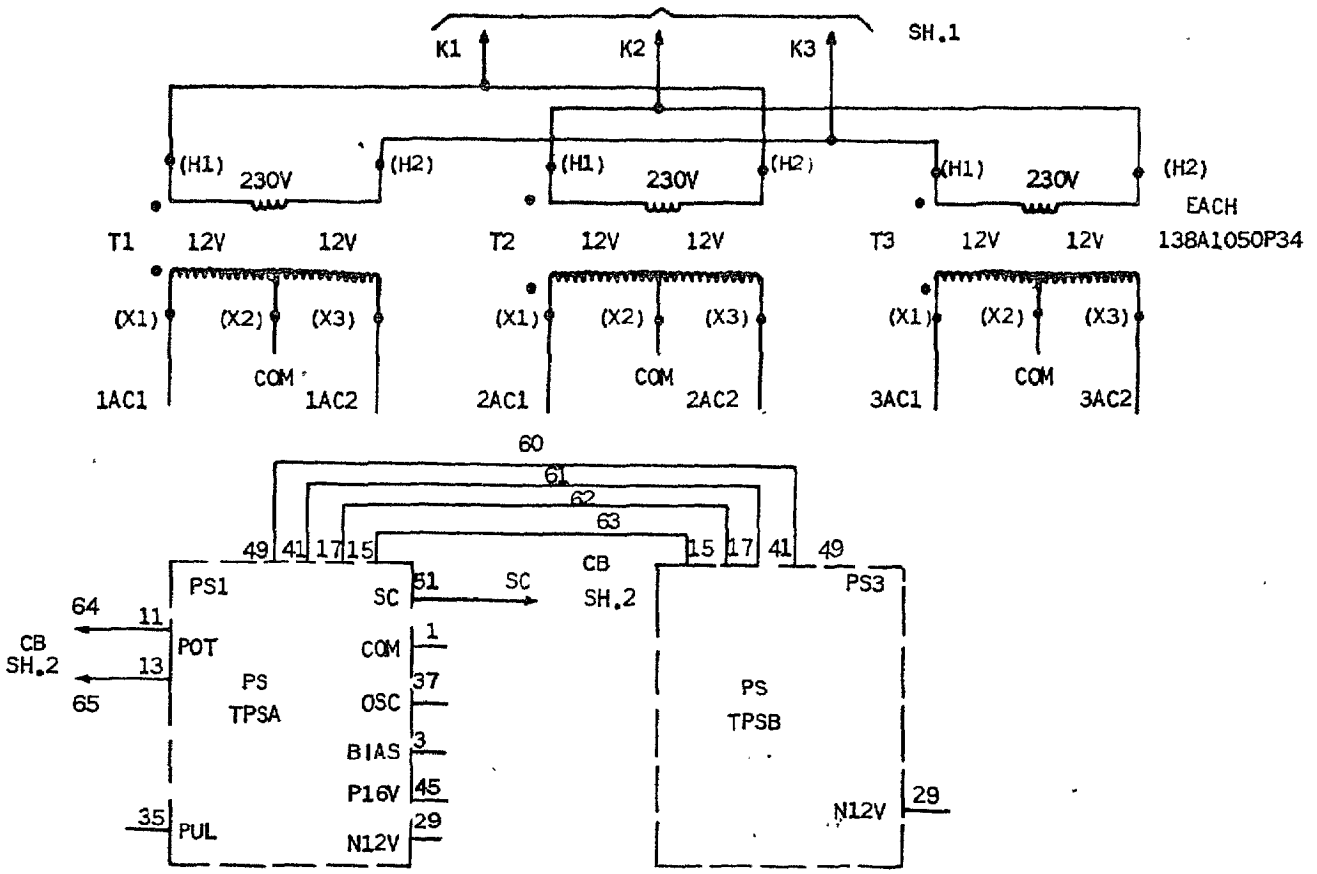
FORM IC 2143B (9-64)



* FC SEE GROUP TABLE (SH.3)

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 50 AMP 250 VOLT				
REV	REV						
MADE BY J. FINCH		APPROVALS	PRINTS TO			219A3405	
ISSUED 6/25/69						CONT ON SHEET 3	SH NO. 2

FORM IC 2143B (9-64)



WIRE TABLE FOR POWER AND REPETITIVE CONNECTIONS

WIRE NO \ DEVICE			F1F		F2F		F3F	PS1	PS3	T1	T2	T3
P16V			45		45		45	45	-	-	-	-
N12V			29		29		29	29	29	-	-	-
OSC			25		25		25	37	-	-	-	-
BIAS			13		13		13	3	-	-	-	-
COM			1_{51}		1_{51}		1_{51}	1_{51}	-	X2	X2	X2
1AC1			33		-			47	-	X1	-	-
1AC2			43		-		-	43	-	X3	-	-
2AC1			-		33		-	-	47	-	X1	-
2AC2			-		43		-	-	43	-	X3	-
3AC1			-		-		33	-	39	-	-	X1
3AC2			-		-		43	-	37	-	-	X3
PUL			27		27		27	35	-	-	-	-

GROUP TABLE

VOLTAGE FEEDBACK	FREQ. CPS	*FC IC3600	▲CB PWB219A 3403	R1 10WATT	R2 10WATT	R3 .5WATT	C1 50V
YES	60	TFCU	G1	12K	12K	1.0K	20 MFD
YES	50	TFCV	G1	12K	12K	1.0K	20 MFD

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 50 AMP 250 VOLT			
REV	REV					
MADE BY J.A. FINCH		APPROVALS	PRINTS TO			219A3405
ISSUED 6/25/69						CONT ON SHEET 4 SH NO. 3

FORM IC2143B (9-64)

INPUT:

A.C. POWER: 230V ±10% AMP = 82% D.C. LOAD AMPS *6, 7 TERMINALS: L1, L2, L3 *4

OUTPUT:

D.C. 250V 50 AMPS 55°C AMBIENT AT PACKAGE P+, N-
250V 80 AMPS FAN COOLED WITH CFM
TRANSFER FUNCTION: (INCLUDES EFFECT OF TRANSPORT LAG)
RESPONSE TO 100% LOAD STEP .1 SEC.
REGULATION WITH LINE, LOAD, TEMP., ETC. ±1%

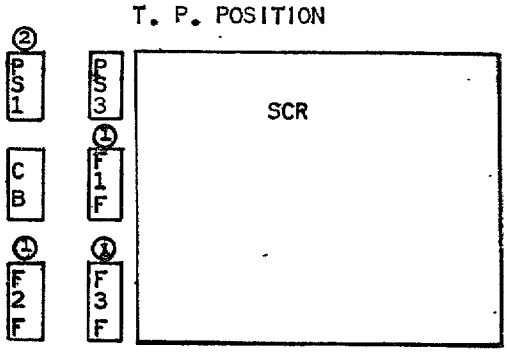
PACKAGE DIMENSIONS: 30"H x 24"W x 12"D (APPROX.)

OUTLINE: 157B6007P1
DULSEAL: 3851G

TEST POINTS

TP	① TFCU/TFCV.	② TPSA
1	--	OSC
2	PA	GND
3	PB	--
4	AC1	+16V
5	--	BIAS
6	UNIJ.EMITTER	-12V
7	INPUT SIG.	--

T. P. POSITION



*NOTES: REFER TO GEK 8074 FOR DETAILED INFORMATION

1. DO NOT GROUND TERMINAL C WHEN INTERNAL FEEDBACK IS USED.
2. GROUND TERMINAL C OR USE AS COMMON WHEN NO FEEDBACK IS USED.
3. POWER TERMINAL P IS POSITIVE WHEN INPUT S IS NEGATIVE WITH RESPECT TO C.
4. PHASE ROTATION 1-2-3
5.
6. INDUCTIVE LOADS WITH .1 SEC. TC OR LONGER; RESISTIVE LOAD OR LESS THAN .1 SEC. TC - CURRENT MUST BE CALCULATED.
7. FREQ. TOLERANCE - ± 1/2 CPS

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE CONSTANT POTENTIAL EXCITER CPE 50 AMP 250 VOLT			
REV	REV					
MADE BY J.A. FINCH		APPROVALS	PRINTS TO			219A3405
ISSUED 6/25/69						CONT ON SHEET FL. SH NO. 4

FORM IC 2143B (9-64)

TAKE ONE IC3605C100CP250FGAN PACKAGE AND REMOVE THE WIRE FROM TERMINAL 25 ON CB TO TERMINAL 29 ON TPSA.

THE ELEMENTARY WILL BE SHEETS 1-4 OF THIS DRAWING. REMOVE EXISTING NP AND ADD A

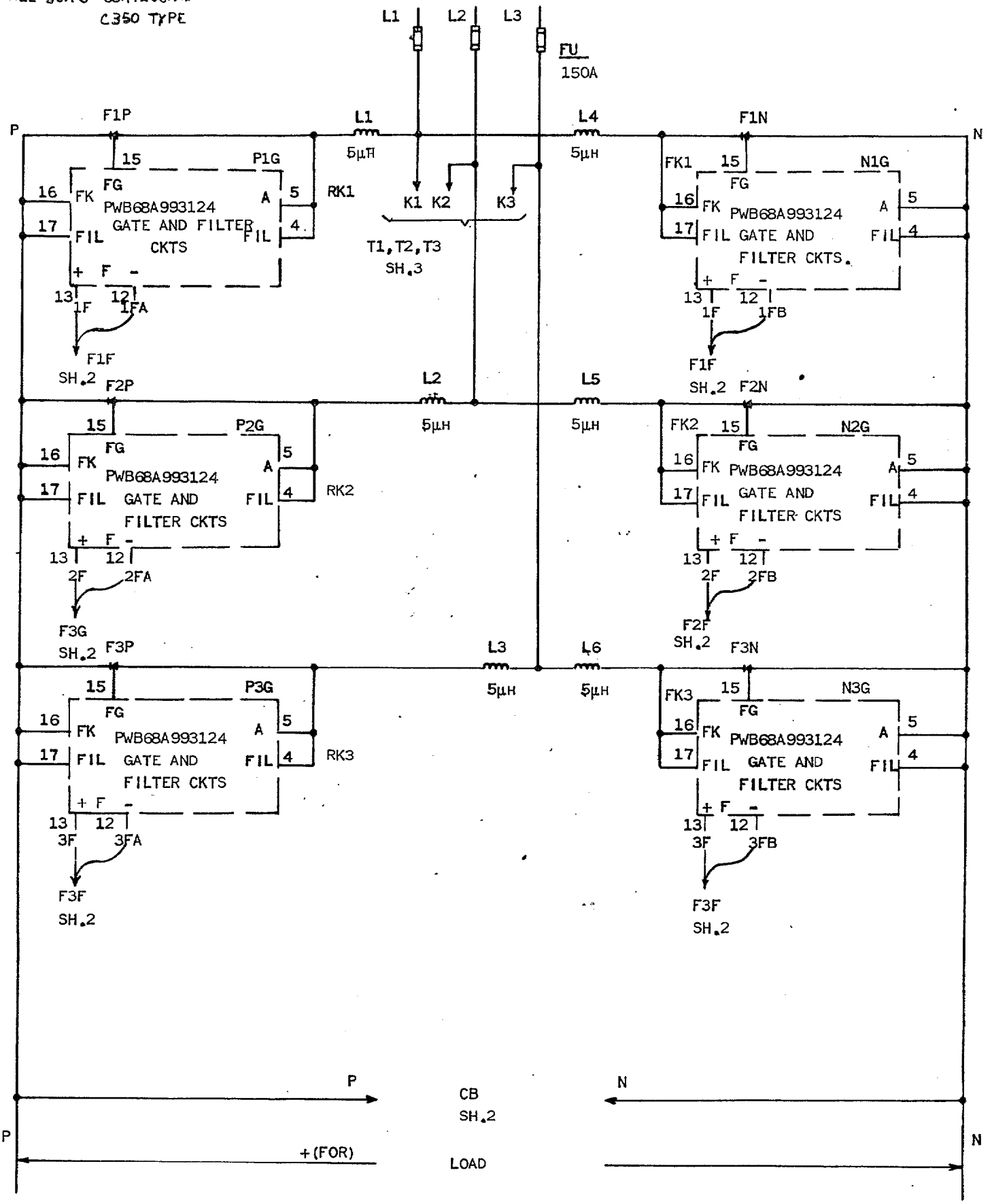
151A4925 NP STAMPED 218A6617P1 IF P1 IS ORDERED.
STAMPED 218A6617P2 IF P2 IS ORDERED.

REV.	REV.	GENERAL ELECTRIC	TITLE	
REV	REV. 1-AB 9-2-71	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	SCR EXCITER 100 AMP 280 VOLT	
MADE BY S. Wamsley	APPROVALS	PRINTS TO	218A6617	
ISSUED 1/17/68			CONT ON SHEET 1	SH. NO 0

FORM IC 2143B (9-64)

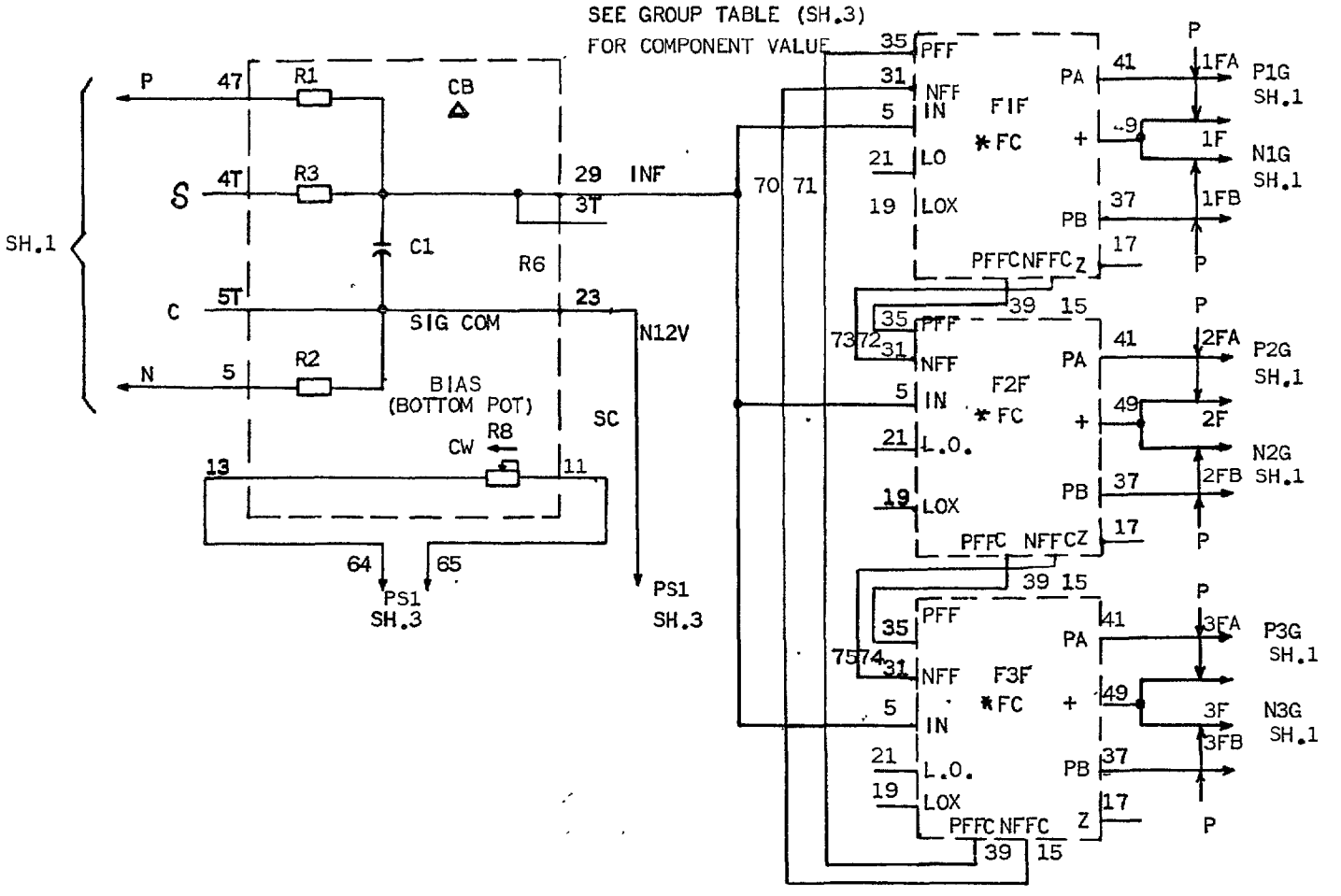
ALL SCR'S 68A7293PXE
C350 TYPE

230V ±10% PHASE ROTATION 1-2-3



REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE	SCR EXCITER 100 AMP 280 VOLT			
REV	REV						
MADE BY S. Wamsley	APPROVALS	PRINTS TO	P6A	DL1	1338	218A6617	
ISSUED 4/17/68			6T			CONT ON SHEET 2	SH NO. 1

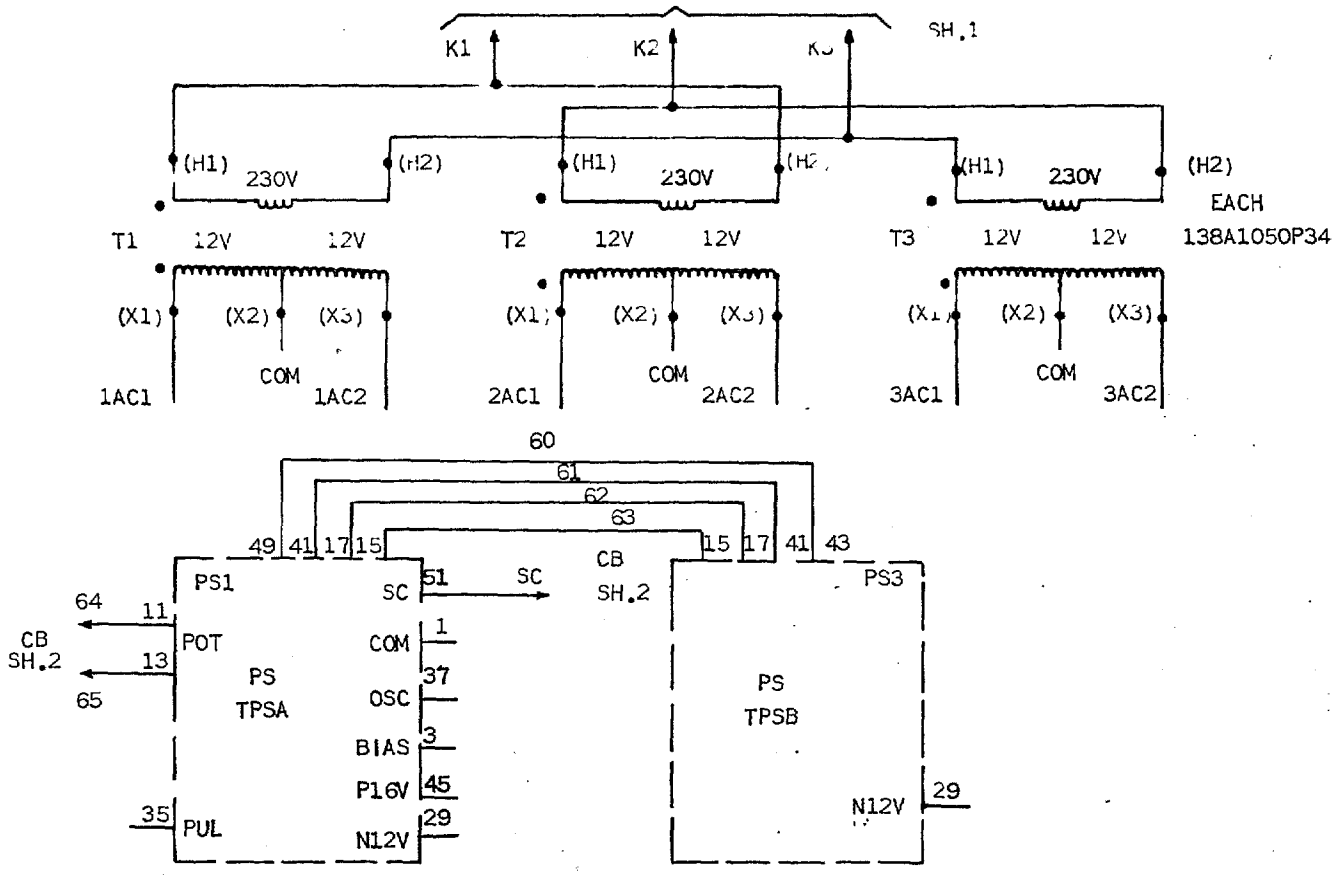
FORM IC 2143B (9-64)



* FC SEE GROUP TABLE (SH. 3)

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE SCR EXCITER 100 AMP 280 VOLT					
REV	REV							
MADE BY S. Wamsley	APPROVALS	PRINTS TO	P6A	DL1	1338	2184A6617		
ISSUE 4/17/68			6T			CONT ON SHEET 3		SH NO. 2

FORM IC 2143B (9-64)



WIRE TABLE FOR POWER AND REPETITIVE CONNECTIONS

WIRE NO \ DEVICE			F1P		F2P		F3P	PS1	PS3	T1	T2	T3
P16V			45		45		45	45	-	-	-	-
N12V			29		29		29	29	29	-	-	-
OSC			25		25		25	37	-	-	-	-
BIAS			13		13		13	3	-	-	-	-
COM			$^{1}_{51}$		$^{1}_{51}$		$^{1}_{51}$	$^{1}_{51}$	-	X2	X2	X2
1AC1			33		-			47	-	X1	-	-
1AC2			43		-		-	43	-	X3	-	-
2AC1			-		33		-	-	47	-	X1	-
2AC2			-		43		-	-	43	-	X3	-
3AC1			-		-		33	-	39	-	-	X1
3AC2			-		-		43	-	37	-	-	X3
PUL			27		27		27	35	-	-	-	-

GROUP TABLE

(P1)	VOLTAGE FEEDBACK	FREQ. CPS	*FC IC3600	▲CB PWB68A 993614	R1 10WATT	R2 10WATT	R3 .5WATT	C1 100V
C100CP250	YES	60	TFCU	G6	12K	12K	1.0K	200MFD
D100CP250	YES	50	TFCV	G6	12K	12K	1.0K	200MFD

(P2) – SAME AS P1 EXCEPT REPLACE PWB68A993614 WITH PWB219A3403G1 AND C1 = 20MFD AT 50V

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE SCR EXCITER 100 AMP 280 VOLT				
REV	REV 1 - AB 9-2-71						
MADE BY S. Wamsley	APPROVALS	PRINTS TO	P6A	1338	DL1	21A46617	
ISSUED 4/17/68			6T			CONT ON SHEET 4	SH NO. 3

FORM IC 2143B (9-64)

INPUT:

A.C. POWER: 230V ±10% AMP= 82% DC LOAD AMPS * 6,7TERMINALS: L1,L2,L3 *4

OUTPUT:

D.C. 250V 100 AMPS 55°C AMBIENT AT PACKAGE P+, N-
250V 150 AMPS FAN COOLED WITH 280 CFM

TRANSFER FUNCTION: (INCLUDES EFFECT OF TRANSPORT LAG)
RESPONSE TO 100% LOAD STEP .1 SEC.
REGULATION WITH LINE, LOAD, TEMP., ETC. ±1%

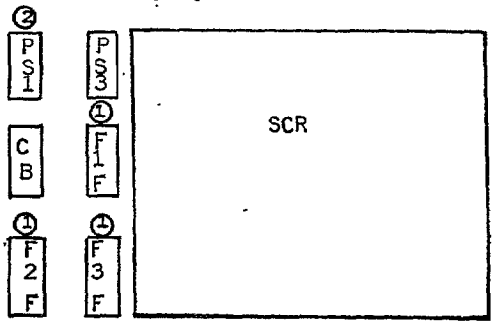
PACKAGE DIMENSIONS: 30"H x 24"W x 12"D (APPROX.)

OUTLINE: 157B6007P1
DULSEAL: 3851G

TEST POINTS

TP	① TFCU/TFCV	② TPSA
1	--	OSC
2	PA	GND
3	PB	--
4	AC1	+16V
5	--	BIAS
6	UNIJ.EMITTER	-12V
7	INPUT SIG.	--

T. P. POSITION

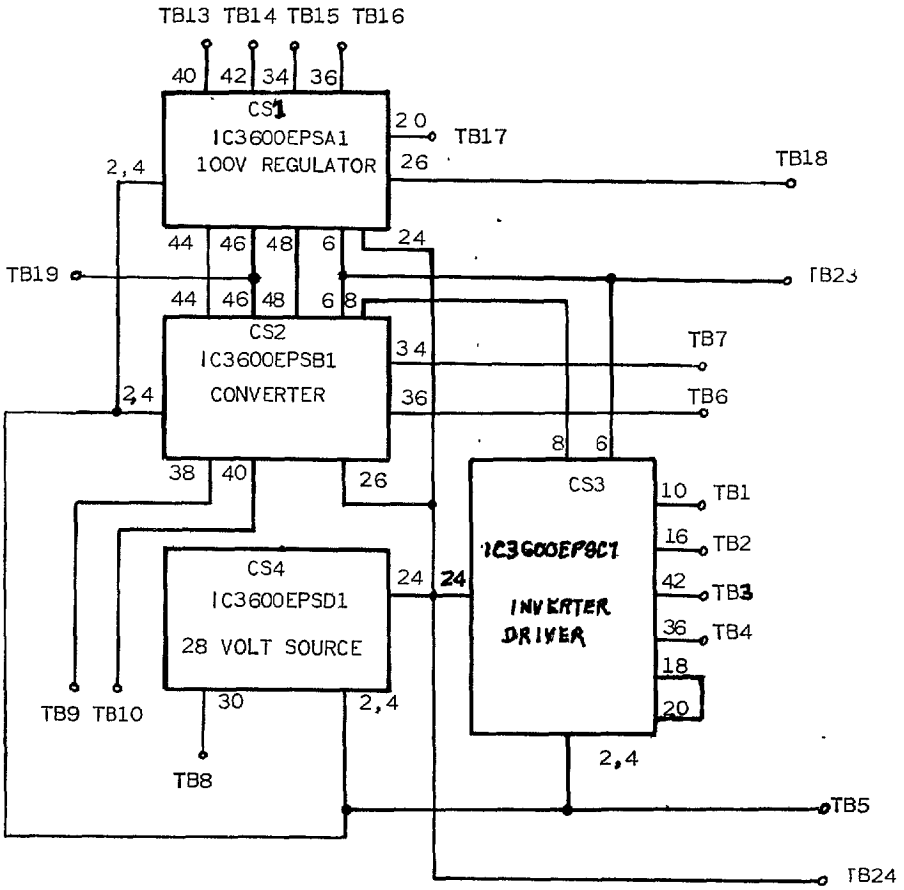


*NOTES: REFER TO GEK 8074 FOR DETAILED INFORMATION

1. DO NOT GROUND TERMINAL C WHEN INTERNAL FEEDBACK IS USED.
2. GROUND TERMINAL C OR USE AS COMMON WHEN NO FEEDBACK IS USED.
3. POWER TERMINAL P IS POSITIVE WHEN INPUT S IS NEGATIVE WITH RESPECT TO C.
4. PHASE ROTATION 1-2-3
5.
6. INDUCTIVE LOADS WITH .1 SEC. TC OR LONGER; RESISTIVE LOAD OR LESS THAN .1 SEC. TC - CURRENT MUST BE CALCULATED.
7. FREQ. TOLERANCE - ± 1/2 CPS.

REV	REV	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE SCR EXCITER 100 AMPS 280 VOLTS			
REV	REV					
MADE BY S. Wamsley	APPROVALS	PRINTS TO	P6A	DL1	1338	218A6617
ISSUED 4/17/68			6T			CONT ON SHEET FL. SH NO. 4

FORM IC 2143B (9-64)



FORM A - 60 HZ

FORM B - 50 HZ

REV J 2-10-69	REV 4	GENERAL ELECTRIC	TITLE		
REV 12 11	REV	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	REGULATOR - INVERTER CARD ASM. 2520		
MADE BY W. C. TUCK	APPROVALS	PRINTS TO	1338	IC3602A124	
DATE 2-19-1968			CONT ON SHEET	FL.	SH NO. 3.

21-38 (9-64)

I. SYSTEM DESCRIPTION

A. General

These instructions describe the operation, adjustment and maintenance of the electric control equipment for a Siltrol GE III mechanical winder. It is assumed that the reader is familiar with the equipment and the general operation of paper mill winders.

The descriptions, illustrations, and circuits used in these instructions are for a typical system. All the functions included here may not be present. See the elementary diagram furnished with the specific equipment for the actual functions provided.

B. Mechanical

Figure 1 shows the sheet flow from unwind to windup stands as well as the basic components of the mechanical system.

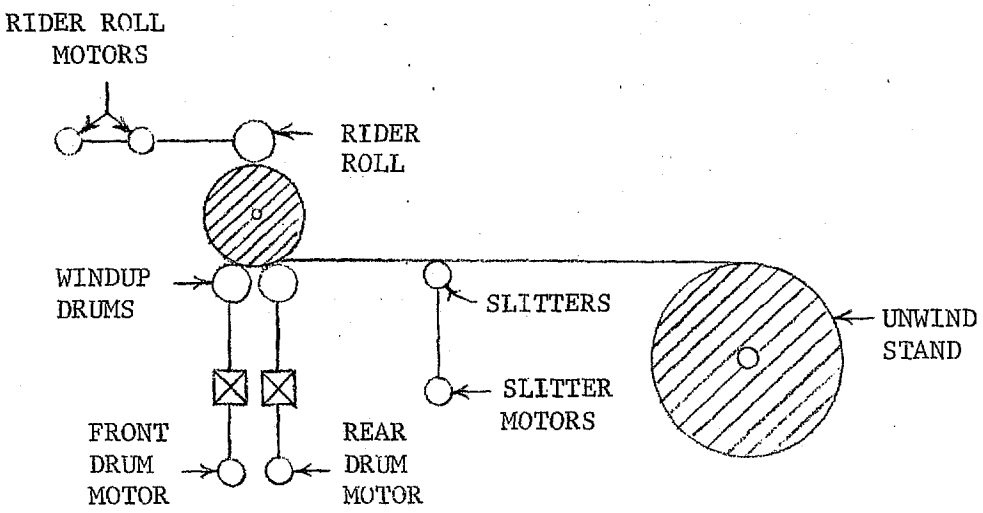


FIGURE 1 - MECHANICAL

C. Electrical

Figure 2 is the system electrical one line diagram. Note that each section (front drum, rear drum, rider rolls, and slitters) has its own static power supply. Front and rear drum power supplies are reversing to allow regenerative stops while maintaining sheet tension. The rear drum motor is speed regulated to control windup speed. The front drum motor is voltage regulated as a function of rear drum voltage with offsetting references to produce a voltage differential (and a corresponding torque differential) between the front and rear drums. The The rider rolls and slitters are strictly voltage regulated with the rear drum armature voltage as reference.

1.0 Regulator Page

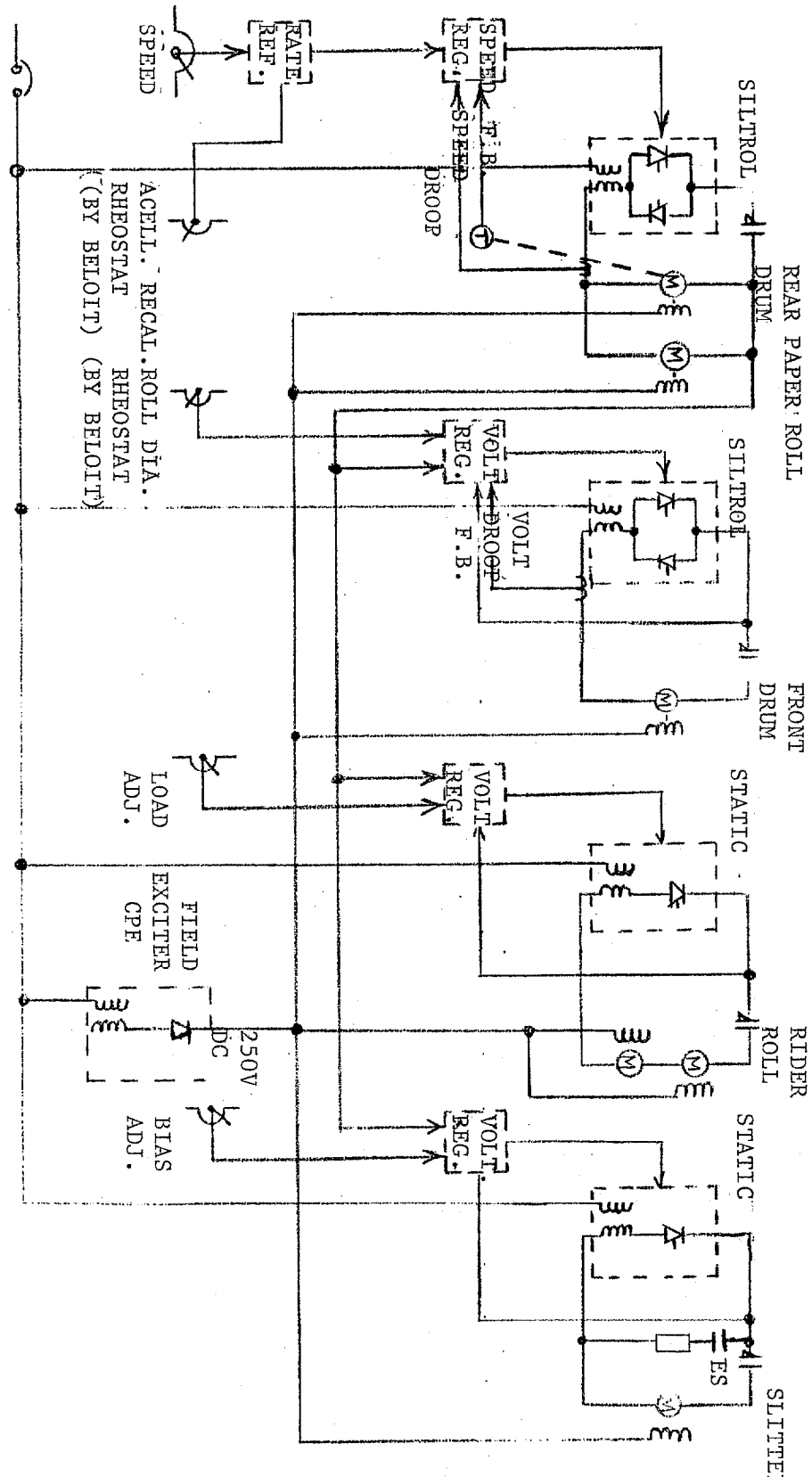
A modular construction concept is used in building the regulator page. Each regulating function has a corresponding module. A module is a combination of amplifiers and associated circuitry (designed for a specific function) mounted and wired in a self-contained housing. To gain electrical entry into the module, a connector card arrangement is used in conjunction with page wiring. All external wiring required to allow the module to function is wired to the right most card slot (S1) of the module. The module is mounted onto the page, and the wiring required for the particular module is brought to an adjacent slot (S2) to the module on the page. Then a connector card electrically connects the required wiring from the page to the module (S1 to S2).

Each slot in a module or page has the availability of accepting 51 wires for all types of applications (only a small number are actually used for a particular application). The connector card is the device which wires each pin of slot S1 to the corresponding pin of slot S2, i.e. (1 to 1, 2 to 2, etc.). It consists of two half cards with 51 wires. The portions of the connector card which plugs into S1 also connects 44 points (22 on either side of the card) of the card to terminations near the front. These terminations become available for monitoring when a diagnostic card is plugged onto this half card. Figure 3 shows this arrangement.

To gain electrical connections from external sources to the page, a ribbon cable is used. The external source is brought to a stab on connection on the base. The ribbon cable, one end of which plugs into the stab on terminal arrangement, carries the continuity of these external sources to the page and, consequently on to the modules. Figures 4A and 4B show the arrangement described above.

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ELECTRICAL ONELINE
FIG. 2

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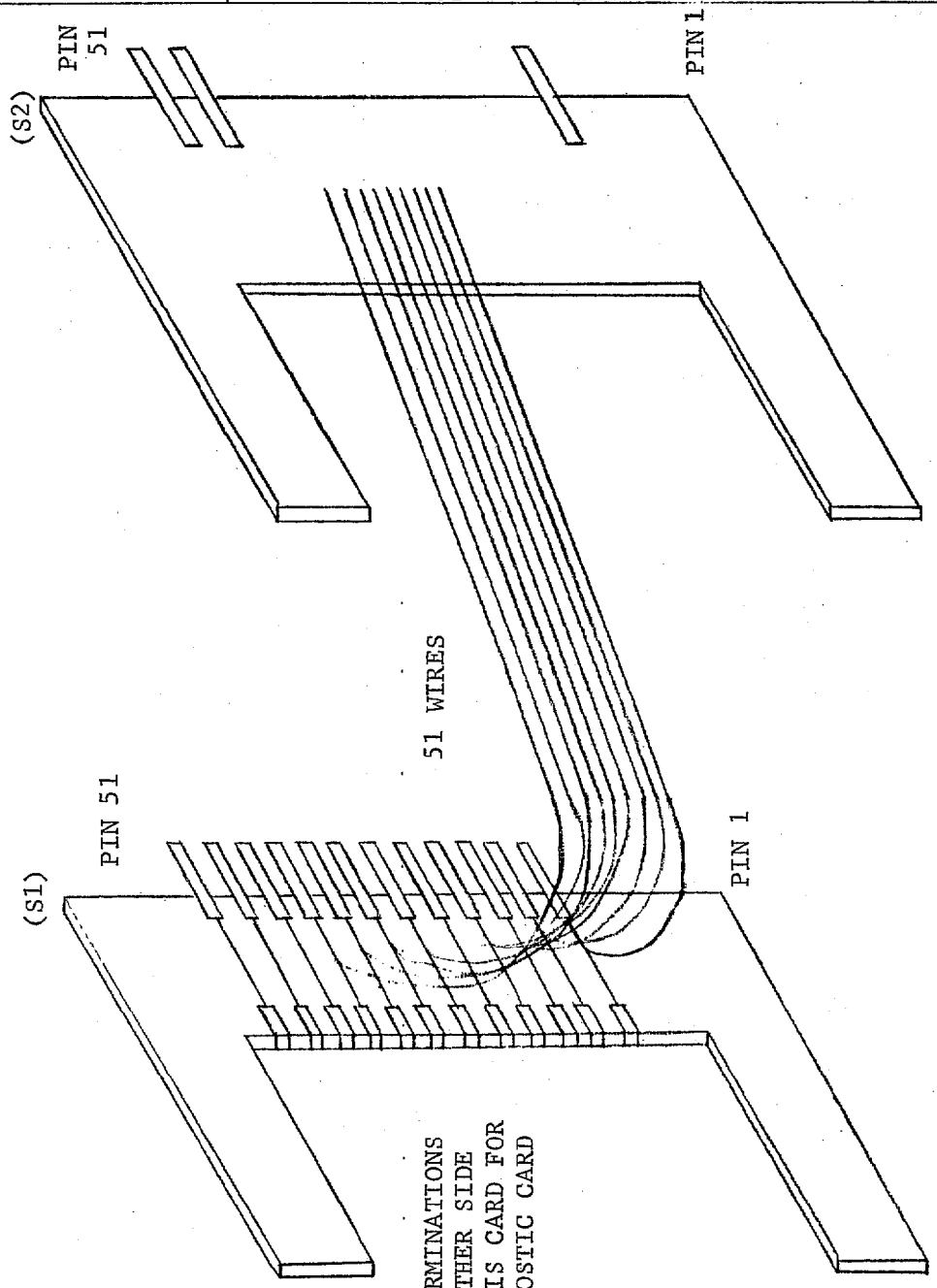


FIGURE 3 - CONNECTOR CARD

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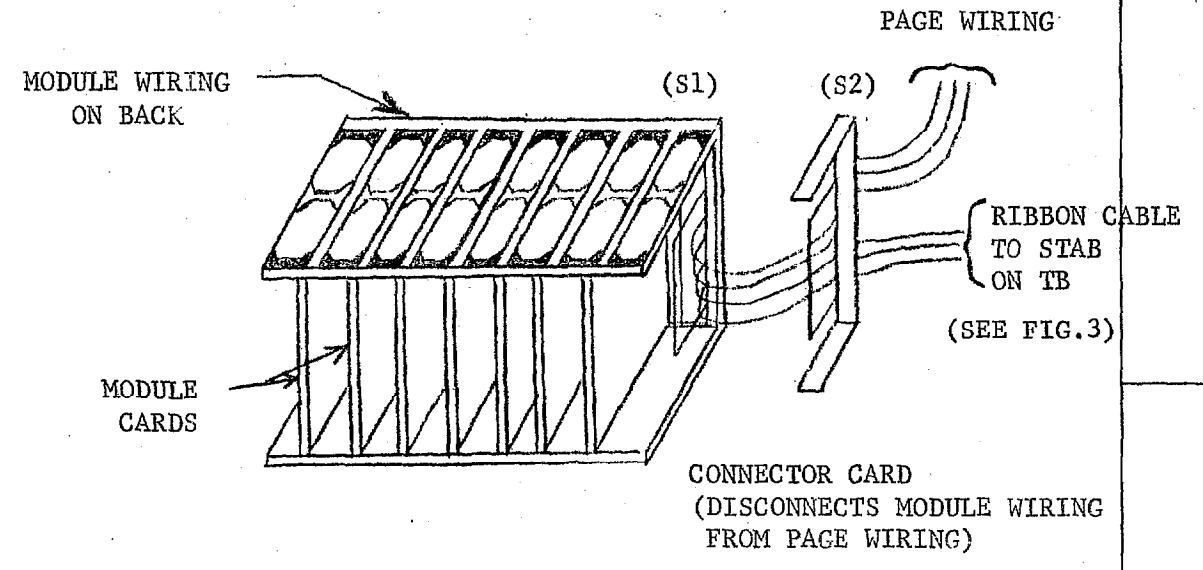


FIGURE 4A - MODULE

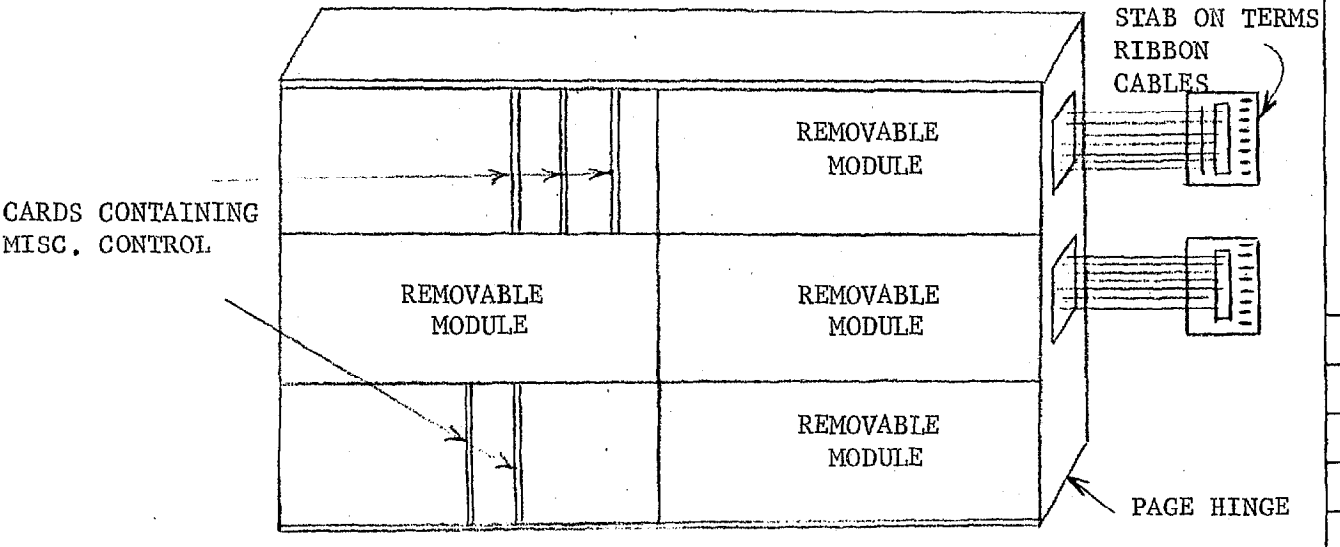


FIGURE 4B - PAGE (TYPICAL)

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2.0 Testing and Monitoring

For test and monitoring, a removable DOM II clip on card is used. It plugs into the diagnostic card slot (S1) of the connector card assembly. This allows 7 module points to be monitored. The output of the module is monitored on a test jack and voltmeter. The other 6 points are sensed by test jacks. This card is cat no. 232A3003G3 and is shown along with a table of typical voltage readings for the monitored points on Sh. 1B of the drive elementary.

In the test mode of operation, a number of steps are required. These steps are necessary to disable equipment operation and set up the test circuitry. A selector switch is provided to make the operator's initiation of the drive sequence inoperative.

As previously described, in the normal mode, the connector card connects all the external signals to the module. These include reference, feedback, switching interlocks, etc. To properly test the module regulating amplifier, most of the external signals must be removed. This is simply done by removing the connector card. However, since the power supply is required for the operational amplifier, a different connector card is used. This is known as the "Test" connector card. This test connector card has only five wires instead of the normal 51 wires. This will allow the following external signals to be connected to the module being tested - +50V, ACOM, -50V, normal input and output. The normal input and output continuity is maintained in the test mode as described later.

3.0 Regulator Test - Quick check.

For ready reference this description is included in the back-sheets of the drive elementary.

4.0 Module Test

The module test consists of checking the regulating amplifier in each module. This is accomplished by reconnecting the amplifier reference and feedback circuitry and introducing external signals to swing the amplifier. The module test card is used for this purpose.

Module Test

The following procedure will allow a test set up:

1. Stop the drive.
2. Select test on the panel selector switch.
3. Interchange the normal connector with the test connector card.

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The above sequence will yield the circuit shown in Figure 5.

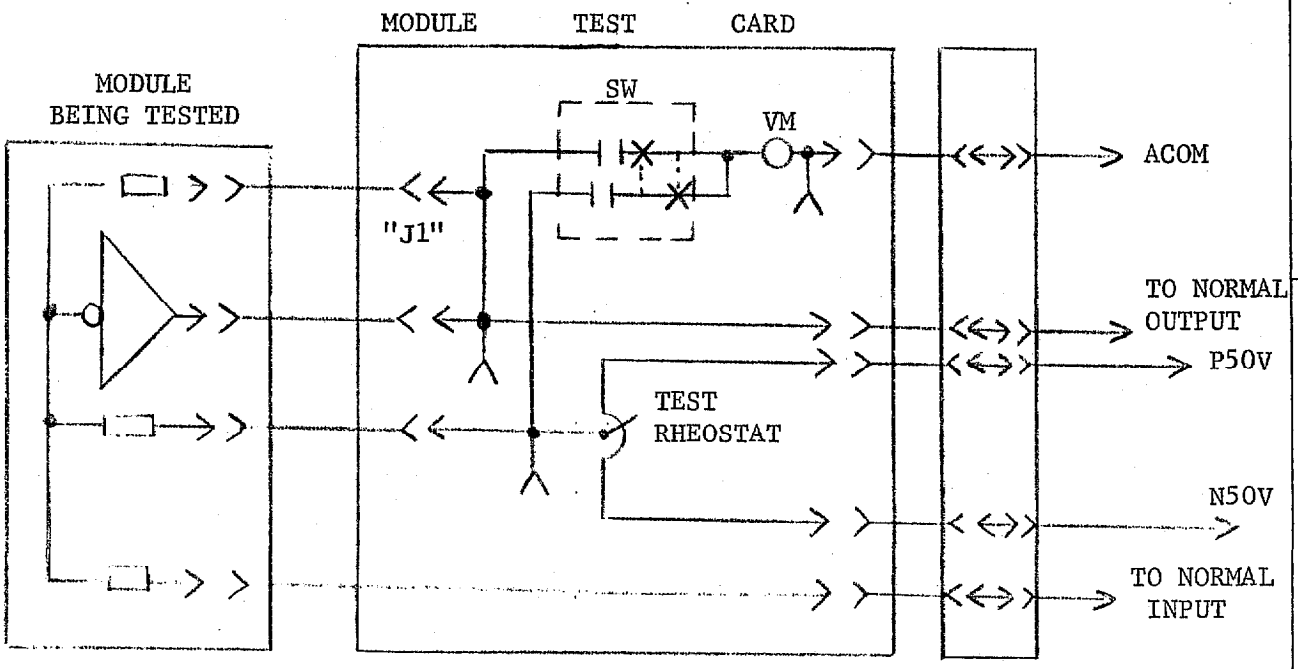


FIGURE 5 - TEST SET UP

With the above arrangement, the normal input will be zero. The amplifier being tested is set up for fixed gain by appropriate wiring in the module test card (J1). The equipment supplied on the module test card is as shown; a switch, to connect the voltmeter to either the test rheostat or the amplifier output, test jacks for monitoring these same two points plus ACOM and a rheostat to swing the amplifier.

The test rheostat is used to swing the output of the operational amplifier.

Since the continuity of the output is maintained, whatever is connected to the output will follow the amplifier output voltage. If the armature power supply is connected to the output, then its output will change.

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5.0 Cascade Test

Since the regulator page consists of a cascade arrangement of modules, a cascade test is available. Each of the modules to be tested in cascade, are set up exactly as described under the Module Test Card description. All the test rheostats are set for zero volts. Then, the test rheostat for one of the modules is adjusted and all the amplifiers in the modules to which this module output is connected will follow the voltage. Since all the modules do not have the same input resistors, the amount of amplifier swing will not necessarily be one for one. The different input impedances will give different gains and different output voltages.

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II. CONTROL OPERATION

The following will describe the reference excitation, speed regulator for the rear drum motor, front drum torque regulator, and voltage control for the rider roll and slitter circuits.

A. Reference Excitation (Refer to Drive Elem. Sh. 1A)

The exciter panel supplies the 250V DC excitation for the shunt field of all motors. This voltage is supplied by an unregulated constant potential exciter. For 110V power requirements, a step down power transformer is mounted and wired on the panel.

B. Rear Drum Control (Refer to Drive Elem. Section 2)

To control the speed of the winder, a speed regulator is used on the rear drum motor. Different modules containing the regulating functions are combined in page form to perform this function.

1.0 Magnetics. The winder control functions thread, run, normal stop, jog, and E-Stop are controlled by pushbuttons. A selector switch is provided for stall tension.

PROTECTION: The UV relay initiates the protective sequence. Whenever the protective interlocks (overcurrent, armature supply fault, etc.) open, the UV relay will drop out causing a coast stop.

THREAD: When the operator pushes the THREAD pushbutton, the drive will accelerate on a ramp to minimum running speed. From the thread position, the operator may go to either the run mode or initiate a normal or E-Stop by operating the corresponding pushbutton.

RUN: When the operator pushes the RUN pushbutton, the run relay (R) will energize closing in the line contactor (M) and the winder will accelerate on a ramp to the speed corresponding to the setting of the master speed rheostat (RHM).

NORMAL STOP: Operating the normal stop pushbutton causes the drive to ramp to zero speed. When the speed nears zero, the zero speed relay drops out opening the M contactor.

E-STOP: This provides the same type of stop as a normal stop except at a faster rate.

JOG: Operation of this pushbutton causes the drive to accelerate to a preset speed for the duration of the time the pushbutton is held down. Note, this reference does not operate through the rate; however, current limit backup is still present.

STALL TENSION: When this selector switch is placed in the ON position, the drive will hold stall tension whenever it is at zero speed. The drive will return to stall tension at the end of a run as soon as the zero speed relay drops out.

NIP GUARD: When the nip guard relay drops out, the speed reference is switched to its minimum value.

2.0 Regulator. Refer to Figure 6 for a block diagram of the unwind speed regulator. This regulator consists of the ARC, SPD, and CL modules.

a. ARC Module

This module provides a rate of change circuit for ramping the speed reference signal. At steady state, the output (net speed reference) will equal the input (operator's setting). If the input is quickly changed from one value to another, the output voltage will ramp to the new value. The module also provides for rounding into and out of the ramp.

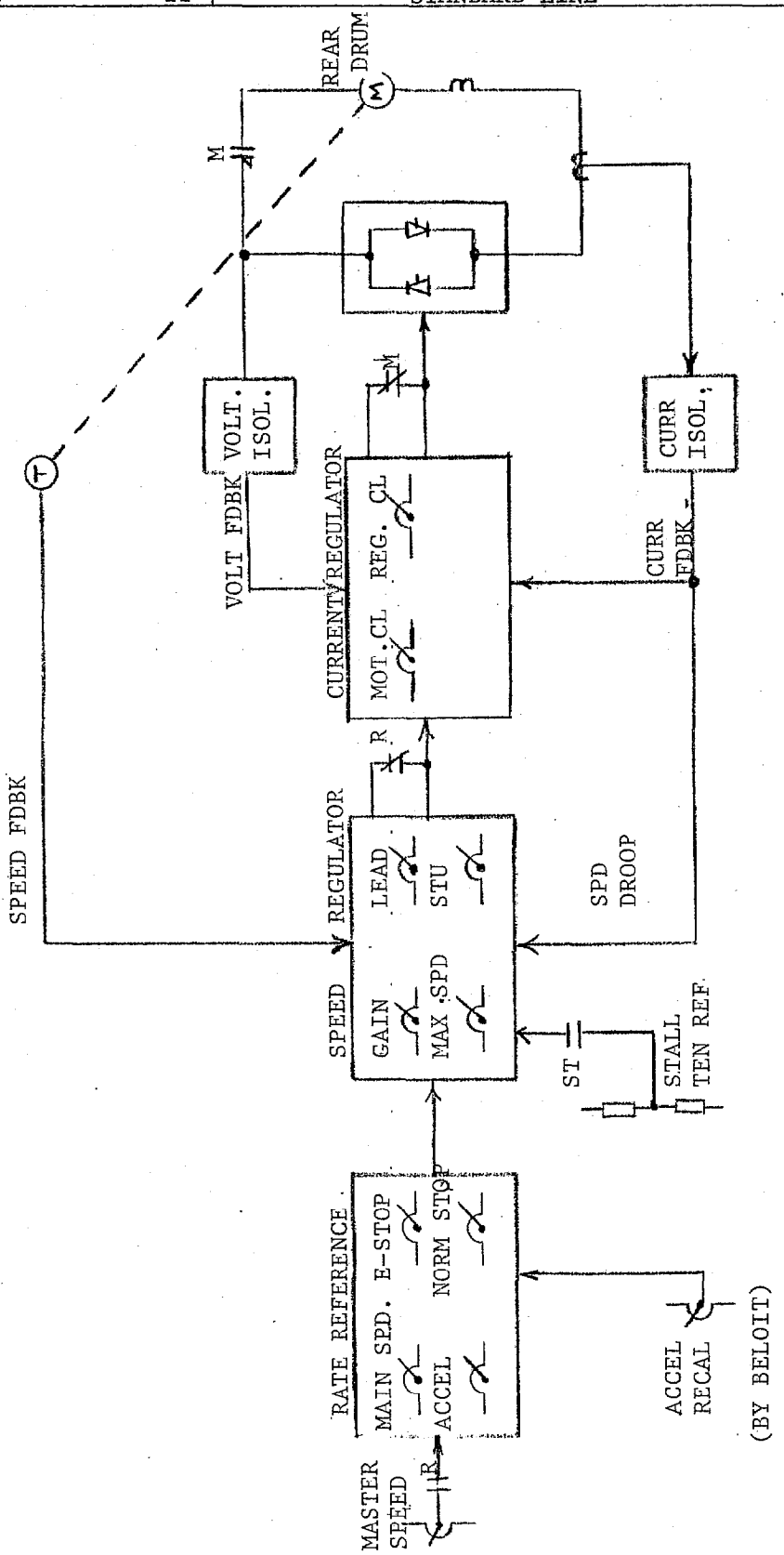
b. SPD Module

The speed module compares the net speed reference signal to the speed feedback (obtained from the tachometer). The difference is amplified and inputted to the CL module. This, as will be described, causes the motor speed to change until at steady state, the speed is held constant and proportional to the net speed reference.

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REAR DRUM REGULATOR

FIG. 6

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c. CL Module

In the RUN mode, the CL module adjusts armature current constant and proportional to its reference (the output of the SPD module). If its input changes, the output of the module will change the reference to the armature power supply. This changes the armature supply output voltage and hence changes the armature current.

d. System Operation

The curves plotted on Figure 7 show the various inputs and outputs of the previously described module during an acceleration period from 10% to rated speed, illustrating the following:

The speed reference is adjusted from 10% to rated. The ramp output of the ARC module causes the output of the SPD module to rise to a fixed level. The CL module output causes armature current to increase and the motor to accelerate. As top speed is reached, the terminal volts and the CEMF approach each other and the output of the SPD module and armature current decrease to the running load level.

The dotted line in the plot of armature supply output is CEMF. The difference between the two plots is the voltage available for the IR drop.

For a detailed description of module operation refer to the module application data.

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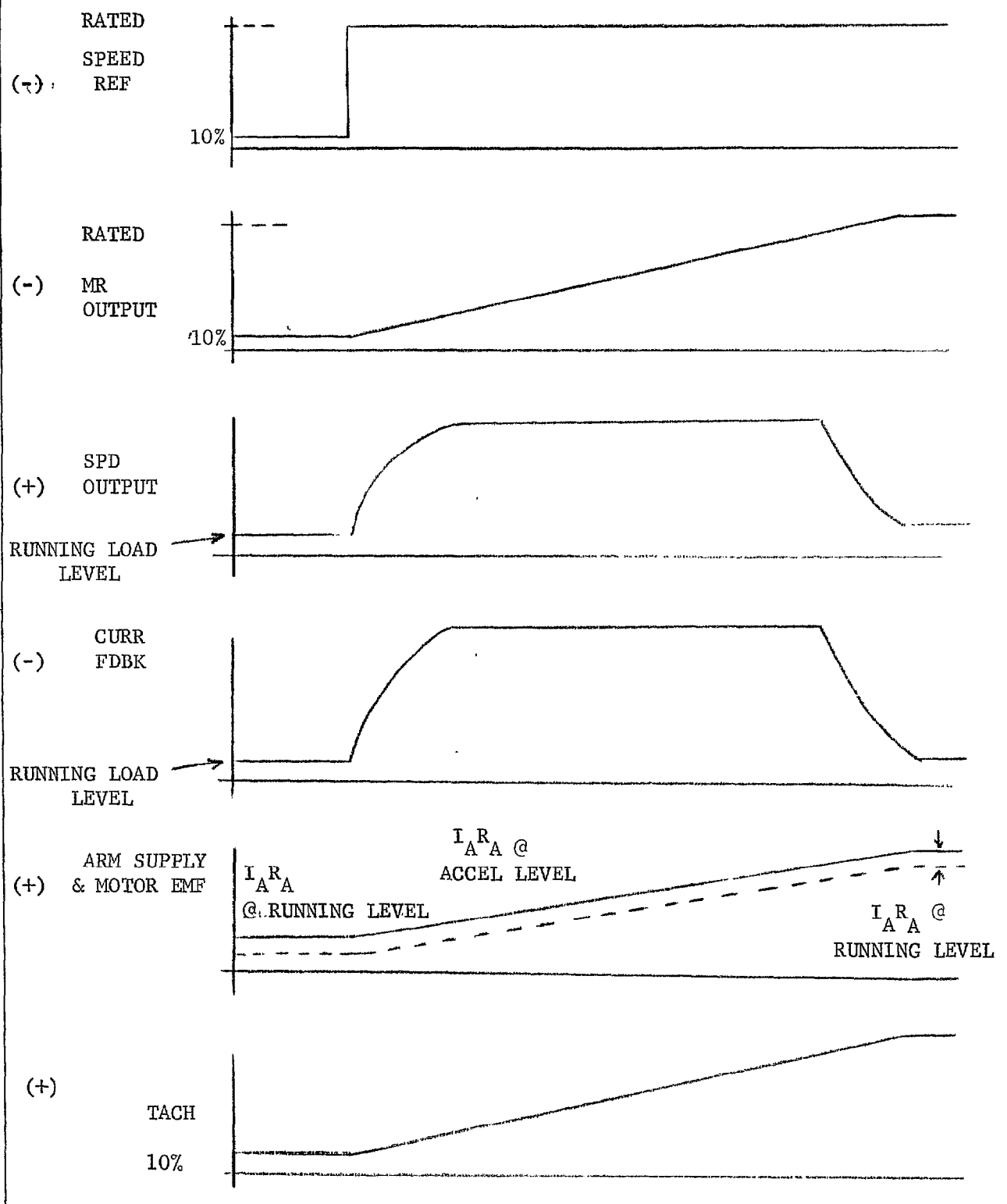


FIGURE 7

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C. Front Drum (Refer to Drive Elem. Section 3)

The front drum regulator is basically a voltage regulator with spillover current limit. A single module combines both these functions.

1.0 Magnetics. The front drum regulator follows the rear drum in all modes of operation. Front drum fault conditions either coast stop or E-stop the winder. For example front drum power supply fault conditions coast stop the winder while an instantaneous overcurrent E-Stops the winder.

2.0 Regulator. Refer to Figure 8 for a block diagram of the front drum voltage regulator.

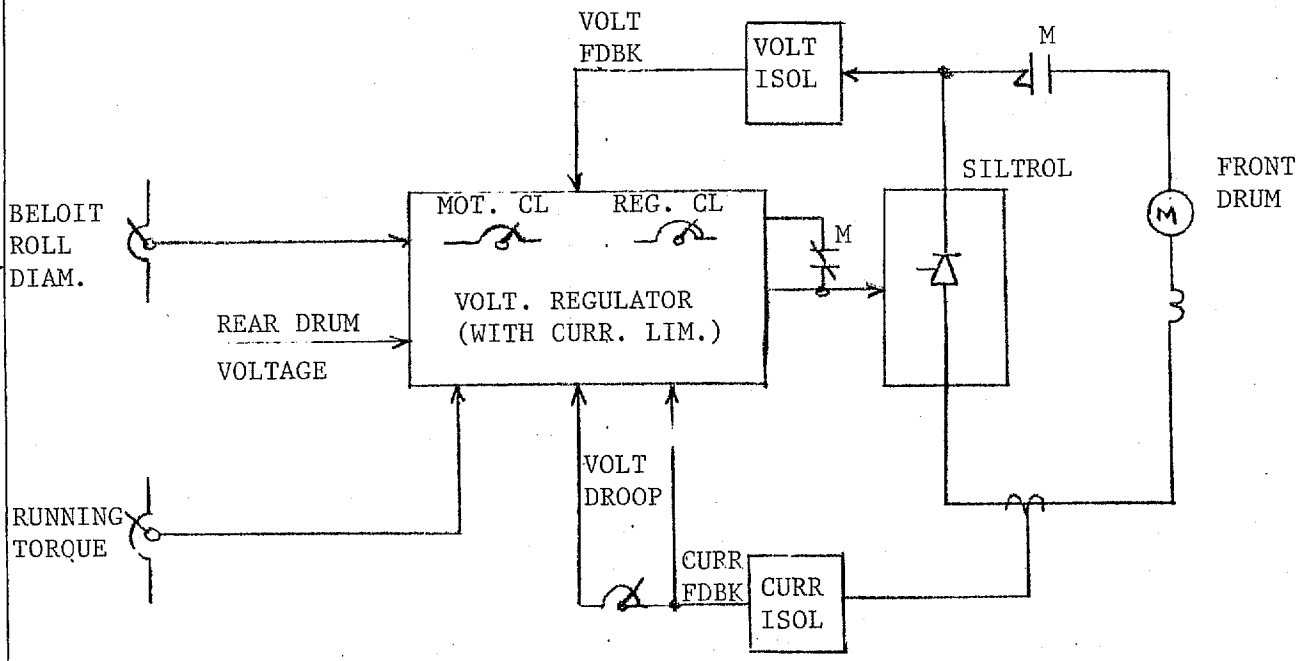


FIGURE 8 - FRONT DRUM VOLTAGE REGULATOR

The purpose of the front drum voltage regulator is to provide a torque differential between the front and rear drums. This differential is provided by the sum of the Beloit Roll Diameter Rheostat and the Running Torque Adjustment Rheostat.

Suppose the two auxiliary references are reduced to zero, then the reference to the front drum voltage regulator is supplied only by the rear drum voltage feedback. Since the front and rear drums are loaded identically and their armature voltages are tracking, the torque differential will be zero. Now suppose the running torque adjustment is increased. The front drum voltage will still follow the rear drum but it

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will have an offset included as shown below, (Figure 9.) This offset increases the armature current of the front drum creating a torque differential between the two drums proportional to the offset reference.

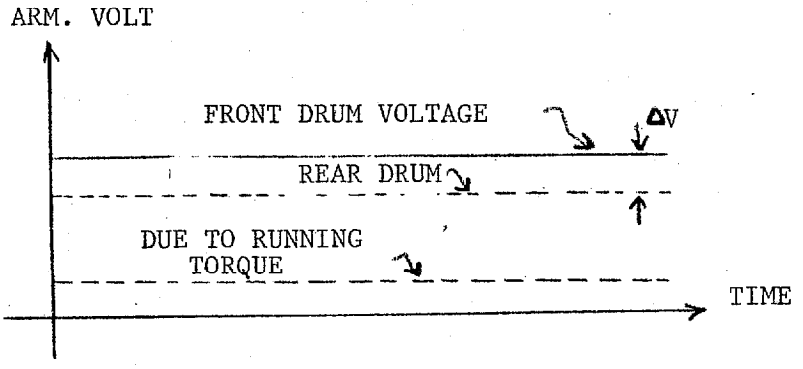


FIGURE 9

Now if the reference from the Beloit Roll Diameter is added the composite front drum voltage will appear as shown below.

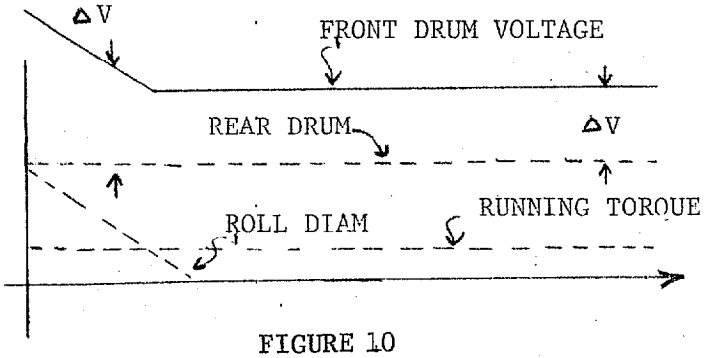


FIGURE 10

This armature voltage profile allows for a greater torque differential when the windup roll is small (resulting in a tighter core) gradually tapering into a fixed value as the roll builds up.

Voltage droop is present on this regulator to counteract a motor characteristic of rising speed with increased load. This function is not needed for front drum regulator stability, but is necessary for system stability. Voltage droop tends to offset the ability of a front drum motor with a rising speed charactistic to tend to take load from the rear drum during heavy load conditions.

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D. Rider Roll (Refer to Drive Elem. Section 4)

The rider roll regulator is a voltage regulator (no current limit) with the rear drum voltage feedback as reference.

1.0 Magnetics. The rider roll control follows the rear drum in all modes of operation providing the rider roll OFF/ON selector switch is in the ON position. If this switch is placed in the OFF position during winder operation, the rider roll only will coast stop. A dynamic braking stop occurs when the rider rolls are fully raised and the extreme travel limit switch picks up. Interlocking is provided to prevent starting the rider roll on excessive voltage.

Static armature supply protective devices (if present) and rider roll over current (OC) will E-stop the winder. Rider roll over load (OL) will normal stop the winder.

2.0 Regulator. The rider roll voltage regulator consists of a voltage regulating module (VR) and a non-reversing static armature supply.

Adjustments are provided (in the VR module) for maximum volts, stability and load compensation. During standby operation, the reference to the VR module is suicided in order to regulate 0 volts on the armature supply.

The block diagram for the rider roll regulator is shown below.

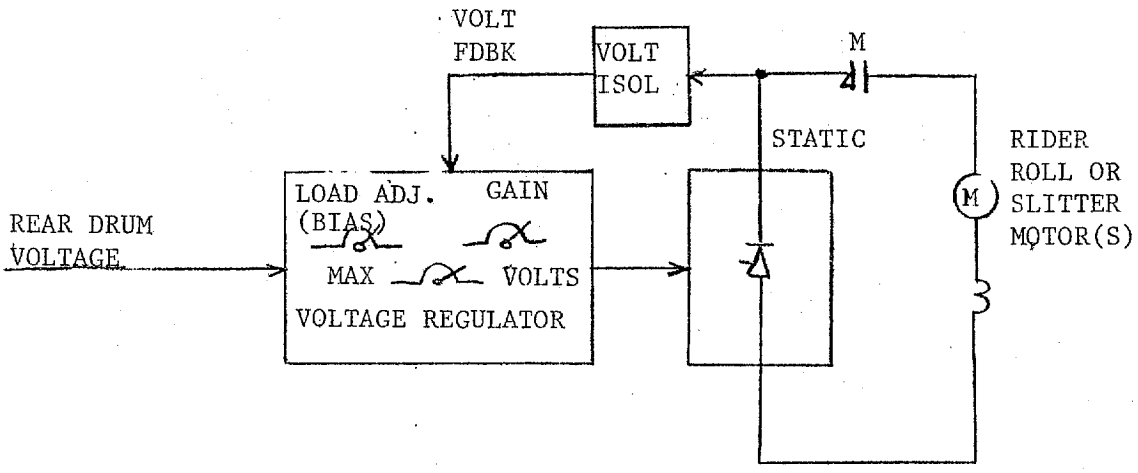


FIGURE 11 - RIDER ROLL OR SLITTER BLOCK DIAGRAM

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E. Slitters (Refer to Drive Elem. Section 5)

The slitter regulator is a voltage regulator with features similar to that of the rider roll.

1.0 Magnetics. The slitters follow the rear drum in all modes of operation providing the "Slitter OFF/ON Selector Switch" is in the ON position.

The slitter control is interlocked to prevent starting under excessive voltage by requiring the winder to be at zero speed to select slitter ON.

Slitter instantaneous over current (OC) causes the winder to E-Stop. Slitter thermal over load (OL) causes the winder to normal stop.

A pump-back resistor is present in the armature circuit so that during winder E-Stop, the slitter inertial energy can be dissipated to allow the slitter speed to track that of the rear drum. If this resistor were not present the slitters would simply coast stop causing tracking error.

2.0 Regulator. The slitter voltage regulator consists of a voltage regulating module (VR) and a non-reversing static supply.

Adjustments are provided for maximum volts, stability, and load compensation. The load compensation pot is used in this case to set the slitter minimum speed.

See Figure 11 for the regulator block diagram.

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III. INSTALLATION AND TUNE-UP

A. Preliminary Inspection

1. Visually inspect the equipment for shipment damages.

2. If equipment is to be stored before use, provide adequate protection against the elements and other contaminating conditions.

3. If the equipment has been stored or unused for prolonged periods, be sure all equipment is dry and clean before energizing.

4. Manually move all mechanical devices to check for binding or misalignment.

5. Check to see that all interconnecting wires and remote devices agree with the elementary diagram.

6. Check shielded wires to confirm that the shields have been tied down in accordance with the interconnection drawings.

7. Check terminals for loose connections, electrical clearance, etc.

8. Make sure all contacts are clean and properly positioned.

9. Check motor armatures and fields for proper ohms and grounds. In checking for grounds, avoid subjecting the control equipment to voltages which might cause damage to the semiconductor components.

10. Insure that the voltage and current feedback signals are being extracted from the proper points on the motors and armature power supplies.

11. Assure adequate electrical, thermal and mechanical clearances have been maintained.

12. Establish effective communication between control panels and operator's desk, prior to actual tune-up of equipment.

B. General Tune-up Procedure

Experience has indicated that by following certain procedures, a more trouble-free and less expensive installation will result. Some of these procedures are listed below. The following instruction should not be interpreted to represent all conditions which might be encountered, or need be considered during any start-up. The actual start-up procedure should be established by those responsible, giving due regard to safety of personnel and

protection of the electrical and mechanical equipments involved. The following instructions assume that the remote-mounted equipment is in proper working order. It is also assumed that the preliminary inspection has been completed and the system is free from grounds, haywires, etc.

C. Initial Conditions

Before applying power to the control equipment, open the control power disconnect switches and disconnect one motor armature circuit cable to each motor at the control panel. Also disconnect one motor field lead to each motor at the control panel.

D. Exciter Section

Close the AC disconnect device to apply power to the power transformers on the exciter panel. Check that each transformer has the proper output voltage. Also verify that the CPE output voltage is 250V when the motor fields are reconnected.

E. Magnetics Check

1.0 With the relay excitation applied, check the protective relaying by opening the protective circuits in the UV relay coil circuit. With the UV relay dropped out, operation of the thread, run, and jog functions should not be effective.

2.0 Disconnect the 67V relay excitation to the regulator relays. Initiate a run and jog and observe that the proper sequence occurs with the line contactor energizing. While in each of these modes initiate a normal and E-Stop and check that the stopping sequence occurs with the appropriate seal-in-circuits being opened. Reconnect the 67V relay supply.

F. Front Drum Voltage Regulator

1.0 Module Test.

a. Set up the CL module in test mode and check for proper module operation. NOTE: Pickup M contactor.

b. Set the test rheostat for (+) 15V out of the CL module. Check for rated siltrol voltage (+500V).

2.0 Voltage Regulator

a. Disconnect the winder voltage reference to the CL module. Jumper res. R71 location ROF(7-9) in regulator page. Use R9 (in front drum page) as a voltage reference.

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b. Reconnect the armature and field leads for the front drum motor only,

c. Beloit roll diameter should be fully clockwise.

d. With drive unloaded, pick up the M contactor. Slowly increase the reference and note that voltage is being regulated and the drive is stable. (Full clockwise on RH9 should give approximately 190V. out of Siltrol).

e. Set RH31 and RH32 for the desired values for regenerating and motoring current limit respectively, 0-200% is the adjustment range.

f. Remove jumper resistor R71 and reconnect the winder voltage to the CL module. Disconnect the front drum armature and field leads.

G. Rider Roll Voltage Regulator

1.0 Module test

a. Set RH4 of VR module at a setting of 5.

b. Pickup the Rider Roll Contactor, using RH2(VR) as a reference, swing the output of the VR module from 0 to (-) 16 volts.

2.0 Voltage Regulator

a. It is not necessary to reconnect the armature and field leads of the rider roll motors for this test.

b. With RH2 (VR) fully CW, pickup the rider roll M contactor. Increase the rider roll armature voltage by turning RH2 (VR) CCW. Note that voltage is being regulated.

c. It should not be necessary to check the step response of this regulator. If it is desired however, the regulator should be set up for an exponential response with a time to 63% of the final value in approximately .05 to .1 seconds. See the VR module instructions for this adjustment.

H. Slitter Voltage Regulator (If Present).

Follow procedure given for Rider Roll Tune-up, Section III G. Substituting slitters for Rider Rolls.

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I. Rear Drum Speed Regulator

1.0 Module Test

Set up each module in the test mode and check for proper module operation both individually and in cascade. Note this is not necessary for the ARC module.

2.0 ARC Module

Determine the minimum speed level and the time for acceleration and deceleration. Remove the connector card from the SPD and CL modules. Initiate a run to the minumum speed level. Adjust the minimum speed pot (located in ARC module) for the desired thread speed output of the ARC module, i.e., 3V for 10%, 0.3V for 1% speed.

With the Beloit Robinson Helpern Rheostat fully CCW, rotate the operator's speed adjustment to rated. Adjust the ACCEL rate pot (ARC module) for the maximum accelerating rate. Set the speed reference at zero. Adjust the normal stop pot (ARC Module) for the required deceleration time. With the speed reference at rated, E-Stop the winder and set the E-Stop pot (ARC Module) for the required emergency stop time. For a more detailed description of these settings, refer to the ARC module instructions.

3.0 CL Module (Replace the CL Module Connector Card)

Set the SPD module in the test mode and adjust the test rheostat for zero volts output of the SPD module. The motor field of the windup motors should still be disconnected. Reconnect the armature lead of the rear drum motor and initiate a RUN.

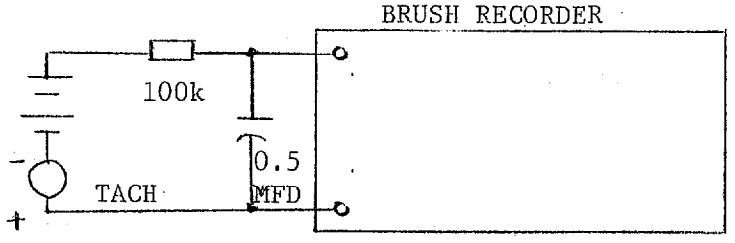
In the following check, it should be remembered that armature current will be circulating at locked rotor and the current should not exceed 50% of rated and not for an extended period. Adjust the output of the SPD module with test rheostat for +1.8V, the armature current of the rear drum motor should increase to approximately 25% of rated. The current feedback should be -1.9V corresponding to -7.5 volts at rated armature current. Observe that the armature current is not oscillating violently, this would indicate an unstable system. Adjust the SPD module output for -1.8V and the armature current should go to approximately -25% of rated. Stop the drive. Reconnect SPD module connector card.

4.0 SPD Module

As initial settings, set the LEAD pot in the SPD module for half rotation and the GAIN pot full CCW. Reconnect the motor field, initiate a run to minimum speed. Observe stable operation at minimum speed.

The +CL and -CL pots in the CL module set the maximum values of motoring and regenerating currents respectively. Full CW rotation will limit current to approximately 200%. Set these for the desired level of maximum armature current.

Slowly increase the speed to rated. When maximum speed is reached, adjust the maximum speed pot for gear in speed. Adjust the rear drum motor field resistor so that the armature power supply is just below rated voltage. Bring the speed down to approximately 30% of rated and connect the tach feedback to a brush recorder and buck out the tach voltage as shown in Figure 13. This will allow the response of the speed regulator to be analyzed.



The jog circuit will be used to provide a step in speed reference. Rotate the STU pot in the SPD module to 10% rotation. (Sometimes, capacitors are added in the jog reference circuit to provide a lag in reference. If this is the case, remove them for this setup.)

The response of the speed regulator should be adjusted (when paper becomes available) with 60% of maximum paper on the wind-up rolls. The inertia time constant varies as the roll builds up. With 60% of maximum paper the inertia time constant is about at an average value. The response should be set for a critically damped response. Then, when the roll builds to maximum, the overshoot won't be excessive and conversely, at core the undershoot is acceptable.

To achieve a critically damped response to the step in reference (JOG), the GAIN and LEAD pots are adjusted. The time to 63% of the final value should be adjusted for approximately 1 to 2 sec.

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The GAIN pot effects the time of the response and the LEAD pot adjusts the overshoot. There is interaction between the two adjustments so a compromise setting is required.

If the drive is difficult to stabilize at high speeds, the problem may be a lack of inherent motor speed droop. A speed droop circuit is present in this regulator (see ARC module). To counteract this condition, start with 0 speed droop and add droop (with jumpers) until stable operation at all speeds is attained.

Stop the drive and initiate a JOG, set the STU pot in the SPD module for the desired jog speed.

J. System Tune Up

1.0 Reconnect the armature and field leads for the front and rear drum motors.

2.0 With the Beloit Roll Diameter Rheostat and the Running Torque Rheostat at 0 volts, initiate a run to thread speed with the drive unloaded. Observe that the front drum voltage is approximately tracking the rear drum voltage.

3.0 Gradually take the drive to rated speed. Adjust the front drum motor field resistor so that front drum speed matches the rear drum speed.

4.0 Reconnect the rider roll motors armature and field leads.

5.0 With the load adjust pot (RH2 of VR) at 0 volts, initiate a run to rated speed. Set the max volts pot (RH3 of VR) for rated armature voltage on the rider roll armature supply. Set the rider roll motor field resistors for rated speed at rated voltage.

6.0 Reconnect the slitters (if present), armature and field leads.

7.0 With the load adjust pot (RH2 of VR) at 0 volts, initiate a run to rated speed. Set the maximum volts pot (RH3 of VR) for rated armature voltage on the slitter armature supply. Set the slitter motor field resistors for rated speed. Set RH2 of the VR module for the required slitter speed differential.

8.0 High speed / Low speed operation (if supplied). Select low speed mode of operation. Set the unshorted field resistors of the rear drum and front drum for rated low speed so that rated low speed is attained with the armature power supply just below rated. (Note: Readjust the high speed field resistors if necessary at the completion of this section).

Rated low speed reference is set by adding a resistor in slot H (8-9) of the ARC module. This recalibrates the input reference to the rate reference. This resistor has already been installed in the factory. If necessary it may be altered to obtain the correct results.

Reduce the reference to thread speed. Thread speed in low speed operation should be the same as thread in high speed operation. The resistor located in slot H (7-10) (ARC) recalibrates the thread speed.

Increase the drive to rated low speed. Adjust the resistor located in slot H (17-20) (ARC) for rated rider roll and slitter speed if necessary.

9.0 Stall Tension

With paper on the winder, select stall tension with the drive at zero speed. Approximately 10% rated armature current should circulate in the front and rear drum motors.

If necessary to modify the required current adjust the value of resistor R2 mounted in the rear drum control panel.

10.0 Running Torque

With the winder operating with paper, adjust the running torque pot (located in the front drum regulator page) for the torque differential required for a tight roll.

11.0 Rated Core Torque

With the Beloit Roll Diameter Rheostat fully CCW, adjust the rated core torque pot (located in front drum regulator page) for the required starting torque differential between the front and rear drums.

12.0 Rider Roll Load Adjust

Adjust the Rider Roll Running load with RH2 of the VR module.

13.0 Front drum voltage droop. If problems are encountered with slightly oscillating armature current in the rear drum and front drum motors while operating under heavy load, the cause could be a rising speed with load characteristic of the front drum motor, This condition may be corrected by adding voltage droop on the front drum regulator, RH10 (located in the front drum regulator page) adjusts the amount of voltage droop.

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IV. TROUBLE SHOOTING

A. General

The trouble shooting discussion is divided into two sections. One section deals with system type problems; the other section deals with regulator problems.

B. System

Rather than list a procedure for system trouble shooting, which is impossible, a discussion of possible system problem areas is undertaken.

1. Ripple in the Tachometer Generator Voltage

The speed-regulating preamplifier (SPD) can operate satisfactorily with a moderate amount of ripple in the tachometer voltage; however, excessive ripple will cause "saturation" of the amplifier (especially at high operating speeds) and will result in poor speed regulation.

When properly connected to the drive, the tachometer generator will normally have a ripple voltage of less than 2 percent of its operating d-c voltage, and the major or primary frequency of the ripple will be two times the tachometer revolutions per second. If the ripple exceeds this level, check the tachometer for shorted windings or commutator bars, and check to see that the tachometer is properly aligned with the shaft that drives it. Also, check to see that the tachometer is not being subjected to end thrust through the shaft coupling and that there is no backlash in the tachometer coupling.

In any event the tachometer ripple voltage must not exceed the following value or the operation of the regulator will be impaired:

Ripple peak-to-peak volts = 0.0017 x ripple frequency

x tach volts at rated machine speed

2. Backlash in the Drive Gears or Couplings and Shaft Twist Between Motor and Load

Backlash in the gears or couplings between the motor and the machine may effect the stability of the speed regulator. As the response of the regulator is increased, a point may be reached where the motor will "rattle" or oscillate back and forth in the backlash region of the gears and/or coupling. The oscillation will be most noticeable on release of slack takeup. Continued operation in this condition can cause damage to the gears.

Shaft twist or windup between the motor and the load may also introduce an oscillation into the system that will effect the stability of the speed regulator.

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2. Continued

Typical oscillation frequencies for the above two conditions are in the 5 to 20 cycles per second range and this frequency is not normally influenced by machine operating speed.

If either of these conditions occur, it is helpful to set up the regulator with some "undershoot." It may also be necessary to reduce the overall response of the regulator and adding a capacitor in parallel with the LEAD pot in the SPD module so that the impedance of the LEAD pot setting times the capacitance will give a time constant of approximately 0.2 sec.

3. Inherent Motor Speed Droop

For proper operation of the speed regulator, the motor speed must decrease (droop) with an increase in load. Normally this droop should be between 1 percent and 5 percent (of rated speed) for a 100 percent (of rated load) increase in load with constant voltage applied to the motor armature. If the speed increases with increasing load, it may be difficult or impossible to "stabilize" the drive.

Usually this effect is most noticeable at heavy load conditions.

If the motor droop is not present, check to see that the motor series field (if provided) is connected properly. (See rear drum tune up also.)

4. Electrical "Noise" Pickup from Remote Devices

When an electrical circuit experiences a rapid change in the value of current flow, it may also experience a short but violent voltage surge. This rapid current change and voltage surge can result in electrical signals being transmitted to nearby wires through capacitive or inductive coupling effects. The resultant signal in the nearby wires is called "noise."

If the "noise" gets into the operational amplifiers in the speed regulators, the amplifiers will amplify this "noise" signal and the section speed will change. Such speed changes are undesirable so the "noise" must be eliminated or reduced to a very low value.

Interruption of the coil circuits on contactors and relays causes sudden current changes and voltage surges. Therefore all the relays and contactors on the control panels have noise suppression devices on their coils. The relays and contactors have a resistor-capacitor network. When interlocks on the panels are used to operate relays in the purchaser's remote control panels (such as a-c motor starters, etc.) it may be necessary to equip the coils in that control equipment with noise suppressors. Typical acceptable suppressors are as follows:

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Circuit Rating	Suppressor
0-300ma, 110V A-C	Thyrector Diode 6RS20SP5B5
0-300ma, 220V A-C	Thyrector Diode 6RS20SP10B10
0-100ma, 440V A-C	Thyrector Diode 6RS5SP20B20
0-300ma, 440V A-C	Thyrector Diode 6RS20SP20B20
0-100ma, 250V D-C	Thyrector Diode 6RS5AP12B2
more than 100ma, 250V D-C	1-mfd, 600-volt capacitor in series with 100-ohm, 2-watt resistor

Electrical "noise" can be generated by many other devices. These signals are very difficult to detect because normally they are of extremely short duration. To guard against pickup of this noise, it it advisable to keep the control wiring (to operator's devices, etc.) away as far as possible from other wiring. There should be a grounded metal barrier between the control wiring and other circuits. For example, run the wiring in well-grounded metal conduit. For specific information on which wires are susceptible to noise pickup and for recommendations on how to shield them, refer to the interconnection diagram furnished with the control panel.

C. Regulator Problems

1. General

During the design and manufacture of this control, everything practical has been done to make it operate consistently and without failure. However, eventually some trouble may develop. To be prepared for this eventuality, it is suggested that the user:

a. Keep the elementary diagram and instruction books handy.

b. Become familiar with the overall concept of the drive and insofar as possible with the various circuits involved.

c. Keep portable instruments handy for checking voltage, current, and resistance. A portable industrial volt-ohm milliammeter with a 20,000 ohms per volt d-c sensitivity is the minimum recommended. A fast dual-trace oscilloscope (such as the Tektronix Type 502) and a dual-pen strip-chart recorder (such as the Brush Mark II) are desirable but not absolutely necessary.

d. Keep on hand an adequate supply of spare parts.

Should the drive become inoperative or its performance change, care should be exercised during trouble-shooting in order not to get the drive out of adjustment. Generally, the difficulty stems from a loose connection, broken wire, or failed component and it can be corrected without making additional adjustments to the control. Considerable time will be saved if the difficulty can be corrected without changing the component settings necessitating retuning the drive.

The first step in the trouble-shooting procedure should be to identify the problem with a specific area of the system, then systematically reduce this area to a specific module or component. The next step is to repair or replace the faulty component. The final step is to check the drive operation to be sure that no other secondary or contributing problems exist.

The following trouble-shooting guide may be of assistance in tracking down the problem to a general area of the control.

D. Trouble Shooting Guide

1. Winder will not start

   a. Check protective relaying.

   b. Check that the proper relays are energized.

2. Winder starts but is not held to the set reference.

   a. Check that the regulator relays are normal.

   b. Check speed reference signal from ARC module.

      (1) If input is not equal to output, check module instructions.

   c. If ARC module is operating properly, check that the SPD module is not in clamp:

      (1) If it is, then the drive is in current limit.
      (2) If not, then check SPD module in test mode.

   d. If the ARC and SPD modules are functioning properly, check that the CL module is holding a current proportional to the input.

      (1) If it is, then check the armature power supply.
      (2) If not, check CL module in the test mode.

3. Winder will not start in thread or jog.

   If drive will not start during thread or jog, the problem may be due to the speed droop circuit. Symptoms would be on initiation of thread closes "M" contactor. Current goes up to some level (less than current limit) but the drive does not move. Possible solutions are:

      (1) Reduce the amount of speed droop.
      (2) Increase the thread or jog speed.
      (3) Decrease the amount of sheet tension during starting.

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CONT ON SHEET FL. SH NO. 30

TITLE

SILTROL GE TYPE III
MECHANICAL WINDER
DRIVE INSTRUCTIONS

FIRST MADE FOR STANDARD LINE

REVISIONS

V. MAINTENANCE

A. General

Periodic inspections and maintenance are important to continuous operation. Regular inspections should be performed so that replacements or repairs can be made before a failure occurs. Costly downtime can be reduced by following three basic rules:

1. Prepare a preventive-maintenance program consisting of items to be checked and recorded at each scheduled inspection. Sometimes frequent inspections are desirable on new equipments. The interval between inspections can then be increased at the user's discretion as he becomes more familiar with the equipment and particular operating conditions.
2. Keep on hand an adequate supply of renewal and repair parts.
3. Much time and trouble can be saved by buying renewal parts from the manufacturer who originally supplied the equipment. You are then certain such renewal parts will conform with the specifications for the original parts.

B. Maintenance Schedule

The maintenance schedule and activity outlined in publication GET-1195, "How to Maintain Industrial Control," applies to this control.

C. Renewal Parts

When ordering renewal parts, specify the complete catalog number of the part. If there is any doubt, include the panel catalog number and the device functional nomenclature.

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